

|                                                 |                                   |                     |
|-------------------------------------------------|-----------------------------------|---------------------|
| SCOPE OF APPLICATION<br>All Project/Engineering | <b>HYUNDAI</b><br><b>AutoEver</b> | SHT/SHTS<br>1 / 154 |
| Responsibility:<br>차량전장SW사업부                    | AUTOSAR Csm User Manual           | DOC. NO             |
| <b>AUTOSAR Csm User Manual</b>                  |                                   |                     |

| Document Change History |      |        |                         |                                                                                                                                                                                                               |
|-------------------------|------|--------|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Date<br>(YYYY-MM-DD)    | Ver. | Editor | Chap                    | 내용(개정 전 -> 개정 후)                                                                                                                                                                                              |
| 2016-05-04              | 1.0  | 강신일    | 전체                      | • 신규제정                                                                                                                                                                                                        |
| 2016-06-16              | 1.1  | 강신일    | 6.1.1,<br>6.2.4<br>4    | • Csm_ReturnType 제거<br>• Release 정보, Deviation 갱신                                                                                                                                                             |
| 2016-07-29              | 1.2  | 강신일    | 4                       | • Release 정보 갱신<br>Change Log 삭제                                                                                                                                                                              |
| 2016-10-17              | 1.3  | 강신일    | 4<br>5<br>6.2<br>7.2.1  | • Release 정보 갱신<br>Change Log 추가<br>Mac 지원 알고리즘 추가<br>• Init configuration 변경 사항 갱신<br>Max Data Buffer Size 설정 사항 변경<br>CalMacGenerate/Verify Container 정보 추가<br>• Mac interface Api 추가<br>• Error Message 추가 |
| 2017-01-11              | 1.4  | 강신일    | 4<br>5.1<br>7.2.2       | • Release 정보 갱신<br>• CsmSeedRefAddr 설정 추가<br>• Generator warning 추가                                                                                                                                           |
| 2017-03-21              | 1.5  | 강신일    | 4<br>5                  | • Release 정보 갱신, Sync/Async 동작 방식 변경 설명 추가<br>• Use Port 설정 추가 및 Callback 처리 방식 설명 변경                                                                                                                         |
| 2017-09-27              | 1.6  | 강신일    | 전체                      | • HSM 및 신규 알고리즘 지원                                                                                                                                                                                            |
| 2017-11-13              | 1.8  | 강신일    | 4<br>5.3/5.4<br>8.1/8.2 | • Release 정보 갱신, 지원 알고리즘 제조사 수정<br>• Max key size 제한 제거<br>• 알고리즘 유의 사항, 파라미터 설명 추가                                                                                                                           |
| 2017-11-20              | 1.9  | 강신일    | 4                       | • Release 정보 갱신                                                                                                                                                                                               |
| 2017-12-05              | 2.0  | 강신일    | 5                       | • Release 정보 갱신<br>• Configuration Guide 설정 사항 변경 및 설명 추가                                                                                                                                                     |
| 2018-01-12              | 2.1  | 강신일    | 8                       | • Release 정보 갱신<br>• 알고리즘 유의사항, 테스트 케이스 추가                                                                                                                                                                    |
| 2018-01-13              | 2.2  | 강신일    | 8                       | • 일부 알고리즘 주의 사항 추가                                                                                                                                                                                            |
| 2018-02-09              | 2.3  | 강신일    | 전체                      | • Release 정보 갱신<br>• Configuration Guide 설명 추가<br>• HaeHsmCmacAes 입력 데이터 유의사항 추가                                                                                                                              |
| 2018-03-06              | 2.4  | 강신일    | 4.3<br>8.1.1            | • Release 정보 갱신<br>• Csm API 유의사항 갱신                                                                                                                                                                          |
| 2018-03-19              | 2.5  | 강신일    | 4.3<br>4.4<br>8.1.1     | • Release 정보 갱신<br>• SPC58x HSM AES 기반 알고리즘 제약 사항 추가<br>• 제약 사항 구체화                                                                                                                                           |

|                                   |                                         |                 |              |                 |
|-----------------------------------|-----------------------------------------|-----------------|--------------|-----------------|
| 31th Edition Date :<br>2020-02-24 | File Name:<br>AUTRON_AUTOSAR_Csm_UM.pdf | Creation<br>강신일 | Check<br>윤영진 | Approval<br>백종현 |
| Document<br>Management System     |                                         | 2020-02-24      | 2020-02-24   | 2020-02-24      |

|            |      |     |                                                        |                                                                                                                                                                                                                                              |
|------------|------|-----|--------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2018-04-20 | 2.6  | 강신일 | 4.3<br>8.2                                             | <ul style="list-style-type: none"> <li>• Release 정보 갱신</li> <li>• IAU Key 인덱스 사용 가이드 추가</li> <li>• AuthHsmPseudoRandomGenerate 알고리즘 가이드 추가</li> </ul>                                                                                        |
| 2018-05-11 | 2.7  | 강신일 | 4.3<br>5.16<br>6.1/6.2<br>8.2/8.4                      | <ul style="list-style-type: none"> <li>• Release 정보 갱신</li> <li>• AuthHsmEcdh 알고리즘 관련 가이드 추가</li> </ul>                                                                                                                                      |
| 2018-05-23 | 2.8  | 강신일 | 4.3<br>8.2.5.1                                         | <ul style="list-style-type: none"> <li>• Release 정보 갱신</li> <li>• AuthHsmAesCtr 입력 데이터 길이 제한 제거</li> </ul>                                                                                                                                   |
| 2018-06-12 | 2.9  | 강신일 | 4.3<br>8.2.7.10                                        | <ul style="list-style-type: none"> <li>• Release 정보 갱신</li> <li>• 랜덤값 인자 설명 추가</li> <li>• 오타 수정</li> </ul>                                                                                                                                   |
| 2018-09-04 | 3.0  | 강신일 | 4.3<br>8.1.6                                           | <ul style="list-style-type: none"> <li>• Release 정보 갱신</li> <li>• RH850 F1KM HSM 공유 메모리 주의 사항 추가</li> </ul>                                                                                                                                  |
| 2018-11-07 | 3.1  | 강신일 | 4.3                                                    | <ul style="list-style-type: none"> <li>• Release 정보 갱신</li> </ul>                                                                                                                                                                            |
| 2018-11-16 | 3.2  | 강신일 | 4.3                                                    | <ul style="list-style-type: none"> <li>• Release 정보 갱신</li> </ul>                                                                                                                                                                            |
| 2019-03-28 | 3.3  | 강신일 | 4.3<br>4.7<br>8.2.6.1<br>11.7,<br>12,<br>15.5,<br>15.7 | <ul style="list-style-type: none"> <li>• Release 정보 갱신</li> <li>• 파라미터 설명 오타 수정</li> <li>• 오토에버 SW 라이브러리 추가에 따른 지원 알고리즘 리스트 업데이트</li> <li>• 신규 암호화 알고리즘 설명 추가, Escrypt 알고리즘 제거</li> </ul>                                                    |
| 2019-04-24 | 3.4  | 강신일 | 4.3                                                    | <ul style="list-style-type: none"> <li>• Release 정보 갱신</li> </ul>                                                                                                                                                                            |
| 2019-07-17 | 3.5  | 강신일 | 4.3<br>4.4,<br>5,<br>8.2,<br>8.4                       | <ul style="list-style-type: none"> <li>• Release 정보 갱신</li> <li>• S32K CSEC 알고리즘 가이드 추가</li> </ul>                                                                                                                                           |
| 2019-08-28 | 3.6  | 강신일 | 4.3                                                    | <ul style="list-style-type: none"> <li>• Release 정보 갱신</li> </ul>                                                                                                                                                                            |
| 2019-10-17 | 3.7  | 강신일 | 4.3                                                    | <ul style="list-style-type: none"> <li>• Release 정보 갱신</li> <li>• 파일 구조 변경 (lib 폴더 파일을 delivery로 이동)</li> </ul>                                                                                                                              |
| 2019-12-05 | 3.8  | 강신일 | 4.3<br>5.6.1<br>8                                      | <ul style="list-style-type: none"> <li>• Release 정보 갱신</li> <li>• HaeHsmPseudoRandomGenerate 알고리즘 추가</li> <li>• Hae Hsm Key 사용 방식 변경</li> </ul>                                                                                              |
| 2020-01-06 | 3.9  | 강신일 | 4.3<br>4.5<br>6.2.11<br>8.2.8.1<br>8.4.7.1             | <ul style="list-style-type: none"> <li>• Release 정보 갱신</li> <li>• Csm_KeyDerive deviation 추가</li> <li>• Csm_KeyDeriveStart, Csm_KeyDeriveUpdate 설명 변경</li> <li>• AuthHsmPbkdf2 파라미터 설명 변경</li> <li>• AuthHsmPbkdf2 PSK 테스트 케이스 추가</li> </ul> |
| 2020-02-24 | 3.10 | 강신일 | 4.3                                                    | <ul style="list-style-type: none"> <li>• Release 정보 갱신</li> </ul>                                                                                                                                                                            |
| 2020-03-13 | 3.11 | 임재현 | 4.3                                                    | <ul style="list-style-type: none"> <li>• Release 정보 갱신</li> </ul>                                                                                                                                                                            |
| 2020-03-20 | 3.12 | 임재현 | 4.3                                                    | <ul style="list-style-type: none"> <li>• Release 정보 갱신</li> </ul>                                                                                                                                                                            |

|            |         |     |                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                  |
|------------|---------|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2020-06-26 | 2.6.0.0 | 임재현 | 4.3<br>4.4<br>6.1.18<br>6.1.19<br>6.1.20<br>8.2.4.3<br>8.2.5.7<br>8.2.5.8<br>8.2.8.2<br>8.2.9.2<br>8.4.4.7<br>8.4.4.8<br>8.4.6.15<br>8.4.6.16<br>8.4.7.2<br>8.4.8.2<br>8.5<br>4.4<br>8.2.8.1<br>8.4.7<br>8.4.8<br>8.5 | <ul style="list-style-type: none"> <li>• Release 정보 갱신</li> <li>• HAE HSM 알고리즘 12건 가이드 추가</li> <li>• AuthHsmPbkdf2 설명 수정 (Sha1 -&gt; Sha2)</li> <li>• Limitation 추가 (HSM 및 Key 관리 기능)</li> <li>• ECDH, PBKDF2 TestCase 개선</li> <li>• Autoever HSM 2.x 포팅 관련 가이드 신규 추가</li> </ul> |
| 2020-10-08 | 2.6.1.0 | 임재현 | 4.3                                                                                                                                                                                                                   | <ul style="list-style-type: none"> <li>• Release 정보 갱신</li> </ul>                                                                                                                                                                                                                |
| 2021-01-15 | 2.6.2.0 | 임재현 | 4.3<br>8.5                                                                                                                                                                                                            | <ul style="list-style-type: none"> <li>• Release 정보 갱신</li> <li>• Autoever HSM 2.x 관련 내용 개선</li> </ul>                                                                                                                                                                           |
| 2021-03-31 | 2.6.3.0 | 임재현 | 4.3                                                                                                                                                                                                                   | <ul style="list-style-type: none"> <li>• Release 정보 갱신</li> </ul>                                                                                                                                                                                                                |
| 2021-04-16 | 2.6.4.0 | 임재현 | 4.3<br>4.4                                                                                                                                                                                                            | <ul style="list-style-type: none"> <li>• Release 정보 갱신</li> <li>• Limitation 정보 추가</li> </ul>                                                                                                                                                                                    |

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## 1. Overview

AUTOSAR 표준 SRS/SWS를 기반으로 작성되었으며 모듈 사용시 보다 자세한 기능적인 설명이 필요한 경우, 아래 Reference 문서를 참고한다.

설정관련 Category의 해석은 다음과 같다.

- Changeable (C) : User 에 의해서 설정 가능한 항목
- Fixed (F) : User 에 의한 변경이 불가능한 항목
- NotSupported (N) : 사용되지 않는 항목

## 2. Reference

| Sl. No. | Title                                | Version |
|---------|--------------------------------------|---------|
| 1.      | AUTOSAR_SWS_CryptoServiceManager.pdf | 1.2.0   |
|         |                                      |         |

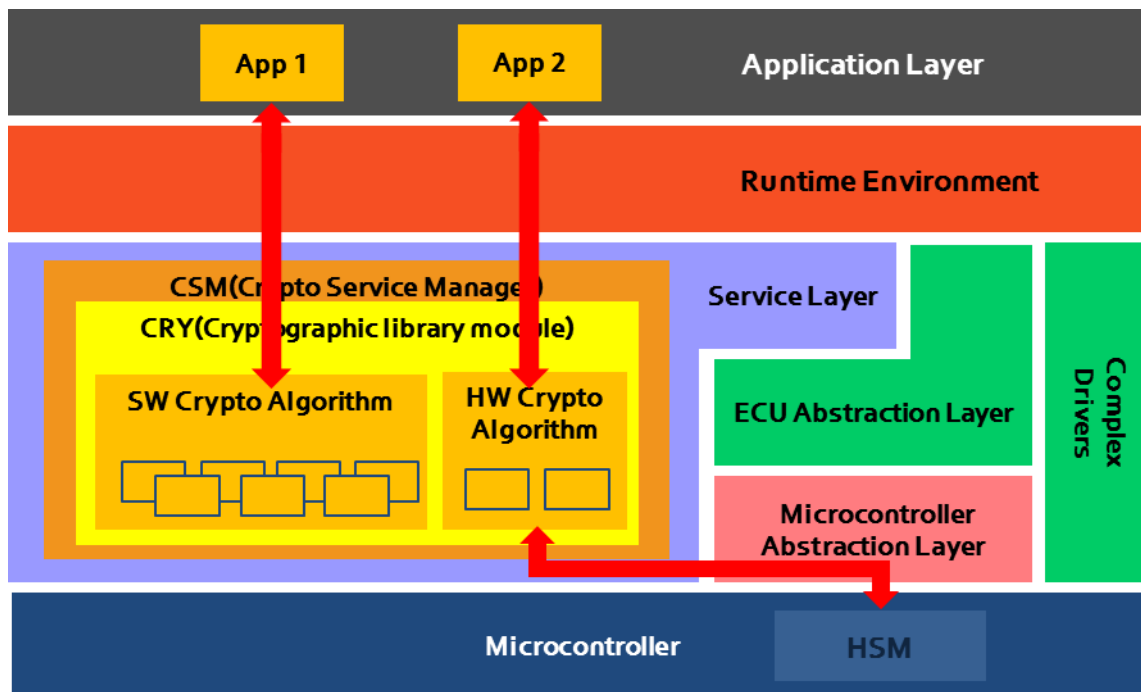
### 3. AUTOSAR System

#### AUTOSAR Csm Module

AUTOSAR 플랫폼의 Layered Architecture 는 아래와 같다. AUTOSAR 플랫폼은, Service Layer, ECU Abstraction Layer, Microcontroller Abstraction Layer, Complex Device Drivers 및 Library 로 구분될 수 있다.

Crypto Service Manager(CSM) 모듈은 AUTOSAR layer에서 Service layer에 속한다. CSM 모듈은 다른 BSW, ASW, CDD 에 SW crypto algorithm 또는 HSM(Hardware Security Module)이 지원되는 MCU의 경우 HSM의 crypto algorithm을 제공한다. Crypto service의 동작은 설정에 따라 호출한 사용자의 context에서 동작하거나 다른 task context에서 동작한다. CSM 모듈의 각 서비스는 독립적으로 사용 가능하다.

CSM 모듈은 CryptoService(암복호화기능)에 대한 HSM 을 사용자가 직접 사용하는 시나리오는 제공하지 않는다.



## 4. Product Release Notes

### Overview

이 Chapter에서는, 현대오트론 Csm Module에 대한 release 관련 내용을 제공하는데 목적이 있으며, Csm Software product release version 에 대한, 제한사항 및 특이사항을 기술하고 있다.

### 4.1 Scope of the release

이 문서에 대한 모든 내용은, 다음의 현대오트론 Csm 모듈에 한정한다.

| Module | Autosar version | Module version |
|--------|-----------------|----------------|
| Csm    | 4.0.3           | 2.6.4.0        |

※ Module version 은 각 모듈의 BswModule Description(Bswmd)파일의 Sw version 을 의미한다.

### Change Log

#### Version 1.0.7

##### ➤ 신규 기능

■ Hmac Sha1 Generate/Verify, Hmac Sha256 Generate/Verify, Cmac Aes Generate/Verify 기능 구현

|              |                    |
|--------------|--------------------|
| 원인           | MAC 암호화 알고리즘 추가 구현 |
| 동작 영향        | 없음                 |
| 설정 영향        | 없음                 |
| ASW 조치 필요 사항 | 없음                 |

##### ➤ 개선 사항

■ 표준 API 파라미터 사용

|              |                                                                   |
|--------------|-------------------------------------------------------------------|
| 원인           | 일부 Csm API 에서 표준에 정의된 const 가 누락되어 사용. 내부 사용 변수의 configuration 설정 |
| 동작 영향        | 없음                                                                |
| 설정 영향        | Init Configuration 변경 (NULL_PTR 로 설정 변경)                          |
| ASW 조치 필요 사항 | 없음                                                                |

#### Version 1.1.0

##### ➤ 신규 기능

■ Autron Random Seed Refernce Address 설정 기능 구현

|              |                                              |
|--------------|----------------------------------------------|
| 원인           | Autron Random 의 Seed 를 참조하는 address 설정 기능 구현 |
| 동작 영향        | 없음                                           |
| 설정 영향        | 없음                                           |
| ASW 조치 필요 사항 | 없음                                           |

■ Csm\_AlignType 의 uint32 지원

|              |                              |
|--------------|------------------------------|
| 원인           | Csm_AlignType 의 uint32 설정 지원 |
| 동작 영향        | 없음                           |
| 설정 영향        | 없음                           |
| ASW 조치 필요 사항 | 없음                           |

Version 1.2.0

4.3.3 > 신규 기능

■ Primitive 별 Sync, Async 동작 지원

|              |                                                   |
|--------------|---------------------------------------------------|
| 원인           | Primitive 별로 선택적으로 Sync, Async 을 설정, 사용할 수 있도록 지원 |
| 동작 영향        | 없음                                                |
| 설정 영향        | 없음                                                |
| ASW 조치 필요 사항 | 없음                                                |

■ Aut Hac Random Seed, Aut Hac Random Generate 기능 구현

|              |                        |
|--------------|------------------------|
| 원인           | HAC Random 암호화 알고리즘 구현 |
| 동작 영향        | 없음                     |
| 설정 영향        | 없음                     |
| ASW 조치 필요 사항 | 없음                     |

■ Use Port 기능 구현

|              |                                 |
|--------------|---------------------------------|
| 원인           | Rte Port Interface 사용 유무를 파라미터화 |
| 동작 영향        | 없음                              |
| 설정 영향        | 없음                              |
| ASW 조치 필요 사항 | 없음                              |

Version 2.0.0

➤ 신규 기능

■ Chorus HSM 지원

|              |                                                        |
|--------------|--------------------------------------------------------|
| 원인           | ST 社 SPC58ECx(Chorus) HSM(Hardware Security Module) 지원 |
| 동작 영향        | 없음                                                     |
| 설정 영향        | 없음                                                     |
| ASW 조치 필요 사항 | 없음                                                     |

■ RH850 F1K ICUS 지원

|              |                                                                   |
|--------------|-------------------------------------------------------------------|
| 원인           | Renesas 社 RH850 F1K ICUS(Intelligent Cryptographic Unit Slave) 지원 |
| 동작 영향        | 없음                                                                |
| 설정 영향        | 없음                                                                |
| ASW 조치 필요 사항 | 없음                                                                |

■ Aurix TC27x HSM 지원

|              |                         |
|--------------|-------------------------|
| 원인           | Infineon 社 TC27x HSM 지원 |
| 동작 영향        | 없음                      |
| 설정 영향        | 없음                      |
| ASW 조치 필요 사항 | 없음                      |

■ PKCS#1 V1.5 SHA256 지원

|              |                               |
|--------------|-------------------------------|
| 원인           | PKCS#1 V1.5 SHA256 인증 알고리즘 지원 |
| 동작 영향        | 없음                            |
| 설정 영향        | 없음                            |
| ASW 조치 필요 사항 | 없음                            |

Version 2.0.1

➤ 개선 사항

■ Bswmd\_Csm 자동 생성 방식 개선

|       |                                                                              |
|-------|------------------------------------------------------------------------------|
| 원인    | 최초 Ecu_Csm 생성시 Csm module description 이 없으므로 Bswmd_Csm 자동 생성 과정에서 불필요한 오류 발생 |
| 동작 영향 | 없음                                                                           |

|              |    |
|--------------|----|
| 설정 영향        | 없음 |
| ASW 조치 필요 사항 | 없음 |

### Version 2.0.2

#### ➤ 개선 사항

##### ■ Swcd\_Bsw\_Csm 사용성 개선

4.3.6

|              |                                                                                                                                  |
|--------------|----------------------------------------------------------------------------------------------------------------------------------|
| 원인           | 각 알고리즘 Use Port 사용 & Callback 설정 시 Csm Service Sw Component 의 Runnable 에 Callback 을 위한 Synchronous Server Call Point 자동 생성 기능 추가 |
| 동작 영향        | 없음                                                                                                                               |
| 설정 영향        | 없음                                                                                                                               |
| ASW 조치 필요 사항 | 없음                                                                                                                               |

### Version 2.0.3

4.3.7

#### ➤ 개선 사항

##### ■ HaeHsmTrueRandomGenerate 사용성 개선

|              |                                                             |
|--------------|-------------------------------------------------------------|
| 원인           | HaeHsmTrueRandomGenerate 가 생성 가능한 난수 길이를 4 의 배수에서 정수 길이로 수정 |
| 동작 영향        | 없음                                                          |
| 설정 영향        | 없음                                                          |
| ASW 조치 필요 사항 | 없음                                                          |

##### ■ 설정 제한 제거

4.3.8

|              |                                                                        |
|--------------|------------------------------------------------------------------------|
| 원인           | Integrator role 로 고정된 설정 파라미터를 사용자 사용 시나리오에 따라 수정 가능하도록 사용자 설정 가능으로 변경 |
| 동작 영향        | 없음                                                                     |
| 설정 영향        | 없음                                                                     |
| ASW 조치 필요 사항 | Configuration Guide 에 따라 필요시 파라미터를 사용자가 설정함                            |

### Version 2.1.0

#### ➤ 개선 사항

##### ■ 사용자 매뉴얼 업데이트

|              |                                     |
|--------------|-------------------------------------|
| 원인           | 알고리즘 사용시 유의사항 업데이트, 알고리즘 테스트 케이스 추가 |
| 동작 영향        | 없음                                  |
| 설정 영향        | 없음                                  |
| ASW 조치 필요 사항 | 없음                                  |

■ CryptoLib 모듈 정합

|              |                                         |
|--------------|-----------------------------------------|
| 원인           | CryptoLib 모듈의 Generator 생성 output 사용 적용 |
| 동작 영향        | 없음                                      |
| 설정 영향        | 없음                                      |
| ASW 조치 필요 사항 | 없음                                      |

■ HaeHsmCmacAes 입력 데이터 제한

|              |                                                                    |
|--------------|--------------------------------------------------------------------|
| 원인           | HaeHsmCmacAes 알고리즘은 오토에버 HSM 프레임워크의 제한으로 입력 데이터 길이가 4 바이트의 배수여야 함. |
| 동작 영향        | 없음                                                                 |
| 설정 영향        | 없음                                                                 |
| ASW 조치 필요 사항 | 입력 데이터가 4 바이트 배수가 아닌 경우 0 패딩을 해줘야 함                                |

4.3.9

**Version 2.1.1**

➤ 개선 사항

■ MAC generation / verification 처리 분리

|              |                                              |
|--------------|----------------------------------------------|
| 원인           | Chorus HSM 의 MAC service 내부 API 변경에 따른 코드 수정 |
| 동작 영향        | 없음                                           |
| 설정 영향        | 없음                                           |
| ASW 조치 필요 사항 | 없음                                           |

4.3.10

**Version 2.1.2**

➤ 개선 사항

■ 내부 function stack 처리 동작 개선

|       |                                           |
|-------|-------------------------------------------|
| 원인    | 비동기 요청 처리를 위한 내부 function stack 처리 효율성 개선 |
| 동작 영향 | 없음                                        |
| 설정 영향 | 없음                                        |

|              |    |
|--------------|----|
| ASW 조치 필요 사항 | 없음 |
|--------------|----|

**Version 2.2.0**

## ➤ 개선 사항

## ■ IAU 제어기 특화 주입 ECC 키 쌍 PSK 적용

|              |                                                                  |
|--------------|------------------------------------------------------------------|
| 원인           | IAU 제어기 특화 주입한 ECC 키 쌍을 암호화 알고리즘의 키입력시 PSK 와 같이 인덱스로 사용 가능하도록 개선 |
| 동작 영향        | 없음                                                               |
| 설정 영향        | 없음                                                               |
| ASW 조치 필요 사항 | 없음                                                               |

## ■ AuthHsmPseudoRandomGenerate 알고리즘 가이드 추가

|              |                                              |
|--------------|----------------------------------------------|
| 원인           | AuthHsmPseudoRandomGenerate 알고리즘 설정 및 가이드 추가 |
| 동작 영향        | 없음                                           |
| 설정 영향        | 없음                                           |
| ASW 조치 필요 사항 | 없음                                           |

## ■ HAE HSM 드라이버 업데이트

|              |                                    |
|--------------|------------------------------------|
| 원인           | Aurix TC27x HSM 드라이버 관련 인터페이스 업데이트 |
| 동작 영향        | 없음                                 |
| 설정 영향        | 없음                                 |
| ASW 조치 필요 사항 | 없음                                 |

**Version 2.2.1**

## ➤ 신규 기능

## ■ SPC58x HSM 용 ECDH 알고리즘 인터페이스 추가

|              |                                    |
|--------------|------------------------------------|
| 원인           | SPC58x HSM 용 ECDH 알고리즘 인터페이스 추가 구현 |
| 동작 영향        | 없음                                 |
| 설정 영향        | 없음                                 |
| ASW 조치 필요 사항 | 없음                                 |

**Version 2.2.2**

## ➤ 개선 사항

■ SPC58x HSM 용 AES CTR 블록 길이 제한 오류 수정

|              |                                                                                                                             |
|--------------|-----------------------------------------------------------------------------------------------------------------------------|
| 원인           | AES 표준은 CTR 모드 사용시 입력 데이터 길이에 제한이 없으나, SPC58x HSM 용 AES CTR 알고리즘 사용시 입력 데이터 길이에 16 바이트 제한을 두어 문제 발생. 입력 데이터 16 바이트 길이 제한 제거 |
| 동작 영향        | 없음                                                                                                                          |
| 설정 영향        | 없음                                                                                                                          |
| ASW 조치 필요 사항 | 없음                                                                                                                          |

**Version 2.2.3**

➤ 개선 사항

4.3.14 ■ AuthHsmEcdsaSha256Verify Raw Key 사용시 결과 오류 수정

|              |                                                                               |
|--------------|-------------------------------------------------------------------------------|
| 원인           | AuthHsmEcdsaSha256Verify 알고리즘을 Raw Key 사용시 HSM 내부 RAM Key 버퍼 인덱싱 버그로 결과 오류 발생 |
| 동작 영향        | 없음                                                                            |
| 설정 영향        | 없음                                                                            |
| ASW 조치 필요 사항 | 없음                                                                            |

4.3.15

**Version 2.2.4**

➤ 신규 기능

■ RH850 F1KM HSM 지원

|              |                                                                                 |
|--------------|---------------------------------------------------------------------------------|
| 원인           | RH850 F1KM 의 경우 HSM 의 PKCS#1V1.5 인증 알고리즘 사용시 내부 HASH 결과 버퍼를 Global RAM B 영역에 배치 |
| 동작 영향        | 없음                                                                              |
| 설정 영향        | 없음                                                                              |
| ASW 조치 필요 사항 | 없음                                                                              |

4.3.16

**Version 2.2.5**

➤ 개선 사항

■ AuthHsmHmacSha256Verify Update 사용시 리셋 발생 오류 수정

|       |                                           |
|-------|-------------------------------------------|
| 원인    | 해당 알고리즘 Update 수행시 Sync/Async 판별 로직 구현 오류 |
| 동작 영향 | 없음                                        |
| 설정 영향 | 없음                                        |

|              |    |
|--------------|----|
| ASW 조치 필요 사항 | 없음 |
|--------------|----|

### Version 2.2.6

#### ➤ 개선 사항

##### ■ HAE 암호화 알고리즘 인터페이스 구현

|              |                                    |
|--------------|------------------------------------|
| 원인           | HAE 암호화 알고리즘 라이브러리 지원을 위한 인터페이스 구현 |
| 동작 영향        | 없음                                 |
| 설정 영향        | 없음                                 |
| ASW 조치 필요 사항 | 없음                                 |

### Version 2.3.0

#### ➤ 개선 사항

##### ■ 현대오토에버(HAE) 일반 암호화 알고리즘 인터페이스 구현

|              |                                               |
|--------------|-----------------------------------------------|
| 원인           | 현대오토에버(HAE) 일반 암호화 알고리즘 라이브러리 지원을 위한 인터페이스 구현 |
| 동작 영향        | 없음                                            |
| 설정 영향        | 없음                                            |
| ASW 조치 필요 사항 | 없음                                            |

### Version 2.3.1

#### ➤ 개선 사항

##### ■ RH850 F1K ICUS 드라이버 인터페이스 변경 적용

|              |                                                 |
|--------------|-------------------------------------------------|
| 원인           | Csm 모듈 내에서 사용중인 ICUS 드라이버의 인터페이스가 변경된 사항에 대해 적용 |
| 동작 영향        | 없음                                              |
| 설정 영향        | 없음                                              |
| ASW 조치 필요 사항 | 없음                                              |

### Version 2.4.0

#### ➤ 개선 사항

##### ■ S32K CSEC 드라이버 인터페이스 구현

|       |                                  |
|-------|----------------------------------|
| 원인    | CSEC 드라이버 알고리즘을 지원하기 위한 인터페이스 구현 |
| 동작 영향 | 없음                               |

|              |    |
|--------------|----|
| 설정 영향        | 없음 |
| ASW 조치 필요 사항 | 없음 |

### Version 2.4.1

#### ➤ 개선 사항

##### ■ ASK 라이브러리 최신 헤더파일명(HKMC\_ASK\_Server.h) 적용

4.3.21

|              |                                                                                           |
|--------------|-------------------------------------------------------------------------------------------|
| 원인           | 보안 라이브러리 분리 배포로 인해 업체에서 직접 라이브러리 패키지를 통합하게 되어 실제 ASK 라이브러리 배포 패키지에 있는 헤더파일명을 직접 사용하도록 변경함 |
| 동작 영향        | 없음                                                                                        |
| 설정 영향        | 없음                                                                                        |
| ASW 조치 필요 사항 | 없음                                                                                        |

### Version 2.4.1.0

4.3.22

#### ➤ 개선 사항

##### ■ 파일 구조 변경

4.3.23

|              |                         |
|--------------|-------------------------|
| 원인           | Code Open 을 위해 파일 구조 변경 |
| 동작 영향        | 없음                      |
| 설정 영향        | 없음                      |
| ASW 조치 필요 사항 | 없음                      |

### Version 2.5.0.0

#### ➤ 개선 사항

##### ■ 오토에버 HSM 2.0 지원

4.3.24

|              |                       |
|--------------|-----------------------|
| 원인           | 오토에버 HSM 2.0 인터페이스 지원 |
| 동작 영향        | 오토에버 HSM 1.0 미지원      |
| 설정 영향        | 없음                    |
| ASW 조치 필요 사항 | 없음                    |

### Version 2.5.1.0

#### ➤ 개선 사항

##### ■ Autron HSM PBKDF2 알고리즘 PSK 지원

|    |                                                        |
|----|--------------------------------------------------------|
| 원인 | Secure Flash 2.0 구현시 Meta data 를 이용한 복호화 키 유도과정에 HSM 의 |
|----|--------------------------------------------------------|

|              |                                                                                                        |
|--------------|--------------------------------------------------------------------------------------------------------|
|              | PBKDF2 알고리즘을 사용하며 이때 사용되는 key 종류는 PSK(PreSharedKey)이다. 기존의 Autron HSM 의 PBKDF2 알고리즘은 PSK 를 지원하지 않고 있다. |
| 동작 영향        | Autron HSM 의 PBKDF2 알고리즘에서 PSK 지원                                                                      |
| 설정 영향        | 없음                                                                                                     |
| ASW 조치 필요 사항 | Deviation 참조                                                                                           |

## Version 2.6.0.0

### ➤ 개선 사항

4.3.25

#### ■ 불필요 Possible Error 제거

|              |                                                                                                                       |
|--------------|-----------------------------------------------------------------------------------------------------------------------|
| 원인           | 자동 생성되는 CsmRandomSeed ClientServerInterface 의 RandomSeedFinish operation 의 불필요한 CSM_E_SMALL_BUFFER possible error 제거함 |
| 동작 영향        | 없음                                                                                                                    |
| 설정 영향        | 없음                                                                                                                    |
| ASW 조치 필요 사항 | 없음                                                                                                                    |

### ➤ 개선 사항

#### ■ Autoever HSM 2.x Crypto Algorithm 12 건 지원

|              |                                                                                                                                                                   |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 원인           | HSM 1.0 에서 지원했던 Csm Algorithm Interface 에 대해 HSM 2.0 에서도 지원될 수 있도록 하기 12 건에 대한 Cry Interface 신규 개발 (4.4 Limitation 지원되는 Crypto Service 목록 A.HSM 2.0 적용 MCU 부분 참조) |
| 동작 영향        | 없음                                                                                                                                                                |
| 설정 영향        | 없음                                                                                                                                                                |
| ASW 조치 필요 사항 | 없음                                                                                                                                                                |

### ➤ 오류 수정

#### ■ Autoever HSM 2.x Pkcs1Pss 인터페이스 개선

|    |                                                             |
|----|-------------------------------------------------------------|
| 원인 | Cry_HaeHsmPkcs1Pss 내부 로직 오류 개선 (Tasking 컴파일러에서 미동작하는 오류 해결) |
|----|-------------------------------------------------------------|

|              |    |
|--------------|----|
| 동작 영향        | 없음 |
| 설정 영향        | 없음 |
| ASW 조치 필요 사항 | 없음 |

➤ 개선 사항

- Autoever HSM 2.x CryptoService 이외 기능에 대해 Csm 모듈 의존성 제거

|              |                                                                                 |
|--------------|---------------------------------------------------------------------------------|
| 원인           | HSM에서 사용하는 OS ISR 및 Interrupt는 HSM 정책에 따라 바뀔 수 있기 때문에 CSM 의존성 제거가 필요.           |
| 동작 영향        | 없음                                                                              |
| 설정 영향        | 없음                                                                              |
| ASW 조치 필요 사항 | HSM Crypto Service 기능을 제외한 HSM 포팅 및 ISR 관련 사항은 SAG_Crypto 문서를 통해 ASW 담당자가 직접 설정 |

Version 2.6.1.0

4.3.26

➤ 오류 수정

- HaeHsmCmacAes 인터페이스 오류 수정 (오토에버 HSM 2.0 CmacAes)

|              |                                                                                                            |
|--------------|------------------------------------------------------------------------------------------------------------|
| 원인           | Autoever HSM 2.0 CmacAes 가 1Byte 단위 처리를 지원하지만 Csm의 Cry 인터페이스가 4Byte 배수가 아니면 에러를 리턴하게 구성된 것이 원인. 해당 원인 개선함. |
| 동작 영향        | 없음                                                                                                         |
| 설정 영향        | 없음                                                                                                         |
| ASW 조치 필요 사항 | 없음                                                                                                         |

➤ 개선 사항

- 동일한 Input file 및 설정 변경점 없는 경우에도 Generated 파일 내용 변경되는 사항 개선

|              |                                                         |
|--------------|---------------------------------------------------------|
| 원인           | Csm Generator의 generated code 순서 정렬 기능 개선이 필요하여 이를 개선함. |
| 동작 영향        | 없음                                                      |
| 설정 영향        | 없음                                                      |
| ASW 조치 필요 사항 | 없음                                                      |

**Version 2.6.2.0**

## ➤ 개선 사항

## ■ R44 RTE 와의 모듈 정합성 업데이트

|              |                                                         |
|--------------|---------------------------------------------------------|
| 원인           | Reentrant 와 Can be invoked Concurrently 설정이 동기화 되도록 개선함 |
| 동작 영향        | 없음                                                      |
| 설정 영향        | 없음                                                      |
| ASW 조치 필요 사항 | 없음                                                      |

## ■ Swcd\_Bsw\_Csm.arxml 생성시 순서 정렬 기능 개선

|              |                                                              |
|--------------|--------------------------------------------------------------|
| 원인           | Generation 시 Swcd_Bsw_Csm.arxml 파일이 동일하게 생성되도록 순서 정렬기능을 개선함. |
| 동작 영향        | 없음                                                           |
| 설정 영향        | 없음                                                           |
| ASW 조치 필요 사항 | 없음                                                           |

## ■ MISRA 개선

|              |               |
|--------------|---------------|
| 원인           | MISRA 위반사항 개선 |
| 동작 영향        | 없음            |
| 설정 영향        | 없음            |
| ASW 조치 필요 사항 | 없음            |

**Version 2.6.3.0**

## ➤ 개선 사항

## ■ 정적검증 결과 개선 반영

|              |                   |
|--------------|-------------------|
| 원인           | 정적검증 결과 개선 반영이 필요 |
| 동작 영향        | 없음                |
| 설정 영향        | 없음                |
| ASW 조치 필요 사항 | 없음                |

**Version 2.6.4.0**

➤ 오류 수정

- Async API 수행을 위한 BG Task 가 종료되지 않은 상태에서 BG Task 재요청하여 BG Task 구동되지 못하는 문제 개선

|              |                                                                                                                                                                                   |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 원인           | 생성되는 BG Task 함수 원형의 Terminate Task 호출 직전에 BG Task 재요청이 Preemption 되면 BG Task에 Os Limit이 발생하게 되고 Callback 이 호출되지 않는다. 이러한 경우는 CPU 로드가 매우높아져서 BG Task의 수행 시간이 짧아지는 환경에서 간헐적으로 발생한다. |
| 동작 영향        | 없음                                                                                                                                                                                |
| 설정 영향        | 없음                                                                                                                                                                                |
| ASW 조치 필요 사항 | 없음                                                                                                                                                                                |

## Limitations

4.4

- Csm 모듈은 각 서비스에 대해 한번에 하나의 인스턴스만 사용 가능하다(CSM0016)  
[인스턴스 제약 사항](#) 참고
- SPC58x HSM 의 AES 기반 알고리즘은 단일 HW AES 모듈을 사용하므로 한번에 하나의 요청만 처리 가능하다.  
즉 다음의 알고리즘 중 하나의 요청 처리 중 다른 알고리즘 요청 시 E\_NOT\_OK 리턴이 회신된다.  
- AES ECB Encrypt/Decrypt, AES CBC Encrypt/Decrypt, AES CTR Encrypt/Decrypt, CMAC-AES Generation/Verification  
[SPC58x HSM AES 기반 알고리즘 제약 사항](#) 참고
- 지원되는 Crypto Service 목록  
A. HSM 2.0 적용 MCU (F1KM, SPC58x, TC3xx)

| Category | Algorithm                                                                            | Host Core /HSM Core | Algorithm 제조사 |
|----------|--------------------------------------------------------------------------------------|---------------------|---------------|
| Hash     | Seed Key                                                                             | Host Core           | Autron        |
|          | Advanced SeedKey                                                                     | Host Core           | Autoever      |
|          | Sha1<br>Sha256                                                                       | Host Core           | Autoever      |
|          | Sha1<br>Sha256                                                                       | HSM Core            | Autoever      |
| MAC      | Hmac-Sha1 Generate/Verify<br>Hmac-Sha256 Generate/Verify<br>Cmac-Aes Generate/Verify | Host Core           | Autoever      |
|          | Hmac-Sha256 Generate/Verify<br>Cmac-Aes Generate/Verify                              | HSM Core            | Autoever      |
| Random   | Pseudo Random Seed/Generate                                                          | Host Core           | Autoever      |
|          | True Random Generate                                                                 | HSM Core            | Autoever      |

| Category     | Algorithm                                  | Host Core /HSM Core | Algorithm 제조사 |
|--------------|--------------------------------------------|---------------------|---------------|
|              | Pseudo Random Generate                     |                     |               |
| Symmetric    | AES 128bit (ECB, CBC, CTR) Encrypt/Decrypt | Host Core           | Autoever      |
|              | AES 128bit (ECB, CBC, CTR) Encrypt/Decrypt | HSM Core            | Autoever      |
| Asymmetric   | RSA 2048bit Encrypt/Decrypt                | Host Core           | Autoever      |
| Signature    | PKCS#1 V1.5 Sha1 Generate/Verify           | Host Core           | Autoever      |
|              | PKCS#1 V1.5 Sha256 Generate/Verify         | HSM Core            | Autoever      |
|              | PKCS#1 PSS Sha256 Generate/Verify          | HSM Core            | Autoever      |
|              | ECDSA Sha256 Generate/Verify               | HSM Core            | Autoever      |
| Key Derive   | PBKDF2(PKCS#5)                             | HSM Core            | Autoever      |
| Key Exchange | ECDH Secret Calculation                    | HSM Core            | Autoever      |

#### B. MPC560xB

| Category   | Algorithm                                                                            | Host Core /HSM Core | Algorithm 제조사 |
|------------|--------------------------------------------------------------------------------------|---------------------|---------------|
| Hash       | Seed Key                                                                             | Host(SW)            | Autron        |
|            | Advanced SeedKey                                                                     | Host(SW)            | Autoever      |
|            | Sha1<br>Sha256                                                                       | Host(SW)            | Autoever      |
| MAC        | Hmac-Sha1 Generate/Verify<br>Hmac-Sha256 Generate/Verify<br>Cmac-Aes Generate/Verify | Host(SW)            | Autoever      |
| Random     | Pseudo Random Seed/Generate                                                          | Host(SW)            | Autoever      |
| Symmetric  | AES 128bit (ECB, CBC, CTR) Encrypt/Decrypt                                           | Host(SW)            | Autoever      |
| Asymmetric | RSA 2048bit Encrypt/Decrypt                                                          | Host(SW)            | Autoever      |
| Signature  | PKCS#1 V1.5 Sha1 Generate/Verify<br>PKCS#1 V1.5 Sha256 Generate/Verify               | Host(SW)            | Autoever      |

#### B. RH850 F1L

| Category   | Algorithm                                                                            | Host Core /HSM Core | Algorithm 제조사 |
|------------|--------------------------------------------------------------------------------------|---------------------|---------------|
| Hash       | Seed Key                                                                             | Host(SW)            | Autron        |
|            | Advanced SeedKey                                                                     | Host(SW)            | Autoever      |
|            | Sha1<br>Sha256                                                                       | Host(SW)            | Autoever      |
| MAC        | Hmac-Sha1 Generate/Verify<br>Hmac-Sha256 Generate/Verify<br>Cmac-Aes Generate/Verify | Host(SW)            | Autoever      |
| Random     | Pseudo Random Seed/Generate                                                          | Host(SW)            | Autoever      |
| Symmetric  | AES 128bit (ECB, CBC, CTR) Encrypt/Decrypt                                           | Host(SW)            | Autoever      |
| Asymmetric | RSA 2048bit Encrypt/Decrypt                                                          | Host(SW)            | Autoever      |
| Signature  | PKCS#1 V1.5 Sha1 Generate/Verify<br>PKCS#1 V1.5 Sha256 Generate/Verify               | Host(SW)            | Autoever      |

C. RH850 F1K

| Category   | Algorithm                                  | Host Core /HSM Core | Algorithm 제조사 |
|------------|--------------------------------------------|---------------------|---------------|
| Hash       | Seed Key                                   | Host(SW)            | Autron        |
|            | Advanced SeedKey                           | Host(SW)            | Autoever      |
|            | Sha1                                       | Host(SW)            | Autoever      |
|            | Sha256                                     |                     |               |
| Mac        | Hmac-Sha1 Generate/Verify                  | Host(SW)            | Autoever      |
|            | Hmac-Sha256 Generate/Verify                |                     |               |
|            | Cmac-Aes Generate/Verify                   |                     |               |
|            | Cmac-Aes Generate/Verify                   | HSM(HW)             | Autron        |
| Random     | Pseudo Random Seed/Generate                | Host(SW)            | Autoever      |
|            | Random Seed/Generate                       | HSM(HW)             | Autron        |
| Symmetric  | AES 128bit (ECB, CBC, CTR) Encrypt/Decrypt | Host(SW)            | Autoever      |
|            | AES 128bit (ECB, CBC) Encrypt/Decrypt      | HSM(HW)             | Autron        |
| Asymmetric | RSA 2048bit Encrypt/Decrypt                | Host(SW)            | Autoever      |
| Signature  | PKCS#1 V1.5 Sha1 Generate/Verify           | Host(SW)            | Autoever      |
|            | PKCS#1 V1.5 Sha256 Generate/Verify         |                     |               |

D. RH850 F1KM (HSM 1.0)

| Category   | Algorithm                                  | Host Core /HSM Core | Algorithm 제조사 |
|------------|--------------------------------------------|---------------------|---------------|
| Hash       | Seed Key                                   | Host(SW)            | Autron        |
|            | Advanced SeedKey                           | Host(SW)            | Autoever      |
|            | Sha1                                       | Host(SW)            | Autoever      |
|            | Sha256                                     |                     |               |
| MAC        | Sha1                                       | HSM(SW)             | Autoever      |
|            | Sha256                                     |                     |               |
|            | Hmac-Sha1 Generate/Verify                  | Host(SW)            | Autoever      |
|            | Hmac-Sha256 Generate/Verify                |                     |               |
|            | Cmac-Aes Generate/Verify                   |                     |               |
| MAC        | Hmac-Sha1 Generate/Verify                  | HSM(SW)             | Autoever      |
|            | Hmac-Sha256 Generate/Verify                |                     |               |
|            | Cmac-Aes Generate/Verify                   | HSM(HW)             | Autron        |
|            | Cmac-Aes Generate/Verify                   | HSM(HW)             | Autron        |
| Random     | Pseudo Random Seed/Generate                | Host(SW)            | Autoever      |
|            | True Random Generate                       | HSM(HW)             | Autron        |
|            | Pseudo Random Generate                     |                     |               |
| Symmetric  | AES 128bit (ECB, CBC, CTR) Encrypt/Decrypt | Host(SW)            | Autoever      |
|            | AES 128bit (ECB, CBC, CTR) Encrypt/Decrypt | HSM(HW)             | Autron        |
| Asymmetric | RSA 2048bit Encrypt/Decrypt                | Host(SW)            | Autoever      |
|            | RSA 2048bit Encrypt/Decrypt                | HSM(SW)             | Autoever      |
|            | ECIES Encrypt/Decrypt                      | HSM(SW)             | Autoever      |
|            | - secp256r1                                |                     |               |
| Signature  | - SHA256, HMAC-SHA256                      | HSM(SW)             | Autoever      |
|            | - XOR, AES_128_CBC_IV0, AES_128_CTR_IV0    |                     |               |
| Signature  | PKCS#1 V1.5 Sha1 Generate/Verify           | Host(SW)            | Autoever      |

| Category     | Algorithm                          | Host Core /HSM Core | Algorithm 제조사 |
|--------------|------------------------------------|---------------------|---------------|
|              | PKCS#1 V1.5 Sha256 Generate/Verify |                     |               |
|              | PKCS#1 V1.5 Sha1 Generate/Verify   | HSM(SW)             | Autoever      |
|              | PKCS#1 V1.5 Sha256 Generate/Verify | HSM(SW)             | Autoever      |
|              | PKCS#1 PSS Sha256 Generate/Verify  | HSM(SW)             | Autoever      |
|              | ECDSA Sha256 Generate/Verify       | HSM(SW)             | Autoever      |
| Key Derive   | PBKDF2(PKCS#5)                     | HSM(SW)             | Autoever      |
| Key Exchange | ECDH Secret Calculation            | HSM(SW)             | Autoever      |

#### E. AURIX TC27x / TC29x

| Category   | Algorithm                                                                            | Host Core /HSM Core | Algorithm 제조사 |
|------------|--------------------------------------------------------------------------------------|---------------------|---------------|
| Hash       | Seed Key                                                                             | Host(SW)            | Autron        |
|            | Advanced SeedKey                                                                     | Host(SW)            | Autoever      |
|            | Sha1<br>Sha256                                                                       | Host(SW)            | Autoever      |
|            | Sha1<br>Sha256                                                                       | HSM                 | Autoever      |
| MAC        | Hmac-Sha1 Generate/Verify<br>Hmac-Sha256 Generate/Verify<br>Cmac-Aes Generate/Verify | Host(SW)            | Autoever      |
|            | Hmac-Sha256 Generate/Verify<br>Cmac-Aes Generate/Verify                              | HSM                 | Autoever      |
| Random     | Pseudo Random Seed/Generate                                                          | Host(SW),<br>HSM    | Autoever      |
|            | True Random Generate                                                                 | HSM                 | Autoever      |
| Symmetric  | AES 128bit (ECB, CBC, CTR) Encrypt/Decrypt                                           | Host(SW)            | Autoever      |
| Asymmetric | RSA 2048bit Encrypt/Decrypt                                                          | Host(SW)            | Autoever      |
| Signature  | PKCS#1 V1.5 Sha1 Generate/Verify<br>PKCS#1 V1.5 Sha256 Generate/Verify               | Host(SW)            | Autoever      |
|            | PKCS#1 V1.5 Sha1 Verify<br>PKCS#1 V1.5 Sha256 Generate/Verify                        | HSM                 | Autoever      |

#### C. SPC584Bx / SPC58ECx (HSM 1.0)

| Category | Algorithm                                                | Host Core /HSM Core | Algorithm 제조사 |
|----------|----------------------------------------------------------|---------------------|---------------|
| Hash     | Seed Key                                                 | Host(SW)            | Autron        |
|          | Advanced SeedKey                                         | Host(SW)            | Autoever      |
|          | Sha1<br>Sha256                                           | Host(SW)            | Autoever      |
|          | Sha1<br>Sha256                                           | HSM(SW)             | Autoever      |
| MAC      | Hmac-Sha1 Generate/Verify<br>Hmac-Sha256 Generate/Verify | Host(SW)            | Autoever      |

| Category     | Algorithm                                                                                                | Host Core /HSM Core | Algorithm 제조사 |
|--------------|----------------------------------------------------------------------------------------------------------|---------------------|---------------|
|              | Cmac-Aes Generate/Verify                                                                                 |                     |               |
|              | Hmac-Sha1 Generate/Verify                                                                                | HSM(SW)             | Autoever      |
|              | Hmac-Sha256 Generate/Verify                                                                              |                     |               |
|              | Cmac-Aes Generate/Verify                                                                                 | HSM(HW)             | Autron        |
| Random       | Pseudo Random Seed/Generate                                                                              | Host(SW)            | Autoever      |
|              | True Random Generate                                                                                     | HSM(HW)             | Autron        |
|              | Pseudo Random Generate                                                                                   | HSM(SW)             | Autoever      |
| Symmetric    | AES 128bit (ECB, CBC, CTR) Encrypt/Decrypt                                                               | Host(SW)            | Autoever      |
|              | AES 128bit (ECB, CBC, CTR) Encrypt/Decrypt                                                               | HSM(HW)             | Autron        |
| Asymmetric   | RSA 2048bit Encrypt/Decrypt                                                                              | Host(SW)            | Autoever      |
|              | RSA 2048bit Encrypt/Decrypt                                                                              | HSM(SW)             | Autoever      |
|              | ECIES Encrypt/Decrypt<br>- secp256r1<br>- SHA256, HMAC-SHA256<br>- XOR, AES_128_CBC_IV0, AES_128_CTR_IV0 | HSM(SW)             | Autoever      |
|              |                                                                                                          |                     |               |
| Signature    | PKCS#1 V1.5 Sha1 Generate/Verify                                                                         | Host(SW)            | Autoever      |
|              | PKCS#1 V1.5 Sha256 Generate/Verify                                                                       |                     |               |
|              | PKCS#1 V1.5 Sha1 Generate/Verify                                                                         | HSM(SW)             | Autoever      |
|              | PKCS#1 V1.5 Sha256 Generate/Verify                                                                       |                     |               |
|              | PKCS#1 PSS Sha256 Generate/Verify                                                                        | HSM(SW)             | Autoever      |
|              | ECDSA Sha256 Generate/Verify                                                                             | HSM(SW)             | Autoever      |
| Key Derive   | PBKDF2(PKCS#5)                                                                                           | HSM(SW)             | Autoever      |
| Key Exchange | ECDH Secret Calculation                                                                                  | HSM(SW)             | Autoever      |

#### D. S32K

| Category   | Algorithm                                  | Host Core /HSM Core | Algorithm 제조사 |
|------------|--------------------------------------------|---------------------|---------------|
| Hash       | Seed Key                                   | Host(SW)            | Autron        |
|            | Advanced SeedKey                           | Host(SW)            | Autoever      |
|            | Sha1<br>Sha256                             | Host(SW)            | Autoever      |
| Mac        | Hmac-Sha1 Generate/Verify                  | Host(SW)            | Autoever      |
|            | Hmac-Sha256 Generate/Verify                |                     |               |
|            | Cmac-Aes Generate/Verify                   |                     |               |
| Random     | Cmac-Aes Generate/Verify                   | HSM(HW)             | Autron        |
|            | Pseudo Random Seed/Generate                | Host(SW)            | Autoever      |
|            | Random Seed/Generate                       | HSM(HW)             | Autron        |
| Symmetric  | AES 128bit (ECB, CBC, CTR) Encrypt/Decrypt | Host(SW)            | Autoever      |
|            | AES 128bit (ECB, CBC) Encrypt/Decrypt      | HSM(HW)             | Autron        |
| Asymmetric | RSA 2048bit Encrypt/Decrypt                | Host(SW)            | Autoever      |
| Signature  | PKCS#1 V1.5 Sha1 Generate/Verify           | Host(SW)            | Autoever      |
|            | PKCS#1 V1.5 Sha256 Generate/Verify         |                     |               |

- HSM CryptoService(암호서비스)가 아닌 HSM Crypto Application 의 경우 (ex. SecureBoot, Firmware Temper Detect, HSM Driver Init 등) 는 CSM 추상화 대상이 아니므로 직접 API 를 호출하여야 한다. 특히, 본 AUTOSAR 버전의 CSM에서는 Key Store, Key Genere 등과 같은 Key 관리 기능에 대해서는 지원하지 않는다.
- OsTask\_BSW\_BG\_Csm 는 Csm 에서 Async Job Processing 을 위해서만 사용하는 BG Task 이므로 다른용도로 활용되면 안된다.
- Csm Async API 사용의 경우에 등록된 CallBack Function 은 OsTask\_BSW\_BG\_Csm 기반이기 때문에 CPU Load 증가 시 다른 Task 들에 의해 지속적으로 선점 된다. 따라서 Callback 에서는 변수 등을 업데이트 하는 정도로 활용하고, 나머지 Csm Async API 는 CallBack 함수 에서 호출하지 않는 것을 권장한다.

## Deviations

### 4.5

#### ➤ Csm\_ReturnType 변경

AUTOSAR R4.2.2 Csm 모듈의 Csm\_ReturnType 삭제 내용을 선반영함

Csm\_ReturnType → Std\_ReturnType 으로 변경

CSM\_E\_OK → E\_OK, CSM\_E\_NOT\_OK → E\_NOT\_OK 로 변경됨

#### ➤ Sync/Async 동작 방식 변경

표준 : CsmUseSyncJobProcessing 파라미터 설정에 따라 모듈 전체의 Sync/Async Job 처리 방식을 결정

변경 : 개별 Primitive Config 컨테이너의 Callback Function 등록 유무에 따라 해당 Primitive 의 Sync/Async 동작 방식 결정. 즉, Callback Function 설정시 해당 Primitive 는 Async 로 처리, 미설정시 Sync 로 처리

#### ➤ Csm Service component 의 runnable 속성 변경

표준 : Runnable 의 속성 중 Can Be Invoked Concurrently 의 값이 false 임

변경 : Csm 모듈에서 개별 service 요청을 queueing 처리하여 background task 에서 처리하므로 결과적으로 task mapping 이 필요없음. 따라서 Can Be Invoked Concurrently 의 값을 true 로 생성하도록 변경

#### ➤ Csm\_KeyDeriveStart API 의 keyLength 파라미터 용도 변경

표준 : Key Derive 결과로 유도되는 key 의 길이

변경 : Csm\_KeyDeriveUpdate API 의 password 종류 결정.

keyLength 가 0 일 경우 사용자 password 사용, 0 이외의 값일 경우 PSK 의 key index 로 사용

## 5. Configuration Guide

Container 파라미터 중 Key, Buffer 크기 관련 설정은 [알고리즘 파라미터 허용 범위](#)를 참고하여 설정하도록 한다.

### CsmGeneral Container

| Parameter Name                                               | Value                  | Category |
|--------------------------------------------------------------|------------------------|----------|
| Short Name                                                   | User Defined           | C        |
| Version Info Api                                             | User Defined           | C        |
| Dev Error Detect                                             | User Defined           | C        |
| Max Align SCalar Type <sup>1)</sup>                          | uint8                  | F        |
| Maximum Blocking Time <sup>2)</sup>                          |                        | N        |
| Use Interruption <sup>3)</sup>                               |                        | N        |
| Use Sync Job Processing <sup>4)</sup>                        |                        | N        |
| Main Function Period <sup>5)</sup>                           | RTE Task mapping 설정 참조 | C        |
| Use Common Fbl Crypto Lib <sup>6)</sup><br>(AUTRON specific) |                        | N        |
| Seed Ref Addr <sup>7)</sup><br>(AUTRON specific)             | 플랫폼 배포시의 설정 사용         | F        |

1) 모듈 내부에서 사용하는 align 단위  
uint8 값 이외 미지원

2) 미지원

3) 미지원

4) 미지원

각각의 primitive의 Sync / Async 동작 방식은 Callback function 등록 유무에 따라 동작한다. [Deviations](#) 참조

5) Csm\_MainFunction의 주기

6) 미사용

7) AutRandom 알고리즘 사용시 Seed 값을 얻기 위한 주소로 MCU 별로 배포담당자가 결정하여 고정함  
AutRandom 알고리즘 미사용시 실제 적용되지 않음

### CsmHash Container

| Parameter Name                       | Value        | Category |
|--------------------------------------|--------------|----------|
| Short Name                           | User Defined | C        |
| Max Result Buffer Size <sup>1)</sup> | User Defined | C        |
| Max Data Buffer Size <sup>2)</sup>   | User Defined | C        |

1) CsmHashConfig 설정 암호화 알고리즘 중 결과 버퍼의 최대 크기(byte)

- Rte\_Type.h 파일에 typedef uint8 HashDataBuffer[xxx] 정의 생성시 배열 크기를 결정하며 해당 data type 은 CsmHash Interface 의 HashUpdate operation 의 인자로 사용된다.

## 2) CsmHashConfig 설정 암호화 알고리즘 중 입력 데이터 버퍼 최대 크기(byte)

- Rte\_Type.h 파일에 typedef uint8 HashResultBuffer [xxx] 정의 생성시 배열 크기를 결정하며 해당 data type 은 CsmHash Interface 의 HashFinish operation 의 인자로 사용된다.

## CsmHashConfig Container

| Parameter Name                   | Value        | Category |
|----------------------------------|--------------|----------|
| Short Name <sup>1)</sup>         | User Defined | C        |
| Callback Hash <sup>2)</sup>      | User Defined | C        |
| Init Configuration <sup>3)</sup> |              | N        |
| Primitive Name <sup>4)</sup>     | 아래 설명 사항 참조  | C        |
| Use Port <sup>5)</sup>           | User Defined | C        |
| Use External Task <sup>6)</sup>  | false        | N        |

1) Short Name 은 Csm 모듈 내에서 unique 해야 함

2) 암호화 알고리즘 동작(ex. Start, Update, Finish, ...) 요청시 처리 동작 방식 결정

Callback function 등록시 : Async 동작 (Csm 내부 Task 에서 요청 처리 후 callback function 호출)

Use Port 사용시 [ShortName]\_Callback R-Port 와 연결하여 생성된 함수 이름을 등록해야 한다.

Callback function 미등록시 : Sync 동작 (User context 에서 해당 요청 처리)

3) 암호화 알고리즘 Start 시 전달되는 구조체 포인터.

NULL\_PTR 혹은 미설정

4) 모듈에서 지원되는 암호화 알고리즘 명칭

※ 설정 가능값 : AutSeedKey, HaeAdvSeedKey, HaeSha1, HaeSha256, AuthHsmSha1, AuthHsmSha256, HaeHsmSha1, HaeHsmSha256

MCU 별 가능 알고리즘은 [Limitations](#) 참조

5) Rte Port Interface 지원 유무

true 설정시 Swcd\_Bsw\_Csm.arxml 파일의 Csm component 에 다음과 같은 설정이 자동 생성된다.

a) [ShortName]\_Hash 이름의 P-Port

b) Callback function 등록시 [ShortName]\_Callback 이름의 R-Port,

SynchronousServerCallPoint\_[ShortName]\_Callback\_JobFinished 이벤트

c) OperationInvokedEvent\_[ShortName]\_Hash\_HashStart/Update/Finish 이벤트

d) [ShortName]\_Hash 이름의 P-Port 와 연계된 Port API Option

6) 미사용

## CsmMacGenerate Container

| Parameter Name                       | Value        | Category |
|--------------------------------------|--------------|----------|
| Short Name                           | User Defined | C        |
| Max Key Size <sup>1)</sup>           | User Defined | C        |
| Max Data Buffer Size <sup>2)</sup>   | User Defined | C        |
| Max Result Buffer Size <sup>3)</sup> | User Defined | C        |

### 1) CsmMacGenerateConfig 설정 암호화 알고리즘 중 입력 가능 최대 Key 크기(byte)

- Rte\_Type.h 파일에 typedef struct Csm\_SymKeyType 구조체의 Csm\_AlignType data[xxx] 배열 생성시 최대 크기를 결정하며 해당 data type 은 CsmMacGenerate Interface 의 MacGenerateStart operation 의 인자로 사용된다.

Ex) CMAC-AES 알고리즘 : 16

[Mac](#) 의 keyPtr→length 참조

### 2) CsmMacGenerateConfig 설정 암호화 알고리즘 중 입력 데이터 버퍼 최대 크기(byte)

- Rte\_Type.h 파일에 typedef uint8 MacGenerateDataBuffer[xxx] 정의 생성시 배열 크기를 결정하며 해당 data type 은 CsmMacGenerate Interface 의 MacGenerateUpdate operation 의 인자로 사용된다.

### 3) CsmMacGenerateConfig 설정 암호화 알고리즘 중 결과 버퍼 최대 크기(byte)

- Rte\_Type.h 파일에 typedef uint8 MacGenerateResultBuffer[xxx] 정의 생성시 배열 크기를 결정하며 해당 data type 은 CsmMacGenerate Interface 의 MacGenerateFinish operation 의 인자로 사용된다.

Ex) CMAC-AES 알고리즘 : 16

HMAC-SHA1 : 20

HMAC-SHA256 : 32

[Mac](#) 의 \*resultLengthPtr 참조

## CsmMacGenerateConfig Container

| Parameter Name                      | Value        | Category |
|-------------------------------------|--------------|----------|
| Short Name <sup>1)</sup>            | User Defined | C        |
| Callback Mac Generate <sup>2)</sup> | User Defined | C        |
| Init Configuration <sup>3)</sup>    |              | N        |
| Primitive Name <sup>4)</sup>        | 아래 설명 사항 참조  | C        |
| Use Port <sup>5)</sup>              | User Defined | C        |
| Use External Task <sup>6)</sup>     | false        | N        |

### 1) Short Name 은 Csm 모듈 내에서 unique 해야 함

### 2) 암호화 알고리즘 동작(ex. Start, Update, Finish, ...) 종료시 호출될 callback function

Callback function 등록시 : Async 동작 (Csm 내부 Task 에서 요청 처리 후 callback function 호출)

Use Port 사용시 [ShortName]\_Callback R-Port 와 연결하여 생성된 함수 이름을 등록해야 한다.

Callback function 미등록시 : Sync 동작 (User context 에서 해당 요청 처리)

3) 암호화 알고리즘 Start 시 전달되는 구조체 포인터

NULL\_PTR 혹은 미설정

4) 모듈에서 지원되는 암호화 알고리즘 명칭

※ 설정 가능값 : HaeHmacSha1Generate, HaeHmacSha256Generate, HaeCmacAesGenerate, AuthsmHmacSha1Generate, AuthsmHmacSha256Generate, AuthsmCmacAesGenerate, HaeHsmHmacSha256Generate, HaeHsmCmacAesGenerate, AutIcusCmacAesGenerate, AutCsecCmacAesGenerate

MCU 별 가능 알고리즘은 [Limitations](#) 참조

5) Rte Port Interface 지원 유무

true 설정시 Swcd\_Bsw\_Csm.arxml 파일의 Csm component 에 다음과 같은 설정이 자동 생성된다.

a) [ShortName]\_MacGenerate 이름의 P-Port

b) Callback function 등록시 [ShortName]\_Callback 이름의 R-Port,

SynchronousServerCallPoint\_[ShortName]\_Callback\_JobFinished 이벤트

c) OperationInvokedEvent\_[ShortName]\_MacGenerate\_MacGenerateStart/Update/Finish 이벤트

d) [ShortName]\_MacGenerate 이름의 P-Port 와 연계된 Port API Option

6) 미사용

5.4

## CsmMacVerify Container

| Parameter Name                        | Value        | Category |
|---------------------------------------|--------------|----------|
| Short Name                            | User Defined | C        |
| Max Key Size <sup>1)</sup>            | User Defined | C        |
| Max Data Buffer Size <sup>2)</sup>    | User Defined | C        |
| Max Compare Buffer Size <sup>3)</sup> | User Defined | C        |

1) CsmMacVerifyConfig 설정 암호화 알고리즘 중 입력 가능 최대 Key 크기(byte)

- Rte\_Type.h 파일에 typedef struct Csm\_SymKeyType 구조체의 Csm\_AlignType data[xxx] 배열 생성시 최대 크기를 결정하며 해당 data type 은 CsmMacVerify Interface 의 MacVerifyStart operation 의 인자로 사용된다.

Ex) CMAC-AES 알고리즘 : 16

[Mac](#) 의 keyPtr→length 참조

2) CsmMacVerifyConfig 설정 암호화 알고리즘 중 입력 데이터 버퍼 최대 크기(byte)

- Rte\_Type.h 파일에 typedef uint8 MacVerifyDataBuffer[xxx] 정의 생성시 배열 크기를 결정하며 해당 data

type 은 CsmMacVerify Interface 의 MacVerifyUpdate operation 의 인자로 사용된다.

### 3) CsmMacVerifyConfig 설정 암호화 알고리즘 중 결과 비교 버퍼 최대 크기(byte)

- Rte\_Type.h 파일에 typedef uint8 MacVerifyCompareBuffer[xxx] 정의 생성시 배열 크기를 결정하며 해당 data type 은 CsmMacVerify Interface 의 MacVerifyFinish operation 의 인자로 사용된다.

Ex) CMAC-AES 알고리즘 : 16

HMAC-SHA1 : 20

HMAC-SHA256 : 32

[Mac](#) 의 MacLength 참조

## CsmMacVerifyConfig Container

| Parameter Name                    | Value        | Category |
|-----------------------------------|--------------|----------|
| Short Name <sup>1)</sup>          | User Defined | C        |
| Callback Mac Verify <sup>2)</sup> | User Defined | C        |
| Init Configuration <sup>3)</sup>  |              | N        |
| Primitive Name <sup>4)</sup>      | 아래 설명 사항 참조  | C        |
| Use Port <sup>5)</sup>            | User Defined | C        |
| Use External Task <sup>6)</sup>   | false        | N        |

1) Short Name 은 Csm 모듈 내에서 unique 해야 함

2) 암호화 알고리즘 동작(ex. Start, Update, Finish, ...) 요청시 처리 동작 방식 결정

Callback function 등록시 : Async 동작 (Csm 내부 Task 에서 요청 처리 후 callback function 호출)

Use Port 사용시 [ShortName]\_Callback R-Port 와 연결하여 생성된 함수 이름을 등록해야 한다.

Callback function 미등록시 : Sync 동작 (User context 에서 해당 요청 처리)

3) 암호화 알고리즘 Start 시 전달되는 구조체 포인터

NULL\_PTR 혹은 미설정

4) 모듈에서 지원되는 암호화 알고리즘 명칭

※ 설정 가능값 : HaeHmacSha1Verify, HaeHmacSha256Verify, HaeCmacAesVerify,

AuthsmHmacSha1Verify, AuthsmHmacSha256Verify, AuthsmCmacAesVerify,

HaeHsmHmacSha256Verify, HaeHsmCmacAesVerify, AutIcusCmacAesVerify, AutCsecCmacAesVerify

MCU 별 가능 알고리즘은 [Limitations](#) 참조

5) Rte Port Interface 지원 유무

true 설정시 Swcd\_Bsw\_Csm.arxml 파일의 Csm component 에 다음과 같은 설정이 자동 생성된다.

a) [ShortName]\_MacVerify 이름의 P-Port

b) Callback function 등록시 [ShortName]\_Callback 이름의 R-Port,

SynchronousServerCallPoint\_[ShortName]\_Callback\_JobFinished 이벤트

c) OperationInvokedEvent\_[ShortName]\_MacVerify\_MacVerifyStart/Update/Finish 이벤트

d) [ShortName]\_MacVerify 이름의 P-Port 와 연계된 Port API Option

6) 미사용

## CsmRandomSeed Container

| Parameter Name                     | Value        | Category |
|------------------------------------|--------------|----------|
| Short Name                         | User Defined | C        |
| Max Data Buffer Size <sup>1)</sup> | User Defined | C        |

1) CsmRandomSeedConfig 설정 암호화 알고리즘 중 입력 데이터 버퍼 최대 크기(byte)

- Rte\_Type.h 파일에 typedef uint8 RandomSeedDataBuffer[xxx] 정의 생성시 배열 크기를 결정하며 해당 data type 은 CsmRandomSeed Interface 의 RandomSeedUpdate operation 의 인자로 사용된다.

## CsmRandomSeedConfig Container

| Parameter Name                     | Value        | Category |
|------------------------------------|--------------|----------|
| Short Name <sup>1)</sup>           | User Defined | C        |
| Callback Random Seed <sup>2)</sup> | User Defined | C        |
| Init Configuration <sup>3)</sup>   |              | N        |
| Primitive Name <sup>4)</sup>       | 아래 설명 사항 참조  | C        |
| Use Port <sup>5)</sup>             | User Defined | C        |
| Use External Task <sup>6)</sup>    | false        | N        |

1) Short Name 은 Csm 모듈 내에서 unique 해야 함

2) 암호화 알고리즘 동작(ex. Start, Update, Finish, ...) 요청시 처리 동작 방식 결정

Callback function 등록시 : Async 동작 (Csm 내부 Task 에서 요청 처리 후 callback function 호출)

Use Port 사용시 [ShortName]\_Callback R-Port 와 연결하여 생성된 함수 이름을 등록해야 한다.

Callback function 미등록시 : Sync 동작 (User context 에서 해당 요청 처리)

3) 암호화 알고리즘 Start 시 전달되는 구조체 포인터

NULL\_PTR 혹은 미설정

4) 모듈에서 지원되는 암호화 알고리즘 명칭

※ 설정 가능값 : HaeAdvRandomSeed, HaePseudoRandomSeed, AutIcusRandomSeed,

AutCsecRandomSeed

MCU 별 가능 알고리즘은 [Limitations](#) 참조

5) Rte Port Interface 지원 유무

true 설정시 Swcd\_Bsw\_Csm.arxml 파일의 Csm component 에 다음과 같은 설정이 자동 생성된다.

- a) [ShortName]\_RandomSeed 이름의 P-Port
  - b) Callback function 등록시 [ShortName]\_Callback 이름의 R-Port,  
SynchronousServerCallPoint\_[ShortName]\_Callback\_JobFinished 이벤트
  - c) OperationInvokedEvent\_[ShortName]\_RandomSeed\_RandomSeedStart/Update/Finish 이벤트
  - d) [ShortName]\_RandomSeed 이름의 P-Port 와 연계된 Port API Option
- 6) 미사용

## CsmRandomGenerate Container

| Parameter Name                     | Value        | Category |
|------------------------------------|--------------|----------|
| Short Name                         | User Defined | C        |
| Max Data Buffer Size <sup>1)</sup> | User Defined | C        |

- 1) CsmRandomGenerateConfig 설정 암호화 알고리즘 중 출력 데이터 버퍼 최대 크기(byte)
  - Rte\_Type.h 파일에 typedef uint8 RandomGenerateResultBuffer[xxx] 정의 생성시 배열 크기를 결정하며 해당 data type 은 CsmRandomGenerate Interface 의 RandomGenerate operation 의 인자로 사용된다.

## CsmRandomGenerateConfig Container

| Parameter Name                         | Value        | Category |
|----------------------------------------|--------------|----------|
| Short Name <sup>1)</sup>               | User Defined | C        |
| Callback Random Generate <sup>2)</sup> | User Defined | C        |
| Init Configuration <sup>3)</sup>       |              | N        |
| Primitive Name <sup>4)</sup>           | 아래 설명 사항 참조  | C        |
| Use Port <sup>5)</sup>                 | User Defined | C        |
| Use External Task <sup>6)</sup>        | false        | N        |

- 1) Short Name 은 Csm 모듈 내에서 unique 해야 함
- 2) 암호화 알고리즘 동작 요청시 처리 동작 방식 결정
 

Callback function 등록시 : Async 동작 (Csm 내부 Task 에서 요청 처리 후 callback function 호출)

Use Port 사용시 [ShortName]\_Callback R-Port 와 연결하여 생성된 함수 이름을 등록해야 한다.

Callback function 미등록시 : Sync 동작 (User context 에서 해당 요청 처리)
- 3) 암호화 알고리즘 요청시 전달되는 구조체 포인터
 

NULL\_PTR 혹은 미설정
- 4) 모듈에서 지원되는 암호화 알고리즘 명칭
 

※ 설정 가능값 : AutRandomGenerate, HaeAdvRandomGenerate, HaePseudoRandomGenerate,

AuthSmTrueRandomGenerate, AuthSmPseudoRandomGenerate, HaeHsmTrueRandomGenerate, HaeHsmPseudoRandomGenerate, AutIcusRandomGenerate, AutCsecRandomGenerate

MCU 별 가능 알고리즘은 [Limitations](#) 참조

#### 5) Rte Port Interface 지원 유무

true 설정시 Swcd\_Bsw\_Csm.arxml 파일의 Csm component 에 다음과 같은 설정이 자동 생성된다.

- a) [ShortName]\_RandomGenerate 이름의 P-Port
- b) Callback function 등록시 [ShortName]\_Callback 이름의 R-Port,  
SynchronousServerCallPoint\_[ShortName]\_Callback\_JobFinished 이벤트
- c) OperationInvokedEvent\_[ShortName]\_RandomGenerate\_RandomGenerate 이벤트
- d) [ShortName]\_RandomGenerate 이름의 P-Port 와 연계된 Port API Option

#### 6) 미사용

## CsmSymBlockEncrypt Container

| Parameter Name                       | Value        | Category |
|--------------------------------------|--------------|----------|
| Short Name                           | User Defined | C        |
| Max Key Size <sup>1)</sup>           | User Defined | C        |
| Max Data Buffer Size <sup>2)</sup>   | User Defined | C        |
| Max Result Buffer Size <sup>3)</sup> | User Defined | C        |

#### 1) CsmSymBlockEncryptConfig 설정 암호화 알고리즘 중 입력 가능 최대 Key 크기(byte)

- Rte\_Type.h 파일에 typedef struct Csm\_SymKeyType 구조체의 Csm\_AlignType data[xxx] 배열 생성시 최대 크기를 결정하며 해당 data type 은 CsmSymBlockEncrypt Interface 의 SymBlockEncryptStart operation 의 인자로 사용된다.

Ex) AES 알고리즘 : 16

[Symmetric Block](#) 의 keyPtr→length 참조

#### 2) CsmSymBlockEncryptConfig 설정 암호화 알고리즘 중 입력 데이터 버퍼 최대 크기(byte)

- Rte\_Type.h 파일에 typedef uint8 SymBlockEncryptDataBuffer[xxx] 정의 생성시 배열 크기를 결정하며 해당 data type 은 CsmSymBlockEncrypt Interface 의 SymBlockEncryptUpdate operation 의 인자로 사용된다.

#### 3) CsmSymBlockEncryptConfig 설정 암호화 알고리즘 중 결과 버퍼 최대 크기(byte)

- Rte\_Type.h 파일에 typedef uint8 SymBlockEncryptResultBuffer[xxx] 정의 생성시 배열 크기를 결정하며 해당 data type 은 CsmSymBlockEncrypt Interface 의 SymBlockEncryptUpdate operation 의 인자로 사용된다.

## CsmSymBlockEncryptConfig Container

| Parameter Name                           | Value        | Category |
|------------------------------------------|--------------|----------|
| Short Name <sup>1)</sup>                 | User Defined | C        |
| Callback Sym Block Encrypt <sup>2)</sup> | User Defined | C        |
| Init Configuration <sup>3)</sup>         |              | N        |
| Primitive Name <sup>4)</sup>             | 아래 설명 사항 참조  | C        |
| Use Port <sup>5)</sup>                   | User Defined | C        |
| Use External Task <sup>6)</sup>          | false        | N        |

- Short Name 은 Csm 모듈 내에서 unique 해야 함
- 암호화 알고리즘 동작(ex. Start, Update, Finish, ...) 종료시 호출될 callback function  
 Callback function 등록시 : Async 동작 (Csm 내부 Task 에서 요청 처리 후 callback function 호출)  
 Use Port 사용시 [ShortName]\_Callback R-Port 와 연결하여 생성된 함수 이름을 등록해야 한다.  
 Callback function 미등록시 : Sync 동작 (User context 에서 해당 요청 처리)
- 암호화 알고리즘 Start 시 전달되는 구조체 포인터  
 NULL\_PTR 혹은 미설정
- 모듈에서 지원되는 암호화 알고리즘 명칭  
 ※ 설정 가능값 : HaeAesEcbEncrypt, AuthHsmAesEcbEncrypt, AutIcusAesEcbEncrypt,  
 AutCsecAesEcbEncrypt, HaeHsmAesEcbEncryption  
 MCU 별 가능 알고리즘은 [Limitations](#) 참조
- Rte Port Interface 지원 유무  
 true 설정시 Swcd\_Bsw\_Csm.arxml 파일의 Csm component 에 다음과 같은 설정이 자동 생성된다.
  - [ShortName]\_SymBlockEncrypt 이름의 P-Port
  - Callback function 등록시 [ShortName]\_Callback 이름의 R-Port,  
 SynchronousServerCallPoint\_[ShortName]\_Callback\_JobFinished 이벤트
  - OperationInvokedEvent\_[ShortName]\_SymBlockEncrypt\_SymBlockEncryptStart/Update/Finish 이벤트
  - [ShortName]\_SymBlockEncrypt 이름의 P-Port 와 연계된 Port API Option
- 미사용

## CsmSymBlockDecrypt Container

| Parameter Name                       | Value        | Category |
|--------------------------------------|--------------|----------|
| Short Name                           | User Defined | C        |
| Max Key Size <sup>1)</sup>           | User Defined | C        |
| Max Data Buffer Size <sup>2)</sup>   | User Defined | C        |
| Max Result Buffer Size <sup>3)</sup> | User Defined | C        |

- 1) CsmSymBlockDecryptConfig 설정 암호화 알고리즘 중 입력 가능 최대 Key 크기(byte)
  - Rte\_Type.h 파일에 typedef struct Csm\_SymKeyType 구조체의 Csm\_AlignType data[xxx] 배열 생성시 최대 크기를 결정하며 해당 data type 은 CsmSymBlockDecrypt Interface 의 SymBlockDecryptStart operation 의 인자로 사용된다.
  - Ex) AES 알고리즘 : 16
  - [Symmetric Block](#) 의 keyPtr→length 참조
- 2) CsmSymBlockDecryptConfig 설정 암호화 알고리즘 중 입력 데이터 버퍼 최대 크기(byte)
  - Rte\_Type.h 파일에 typedef uint8 SymBlockDecryptDataBuffer[xxx] 정의 생성시 배열 크기를 결정하며 해당 data type 은 CsmSymBlockDecrypt Interface 의 SymBlockDecryptUpdate operation 의 인자로 사용된다.
- 3) CsmSymBlockDecryptConfig 설정 암호화 알고리즘 중 결과 버퍼 최대 크기(byte)
  - Rte\_Type.h 파일에 typedef uint8 SymBlockDecryptResultBuffer[xxx] 정의 생성시 배열 크기를 결정하며 해당 data type 은 CsmSymBlockDecrypt Interface 의 SymBlockDecryptUpdate operation 의 인자로 사용된다.

### CsmSymBlockDecryptConfig Container

| Parameter Name                           | Value        | Category |
|------------------------------------------|--------------|----------|
| Short Name <sup>1)</sup>                 | User Defined | C        |
| Callback Sym Block Decrypt <sup>2)</sup> | User Defined | C        |
| Init Configuration <sup>3)</sup>         |              | N        |
| Primitive Name <sup>4)</sup>             | 아래 설명 사항 참조  | C        |
| Use Port <sup>5)</sup>                   | User Defined | C        |
| Use External Task <sup>6)</sup>          | false        | N        |

- 1) Short Name 은 Csm 모듈 내에서 unique 해야 함
- 2) 암호화 알고리즘 동작(ex. Start, Update, Finish, ...) 종료시 호출될 callback function  
 Callback function 등록시 : Async 동작 (Csm 내부 Task 에서 요청 처리 후 callback function 호출)  
 Use Port 사용시 [ShortName]\_Callback R-Port 와 연결하여 생성된 함수 이름을 등록해야 한다.  
 Callback function 미등록시 : Sync 동작 (User context 에서 해당 요청 처리)
- 3) 암호화 알고리즘 Start 시 전달되는 구조체 포인터  
 NULL\_PTR 혹은 미설정
- 4) 모듈에서 지원되는 암호화 알고리즘 명칭  
 ※ 설정 가능값 : HaeAesEcbDecrypt, AuthHsmAesEcbDecrypt, AutIcusAesEcbDecrypt,  
 AutCsecAesEcbDecrypt, HaeHsmAesEcbDecrypt  
 MCU 별 가능 알고리즘은 [Limitations](#) 참조
- 5) Rte Port Interface 지원 유무

true 설정시 Swcd\_Bsw\_Csm.arxml 파일의 Csm component 에 다음과 같은 설정이 자동 생성된다.

- a) [ShortName]\_SymBlockDecrypt 이름의 P-Port
  - b) Callback function 등록시 [ShortName]\_Callback 이름의 R-Port,  
SynchronousServerCallPoint\_[ShortName]\_Callback\_JobFinished 이벤트
  - c) OperationInvokedEvent\_[ShortName]\_SymBlockDecrypt\_SymBlockDecryptStart/Update/Finish 이벤트
  - d) [ShortName]\_SymBlockDecrypt 이름의 P-Port 와 연계된 Port API Option
- 6) 미사용

## CsmSymEncrypt Container

| Parameter Name                            | Value        | Category |
|-------------------------------------------|--------------|----------|
| Short Name                                | User Defined | C        |
| Max Key Size <sup>1)</sup>                | User Defined | C        |
| Max Init Vector Buffer Size <sup>2)</sup> | User Defined | C        |
| Max Data Buffer Size <sup>3)</sup>        | User Defined | C        |
| Max Result Buffer Size <sup>4)</sup>      | User Defined | C        |

### 1) CsmSymEncryptConfig 설정 암호화 알고리즘 중 입력 가능 최대 Key 크기(byte)

- Rte\_Type.h 파일에 typedef struct Csm\_SymKeyType 구조체의 Csm\_AlignType data[xxx] 배열 생성시 최대 크기를 결정하며 해당 data type 은 CsmSymEncrypt Interface 의 SymEncryptStart operation 의 인자로 사용된다.

Ex) AES 알고리즘 : 16

[Symmetric](#) 의 keyPtr→length 참조

### 2) CsmSymEncryptConfig 설정 암호화 알고리즘 중 Init Vector 버퍼 최대 크기(byte)

- Rte\_Type.h 파일에 typedef uint8 SymEncryptInitVectorBuffer[xxx] 정의 생성시 배열 크기를 결정하며 해당 data type 은 CsmSymEncrypt Interface 의 SymEncryptStart operation 의 인자로 사용된다.

Ex) AES 알고리즘 : 16

[Symmetric](#) 의 InitVectorLength 참조

### 3) CsmSymEncryptConfig 설정 암호화 알고리즘 중 입력 데이터 버퍼 최대 크기(byte)

- Rte\_Type.h 파일에 typedef uint8 SymEncryptDataBuffer[xxx] 정의 생성시 배열 크기를 결정하며 해당 data type 은 CsmSymEncrypt Interface 의 SymEncryptUpdate operation 의 인자로 사용된다.

### 4) CsmSymEncryptConfig 설정 암호화 알고리즘 중 결과 버퍼 최대 크기(byte)

- Rte\_Type.h 파일에 typedef uint8 SymEncryptResultBuffer[xxx] 정의 생성시 배열 크기를 결정하며 해당 data type 은 CsmSymEncrypt Interface 의 SymEncryptUpdate operation 의 인자로 사용된다.

## CsmSymEncryptConfig Container

| Parameter Name                     | Value        | Category |
|------------------------------------|--------------|----------|
| Short Name <sup>1)</sup>           | User Defined | C        |
| Callback Sym Encrypt <sup>2)</sup> | User Defined | C        |
| Init Configuration <sup>3)</sup>   |              | N        |
| Primitive Name <sup>4)</sup>       | 아래 설명 사항 참조  | C        |
| Use Port <sup>5)</sup>             | User Defined | C        |
| Use External Task <sup>6)</sup>    | false        | N        |

- Short Name 은 Csm 모듈 내에서 unique 해야 함
- 암호화 알고리즘 동작(ex. Start, Update, Finish, ...) 종료시 호출될 callback function  
 Callback function 등록시 : Async 동작 (Csm 내부 Task 에서 요청 처리 후 callback function 호출)  
 Use Port 사용시 [ShortName]\_Callback R-Port 와 연결하여 생성된 함수 이름을 등록해야 한다.  
 Callback function 미등록시 : Sync 동작 (User context 에서 해당 요청 처리)
- 암호화 알고리즘 Start 시 전달되는 구조체 포인터  
 NULL\_PTR 혹은 미설정
- 모듈에서 지원되는 암호화 알고리즘 명칭  
 ※ 설정 가능값 : HaeAesCbcEncrypt, HaeAesCtrEncrypt, AuthHsmAesCbcEncrypt, AuthHsmAesCtrEncrypt, AutIcusAesCbcEncrypt, AutCsecAesCbcEncrypt, HaeHsmAesCbcEncrypt  
 MCU 별 가능 알고리즘은 [Limitations](#) 참조
- Rte Port Interface 지원 유무  
 true 설정시 Swcd\_Bsw\_Csm.arxml 파일의 Csm component 에 다음과 같은 설정이 자동 생성된다.
  - [ShortName]\_SymEncrypt 이름의 P-Port
  - Callback function 등록시 [ShortName]\_Callback 이름의 R-Port,  
 SynchronousServerCallPoint\_[ShortName]\_Callback\_JobFinished 이벤트
  - OperationInvokedEvent\_[ShortName]\_SymEncrypt\_SymEncryptStart/Update/Finish 이벤트
  - [ShortName]\_SymEncrypt 이름의 P-Port 와 연계된 Port API Option
- 미사용

## CsmSymDecrypt Container

| Parameter Name                            | Value        | Category |
|-------------------------------------------|--------------|----------|
| Short Name                                | User Defined | C        |
| Max Key Size <sup>1)</sup>                | User Defined | C        |
| Max Init Vector Buffer Size <sup>2)</sup> | User Defined | C        |
| Max Data Buffer Size <sup>3)</sup>        | User Defined | C        |
| Max Result Buffer Size <sup>4)</sup>      | User Defined | C        |

## 1) CsmSymDecryptConfig 설정 암호화 알고리즘 중 입력 가능 최대 Key 크기(byte)

- Rte\_Type.h 파일에 typedef struct Csm\_SymKeyType 구조체의 Csm\_AlignType data[xxx] 배열 생성시 최대 크기를 결정하며 해당 data type 은 CsmSymDecrypt Interface 의 SymDecryptStart operation 의 인자로 사용된다.

Ex) AES 알고리즘 : 16

[Symmetric](#) 의 keyPtr→length 참조

## 2) CsmSymDecryptConfig 설정 암호화 알고리즘 중 Init Vector 버퍼 최대 크기(byte)

- Rte\_Type.h 파일에 typedef uint8 SymDecryptInitVectorBuffer[xxx] 정의 생성시 배열 크기를 결정하며 해당 data type 은 CsmSymDecrypt Interface 의 SymDecryptStart operation 의 인자로 사용된다.

Ex) AES 알고리즘 : 16

[Symmetric](#) 의 InitVectorLength 참조

## 3) CsmSymDecryptConfig 설정 암호화 알고리즘 중 입력 데이터 버퍼 최대 크기(byte)

- Rte\_Type.h 파일에 typedef uint8 SymDecryptDataBuffer[xxx] 정의 생성시 배열 크기를 결정하며 해당 data type 은 CsmSymDecrypt Interface 의 SymDecryptUpdate operation 의 인자로 사용된다.

## 4) CsmSymDecryptConfig 설정 암호화 알고리즘 중 결과 버퍼 최대 크기(byte)

- Rte\_Type.h 파일에 typedef uint8 SymDecryptResultBuffer[xxx] 정의 생성시 배열 크기를 결정하며 해당 data type 은 CsmSymDecrypt Interface 의 SymDecryptUpdate operation 의 인자로 사용된다.

5.10.1

## CsmSymDecryptConfig Container

| Parameter Name                     | Value        | Category |
|------------------------------------|--------------|----------|
| Short Name <sup>1)</sup>           | User Defined | C        |
| Callback Sym Decrypt <sup>2)</sup> | User Defined | C        |
| Init Configuration <sup>3)</sup>   |              | N        |
| Primitive Name <sup>4)</sup>       | 아래 설명 사항 참조  | C        |
| Use Port <sup>5)</sup>             | User Defined | C        |
| Use External Task <sup>6)</sup>    | false        | N        |

1) Short Name 은 Csm 모듈 내에서 unique 해야 함

2) 암호화 알고리즘 동작(ex. Start, Update, Finish, ...) 종료시 호출될 callback function

Callback function 등록시 : Async 동작 (Csm 내부 Task 에서 요청 처리 후 callback function 호출)

Use Port 사용시 [ShortName]\_Callback R-Port 와 연결하여 생성된 함수 이름을 등록해야 한다.

Callback function 미등록시 : Sync 동작 (User context 에서 해당 요청 처리)

3) 암호화 알고리즘 Start 시 전달되는 구조체 포인터

NULL\_PTR 혹은 미설정

4) 모듈에서 지원되는 암호화 알고리즘 명칭

※ 설정 가능값 : HaeAesCbcDecrypt, HaeAesCtrDecrypt, AuthSmAesCbcDecrypt, AuthSmAesCtrDecrypt, AutIcusAesCbcDecrypt, AutCsecAesCbcDecrypt, HaeHsmAesCtrDecrypt

MCU 별 가능 알고리즘은 [Limitations](#) 참조

5) Rte Port Interface 지원 유무

true 설정시 Swcd\_Bsw\_Csm.arxml 파일의 Csm component 에 다음과 같은 설정이 자동 생성된다.

a) [ShortName]\_SymDecrypt 이름의 P-Port

b) Callback function 등록시 [ShortName]\_Callback 이름의 R-Port,

SynchronousServerCallPoint\_[ShortName]\_Callback\_JobFinished 이벤트

c) OperationInvokedEvent\_[ShortName]\_SymDecrypt\_SymDecryptStart/Update/Finish 이벤트

d) [ShortName]\_SymDecrypt 이름의 P-Port 와 연계된 Port API Option

6) 미사용

## CsmAsymEncrypt Container

5.11

| Parameter Name                       | Value        | Category |
|--------------------------------------|--------------|----------|
| Short Name                           | User Defined | C        |
| Max Key Size <sup>1)</sup>           | User Defined | C        |
| Max Data Buffer Size <sup>2)</sup>   | User Defined | C        |
| Max Result Buffer Size <sup>3)</sup> | User Defined | C        |

1) CsmAsymEncryptConfig 설정 암호화 알고리즘 중 입력 가능 최대 Key 크기(byte)

- Rte\_Type.h 파일에 typedef struct Csm\_AsymPublicKeyType 구조체의 Csm\_AlignType data[xxx] 배열 생성시 최대 크기를 결정하며 해당 data type 은 CsmAsymEncrypt Interface 의 AsymEncryptStart operation 의 인자로 사용된다.

Ex) RSA 알고리즘 : 260

ECIES 알고리즘 : 164

[Asymmetric](#) 의 keyPtr→length 참조

2) CsmAsymEncryptConfig 설정 암호화 알고리즘 중 입력 데이터 버퍼 최대 크기(byte)

- Rte\_Type.h 파일에 typedef uint8 AsymEncryptDataBuffer[xxx] 정의 생성시 배열 크기를 결정하며 해당 data type 은 CsmAsymEncrypt Interface 의 AsymEncryptUpdate operation 의 인자로 사용된다.

3) CsmAsymEncryptConfig 설정 암호화 알고리즘 중 결과 버퍼 최대 크기(byte)

- Rte\_Type.h 파일에 typedef uint8 AsymEncryptResultBuffer[xxx] 정의 생성시 배열 크기를 결정하며 해당 data type 은 CsmAsymEncrypt Interface 의 AsymEncryptUpdate operation 의 인자로 사용된다.

## CsmSymEncryptConfig Container

| Parameter Name                      | Value        | Category |
|-------------------------------------|--------------|----------|
| Short Name <sup>1)</sup>            | User Defined | C        |
| Callback Asym Encrypt <sup>2)</sup> | User Defined | C        |
| Init Configuration <sup>3)</sup>    |              | N        |
| Primitive Name <sup>4)</sup>        | 아래 설명 사항 참조  | C        |
| Use Port <sup>5)</sup>              | User Defined | C        |
| Use External Task <sup>6)</sup>     | false        | N        |

- Short Name 은 Csm 모듈 내에서 unique 해야 함
- 암호화 알고리즘 동작(ex. Start, Update, Finish, ...) 종료시 호출될 callback function  
 Callback function 등록시 : Async 동작 (Csm 내부 Task 에서 요청 처리 후 callback function 호출)  
 Use Port 사용시 [ShortName]\_Callback R-Port 와 연결하여 생성된 함수 이름을 등록해야 한다.  
 Callback function 미등록시 : Sync 동작 (User context 에서 해당 요청 처리)
- 암호화 알고리즘 Start 시 전달되는 구조체 포인터  
 NULL\_PTR 혹은 미설정
- 모듈에서 지원되는 암호화 알고리즘 명칭  
 ※ 설정 가능값 : HaeRsaEncrypt, AuthHsmRsaEncrypt  
 MCU 별 가능 알고리즘은 [Limitations](#) 참조
- Rte Port Interface 지원 유무  
 true 설정시 Swcd\_Bsw\_Csm.arxml 파일의 Csm component 에 다음과 같은 설정이 자동 생성된다.
  - [ShortName]\_AsymEncrypt 이름의 P-Port
  - Callback function 등록시 [ShortName]\_Callback 이름의 R-Port,  
 SynchronousServerCallPoint\_[ShortName]\_Callback\_JobFinished 이벤트
  - OperationInvokedEvent\_[ShortName]\_AsymEncrypt\_AsymEncryptStart/Update/Finish 이벤트
  - [ShortName]\_AsymEncrypt 이름의 P-Port 와 연계된 Port API Option
- 미사용

## CsmAsymDecrypt Container

| Parameter Name                       | Value        | Category |
|--------------------------------------|--------------|----------|
| Short Name                           | User Defined | C        |
| Max Key Size <sup>1)</sup>           | User Defined | C        |
| Max Data Buffer Size <sup>3)</sup>   | User Defined | C        |
| Max Result Buffer Size <sup>4)</sup> | User Defined | C        |

- CsmAsymDecryptConfig 설정 암호화 알고리즘 중 입력 가능 최대 Key 크기(byte)

- Rte\_Type.h 파일에 typedef struct Csm\_AsymPrivateKeyType 구조체의 Csm\_AlignType data[xxx] 배열 생성시 최대 크기를 결정하며 해당 data type 은 CsmAsymDecrypt Interface 의 AsymDecryptStart operation 의 인자로 사용된다.

Ex) RSA 알고리즘 : 512

ECIES 알고리즘 : 32

[Asymmetric](#) 의 keyPtr→length 참조

## 2) CsmAsymDecryptConfig 설정 암호화 알고리즘 중 입력 데이터 버퍼 최대 크기(byte)

- Rte\_Type.h 파일에 typedef uint8 AsymDecryptDataBuffer[xxx] 정의 생성시 배열 크기를 결정하며 해당 data type 은 CsmAsymDecrypt Interface 의 AsymDecryptUpdate operation 의 인자로 사용된다.

## 3) CsmAsymDecryptConfig 설정 암호화 알고리즘 중 결과 버퍼 최대 크기(byte)

- Rte\_Type.h 파일에 typedef uint8 AsymDecryptResultBuffer[xxx] 정의 생성시 배열 크기를 결정하며 해당 data type 은 CsmAsymDecrypt Interface 의 AsymDecryptUpdate operation 의 인자로 사용된다.

## CsmAsymDecryptConfig Container

5.12.1

| Parameter Name                      | Value        | Category |
|-------------------------------------|--------------|----------|
| Short Name <sup>1)</sup>            | User Defined | C        |
| Callback Asym Decrypt <sup>2)</sup> | User Defined | C        |
| Init Configuration <sup>3)</sup>    |              | N        |
| Primitive Name <sup>4)</sup>        | 아래 설명 사항 참조  | C        |
| Use Port <sup>5)</sup>              | User Defined | C        |
| Use External Task <sup>6)</sup>     | false        | N        |

1) Short Name 은 Csm 모듈 내에서 unique 해야 함

2) 암호화 알고리즘 동작(ex. Start, Update, Finish, ...) 종료시 호출될 callback function

Callback function 등록시 : Async 동작 (Csm 내부 Task 에서 요청 처리 후 callback function 호출)

Use Port 사용시 [ShortName]\_Callback R-Port 와 연결하여 생성된 함수 이름을 등록해야 한다.

Callback function 미등록시 : Sync 동작 (User context 에서 해당 요청 처리)

3) 암호화 알고리즘 Start 시 전달되는 구조체 포인터

NULL\_PTR 혹은 미설정

4) 모듈에서 지원되는 암호화 알고리즘 명칭

※ 설정 가능값 : HaeRsaDecrypt, AuthsmRsaDecrypt

MCU 별 가능 알고리즘은 [Limitations](#) 참조

5) Rte Port Interface 지원 유무

true 설정시 Swcd\_Bsw\_Csm.arxml 파일의 Csm component 에 다음과 같은 설정이 자동 생성된다.

a) [ShortName]\_AsymDecrypt 이름의 P-Port

- b) Callback function 등록시 [ShortName]\_Callback 이름의 R-Port,  
SynchronousServerCallPoint\_[ShortName]\_Callback\_JobFinished 이벤트
- c) OperationInvokedEvent\_[ShortName]\_AsymDecrypt\_AsymDecryptStart/Update/Finish 이벤트
- d) [ShortName]\_AsymDecrypt 이름의 P-Port 와 연계된 Port API Option
- 6) 미사용

## CsmSignatureGenerate Container

| Parameter Name                       | Value        | Category |
|--------------------------------------|--------------|----------|
| Short Name                           | User Defined | C        |
| Max Key Size <sup>1)</sup>           | User Defined | C        |
| Max Data Buffer Size <sup>2)</sup>   | User Defined | C        |
| Max Result Buffer Size <sup>3)</sup> | User Defined | C        |

- 1) CsmSignatureGenerateConfig 설정 암호화 알고리즘 중 사용 키 최대 크기(byte)
  - Rte\_Type.h 파일에 typedef struct Csm\_AsymPrivateKeyType 구조체의 Csm\_AlignType data[xxx] 배열 생성시 최대 크기를 결정하며 해당 data type 은 CsmSignatureGenerate Interface 의 SignatureGenerateStart operation 의 인자로 사용된다.
  - Ex) PKCS#1 알고리즘 : 512  
ECDSA 알고리즘 : 32  
[Signature](#) 의 keyPtr→length 참조
- 2) CsmSignatureGenerateConfig 설정 암호화 알고리즘 중 입력 데이터 버퍼 최대 크기(byte)
  - Rte\_Type.h 파일에 typedef uint8 SignatureGenerateDataBuffer[xxx] 정의 생성시 배열 크기를 결정하며 해당 data type 은 CsmSignatureGenerate Interface 의 SignatureGenerateUpdate operation 의 인자로 사용된다.
- 3) CsmSignatureGenerateConfig 설정 암호화 알고리즘 중 결과 버퍼 최대 크기(byte)
  - Rte\_Type.h 파일에 typedef uint8 SignatureGenerateResultBuffer[xxx] 정의 생성시 배열 크기를 결정하며 해당 data type 은 CsmSignatureGenerate Interface 의 SignatureGenerateFinish operation 의 인자로 사용된다.
  - Ex) PKCS#1 알고리즘 : 256  
ECDSA 알고리즘 : 64  
[Signature](#) 의 \*resultLengthPtr 참조

## CsmSignatureGenerateConfig Container

| Parameter Name                            | Value        | Category |
|-------------------------------------------|--------------|----------|
| Short Name <sup>1)</sup>                  | User Defined | C        |
| Callback Signature Generate <sup>2)</sup> | User Defined | C        |
| Init Configuration <sup>3)</sup>          |              | N        |
| Primitive Name <sup>4)</sup>              | 아래 설명 사항 참조  | C        |
| Use Port <sup>5)</sup>                    | User Defined | C        |
| Use External Task <sup>6)</sup>           | false        | N        |

- Short Name 은 Csm 모듈 내에서 unique 해야 함
- 암호화 알고리즘 동작(ex. Start, Update, Finish, ...) 종료시 호출될 callback function  
 Callback function 등록시 : Async 동작 (Csm 내부 Task 에서 요청 처리 후 callback function 호출)  
 Use Port 사용시 [ShortName]\_Callback R-Port 와 연결하여 생성된 함수 이름을 등록해야 한다.  
 Callback function 미등록시 : Sync 동작 (User context 에서 해당 요청 처리)
- 암호화 알고리즘 Start 시 전달되는 구조체 포인터  
 NULL\_PTR 혹은 미설정
- 모듈에서 지원되는 암호화 알고리즘 명칭  
 ※ 설정 가능값 : HaePkcs1V15Sha1Generate, HaePkcs1V15Sha256Generate,  
 AuthSmPkcs1V15Sha1Generate, AuthSmPkcs1V15Sha256Generate, HaeHsmPkcs1V15Sha256Generate,  
 AuthSmPkcs1PssSha256Generate, AuthHsmEcdsaSha256Generate, HaeHsmPkcs1PssSha256Generate,  
 HaeHsmEcdsaSha256Generate  
 MCU 별 가능 알고리즘은 [Limitations](#) 참조
- Rte Port Interface 지원 유무  
 true 설정시 Swcd\_Bsw\_Csm.arxml 파일의 Csm component 에 다음과 같은 설정이 자동 생성된다.
  - [ShortName]\_SignatureGenerate 이름의 P-Port
  - Callback function 등록시 [ShortName]\_Callback 이름의 R-Port,  
 SynchronousServerCallPoint\_[ShortName]\_Callback\_JobFinished 이벤트
  - OperationInvokedEvent\_[ShortName]\_SignatureGenerate\_SignatureGenerateStart/Update/Finish 이벤트
  - [ShortName]\_SignatureGenerate 이름의 P-Port 와 연계된 Port API Option
- 미사용

## CsmSignatureVerify Container

| Parameter Name             | Value        | Category |
|----------------------------|--------------|----------|
| Short Name                 | User Defined | C        |
| Max Key Size <sup>1)</sup> | User Defined | C        |

| Parameter Name                                  | Value        | Category |
|-------------------------------------------------|--------------|----------|
| Max Data Buffer Size <sup>2)</sup>              | User Defined | C        |
| Max Compare Signature Buffer Size <sup>3)</sup> | User Defined | C        |

1) CsmSignatureVerifyConfig 설정 암호화 알고리즘 중 사용 키 최대 크기(byte)

- Rte\_Type.h 파일에 typedef struct Csm\_AsymPublicKeyType 구조체의 Csm\_AlignType data[xxx] 배열 생성시 최대 크기를 결정하며 해당 data type 은 CsmSignatureVerify Interface 의 SignatureVerifyStart operation 의 인자로 사용된다.

Ex) PKCS#1 알고리즘 : 260

ECDSA 알고리즘 : 64

[Signature](#) 의 keyPtr→length 참조

2) CsmSignatureVerifyConfig 설정 암호화 알고리즘 중 입력 데이터 버퍼 최대 크기(byte)

- Rte\_Type.h 파일에 typedef uint8 SignatureVerifyDataBuffer[xxx] 정의 생성시 배열 크기를 결정하며 해당 data type 은 CsmSignatureVerify Interface 의 SignatureVerifyUpdate operation 의 인자로 사용된다.

3) CsmSignatureVerifyConfig 설정 암호화 알고리즘 중 비교 대상 Signature 버퍼 최대 크기(byte)

- Rte\_Type.h 파일에 typedef uint8 SignatureVerifyCompareSignatureBuffer[xxx] 정의 생성시 배열 크기를 결정하며 해당 data type 은 CsmSignatureVerify Interface 의 SignatureVerifyFinish operation 의 인자로 사용된다.

Ex) PKCS#1 알고리즘 : 256

ECDSA 알고리즘 : 64

[Signature](#) 의 signatureLength 참조

5.14.1

## CsmSignatureVerifyConfig Container

| Parameter Name                          | Value        | Category |
|-----------------------------------------|--------------|----------|
| Short Name <sup>1)</sup>                | User Defined | C        |
| Callback Signature Verify <sup>2)</sup> | User Defined | C        |
| Init Configuration <sup>3)</sup>        |              | N        |
| Primitive Name <sup>4)</sup>            | 아래 설명 사항 참조  | C        |
| Use Port <sup>5)</sup>                  | User Defined | C        |
| Use External Task <sup>6)</sup>         | false        | N        |

1) Short Name 은 Csm 모듈 내에서 unique 해야 함

2) 암호화 알고리즘 동작(ex. Start, Update, Finish, ...) 종료시 호출될 callback function

Callback function 등록시 : Async 동작 (Csm 내부 Task 에서 요청 처리 후 callback function 호출)

Use Port 사용시 [ShortName]\_Callback R-Port 와 연결하여 생성된 함수 이름을 등록해야 한다.

Callback function 미등록시 : Sync 동작 (User context 에서 해당 요청 처리)

3) 암호화 알고리즘 Start 시 전달되는 구조체 포인터

NULL\_PTR 혹은 미설정

4) 모듈에서 지원되는 암호화 알고리즘 명칭

※ 설정 가능값 : HaePkcs1V15Sha1Verify, HaePkcs1V15Sha1SeedVerify, HaePkcs1V15Sha256Verify, HaePkcs1V15Sha256SeedVerify, AuthHsmPkcs1V15Sha1Verify, AuthHsmPkcs1V15Sha1SeedVerify, AuthHsmPkcs1V15Sha256Verify, AuthHsmPkcs1V15Sha256SeedVerify, HaeHsmPkcs1V15Sha1Verify, HaeHsmPkcs1V15Sha1SeedVerify, HaeHsmPkcs1V15Sha256Verify, HaeHsmPkcs1V15Sha256SeedVerify, AuthHsmPkcs1PssSha256Verify, AuthHsmEcdsaSha256Verify, HaeHsmPkcs1PssSha256Verify, HaeHsmEcdsaSha256Verify

MCU 별 가능 알고리즘은 [Limitations](#) 참조

5) Rte Port Interface 지원 유무

true 설정시 Swcd\_Bsw\_Csm.arxml 파일의 Csm component 에 다음과 같은 설정이 자동 생성된다.

a) [ShortName]\_SignatureVerify 이름의 P-Port

b) Callback function 등록시 [ShortName]\_Callback 이름의 R-Port,

SynchronousServerCallPoint\_[ShortName]\_Callback\_JobFinished 이벤트

c) OperationInvokedEvent\_[ShortName]\_SignatureVerify\_SignatureVerifyStart/Update/Finish 이벤트

d) [ShortName]\_SignatureVerify 이름의 P-Port 와 연계된 Port API Option

6) 미사용

5.15

## CsmKeyDerive Container

| Parameter Name                         | Value        | Category |
|----------------------------------------|--------------|----------|
| Short Name                             | User Defined | C        |
| Max Key Size <sup>1)</sup>             | User Defined | C        |
| Max Password Buffer Size <sup>2)</sup> | User Defined | C        |
| Max Salt Buffer Size <sup>3)</sup>     | User Defined | C        |

1) CsmKeyDeriveConfig 설정 암호화 알고리즘 중 출력 키 최대 크기(byte)

- Rte\_Type.h 파일에 typedef struct Csm\_SymKeyType 구조체의 Csm\_AlignType data[xxx] 배열 생성시 최대 크기를 결정하며 해당 data type 은 CsmKeyDerive Interface 의 KeyDeriveFinish operation 의 인자로 사용된다.

[Key Derive](#) 의 keyPtr->length 참조

2) CsmKeyDeriveConfig 설정 암호화 알고리즘 중 입력 Password 버퍼 최대 크기(byte)

- Rte\_Type.h 파일에 typedef uint8 KeyDerivePasswordBuffer[xxx] 정의 생성시 배열 크기를 결정하며 해당 data type 은 CsmKeyDerive Interface 의 KeyDeriveUpdate operation 의 인자로 사용된다.

### 3) CsmKeyDeriveConfig 설정 암호화 알고리즘 중 입력 Salt 버퍼 최대 크기(byte)

- Rte\_Type.h 파일에 typedef uint8 KeyDeriveSaltBuffer[xxx] 정의 생성시 배열 크기를 결정하며 해당 data type 은 CsmKeyDerive Interface 의 KeyDeriveUpdate operation 의 인자로 사용된다.

## CsmKeyDeriveConfig Container

| Parameter Name                    | Value        | Category |
|-----------------------------------|--------------|----------|
| Short Name <sup>1)</sup>          | User Defined | C        |
| Callback Key Derive <sup>2)</sup> | User Defined | C        |
| Init Configuration <sup>3)</sup>  |              | N        |
| Primitive Name <sup>4)</sup>      | 아래 설명 사항 참조  | C        |
| Use Port <sup>5)</sup>            | User Defined | C        |
| Use External Task <sup>6)</sup>   | false        | N        |

1) Short Name 은 Csm 모듈 내에서 unique 해야 함

2) 암호화 알고리즘 동작(ex. Start, Update, Finish, ...) 종료시 호출될 callback function

Callback function 등록시 : Async 동작 (Csm 내부 Task 에서 요청 처리 후 callback function 호출)

Use Port 사용시 [ShortName]\_Callback R-Port 와 연결하여 생성된 함수 이름을 등록해야 한다.

Callback function 미등록시 : Sync 동작 (User context 에서 해당 요청 처리)

3) 암호화 알고리즘 Start 시 전달되는 구조체 포인터

NULL\_PTR 혹은 미설정

4) 모듈에서 지원되는 암호화 알고리즘 명칭

※ 설정 가능값 : AuthHsmPbkdf2, HaeHsmPbkdf2

MCU 별 가능 알고리즘은 [Limitations](#) 참조

5) Rte Port Interface 지원 유무

true 설정시 Swcd\_Bsw\_Csm.arxml 파일의 Csm component 에 다음과 같은 설정이 자동 생성된다.

a) [ShortName]\_KeyDerive 이름의 P-Port

b) Callback function 등록시 [ShortName]\_Callback 이름의 R-Port,

SynchronousServerCallPoint\_[ShortName]\_Callback\_JobFinished 이벤트

c) OperationInvokedEvent\_[ShortName]\_KeyDerive\_KeyDeriveStart/Update/Finish 이벤트

d) [ShortName]\_KeyDerive 이름의 P-Port 와 연계된 Port API Option

6) 미사용

## CsmKeyExchangeCalcSecret Container

| Parameter Name                                     | Value        | Category |
|----------------------------------------------------|--------------|----------|
| Short Name                                         | User Defined | C        |
| Max Base Size <sup>1)</sup>                        |              | N        |
| Max Private Size <sup>2)</sup>                     | User Defined | C        |
| Max Partner Public Value Buffer Size <sup>3)</sup> | User Defined | C        |
| Max Shared Secret Buffer Size <sup>3)</sup>        | User Defined | C        |

- 1) CsmKeyExchangeCalcSecretConfig 설정 암호화 알고리즘 중 Base 정보 최대 크기(byte)
  - 미사용
- 2) CsmKeyExchangeCalcSecretConfig 설정 암호화 알고리즘 중 입력 Private 정보 버퍼 최대 크기(byte)
  - Rte\_Type.h 파일에 typedef struct Csm\_KeyExchangePrivateType 구조체의 Csm\_AlignType data[xxx] 배열 생성시 최대 크기를 결정하며 해당 data type 은 CsmKeyExchangeCalcSecret Interface 의 KeyExchangeCalcSecretStart operation 의 인자로 사용된다.
  - [Key Exchange Calc Secret](#) 의 privateValuePtr->length 참조
- 3) CsmKeyExchangeCalcSecretConfig 설정 암호화 알고리즘 중 입력 상대방 Public 정보 버퍼 최대 크기(byte)
  - Rte\_Type.h 파일에 typedef uint8 KeyExchangePartnerPublicValueBuffer[xxx] 정의 생성시 배열 크기를 결정하며 해당 data type 은 CsmKeyExchangeCalcSecret Interface 의 KeyExchangeCalcSecretUpdate operation 의 인자로 사용된다.
- 4) CsmKeyExchangeCalcSecretConfig 설정 암호화 알고리즘 중 출력 Shared Secret 정보 버퍼 최대 크기(byte)
  - Rte\_Type.h 파일에 typedef uint8 KeyExchangeSharedSecretBuffer[xxx] 정의 생성시 배열 크기를 결정하며 해당 data type 은 CsmKeyExchangeCalcSecret Interface 의 KeyExchangeCalcSecretFinish operation 의 인자로 사용된다.

5.16.1

### CsmKeyExchangeCalcSecretConfig Container

| Parameter Name                    | Value        | Category |
|-----------------------------------|--------------|----------|
| Short Name <sup>1)</sup>          | User Defined | C        |
| Callback Key Derive <sup>2)</sup> | User Defined | C        |
| Init Configuration <sup>3)</sup>  |              | N        |
| Primitive Name <sup>4)</sup>      | 아래 설명 사항 참조  | C        |
| Use Port <sup>5)</sup>            | User Defined | C        |
| Use External Task <sup>6)</sup>   | false        | N        |

- 1) Short Name 은 Csm 모듈 내에서 unique 해야 함
- 2) 암호화 알고리즘 동작(ex. Start, Update, Finish, ...) 종료시 호출될 callback function  
 Callback function 등록시 : Async 동작 (Csm 내부 Task 에서 요청 처리 후 callback function 호출)  
 Use Port 사용시 [ShortName]\_Callback R-Port 와 연결하여 생성된 함수 이름을 등록해야 한다.

Callback function 미등록시 : Sync 동작 (User context 에서 해당 요청 처리)

3) 암호화 알고리즘 Start 시 전달되는 구조체 포인터

NULL\_PTR 혹은 미설정

4) 모듈에서 지원되는 암호화 알고리즘 명칭

※ 설정 가능값 : AuthHsmEcdh, HaeHsmEcdh

MCU 별 가능 알고리즘은 [Limitations](#) 참조

5) Rte Port Interface 지원 유무

true 설정시 Swcd\_Bsw\_Csm.arxml 파일의 Csm component 에 다음과 같은 설정이 자동 생성된다.

a) [ShortName]\_KeyExchangeCalcSecret 이름의 P-Port

b) Callback function 등록시 [ShortName]\_Callback 이름의 R-Port,

SynchronousServerCallPoint\_[ShortName]\_Callback\_JobFinished 이벤트

c) OperationInvokedEvent\_[ShortName]\_

KeyExchangeCalcSecret\_KeyExchangeCalcSecretStart/Update/Finish 이벤트

d) [ShortName]\_KeyExchangeCalcSecret 이름의 P-Port 와 연계된 Port API Option

6) 미사용

## 6. Application Programming Interface (API)

### Type Definitions

#### Csm\_CallbackType

|      |              |                                                                                                                                                             |
|------|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 6.1  | Name:        | Csm_CallbackType                                                                                                                                            |
| 6.1. | Type         | Std_FunctionPointer                                                                                                                                         |
| 6.1. | Description: | Function pointer for the callback function which will be invoked after service completion.<br>Signature: Std_ReturnType (*Csm_CallbackType)(Std_ReturnType) |

#### Csm\_ConfigIdType

|      |              |                                                                                                                                                                                                                                                            |
|------|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 6.1. | Name:        | Csm_ConfigIdType                                                                                                                                                                                                                                           |
| 6.1. | Type         | uint16                                                                                                                                                                                                                                                     |
| 6.1. | Range        | 0..65535                                                                                                                                                                                                                                                   |
| 6.1. | Description: | Identification of a CSM service configuration via a numeric identifier that is unique within a service.<br>The name of a CSM service configuration, i.e. the name of the container Csm_<Service>Config, shall serve as a symbolic name for this parameter. |

#### 6.1.3 Csm\_<Service>ConfigType

|       |                         |                                                    |                                                                                                                                                                                                                                                                           |
|-------|-------------------------|----------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Name: | Csm_<Service>ConfigType |                                                    |                                                                                                                                                                                                                                                                           |
| Type  | Structure               |                                                    |                                                                                                                                                                                                                                                                           |
|       | Csm_CallbackType        | CallbackFct                                        | A pointer to the callback function which shall be called when the configured service has finished. This Element is only available if "CsmUseSyncJobProcessing" is "OFF". (CSM0557_Conf)                                                                                   |
|       | Std_ReturnType          | (*PrimitiveStartFct) (<primitive parameter list>)  | This element shall only exist if the service contains the function Csm_<Service>Start. It is a pointer to the function Cry_<Primitive>Start of the configured cryptographic primitive. For the "primitive parameter list" see the description of Cry_<Primitive>Start.    |
|       | Std_ReturnType          | (*PrimitiveUpdateFct) (<primitive parameter list>) | This element shall only exist if the service contains the function Csm_<Service>Update. It is a pointer to the function Cry_<Primitive>Update of the configured cryptographic primitive. For the "primitive parameter list" see the description of Cry_<Primitive>Update. |
|       | Std_ReturnType          | (*PrimitiveFinishFct) (<primitive parameter list>) | This element shall only exist if the service contains the function Csm_<Service>Finish. It is a pointer to the function Cry_<Primitive>Finish of the configured cryptographic primitive. For the "primitive parameter list" see the description of Cry_<Primitive>Finish. |

|              |                                                                                                                                                                                                                        |                                                  |                                                                                                                                                                                                                                                         |
|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|              |                                                                                                                                                                                                                        | list>>                                           | of the configured cryptographic primitive.<br>For the "primitive parameter list" see the description of Cry_<Primitive>Finish.                                                                                                                          |
|              | Std_Return Type                                                                                                                                                                                                        | (*PrimitiveFct) (<primitive parameter list>)     | This element shall only exist if the service contains the function Csm_<Service>. It is a pointer to the function Cry_<Primitive> of the configured cryptographic primitive. For the "primitive parameter list" see the description of Cry_<Primitive>. |
|              | Std_Return Type                                                                                                                                                                                                        | (*PrimitiveMainFct) (<primitive parameter list>) | A pointer to the function Cry_<Primitive>MainFunction of the configured cryptographic primitive.                                                                                                                                                        |
|              | void                                                                                                                                                                                                                   | *PrimitiveConfigPtr                              | A pointer to the configuration of the underlying cryptographic primitive                                                                                                                                                                                |
| Description: | Data structure which shall encompass all information needed to specify the cryptographic primitives needed for the <Service> cryptographic service. It shall furthermore contain information on the callback function. |                                                  |                                                                                                                                                                                                                                                         |

## Csm\_AlignType

6.1.4

|              |                                                                                                                                                                                                                                                                                                                                         |  |  |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|
| Name:        | Csm_AlignType                                                                                                                                                                                                                                                                                                                           |  |  |
| Type         | <maxAlignScalarType>                                                                                                                                                                                                                                                                                                                    |  |  |
| Description: | <p>A sCalar type which has maximum alignment restrictions on the given platform. This value is configured by "CsmMaxAlignScalarType".<br/>&lt;maxAlignScalarType&gt; can be e.g. uint8, uint16 or uint32.</p> <p>All context buffers shall be aligned according to the maximum alignment of all scalar types on the given platform.</p> |  |  |

6.1.5

## Csm\_VerifyResultType

|              |                                                            |                                                                                                                                               |  |
|--------------|------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|--|
| Name:        | Csm_VerifyResultType                                       |                                                                                                                                               |  |
| Type         | Enumeration                                                |                                                                                                                                               |  |
| Range:       | CSM_E_VER_OK                                               | The result of the verification is "true", i.e. the two compared elements are identical.<br>This return code shall be given as value "0"       |  |
|              | CSM_E_VER_NOT_OK                                           | The result of the verification is "false", i.e. the two compared elements are not identical.<br>This return code shall be given as value "1". |  |
| Description: | Enumeration of the result type of verification operations. |                                                                                                                                               |  |

6.1.5

## Csm\_AsymPublicKeyType

|        |                       |        |                                                       |
|--------|-----------------------|--------|-------------------------------------------------------|
| Name:  | Csm_AsymPublicKeyType |        |                                                       |
| Type   | Structure             |        |                                                       |
| Range: | uint32                | length | This element contains the length of the key stored in |

|              |                                                                                                                                                                                                                                                                                                                                                                                      |      |                                                     |
|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-----------------------------------------------------|
|              |                                                                                                                                                                                                                                                                                                                                                                                      |      | element 'data'                                      |
|              | Csm_AlignType<br>[CSM_ASYM_PUB_KEY_MAX_SIZE]                                                                                                                                                                                                                                                                                                                                         | data | This element contains the key data or a key handle. |
| Description: | Structure for the public asymmetrical key.<br>CSM_ASYM_PUB_KEY_MAX_SIZE shall be chosen such that "CSM_ASYM_PUB_KEY_MAX_SIZE * sizeof(Csm_AlignType)" is greater or equal to the maximum of the configured values CsmAsymEncryptMaxKeySize, CsmSignatureVerifyMaxKeySize, CsmAsymPublicKeyExtractMaxKeySize, CsmSymKeyWrapAsymMaxPubKeySize and CsmAsymPrivateKeyWrapAsymPubKeySize. |      |                                                     |

### Csm\_AsymPrivateKeyType

|       |              |                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                             |
|-------|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| 6.1.1 | Name:        | Csm_AsymPrivateKeyType                                                                                                                                                                                                                                                                                                                                                                                 |                                                                             |
|       | Type         | Structure                                                                                                                                                                                                                                                                                                                                                                                              |                                                                             |
|       | Range:       | uint32                                                                                                                                                                                                                                                                                                                                                                                                 | length This element contains the length of the key stored in element 'data' |
|       |              | Csm_AlignType[CSM_ASYM_PRIV_KEY_MAX_SIZE]                                                                                                                                                                                                                                                                                                                                                              | data This element contains the key data or a key handle.                    |
| 6.1.8 | Description: | Structure for the private asymmetrical key.<br>CSM_ASYM_PRIV_KEY_MAX_SIZE shall be chosen such that "CSM_ASYM_PRIV_KEY_MAX_SIZE * sizeof(Csm_AlignType)" is greater or equal to the maximum of the configured values CsmAsymDecryptMaxKeySize, CsmSignatureGenerateMaxKeySize, CsmAsymPrivateKeyExtractMaxKeySize, CsmAsymPrivateKeyWrapSymMaxPrivKeySize and CsmAsymPrivateKeyWrapAsymMaxPrivKeySize. |                                                                             |

### Csm\_SymKeyType

|  |              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                             |
|--|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
|  | Name:        | Csm_SymKeyType                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                             |
|  | Type         | Structure                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                             |
|  | Range:       | uint32                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | length This element contains the length of the key stored in element 'data' |
|  |              | Csm_AlignType<br>[CSM_SYM_KEY_MAX_SIZE]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | data This element contains the key data or a key handle.                    |
|  | Description: | Structure for the symmetrical key. CSM_SYM_KEY_MAX_SIZE shall be chosen such that "CSM_SYM_KEY_MAX_SIZE * sizeof(Csm_AlignType)" is greater or equal to the maximum of the configured values CsmSymBlockEncryptMaxKeySize, CsmSymBlockDecryptMaxKeySize, CsmSymEncryptMaxKeySize, CsmSymDecryptMaxKeySize, CsmKeyDeriveMaxKeySize, CsmSymKeyExtractMaxKeySize, CsmMacGenerateMaxKeySize and CsmMacVerifyMaxKeySize, CsmSymKeyWrapSymMaxSymKeySize, CsmSymKeyWrapAsymMaxSymKeySize, CsmAsymPrivateKeyWrapSymMaxSymKeySize, CsmKeyExchangeCalcSymKeyMaxSymKeySize and CsmKeyDeriveSymKeyMaxSymKeySize. |                                                                             |

### Csm\_AuthHsmPkcs1PssDataType

|              |                                         |              |                     |
|--------------|-----------------------------------------|--------------|---------------------|
| Name:        | Csm_AuthHsmPkcs1PssDataType             |              |                     |
| Type         | Structure                               |              |                     |
| Range:       | const uint8*                            | pData        | signing message     |
|              | const uint8*                            | pSalt        | salt text           |
|              | uint32                                  | ulDataLength | length of message   |
|              | uint32                                  | ulSaltLength | length of salt text |
| Description: | Structure for the PKCS#1 PSS algorithm. |              |                     |

### Csm\_AuthHsmEcdsaDataType

|              |                                    |                 |                           |
|--------------|------------------------------------|-----------------|---------------------------|
| Name:        | Csm_AuthHsmEcdsaDataType           |                 |                           |
| Type         | Structure                          |                 |                           |
| Range:       | const uint8*                       | pMessage        | signing message           |
|              | const uint8*                       | pSecretNumber   | unpredictable data        |
|              | uint32                             | ulMessageLength | length of signing message |
| Description: | Structure for the ECDSA algorithm. |                 |                           |

### Csm\_AuthHsmEcdsaSignatureType

|              |                                    |             |                      |
|--------------|------------------------------------|-------------|----------------------|
| Name:        | Csm_AuthHsmEcdsaSignatureType      |             |                      |
| Type         | Structure                          |             |                      |
| Range:       | uint8*                             | pSignatureR | r of ECDSA signature |
|              | uint8*                             | pSignatureS | s of ECDSA signature |
| Description: | Structure for the ECDSA algorithm. |             |                      |

### Csm\_AuthHsmEciesPduType

|              |                                    |            |                             |
|--------------|------------------------------------|------------|-----------------------------|
| Name:        | Csm_AuthHsmEciesPduType            |            |                             |
| Type         | Structure                          |            |                             |
| Range:       | uint8 [32]                         | publicKeyX | x of public key             |
|              | uint8 [32]                         | publicKeyY | y of public key             |
|              | uint8 [32]                         | encMsg     | encrypted message           |
|              | uint8 [16]                         | macTag     | MAC tag                     |
|              | uint32                             | encMsgLen  | length of encrypted message |
| Description: | Structure for the ECIES algorithm. |            |                             |

### Csm\_AuthHsmEciesEncryptKeyType

|       |                                |  |  |
|-------|--------------------------------|--|--|
| Name: | Csm_AuthHsmEciesEncryptKeyType |  |  |
| Type  | Structure                      |  |  |

|              |                                    |             |                               |
|--------------|------------------------------------|-------------|-------------------------------|
| Range:       | uint8 [4]                          | length      | information of encryption key |
|              | uint8 [32]                         | privateKeyA | private key of A              |
|              | uint8 [64]                         | publicKeyA  | public key of A               |
|              | uint8 [64]                         | publicKeyB  | public key of B               |
| Description: | Structure for the ECIES algorithm. |             |                               |

### Csm\_AuthHsmEciesEncryptDataType

|              |                                    |                 |                                                          |
|--------------|------------------------------------|-----------------|----------------------------------------------------------|
| Name:        | Csm_AuthHsmEciesEncryptDataType    |                 |                                                          |
| Type         | Structure                          |                 |                                                          |
| Range:       | const uint8*                       | pMessage        | message to be encrypted                                  |
|              | const uint8*                       | pParam1         | output of SHA256, used to calculate KDF                  |
|              | const uint32                       | ulMessageLength | length of message                                        |
|              | const uint8                        | ucEncMode       | encryption mode<br>XOR : 1<br>AES_CTR : 2<br>AES_CBC : 3 |
| Description: | Structure for the ECIES algorithm. |                 |                                                          |

### Csm\_AuthHsmEciesDecryptDataType

|              |                                    |           |                                                          |
|--------------|------------------------------------|-----------|----------------------------------------------------------|
| Name:        | Csm_AuthHsmEciesDecryptDataType    |           |                                                          |
| Type         | Structure                          |           |                                                          |
| Range:       | Csm_AuthHsmEciesPduType*           | pPdu      | Pdu to be decrypted                                      |
|              | const uint8*                       | pParam1   | output of SHA256, used to calculate KDF                  |
|              | const uint8                        | ucEncMode | encryption mode<br>XOR : 1<br>AES_CTR : 2<br>AES_CBC : 3 |
| Description: | Structure for the ECIES algorithm. |           |                                                          |

### Csm\_KeyExchangeBaseType

|              |                                                                                                                                                                                                                                         |        |                                                                      |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|----------------------------------------------------------------------|
| Name:        | Csm_KeyExchangeBaseType                                                                                                                                                                                                                 |        |                                                                      |
| Type         | Structure                                                                                                                                                                                                                               |        |                                                                      |
| Range:       | uint32                                                                                                                                                                                                                                  | length | This element contains the length of the key stored in element 'data' |
|              | Csm_AlignType[CSM_KEY_EX_BASE_MAX_SIZE]                                                                                                                                                                                                 | data   | This element contains the key data or a key handle.                  |
| Description: | Structure with base type information of the key exchange protocol.<br>CSM_KEY_EX_BASE_MAX_SIZE shall be chosen such that "CSM_KEY_EX_BASE_MAX_SIZE * sizeof(Csm_AlignType)" is greater or equal to the maximum of the configured values |        |                                                                      |

|                                                                                                                             |     |
|-----------------------------------------------------------------------------------------------------------------------------|-----|
| CsmKeyExchangeCalcPubValMaxBaseTypeSize,<br>CsmKeyExchangeCalcSecretMaxBaseSize<br>CsmKeyExchangeCalcSymKeyMaxBaseTypeSize. | and |
|-----------------------------------------------------------------------------------------------------------------------------|-----|

### Csm\_KeyExchangePrivateType

|              |                                                                                                                                                                                                                                                                                                                                                                     |        |                                                                      |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|----------------------------------------------------------------------|
| Name:        | Csm_KeyExchangePrivateType                                                                                                                                                                                                                                                                                                                                          |        |                                                                      |
| Type         | Structure                                                                                                                                                                                                                                                                                                                                                           |        |                                                                      |
| Range:       | uint32                                                                                                                                                                                                                                                                                                                                                              | length | This element contains the length of the key stored in element 'data' |
|              | Csm_AlignType[CSM_KEY_EX_PRIV_MAX_SIZE]                                                                                                                                                                                                                                                                                                                             | data   | This element contains the key data or a key handle.                  |
| Description: | Structure with the private information of the key exchange protocol only known to the current user.<br>CSM_KEY_EX_PRIV_MAX_SIZE shall be chosen such that "CSM_KEY_EX_PRIV_MAX_SIZE * sizeof(Csm_AlignType)" is greater or equal to the maximum of the configured values CsmKeyExchangeCalcSecretMaxPrivateTypeSize and CsmKeyExchangeCalcSymKeyMaxPrivateTypeSize. |        |                                                                      |

### Csm\_HaeHsmPkcs1PssDataType

|              |                                         |              |                     |
|--------------|-----------------------------------------|--------------|---------------------|
| Name:        | Csm_HaeHsmPkcs1PssDataType              |              |                     |
| Type         | Structure                               |              |                     |
| Range:       | const uint8*                            | pData        | signing message     |
|              | const uint8*                            | pSalt        | salt text           |
|              | uint32                                  | ulDataLength | length of message   |
|              | uint32                                  | ulSaltLength | length of salt text |
| Description: | Structure for the PKCS#1 PSS algorithm. |              |                     |

### Csm\_HaeHsmEcdsaDataType

|              |                                    |                 |                           |
|--------------|------------------------------------|-----------------|---------------------------|
| Name:        | Csm_HaeHsmEcdsaDataType            |                 |                           |
| Type         | Structure                          |                 |                           |
| Range:       | const uint8*                       | pMessage        | signing message           |
|              | const uint8*                       | pSecretNumber   | unpredictable data        |
|              | uint32                             | ulMessageLength | length of signing message |
| Description: | Structure for the ECDSA algorithm. |                 |                           |

### Csm\_HaeHsmEcdsaSignatureType

|       |                              |
|-------|------------------------------|
| Name: | Csm_HaeHsmEcdsaSignatureType |
| Type  | Structure                    |

|              |                                    |             |                       |
|--------------|------------------------------------|-------------|-----------------------|
| Range:       | uint8*                             | pSignatureR | r of ECDSA signatuere |
|              | uint8*                             | pSignatureS | s of ECDSA signatuere |
| Description: | Structure for the ECDSA algorithm. |             |                       |

## Functions

### Initialization/Deinitialization Related Api

|                                 |                                                           |                  |
|---------------------------------|-----------------------------------------------------------|------------------|
| <b>Function Name</b>            | Csm_Init                                                  |                  |
| <b>Syntax</b>                   | void Csm_Init(void)                                       |                  |
| <b>Service ID</b>               | 0x00                                                      |                  |
| <b>Sync/Async</b>               | Synchronous                                               |                  |
| <b>Reentrancy</b>               | Non Re-entrant                                            |                  |
| <b>Parameters (In)</b>          | None                                                      |                  |
| <b>Parameters (Inout)</b>       | None                                                      |                  |
| <b>Parameters (Out)</b>         | None                                                      | Parameters (Out) |
| <b>Return Value</b>             | None                                                      |                  |
| <b>Description</b>              | This function shall be used to initialize the CSM module. |                  |
| <b>Preconditions</b>            | None                                                      |                  |
| <b>Configuration Dependency</b> |                                                           |                  |

### Version related Api

|                                 |                                                                                                                                                            |                                                                   |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|
| <b>Function Name</b>            | Csm_GetVersionInfo                                                                                                                                         |                                                                   |
| <b>Syntax</b>                   | void Csm_GetVersionInfo( Std_VersionInfoType* versioninfo )                                                                                                |                                                                   |
| <b>Service ID</b>               | 0x3B                                                                                                                                                       |                                                                   |
| <b>Sync/Async</b>               | Synchronous                                                                                                                                                |                                                                   |
| <b>Reentrancy</b>               | Reentrant                                                                                                                                                  |                                                                   |
| <b>Parameters (In)</b>          | None                                                                                                                                                       |                                                                   |
| <b>Parameters (Inout)</b>       | None                                                                                                                                                       |                                                                   |
| <b>Parameters (Out)</b>         | versioninfo                                                                                                                                                | Pointer to where to store the version information of this module. |
| <b>Return Value</b>             | None                                                                                                                                                       |                                                                   |
| <b>Description</b>              | This service gives version information of the Csm module. The version information includes module Id, vendor Id, InstanceId & vendor specific version no's |                                                                   |
| <b>Preconditions</b>            | None                                                                                                                                                       |                                                                   |
| <b>Configuration Dependency</b> | CSM_VERSION_INFO_API should be STD_ON                                                                                                                      |                                                                   |

### Main Function

|                                 |                                                                  |  |
|---------------------------------|------------------------------------------------------------------|--|
| <b>Function Name</b>            | Csm_MainFunction                                                 |  |
| <b>Syntax</b>                   | void Csm_MainFunction(void)                                      |  |
| <b>Service ID</b>               | 0x01                                                             |  |
| <b>Sync/Async</b>               | Synchronous                                                      |  |
| <b>Reentrancy</b>               | Non Re-entrant                                                   |  |
| <b>Parameters (In)</b>          | None                                                             |  |
| <b>Parameters (Inout)</b>       | None                                                             |  |
| <b>Parameters (Out)</b>         | None                                                             |  |
| <b>Return Value</b>             | None                                                             |  |
| <b>Description</b>              | This API is called cyclically to process the requested services. |  |
| <b>Preconditions</b>            | The CSM module must be initialized before call of this API       |  |
| <b>Configuration Dependency</b> |                                                                  |  |

## Hash interface Api

|                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                             |
|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| <b>Function Name</b>            | Csm_HashStart                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                             |
| <b>Syntax</b>                   | FUNC(Std_ReturnType, CSM_CODE) Csm_HashStart( Csm_ConfigIdType cfgId )                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                             |
| <b>Service ID</b>               | 0x03                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                             |
| <b>Sync/Async</b>               | Sync or Async, dependent on configuration                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                             |
| <b>Reentrancy</b>               | Non Re-entrant                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                             |
| <b>Parameters (In)</b>          | cfgId                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | holds the identifier of the CSM module configuration that has to be used during the hash value computation. |
| <b>Parameters (Inout)</b>       | None                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                             |
| <b>Parameters (Out)</b>         | None                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                             |
| <b>Return Value</b>             | Std_ReturnType                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | E_OK: request successful<br>E_NOT_OK: request failed<br>CSM_E_BUSY: request failed, service is still busy   |
| <b>Description</b>              | This interface shall be used to initialize the hash service of the CSM module. If the service state is "active", the function shall return with "CSM_E_BUSY". Otherwise, this function shall store the given configuration information which is identified by "cfgId", call the function Cry_<Primitive>Start of the primitive which is identified by the "cfgId" and return the value returned by that function. If Cry_<Primitive>Start returned successfully, the service state has to be set to "active" |                                                                                                             |
| <b>Preconditions</b>            | CSM Hash interface should have been initialized                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                             |
| <b>Configuration Dependency</b> | CSM_HASH_STATUS should be STD_ON                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                             |

|                                 |                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                           |
|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| <b>Function Name</b>            | Csm_HashUpdate                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                           |
| <b>Syntax</b>                   | FUNC(Std_ReturnType, CSM_CODE) Csm_HashUpdate(Csm_ConfigIdType cfgId, const uint8* dataPtr, uint32 dataLength)                                                                                                                                                                                                                                                                                                       |                                                                                                           |
| <b>Service ID</b>               | 0x04                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                           |
| <b>Sync/Async</b>               | Sync or Async, dependent on configuration                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                           |
| <b>Reentrancy</b>               | Non-Re-entrant                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                           |
| <b>Parameters (In)</b>          | cfgId                                                                                                                                                                                                                                                                                                                                                                                                                | Holds the identifier of the CSM module configuration that has to be used during the operation.            |
|                                 | dataPtr                                                                                                                                                                                                                                                                                                                                                                                                              | Holds a pointer to the data to be hashed                                                                  |
|                                 | dataLength                                                                                                                                                                                                                                                                                                                                                                                                           | Contains the number of bytes to be hashed.                                                                |
| <b>Parameters (Inout)</b>       | None                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                           |
| <b>Parameters (Out)</b>         | None                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                           |
| <b>Return Value</b>             | Std_ReturnType                                                                                                                                                                                                                                                                                                                                                                                                       | E_OK: request successful<br>E_NOT_OK: request failed<br>CSM_E_BUSY: request failed, service is still busy |
| <b>Description</b>              | This interface shall be used to feed the hash service with the input data. If the service state is "idle", the function has to return with "E_NOT_OK". Otherwise, this function shall call the function Cry_<Primitive>Update of the primitive which is identified by the stored configuration information and return the value returned by that function. The hash computation is done by the underlying primitive. |                                                                                                           |
| <b>Preconditions</b>            | CSM Hash interface should have been initialized.                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                           |
| <b>Configuration Dependency</b> | CSM_HASH_STATUS should be STD_ON                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                           |

|                                 |                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                               |
|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Function Name</b>            | Csm_HashFinish                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                               |
| <b>Syntax</b>                   | FUNC(Std_ReturnType, CSM_CODE) Csm_HashFinish( Csm_ConfigIdType cfgId, uint8* resultPtr, uint32* resultLengthPtr, boolean TruncationIsAllowed )                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                               |
| <b>Service ID</b>               | 0x05                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                               |
| <b>Sync/Async</b>               | Sync or Async, dependent on configuration                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                               |
| <b>Reentrancy</b>               | Non-Re-entrant                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                               |
| <b>Parameters (In)</b>          | cfgId                                                                                                                                                                                                                                                                                                                                                                                                                | Holds the identifier of the CSM module configuration that has to be used during the operation.                                                                                                                                                                                |
|                                 | TruncationIsAllowed                                                                                                                                                                                                                                                                                                                                                                                                  | This parameter states whether a truncation of the result is allowed or not. TRUE: truncation is allowed. FALSE: truncation is not allowed.                                                                                                                                    |
| <b>Parameters (Inout)</b>       | resultLengthPtr                                                                                                                                                                                                                                                                                                                                                                                                      | holds a pointer to the memory location in which the length information is stored. On calling this function this parameter shall contain the size of the buffer provided by resultPtr. When the request has finished, the actual length of the returned value shall be stored. |
| <b>Parameters (Out)</b>         | resultPtr                                                                                                                                                                                                                                                                                                                                                                                                            | holds a pointer to the memory location which will hold the result of the hash value computation. If the result does not fit into the given buffer, and truncation is allowed, the result shall be truncated.                                                                  |
| <b>Return Value</b>             | Std_ReturnType                                                                                                                                                                                                                                                                                                                                                                                                       | E_OK: request successful<br>E_NOT_OK: request failed<br>CSM_E_BUSY: request failed, service is still busy<br>CSM_E_SMALL_BUFFER: the provided buffer is too small to store the result, and truncation was not allowed                                                         |
| <b>Description</b>              | This interface shall be used to finish the hash service of the CSM module. If the service state is "idle", the function has to return with "E_NOT_OK". Otherwise, this function shall call the function Cry_<Primitive>Finish of the primitive which is identified by the stored configuration information and return the value returned by that function. The hash computation is done by the underlying primitive. |                                                                                                                                                                                                                                                                               |
| <b>Preconditions</b>            | CSM Hash interface should have been initialized.                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                               |
| <b>Configuration Dependency</b> | CSM_HASH_STATUS should be STD_ON                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                               |

### Mac interface Api

|                        |                                                                                                             |                                                                                                       |
|------------------------|-------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| <b>Function Name</b>   | Csm_MacGenerateStart                                                                                        |                                                                                                       |
| <b>Syntax</b>          | FUNC(Csm_ReturnType, CSM_CODE) Csm_MacGenerateStart( Csm_ConfigIdType cfgId, const Csm_SymKeyType* keyPtr ) |                                                                                                       |
| <b>Service ID</b>      | 0x06                                                                                                        |                                                                                                       |
| <b>Sync/Async</b>      | Sync or Async, dependent on configuration                                                                   |                                                                                                       |
| <b>Reentrancy</b>      | Non Re-entrant                                                                                              |                                                                                                       |
| <b>Parameters (In)</b> | cfgId                                                                                                       | Holds the identifier of the CSM module configuration which has to be used during the MAC computation. |

|                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                           |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
|                                 | keyPtr                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Holds a pointer to the key necessary for the MAC generation.                                              |
| <b>Parameters (Inout)</b>       | None                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                           |
| <b>Parameters (Out)</b>         | None                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                           |
| <b>Return Value</b>             | Std_ReturnType                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | E_OK: request successful<br>E_NOT_OK: request failed<br>CSM_E_BUSY: request failed, service is still busy |
| <b>Description</b>              | This interface shall be used to initialize the MAC generate service of the CSM module. If the service state is "active", the function shall return with "CSM_E_BUSY". Otherwise, this function shall store the given configuration information which is identified by "cfgId", call the function Cry_<Primitive>Start of the primitive which is identified by the "cfgId" and return the value returned by that function. If Cry_<Primitive>Start returned successfully, the service state has to be set to "active". |                                                                                                           |
| <b>Preconditions</b>            | CSM Mac interface should have been initialized                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                           |
| <b>Configuration Dependency</b> | CSM_MAC_STATUS should be STD_ON                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                           |

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| <b>Function Name</b>            | Csm_MacGenerateUpdate                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                           |
| <b>Syntax</b>                   | FUNC(Csm_ReturnType, CSM_CODE)<br>Csm_MacGenerateUpdate(Csm_ConfigIdType cfgId, const uint8* dataPtr, uint32 dataLength)                                                                                                                                                                                                                                                                                                    |                                                                                                           |
| <b>Service ID</b>               | 0x07                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                           |
| <b>Sync/Async</b>               | Sync or Async, dependent on configuration                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                           |
| <b>Reentrancy</b>               | Non-Re-entrant                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                           |
| <b>Parameters (In)</b>          | cfgId                                                                                                                                                                                                                                                                                                                                                                                                                       | Holds the identifier of the CSM module configuration that has to be used during the operation.            |
|                                 | dataPtr                                                                                                                                                                                                                                                                                                                                                                                                                     | Holds a pointer to the data for which a MAC shall be computed.                                            |
|                                 | dataLength                                                                                                                                                                                                                                                                                                                                                                                                                  | Contains the number of bytes for which the MAC shall be computed.                                         |
| <b>Parameters (Inout)</b>       | None                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                           |
| <b>Parameters (Out)</b>         | None                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                           |
| <b>Return Value</b>             | Std_ReturnType                                                                                                                                                                                                                                                                                                                                                                                                              | E_OK: request successful<br>E_NOT_OK: request failed<br>CSM_E_BUSY: request failed, service is still busy |
| <b>Description</b>              | This interface shall be used to feed the MAC generate service with the input data. If the service state is "idle", the function has to return with "E_NOT_OK". Otherwise, this function shall call the function Cry_<Primitive>Update of the primitive which is identified by the stored configuration information and return the value returned by that function. The MAC computation is done by the underlying primitive. |                                                                                                           |
| <b>Preconditions</b>            | CSM Mac interface should have been initialized.                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                           |
| <b>Configuration Dependency</b> | CSM_MAC_STATUS should be STD_ON                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                           |

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| <b>Function Name</b> | Csm_MacGenerateFinish                                                                                                                                     |  |
| <b>Syntax</b>        | FUNC(Csm_ReturnType, CSM_CODE)<br>Csm_MacGenerateFinish( Csm_ConfigIdType cfgId, uint8* resultPtr, uint32* resultLengthPtr, boolean TruncationIsAllowed ) |  |

|                                 |                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                             |
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| <b>Service ID</b>               | 0x08                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                             |
| <b>Sync/Async</b>               | Sync or Async, dependent on configuration                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                             |
| <b>Reentrancy</b>               | Non-Re-entrant                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                             |
| <b>Parameters (In)</b>          | cfgId                                                                                                                                                                                                                                                                                                                                                                                                       | Holds the identifier of the CSM module configuration that has to be used during the operation.                                                                                                                                                                              |
|                                 | TruncationIsAllowed                                                                                                                                                                                                                                                                                                                                                                                         | This parameter states whether a truncation of the result is allowed or not. TRUE: truncation is allowed. FALSE: truncation is not allowed.                                                                                                                                  |
| <b>Parameters (Inout)</b>       | resultLengthPtr                                                                                                                                                                                                                                                                                                                                                                                             | Holds a pointer to the memory location in which the length information is stored. On calling this function this parameter shall contain the size of the buffer provided by resultPtr. When the request has finished, the actual length of the returned MAC shall be stored. |
| <b>Parameters (Out)</b>         | resultPtr                                                                                                                                                                                                                                                                                                                                                                                                   | Holds a pointer to the memory location which will hold the result of the MAC generation. If the result does not fit into the given buffer, and truncation is allowed, the result shall be truncated                                                                         |
| <b>Return Value</b>             | Std_ReturnType                                                                                                                                                                                                                                                                                                                                                                                              | E_OK: request successful<br>E_NOT_OK: request failed<br>CSM_E_BUSY: request failed, service is still busy<br>CSM_E_SMALL_BUFFER: the provided buffer is too small to store the result, and truncation was not allowed                                                       |
| <b>Description</b>              | This interface shall be used to finish the MAC generation service. If the service state is "idle", the function has to return with "E_NOT_OK". Otherwise, this function shall call the function Cry_<Primitive>Finish of the primitive which is identified by the stored configuration information and return the value returned by that function. The MAC computation is done by the underlying primitive. |                                                                                                                                                                                                                                                                             |
| <b>Preconditions</b>            | CSM Mac interface should have been initialized.                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                             |
| <b>Configuration Dependency</b> | CSM_MAC_STATUS should be STD_ON                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                             |

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| <b>Function Name</b>      | Csm_MacVerifyStart                                                                                      |                                                                                                           |
| <b>Syntax</b>             | FUNC(Csm_ReturnType, CSM_CODE) Csm_MacVerifyStart(Csm_ConfigIdType cfgId, const Csm_SymKeyType* keyPtr) |                                                                                                           |
| <b>Service ID</b>         | 0x09                                                                                                    |                                                                                                           |
| <b>Sync/Async</b>         | Sync or Async, dependent on configuration                                                               |                                                                                                           |
| <b>Reentrancy</b>         | Non Re-entrant                                                                                          |                                                                                                           |
| <b>Parameters (In)</b>    | cfgId                                                                                                   | Holds the identifier of the CSM module configuration which has to be used during the MAC verification.    |
|                           | keyPtr                                                                                                  | Holds a pointer to the key necessary for the MAC verification.                                            |
| <b>Parameters (Inout)</b> | None                                                                                                    |                                                                                                           |
| <b>Parameters (Out)</b>   | None                                                                                                    |                                                                                                           |
| <b>Return Value</b>       | Std_ReturnType                                                                                          | E_OK: request successful<br>E_NOT_OK: request failed<br>CSM_E_BUSY: request failed, service is still busy |

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| <b>Description</b>              | This interface shall be used to initialize the MAC verify service of the CSM module. If the service state is "active", the function shall return with "CSM_E_BUSY". Otherwise, this function shall store the given configuration information which is identified by "cfgId", call the function Cry_<Primitive>Start of the primitive which is identified by the "cfgId" and return the value returned by that function. If Cry_<Primitive>Start returned successfully, the service state has to be set to "active". |
| <b>Preconditions</b>            | CSM Mac interface should have been initialized.                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Configuration Dependency</b> | CSM_MAC_STATUS should be STD_ON                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

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| <b>Function Name</b>            | Csm_MacVerifyUpdate                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                           |
| <b>Syntax</b>                   | FUNC(Csm_ReturnType, CSM_CODE) Csm_MacVerifyUpdate(Csm_ConfigIdType cfgId, const uint8* dataPtr, uint32 dataLength)                                                                                                                                                                                                                                                                                                             |                                                                                                           |
| <b>Service ID</b>               | 0x0a                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                           |
| <b>Sync/Async</b>               | Sync or Async, dependent on configuration                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                           |
| <b>Reentrancy</b>               | Non-Re-entrant                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                           |
| <b>Parameters (In)</b>          | cfgId                                                                                                                                                                                                                                                                                                                                                                                                                           | Holds the identifier of the CSM module configuration that has to be used during the operation.            |
|                                 | dataPtr                                                                                                                                                                                                                                                                                                                                                                                                                         | Holds a pointer to the data for which a MAC shall be verified                                             |
|                                 | dataLength                                                                                                                                                                                                                                                                                                                                                                                                                      | Contains the number of bytes for which the MAC shall be verified.                                         |
| <b>Parameters (Inout)</b>       | None                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                           |
| <b>Parameters (Out)</b>         | None                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                           |
| <b>Return Value</b>             | Std_ReturnType                                                                                                                                                                                                                                                                                                                                                                                                                  | E_OK: request successful<br>E_NOT_OK: request failed<br>CSM_E_BUSY: request failed, service is still busy |
| <b>Description</b>              | This interface shall be used to feed the MAC verification service with the input data. If the service state is "idle", the function has to return with "E_NOT_OK". Otherwise, this function shall call the function Cry_<Primitive>Update of the primitive which is identified by the stored configuration information and return the value returned by that function. The MAC computation is done by the underlying primitive. |                                                                                                           |
| <b>Preconditions</b>            | CSM Mac interface should have been initialized.                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                           |
| <b>Configuration Dependency</b> | CSM_MAC_STATUS should be STD_ON                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                           |

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|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| <b>Function Name</b>   | Csm_MacVerifyFinish                                                                                                                                |                                                                                                |
| <b>Syntax</b>          | FUNC(Csm_ReturnType, CSM_CODE) Csm_MacVerifyFinish(Csm_ConfigIdType cfgId, const uint8* MacPtr, uint32 MacLength, Csm_VerifyResultType* resultPtr) |                                                                                                |
| <b>Service ID</b>      | 0x0b                                                                                                                                               |                                                                                                |
| <b>Sync/Async</b>      | Sync or Async, dependent on configuration                                                                                                          |                                                                                                |
| <b>Reentrancy</b>      | Non-Re-entrant                                                                                                                                     |                                                                                                |
| <b>Parameters (In)</b> | cfgId                                                                                                                                              | Holds the identifier of the CSM module configuration that has to be used during the operation. |

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|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
|                                 | MacPtr                                                                                                                                                                                                                                                                                                                                                                                                        | Holds a pointer to the memory location which will hold the MAC to verify.                                       |
|                                 | MacLength                                                                                                                                                                                                                                                                                                                                                                                                     | Holds the length of the MAC to be verified. Note: the computed MAC will be internally truncated to this length. |
| <b>Parameters (Inout)</b>       | None                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                 |
| <b>Parameters (Out)</b>         | resultPtr                                                                                                                                                                                                                                                                                                                                                                                                     | Holds a pointer to the memory location which will hold the result of the MAC verification.                      |
| <b>Return Value</b>             | Std_ReturnType                                                                                                                                                                                                                                                                                                                                                                                                | E_OK: request successful<br>E_NOT_OK: request failed<br>CSM_E_BUSY: request failed, service is still busy       |
| <b>Description</b>              | This interface shall be used to finish the MAC verification service. If the service state is "idle", the function has to return with "E_NOT_OK". Otherwise, this function shall call the function Cry_<Primitive>Finish of the primitive which is identified by the stored configuration information and return the value returned by that function. The MAC computation is done by the underlying primitive. |                                                                                                                 |
| <b>Preconditions</b>            | CSM Mac interface should have been initialized.                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                 |
| <b>Configuration Dependency</b> | CSM_MAC_STATUS should be STD_ON                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                 |

### Random interface Api

|                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                              |
|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| <b>Function Name</b>            | Csm_RandomSeedStart                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                              |
| <b>Syntax</b>                   | FUNC(Std_ReturnType, CSM_CODE) Csm_RandomSeedStart( Csm_ConfigIdType cfgId )                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                              |
| <b>Service ID</b>               | 0x0c                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                              |
| <b>Sync/Async</b>               | Sync or Async, dependent on configuration                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                              |
| <b>Reentrancy</b>               | Non Re-entrant                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                              |
| <b>Parameters (In)</b>          | cfgId                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | holds the identifier of the CSM module configuration which has to be used during the seeding of the random number generator. |
| <b>Parameters (Inout)</b>       | None                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                              |
| <b>Parameters (Out)</b>         | None                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                              |
| <b>Return Value</b>             | Std_ReturnType                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | E_OK: request successful<br>E_NOT_OK: request failed<br>CSM_E_BUSY: request failed, service is still busy                    |
| <b>Description</b>              | This interface shall be used to initialize the random seed service of the CSM module. If the service state is "active", the function shall return with "CSM_E_BUSY". Otherwise, this function shall store the given configuration information which is identified by "cfgId", call the function Cry_<Primitive>Start of the primitive which is identified by the "cfgId" and return the value returned by that function. If Cry_<Primitive>Start returned successfully, the service state has to be set to "active". |                                                                                                                              |
| <b>Preconditions</b>            | CSM Random interface should have been initialized                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                              |
| <b>Configuration Dependency</b> | CSM_RANDOM_SEED_STATUS should be STD_ON                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                              |

|                      |                      |
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| <b>Function Name</b> | Csm_RandomSeedUpdate |
|----------------------|----------------------|

|                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                           |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| <b>Syntax</b>                   | FUNC(Std_ReturnType, CSM_CODE)<br>Csm_RandomSeedUpdate(Csm_ConfigIdType cfgId, const uint8* seedPtr, uint32 seedLength)                                                                                                                                                                                                                                                                                                                 |                                                                                                           |
| <b>Service ID</b>               | 0x0d                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                           |
| <b>Sync/Async</b>               | Sync or Async, dependent on configuration                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                           |
| <b>Reentrancy</b>               | Non-Re-entrant                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                           |
| <b>Parameters (In)</b>          | cfgId                                                                                                                                                                                                                                                                                                                                                                                                                                   | Holds the identifier of the CSM module configuration that has to be used during the operation.            |
|                                 | seedPtr                                                                                                                                                                                                                                                                                                                                                                                                                                 | holds a pointer to the seed for the random number generator.                                              |
|                                 | seedLength                                                                                                                                                                                                                                                                                                                                                                                                                              | contains the length of the seed in bytes.                                                                 |
| <b>Parameters (Inout)</b>       | None                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                           |
| <b>Parameters (Out)</b>         | None                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                           |
| <b>Return Value</b>             | Std_ReturnType                                                                                                                                                                                                                                                                                                                                                                                                                          | E_OK: request successful<br>E_NOT_OK: request failed<br>CSM_E_BUSY: request failed, service is still busy |
| <b>Description</b>              | This interface shall be used to feed a seed to the random seed service. If the service state is "idle", the function has to return with "E_NOT_OK". Otherwise, this function shall call the function Cry_<Primitive>Update of the primitive which is identified by the stored configuration information and return the value returned by that function. The seeding of the random number generator is done by the underlying primitive. |                                                                                                           |
| <b>Preconditions</b>            | CSM Random interface should have been initialized                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                           |
| <b>Configuration Dependency</b> | CSM_RANDOM_SEED_STATUS should be STD_ON                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                           |

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| <b>Function Name</b>            | Csm_RandomSeedFinish                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                           |
| <b>Syntax</b>                   | FUNC(Std_ReturnType, CSM_CODE)<br>Csm_RandomSeedFinish( Csm_ConfigIdType cfgId )                                                                                                                                                                                                                                                                                                                                                |                                                                                                           |
| <b>Service ID</b>               | 0x0e                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                           |
| <b>Sync/Async</b>               | Sync or Async, dependent on configuration                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                           |
| <b>Reentrancy</b>               | Non-Re-entrant                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                           |
| <b>Parameters (In)</b>          | cfgId                                                                                                                                                                                                                                                                                                                                                                                                                           | Holds the identifier of the CSM module configuration that has to be used during the operation.            |
|                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                           |
|                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                           |
| <b>Parameters (Inout)</b>       | None                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                           |
| <b>Parameters (Out)</b>         | None                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                           |
| <b>Return Value</b>             | Std_ReturnType                                                                                                                                                                                                                                                                                                                                                                                                                  | E_OK: request successful<br>E_NOT_OK: request failed<br>CSM_E_BUSY: request failed, service is still busy |
| <b>Description</b>              | This interface shall be used to finish the random seed service. If the service state is "idle", the function has to return with "E_NOT_OK". Otherwise, this function shall call the function Cry_<Primitive>Finish of the primitive which is identified by the stored configuration information and return the value returned by that function. The seeding of the random number generator is done by the underlying primitive. |                                                                                                           |
| <b>Preconditions</b>            | CSM Random interface should have been initialized                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                           |
| <b>Configuration Dependency</b> | CSM_RANDOM_SEED_STATUS should be STD_ON                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                           |

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|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Function Name</b>            | Csm_RandomGenerate                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                         |
| <b>Syntax</b>                   | FUNC(Std_ReturnType, CSM_CODE) Csm_RandomGenerate (Csm_ConfigIdType cfgId, uint8* resultPtr, uint32 resultLength)                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                         |
| <b>Service ID</b>               | 0x0f                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                         |
| <b>Sync/Async</b>               | Sync or Async, dependent on configuration                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                         |
| <b>Reentrancy</b>               | Non Re-entrant                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                         |
| <b>Parameters (In)</b>          | cfgId                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | holds the identifier of the CSM module configuration which has to be used during random number generation.                                                                                              |
|                                 | resultLength                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | holds the amount of random bytes which should be generated.                                                                                                                                             |
| <b>Parameters (Inout)</b>       | None                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                         |
| <b>Parameters (Out)</b>         | resultPtr                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | holds a pointer to the memory location which will hold the result of the random number generation. The memory location must have at least the size "resultLength".                                      |
| <b>Return Value</b>             | Std_ReturnType                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | E_OK: request successful<br>E_NOT_OK: request failed<br>CSM_E_BUSY: request failed, service is still busy<br>CSM_E_ENTROPY_EXHAUSTION: request failed, entropy of random number generator is exhausted. |
| <b>Description</b>              | This interface shall be used to start the random number generation service of the CSM module. If the service state is "active", the function shall return with "CSM_E_BUSY". Otherwise, this function shall store the given configuration information which is identified by "cfgId", call the function Cry_<Primitive> of the primitive which is identified by the "cfgId" and return the value returned by that function. If Cry_<Primitive> returned successfully, the service state has to be set to "active". The random number generation is done by the underlying primitive. |                                                                                                                                                                                                         |
| <b>Preconditions</b>            | CSM Random interface should have been initialized                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                         |
| <b>Configuration Dependency</b> | CSM_RANDOM_GENERATE_STATUS should be STD_ON                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                         |

### SymBlock interface Api

|                           |                                                                                                                    |                                                                                                                                |
|---------------------------|--------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| <b>Function Name</b>      | Csm_SymBlockEncryptStart                                                                                           |                                                                                                                                |
| <b>Syntax</b>             | FUNC(Csm_ReturnType, CSM_CODE)<br>Csm_SymBlockEncryptStart( Csm_ConfigIdType cfgId, const Csm_SymKeyType* keyPtr ) |                                                                                                                                |
| <b>Service ID</b>         | 0x10                                                                                                               |                                                                                                                                |
| <b>Sync/Async</b>         | Sync or Async, dependent on configuration                                                                          |                                                                                                                                |
| <b>Reentrancy</b>         | Non Re-entrant                                                                                                     |                                                                                                                                |
| <b>Parameters (In)</b>    | cfgId                                                                                                              | holds the identifier of the CSM module configuration which has to be used during the symmetrical block encryption computation. |
|                           | keyPtr                                                                                                             | holds a pointer to the key which has to be used during the symmetrical block encryption computation.                           |
| <b>Parameters (Inout)</b> | None                                                                                                               |                                                                                                                                |

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|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| <b>Parameters (Out)</b>         | None                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                           |
| <b>Return Value</b>             | Std_ReturnType                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | E_OK: request successful<br>E_NOT_OK: request failed<br>CSM_E_BUSY: request failed, service is still busy |
| <b>Description</b>              | This interface shall be used to initialize the symmetrical block encrypt service of the CSM module. If the service state is "active", the function shall return with "CSM_E_BUSY". Otherwise, this function shall store the given configuration information which is identified by "cfgld", call the function Cry_<Primitive>Start of the primitive which is identified by the "cfgld" and return the value returned by that function. If Cry_<Primitive>Start returned successfully, the service state has to be set to "active". |                                                                                                           |
| <b>Preconditions</b>            | CSM SymBlock interface should have been initialized.                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                           |
| <b>Configuration Dependency</b> | CSM_SYM_BLOCK_ENCRYPT_STATUS should be STD_ON                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                           |

|                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                    |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Function Name</b>            | Csm_SymBlockEncryptUpdate                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                    |
| <b>Syntax</b>                   | FUNC(Csm_ReturnType, CSM_CODE)<br>Csm_SymBlockEncryptUpdate(Csm_ConfigIdType cfgld, const uint8* plainTextPtr, uint32 plainTextLength, uint8* cipherTextPtr, uint32* cipherTextLengthPtr )                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                    |
| <b>Service ID</b>               | 0x11                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                    |
| <b>Sync/Async</b>               | Sync or Async, dependent on configuration                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                    |
| <b>Reentrancy</b>               | Non-Re-entrant                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                    |
| <b>Parameters (In)</b>          | cfgld                                                                                                                                                                                                                                                                                                                                                                                                                                          | Holds the identifier of the CSM module configuration that has to be used during the operation.                                                                                                                                                                                     |
|                                 | plainTextPtr                                                                                                                                                                                                                                                                                                                                                                                                                                   | holds a pointer to the plain text that shall be encrypted.                                                                                                                                                                                                                         |
|                                 | plainTextLength                                                                                                                                                                                                                                                                                                                                                                                                                                | contains the length of the plain text in bytes                                                                                                                                                                                                                                     |
| <b>Parameters (Inout)</b>       | cipherTextLengthPtr                                                                                                                                                                                                                                                                                                                                                                                                                            | holds a pointer to a memory location in which the length information is stored. On calling this function this parameter shall contain the size of the buffer provided by cipherTextPtr. When the request has finished, the amount of data that has been encrypted shall be stored. |
| <b>Parameters (Out)</b>         | cipherTextPtr                                                                                                                                                                                                                                                                                                                                                                                                                                  | holds a pointer to the memory location which will hold the encrypted text.                                                                                                                                                                                                         |
| <b>Return Value</b>             | Std_ReturnType                                                                                                                                                                                                                                                                                                                                                                                                                                 | E_OK: request successful<br>E_NOT_OK: request failed<br>CSM_E_BUSY: request failed, service is still busy<br>CSM_E_SMALL_BUFFER: the provided buffer is too small to store the result                                                                                              |
| <b>Description</b>              | This interface shall be used to feed the symmetrical block encryption service with the input data. If the service state is "idle", the function has to return with "E_NOT_OK". Otherwise, this function shall call the function Cry_<Primitive>Update of the primitive which is identified by the stored configuration information and return the value returned by that function. The encryption process is done by the underlying primitive. |                                                                                                                                                                                                                                                                                    |
| <b>Preconditions</b>            | CSM SymBlock interface should have been initialized.                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                    |
| <b>Configuration Dependency</b> | CSM_SYM_BLOCK_ENCRYPT_STATUS should be STD_ON                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                    |

|                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                           |
|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| <b>Function Name</b>            | Csm_SymBlockEncryptFinish                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                           |
| <b>Syntax</b>                   | FUNC(Csm_ReturnType, Csm_SymBlockEncryptFinish( Csm_ConfigIdType cfgId ) CSM_CODE)                                                                                                                                                                                                                                                                                                                                               |                                                                                                           |
| <b>Service ID</b>               | 0x12                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                           |
| <b>Sync/Async</b>               | Sync or Async, dependent on configuration                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                           |
| <b>Reentrancy</b>               | Non-Re-entrant                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                           |
| <b>Parameters (In)</b>          | cfgId                                                                                                                                                                                                                                                                                                                                                                                                                            | Holds the identifier of the CSM module configuration that has to be used during the operation.            |
| <b>Parameters (Inout)</b>       | None                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                           |
| <b>Parameters (Out)</b>         | None                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                           |
| <b>Return Value</b>             | Std_ReturnType                                                                                                                                                                                                                                                                                                                                                                                                                   | E_OK: request successful<br>E_NOT_OK: request failed<br>CSM_E_BUSY: request failed, service is still busy |
| <b>Description</b>              | This interface shall be used to finish the symmetrical block encryption service. If the service state is "idle", the function has to return with "CSM_E_NOT_OK". Otherwise, this function shall call the function Cry_<Primitive>Finish of the primitive which is identified by the stored configuration information and return the value returned by that function. The encryption process is done by the underlying primitive. |                                                                                                           |
| <b>Preconditions</b>            | CSM SymBlock interface should have been initialized.                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                           |
| <b>Configuration Dependency</b> | CSM_SYM_BLOCK_ENCRYPT_STATUS should be STD_ON                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                           |

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| <b>Function Name</b>      | Csm_SymBlockDecryptStart                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                        |
| <b>Syntax</b>             | FUNC(Csm_ReturnType, Csm_SymBlockDecryptStart(Csm_ConfigIdType cfgId, const Csm_SymKeyType* keyPtr) CSM_CODE)                                                                                                                                                                                                                                                                                                                             |                                                                                                                                        |
| <b>Service ID</b>         | 0x09                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                        |
| <b>Sync/Async</b>         | Sync or Async, dependent on configuration                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                        |
| <b>Reentrancy</b>         | Non Re-entrant                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                        |
| <b>Parameters (In)</b>    | cfgId                                                                                                                                                                                                                                                                                                                                                                                                                                     | holds the identifier of the constant CSM module configuration which has to be used during the symmetrical block decryption computation |
|                           | keyPtr                                                                                                                                                                                                                                                                                                                                                                                                                                    | holds a pointer to the key which has to be used during the symmetrical block decryption computation                                    |
| <b>Parameters (Inout)</b> | None                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                        |
| <b>Parameters (Out)</b>   | None                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                        |
| <b>Return Value</b>       | Std_ReturnType                                                                                                                                                                                                                                                                                                                                                                                                                            | E_OK: request successful<br>E_NOT_OK: request failed<br>CSM_E_BUSY: request failed, service is still busy                              |
| <b>Description</b>        | This interface shall be used to initialize the symmetrical block decrypt service of the CSM module.<br>If the service state is "active", the function shall return with "CSM_E_BUSY". Otherwise, this function shall store the given configuration information which is identified by "cfgId", call the function Cry_<Primitive>Start of the primitive which is identified by the "cfgId" and return the value returned by that function. |                                                                                                                                        |

|                                 |                                                                                             |
|---------------------------------|---------------------------------------------------------------------------------------------|
|                                 | If Cry_<Primitive>Start returned successfully, the service state has to be set to "active". |
| <b>Preconditions</b>            | CSM SymBlock interface should have been initialized.                                        |
| <b>Configuration Dependency</b> | CSM_SYM_BLOCK_DECRYPT_STATUS should be STD_ON                                               |

|                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                   |
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| <b>Function Name</b>            | Csm_SymBlockDecryptUpdate                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                   |
| <b>Syntax</b>                   | FUNC(Csm_ReturnType, CSM_CODE)<br>Csm_SymBlockDecryptUpdate(Csm_ConfigIdType cfgId, const uint8* cipherTextPtr, uint32 cipherTextLength, uint8* plainTextPtr, uint32* plainTextLengthPtr)                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                   |
| <b>Service ID</b>               | 0x14                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                   |
| <b>Sync/Async</b>               | Sync or Async, dependent on configuration                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                   |
| <b>Reentrancy</b>               | Non Re-entrant                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                   |
| <b>Parameters (In)</b>          | cfgId                                                                                                                                                                                                                                                                                                                                                                                                                                              | Holds the identifier of the CSM module configuration that has to be used during the operation.                                                                                                                                                                                    |
|                                 | cipherTextPtr                                                                                                                                                                                                                                                                                                                                                                                                                                      | holds a pointer to the constant cipher text that shall be decrypted.                                                                                                                                                                                                              |
|                                 | cipherTextLength                                                                                                                                                                                                                                                                                                                                                                                                                                   | contains the length of the cipher text in bytes.                                                                                                                                                                                                                                  |
| <b>Parameters (Inout)</b>       | plainTextLengthPtr                                                                                                                                                                                                                                                                                                                                                                                                                                 | holds a pointer to a memory location in which the length information is stored. On calling this function this parameter shall contain the size of the buffer provided by plainTextPtr. When the request has finished, the amount of data that has been decrypted shall be stored. |
| <b>Parameters (Out)</b>         | plainTextPtr                                                                                                                                                                                                                                                                                                                                                                                                                                       | holds a pointer to the memory location which will hold the decrypted text.                                                                                                                                                                                                        |
| <b>Return Value</b>             | Std_ReturnType                                                                                                                                                                                                                                                                                                                                                                                                                                     | E_OK: request successful<br>E_NOT_OK: request failed<br>CSM_E_BUSY: request failed, service is still busy<br>CSM_E_SMALL_BUFFER: the provided buffer is too small to store the result                                                                                             |
| <b>Description</b>              | This interface shall be used to feed the symmetrical block decryption service with the input data. If the service state is "idle", the function has to return with "CSM_E_NOT_OK". Otherwise, this function shall call the function Cry_<Primitive>Update of the primitive which is identified by the stored configuration information and return the value returned by that function. The decryption process is done by the underlying primitive. |                                                                                                                                                                                                                                                                                   |
| <b>Preconditions</b>            | CSM SymBlock interface should have been initialized.                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                   |
| <b>Configuration Dependency</b> | CSM_SYM_BLOCK_DECRYPT_STATUS should be STD_ON                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                   |

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|----------------------|--------------------------------------------------------------------------------------|--|
| <b>Function Name</b> | Csm_SymBlockDecryptFinish                                                            |  |
| <b>Syntax</b>        | FUNC(Csm_ReturnType, CSM_CODE)<br>Csm_SymBlockDecryptFinish(Csm_ConfigIdType cfgId ) |  |
| <b>Service ID</b>    | 0x15                                                                                 |  |
| <b>Sync/Async</b>    | Sync or Async, dependent on configuration                                            |  |

|                                 |                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                           |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| <b>Reentrancy</b>               | Non-Re-entrant                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                           |
| <b>Parameters (In)</b>          | cfgId                                                                                                                                                                                                                                                                                                                                                                                                                        | Holds the identifier of the CSM module configuration that has to be used during the operation.            |
| <b>Parameters (Inout)</b>       | None                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                           |
| <b>Parameters (Out)</b>         | None                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                           |
| <b>Return Value</b>             | Std_ReturnType                                                                                                                                                                                                                                                                                                                                                                                                               | E_OK: request successful<br>E_NOT_OK: request failed<br>CSM_E_BUSY: request failed, service is still busy |
| <b>Description</b>              | This interface shall be used to finish the symmetrical block decryption service. If the service state is "idle", the function has to return with "E_NOT_OK". Otherwise, this function shall call the function Cry_<Primitive>Finish of the primitive which is identified by the stored configuration information and return the value returned by that function. The decryption process is done by the underlying primitive. |                                                                                                           |
| <b>Preconditions</b>            | CSM SymBlock interface should have been initialized.                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                           |
| <b>Configuration Dependency</b> | CSM_SYM_BLOCK_DECRYPT_STATUS should be STD_ON                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                           |

### Sym interface Api

|                           |                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                          |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| 6.2 <b>Function Name</b>  | Csm_SymEncryptStart                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                          |
| <b>Syntax</b>             | FUNC(Csm_ReturnType, CSM_CODE) Csm_SymEncryptStart( Csm_ConfigIdType cfgId, const Csm_SymKeyType* keyPtr, const uint8* InitVectorPtr, uint32 InitVectorLength )                                                                                                                                                                                                                                                |                                                                                                                          |
| <b>Service ID</b>         | 0x16                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                          |
| <b>Sync/Async</b>         | Sync or Async, dependent on configuration                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                          |
| <b>Reentrancy</b>         | Non Re-entrant                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                          |
| <b>Parameters (In)</b>    | cfgId                                                                                                                                                                                                                                                                                                                                                                                                          | holds the identifier of the CSM module configuration which has to be used during the symmetrical encryption computation. |
|                           | keyPtr                                                                                                                                                                                                                                                                                                                                                                                                         | holds a pointer to the key which has to be used during the symmetrical block encryption computation.                     |
|                           | InitVectorPtr                                                                                                                                                                                                                                                                                                                                                                                                  | holds a pointer to the initialisation vector which has to be used during the symmetrical encryption computation          |
|                           | InitVectorLength                                                                                                                                                                                                                                                                                                                                                                                               | holds the length of the initialisation vector which has to be used during the symmetrical encryption computation         |
| <b>Parameters (Inout)</b> | None                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                          |
| <b>Parameters (Out)</b>   | None                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                          |
| <b>Return Value</b>       | Std_ReturnType                                                                                                                                                                                                                                                                                                                                                                                                 | E_OK: request successful<br>E_NOT_OK: request failed<br>CSM_E_BUSY: request failed, service is still busy                |
| <b>Description</b>        | This interface shall be used to initialize the symmetrical encrypt service of the CSM module. If the service state is "active", the function shall return with "CSM_E_BUSY". Otherwise, this function shall store the given configuration information which is identified by "cfgId", call the function Cry_<Primitive>Start of the primitive which is identified by the "cfgId" and return the value returned |                                                                                                                          |

|                                 |                                                                                                               |
|---------------------------------|---------------------------------------------------------------------------------------------------------------|
|                                 | by that function. If Cry_<Primitive>Start returned successfully, the service state has to be set to "active". |
| <b>Preconditions</b>            | CSM_Sym interface should have been initialized.                                                               |
| <b>Configuration Dependency</b> | CSM_SYM_ENCRYPT_STATUS should be STD_ON                                                                       |

|                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                    |
|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Function Name</b>            | Csm_SymEncryptUpdate                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                    |
| <b>Syntax</b>                   | FUNC(Csm_ReturnType, CSM_CODE)<br>Csm_SymEncryptUpdate(Csm_ConfigIdType cfgId, const uint8* plainTextPtr, uint32 plainTextLength, uint8* cipherTextPtr, uint32* cipherTextLengthPtr )                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                    |
| <b>Service ID</b>               | 0x17                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                    |
| <b>Sync/Async</b>               | Sync or Async, dependent on configuration                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                    |
| <b>Reentrancy</b>               | Non-Re-entrant                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                    |
| <b>Parameters (In)</b>          | cfgId                                                                                                                                                                                                                                                                                                                                                                                                                                        | Holds the identifier of the CSM module configuration that has to be used during the operation.                                                                                                                                                                                     |
|                                 | plainTextPtr                                                                                                                                                                                                                                                                                                                                                                                                                                 | holds a pointer to the plain text that shall be encrypted.                                                                                                                                                                                                                         |
|                                 | plainTextLength                                                                                                                                                                                                                                                                                                                                                                                                                              | contains the length of the plain text in bytes                                                                                                                                                                                                                                     |
| <b>Parameters (Inout)</b>       | cipherTextLengthPtr                                                                                                                                                                                                                                                                                                                                                                                                                          | holds a pointer to a memory location in which the length information is stored. On calling this function this parameter shall contain the size of the buffer provided by cipherTextPtr. When the request has finished, the amount of data that has been encrypted shall be stored. |
| <b>Parameters (Out)</b>         | cipherTextPtr                                                                                                                                                                                                                                                                                                                                                                                                                                | holds a pointer to the memory location which will hold the encrypted text.                                                                                                                                                                                                         |
| <b>Return Value</b>             | Std_ReturnType                                                                                                                                                                                                                                                                                                                                                                                                                               | E_OK: request successful<br>E_NOT_OK: request failed<br>CSM_E_BUSY: request failed, service is still busy<br>CSM_E_SMALL_BUFFER: the provided buffer is too small to store the result                                                                                              |
| <b>Description</b>              | This interface shall be used to feed the symmetrical encryption service with the input data. If the service state is "idle", the function has to return with "CSM_E_NOT_OK". Otherwise, this function shall call the function Cry_<Primitive>Update of the primitive which is identified by the stored configuration information and return the value returned by that function. The encryption process is done by the underlying primitive. |                                                                                                                                                                                                                                                                                    |
| <b>Preconditions</b>            | CSM_Sym interface should have been initialized.                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                    |
| <b>Configuration Dependency</b> | CSM_SYM_ENCRYPT_STATUS should be STD_ON                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                    |

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|----------------------|-------------------------------------------------------------------------------------------------------------------------------------|--|
| <b>Function Name</b> | Csm_SymEncryptFinish                                                                                                                |  |
| <b>Syntax</b>        | FUNC(Csm_ReturnType, CSM_CODE)<br>Csm_SymEncryptFinish( Csm_ConfigIdType cfgId, uint8* cipherTextPtr, uint32* cipherTextLengthPtr ) |  |
| <b>Service ID</b>    | 0x18                                                                                                                                |  |
| <b>Sync/Async</b>    | Sync or Async, dependent on configuration                                                                                           |  |
| <b>Reentrancy</b>    | Non-Re-entrant                                                                                                                      |  |

|                                 |                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                    |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Parameters (In)</b>          | cfgId                                                                                                                                                                                                                                                                                                                                                                                                                  | Holds the identifier of the CSM module configuration that has to be used during the operation.                                                                                                                                                                                     |
| <b>Parameters (Inout)</b>       | cipherTextLengthPtr                                                                                                                                                                                                                                                                                                                                                                                                    | holds a pointer to a memory location in which the length information is stored. On calling this function this parameter shall contain the size of the buffer provided by cipherTextPtr. When the request has finished, the amount of data that has been encrypted shall be stored. |
| <b>Parameters (Out)</b>         | cipherTextPtr                                                                                                                                                                                                                                                                                                                                                                                                          | holds a pointer to the memory location which will hold the encrypted text.                                                                                                                                                                                                         |
| <b>Return Value</b>             | Std_ReturnType                                                                                                                                                                                                                                                                                                                                                                                                         | E_OK: request successful<br>E_NOT_OK: request failed<br>CSM_E_BUSY: request failed, service is still busy<br>CSM_E_SMALL_BUFFER: the provided buffer is too small to store the result                                                                                              |
| <b>Description</b>              | This interface shall be used to finish the symmetrical encryption service. If the service state is "idle", the function has to return with "E_NOT_OK". Otherwise, this function shall call the function Cry_<Primitive>Finish of the primitive which is identified by the stored configuration information and return the value returned by that function. The encryption process is done by the underlying primitive. |                                                                                                                                                                                                                                                                                    |
| <b>Preconditions</b>            | CSM Sym interface should have been initialized.                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                    |
| <b>Configuration Dependency</b> | CSM_SYM_ENCRYPT_STATUS should be STD_ON                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                    |

|                           |                                                                                                                                                                |                                                                                                                                  |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| <b>Function Name</b>      | Csm_SymDecryptStart                                                                                                                                            |                                                                                                                                  |
| <b>Syntax</b>             | FUNC(Csm_ReturnType, CSM_CODE) Csm_SymDecryptStart(Csm_ConfigIdType cfgId, const Csm_SymKeyType* keyPtr, const uint8* InitVectorPtr, uint32 InitVectorLength ) |                                                                                                                                  |
| <b>Service ID</b>         | 0x19                                                                                                                                                           |                                                                                                                                  |
| <b>Sync/Async</b>         | Sync or Async, dependent on configuration                                                                                                                      |                                                                                                                                  |
| <b>Reentrancy</b>         | Non Re-entrant                                                                                                                                                 |                                                                                                                                  |
| <b>Parameters (In)</b>    | cfgId                                                                                                                                                          | holds the identifier of the constant CSM module configuration which has to be used during the symmetrical decryption computation |
|                           | keyPtr                                                                                                                                                         | holds a pointer to the key which has to be used during the symmetrical decryption computation                                    |
|                           | InitVectorPtr                                                                                                                                                  | holds a pointer to the initialisation vector which has to be used during the symmetrical decryption computation                  |
|                           | InitVectorLength                                                                                                                                               | holds the length of the initialisation vector which has to be used during the symmetrical decryption computation                 |
| <b>Parameters (Inout)</b> | None                                                                                                                                                           |                                                                                                                                  |
| <b>Parameters (Out)</b>   | None                                                                                                                                                           |                                                                                                                                  |
| <b>Return Value</b>       | Std_ReturnType                                                                                                                                                 | E_OK: request successful<br>E_NOT_OK: request failed<br>CSM_E_BUSY: request failed, service is still busy                        |

|                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b>              | This interface shall be used to initialize the symmetrical decrypt service of the CSM module.<br>If the service state is "active", the function shall return with "CSM_E_BUSY". Otherwise, this function shall store the given configuration information which is identified by "cfgld", call the function Cry_<Primitive>Start of the primitive which is identified by the "cfgld" and return the value returned by that function. If Cry_<Primitive>Start returned successfully, the service state has to be set to "active". |
| <b>Preconditions</b>            | CSM Sym interface should have been initialized.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Configuration Dependency</b> | CSM_SYM_DECRYPT_STATUS should be STD_ON                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

|                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                   |
|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Function Name</b>            | Csm_SymDecryptUpdate                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                   |
| <b>Syntax</b>                   | FUNC(Csm_ReturnType, CSM_CODE)<br>Csm_SymDecryptUpdate(Csm_ConfigIdType cfgld, const uint8* cipherTextPtr, uint32 cipherTextLength, uint8* plainTextPtr, uint32* plainTextLengthPtr)                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                   |
| <b>Service ID</b>               | 0x1a                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                   |
| <b>Sync/Async</b>               | Sync or Async, dependent on configuration                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                   |
| <b>Reentrancy</b>               | Non Re-entrant                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                   |
| <b>Parameters (In)</b>          | cfgld                                                                                                                                                                                                                                                                                                                                                                                                                                        | Holds the identifier of the CSM module configuration that has to be used during the operation.                                                                                                                                                                                    |
|                                 | cipherTextPtr                                                                                                                                                                                                                                                                                                                                                                                                                                | holds a pointer to the constant cipher text that shall be decrypted.                                                                                                                                                                                                              |
|                                 | cipherTextLength                                                                                                                                                                                                                                                                                                                                                                                                                             | contains the length of the cipher text in bytes.                                                                                                                                                                                                                                  |
| <b>Parameters (Inout)</b>       | plainTextLengthPtr                                                                                                                                                                                                                                                                                                                                                                                                                           | holds a pointer to a memory location in which the length information is stored. On calling this function this parameter shall contain the size of the buffer provided by plainTextPtr. When the request has finished, the amount of data that has been decrypted shall be stored. |
| <b>Parameters (Out)</b>         | plainTextPtr                                                                                                                                                                                                                                                                                                                                                                                                                                 | holds a pointer to the memory location which will hold the decrypted text.                                                                                                                                                                                                        |
| <b>Return Value</b>             | Std_ReturnType                                                                                                                                                                                                                                                                                                                                                                                                                               | E_OK: request successful<br>E_NOT_OK: request failed<br>CSM_E_BUSY: request failed, service is still busy<br>CSM_E_SMALL_BUFFER: the provided buffer is too small to store the result                                                                                             |
| <b>Description</b>              | This interface shall be used to feed the symmetrical decryption service with the input data. If the service state is "idle", the function has to return with "CSM_E_NOT_OK". Otherwise, this function shall call the function Cry_<Primitive>Update of the primitive which is identified by the stored configuration information and return the value returned by that function. The decryption process is done by the underlying primitive. |                                                                                                                                                                                                                                                                                   |
| <b>Preconditions</b>            | CSM Sym interface should have been initialized.                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                   |
| <b>Configuration Dependency</b> | CSM_SYM_DECRYPT_STATUS should be STD_ON                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                   |

|                      |                      |
|----------------------|----------------------|
| <b>Function Name</b> | Csm_SymDecryptFinish |
|----------------------|----------------------|

|                                 |                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                   |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Syntax</b>                   | FUNC(Csm_ReturnType, CSM_CODE)<br>Csm_SymDecryptFinish(Csm_ConfigIdType cfgId, uint8* plainTextPtr, uint32* plainTextLengthPtr )                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                   |
| <b>Service ID</b>               | 0x1b                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                   |
| <b>Sync/Async</b>               | Sync or Async, dependent on configuration                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                   |
| <b>Reentrancy</b>               | Non-Re-entrant                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                   |
| <b>Parameters (In)</b>          | cfgId                                                                                                                                                                                                                                                                                                                                                                                                                  | Holds the identifier of the CSM module configuration that has to be used during the operation.                                                                                                                                                                                    |
| <b>Parameters (Inout)</b>       | plainTextLengthPtr                                                                                                                                                                                                                                                                                                                                                                                                     | holds a pointer to a memory location in which the length information is stored. On calling this function this parameter shall contain the size of the buffer provided by plainTextPtr. When the request has finished, the amount of data that has been decrypted shall be stored. |
| <b>Parameters (Out)</b>         | plainTextPtr                                                                                                                                                                                                                                                                                                                                                                                                           | holds a pointer to the memory location which will hold the decrypted text.                                                                                                                                                                                                        |
| <b>Return Value</b>             | Std_ReturnType                                                                                                                                                                                                                                                                                                                                                                                                         | E_OK: request successful<br>E_NOT_OK: request failed<br>CSM_E_BUSY: request failed, service is still busy<br>CSM_E_SMALL_BUFFER: the provided buffer is too small to store the result                                                                                             |
| <b>Description</b>              | This interface shall be used to finish the symmetrical decryption service. If the service state is "idle", the function has to return with "E_NOT_OK". Otherwise, this function shall call the function Cry_<Primitive>Finish of the primitive which is identified by the stored configuration information and return the value returned by that function. The decryption process is done by the underlying primitive. |                                                                                                                                                                                                                                                                                   |
| <b>Preconditions</b>            | CSM Sym interface should have been initialized.                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                   |
| <b>Configuration Dependency</b> | CSM_SYM_DECRYPT_STATUS should be STD_ON                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                   |

### Asym interface Api

|                           |                                                                                                                    |                                                                                                                           |
|---------------------------|--------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| <b>Function Name</b>      | Csm_AsymEncryptStart                                                                                               |                                                                                                                           |
| <b>Syntax</b>             | FUNC(Csm_ReturnType, CSM_CODE) Csm_AsymEncryptStart( Csm_ConfigIdType cfgId, const Csm_AsymPublicKeyType* keyPtr ) |                                                                                                                           |
| <b>Service ID</b>         | 0x1c                                                                                                               |                                                                                                                           |
| <b>Sync/Async</b>         | Sync or Async, dependent on configuration                                                                          |                                                                                                                           |
| <b>Reentrancy</b>         | Non Re-entrant                                                                                                     |                                                                                                                           |
| <b>Parameters (In)</b>    | cfgId                                                                                                              | holds the identifier of the CSM module configuration which has to be used during the asymmetrical encryption computation. |
|                           | keyPtr                                                                                                             | holds a pointer to the key which has to be used during the Asymmetrical block encryption computation.                     |
| <b>Parameters (Inout)</b> | None                                                                                                               |                                                                                                                           |
| <b>Parameters (Out)</b>   | None                                                                                                               |                                                                                                                           |
| <b>Return Value</b>       | Std_ReturnType                                                                                                     | E_OK: request successful<br>E_NOT_OK: request failed                                                                      |

|                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
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|                                 | CSM_E_BUSY: request failed, service is still busy                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Description</b>              | This interface shall be used to initialize the asymmetrical encrypt service of the CSM module. If the service state is "active", the function shall return with "CSM_E_BUSY". Otherwise, this function shall store the given configuration information which is identified by "cfgld", call the function Cry_<Primitive>Start of the primitive which is identified by the "cfgld" and return the value returned by that function. If Cry_<Primitive>Start returned successfully, the service state has to be set to "active". |
| <b>Preconditions</b>            | CSM Asym interface should have been initialized.                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Configuration Dependency</b> | CSM_ASYM_ENCRYPT_STATUS should be STD_ON                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

|                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                    |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Function Name</b>            | Csm_AsymEncryptUpdate                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                    |
| <b>Syntax</b>                   | FUNC(Csm_ReturnType, CSM_CODE)<br>Csm_AsymEncryptUpdate(Csm_ConfigIdType cfgld, const uint8* plainTextPtr, uint32 plainTextLength, uint8* cipherTextPtr, uint32* cipherTextLengthPtr )                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                    |
| <b>Service ID</b>               | 0x1d                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                    |
| <b>Sync/Async</b>               | Sync or Async, dependent on configuration                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                    |
| <b>Reentrancy</b>               | Non-Re-entrant                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                    |
| <b>Parameters (In)</b>          | cfgld                                                                                                                                                                                                                                                                                                                                                                                                                                         | Holds the identifier of the CSM module configuration that has to be used during the operation.                                                                                                                                                                                     |
|                                 | plainTextPtr                                                                                                                                                                                                                                                                                                                                                                                                                                  | holds a pointer to the plain text that shall be encrypted.                                                                                                                                                                                                                         |
|                                 | plainTextLength                                                                                                                                                                                                                                                                                                                                                                                                                               | contains the length of the plain text in bytes                                                                                                                                                                                                                                     |
| <b>Parameters (Inout)</b>       | cipherTextLengthPtr                                                                                                                                                                                                                                                                                                                                                                                                                           | holds a pointer to a memory location in which the length information is stored. On calling this function this parameter shall contain the size of the buffer provided by cipherTextPtr. When the request has finished, the amount of data that has been encrypted shall be stored. |
| <b>Parameters (Out)</b>         | cipherTextPtr                                                                                                                                                                                                                                                                                                                                                                                                                                 | holds a pointer to the memory location which will hold the encrypted text.                                                                                                                                                                                                         |
| <b>Return Value</b>             | Std_ReturnType                                                                                                                                                                                                                                                                                                                                                                                                                                | E_OK: request successful<br>E_NOT_OK: request failed<br>CSM_E_BUSY: request failed, service is still busy<br>CSM_E_SMALL_BUFFER: the provided buffer is too small to store the result                                                                                              |
| <b>Description</b>              | This interface shall be used to feed the asymmetrical encryption service with the input data. If the service state is "idle", the function has to return with "CSM_E_NOT_OK". Otherwise, this function shall call the function Cry_<Primitive>Update of the primitive which is identified by the stored configuration information and return the value returned by that function. The encryption process is done by the underlying primitive. |                                                                                                                                                                                                                                                                                    |
| <b>Preconditions</b>            | CSM Asym interface should have been initialized.                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                    |
| <b>Configuration Dependency</b> | CSM_ASYM_ENCRYPT_STATUS should be STD_ON                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                    |

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|----------------------|-----------------------|
| <b>Function Name</b> | Csm_AsymEncryptFinish |
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|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Syntax</b>                   | FUNC(Csm_ReturnType, CSM_CODE)<br>Csm_AsymEncryptFinish( Csm_ConfigIdType cfgId, uint8* cipherTextPtr, uint32* cipherTextLengthPtr )                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                    |
| <b>Service ID</b>               | 0x1e                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                    |
| <b>Sync/Async</b>               | Sync or Async, dependent on configuration                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                    |
| <b>Reentrancy</b>               | Non-Re-entrant                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                    |
| <b>Parameters (In)</b>          | cfgId                                                                                                                                                                                                                                                                                                                                                                                                                   | Holds the identifier of the CSM module configuration that has to be used during the operation.                                                                                                                                                                                     |
| <b>Parameters (Inout)</b>       | cipherTextLengthPtr                                                                                                                                                                                                                                                                                                                                                                                                     | holds a pointer to a memory location in which the length information is stored. On calling this function this parameter shall contain the size of the buffer provided by cipherTextPtr. When the request has finished, the amount of data that has been encrypted shall be stored. |
| <b>Parameters (Out)</b>         | cipherTextPtr                                                                                                                                                                                                                                                                                                                                                                                                           | holds a pointer to the memory location which will hold the encrypted text.                                                                                                                                                                                                         |
| <b>Return Value</b>             | Std_ReturnType                                                                                                                                                                                                                                                                                                                                                                                                          | E_OK: request successful<br>E_NOT_OK: request failed<br>CSM_E_BUSY: request failed, service is still busy<br>CSM_E_SMALL_BUFFER: the provided buffer is too small to store the result                                                                                              |
| <b>Description</b>              | This interface shall be used to finish the asymmetrical encryption service. If the service state is "idle", the function has to return with "E_NOT_OK". Otherwise, this function shall call the function Cry_<Primitive>Finish of the primitive which is identified by the stored configuration information and return the value returned by that function. The encryption process is done by the underlying primitive. |                                                                                                                                                                                                                                                                                    |
| <b>Preconditions</b>            | CSM Asym interface should have been initialized.                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                    |
| <b>Configuration Dependency</b> | CSM_ASYM_ENCRYPT_STATUS should be STD_ON                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                    |

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|---------------------------|-----------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| <b>Function Name</b>      | Csm_AsymDecryptStart                                                                                                  |                                                                                                                                   |
| <b>Syntax</b>             | FUNC(Csm_ReturnType, CSM_CODE)<br>Csm_AsymDecryptStart(Csm_ConfigIdType cfgId, const Csm_AsymPrivateKeyType* keyPtr ) |                                                                                                                                   |
| <b>Service ID</b>         | 0x1f                                                                                                                  |                                                                                                                                   |
| <b>Sync/Async</b>         | Sync or Async, dependent on configuration                                                                             |                                                                                                                                   |
| <b>Reentrancy</b>         | Non Re-entrant                                                                                                        |                                                                                                                                   |
| <b>Parameters (In)</b>    | cfgId                                                                                                                 | holds the identifier of the constant CSM module configuration which has to be used during the asymmetrical decryption computation |
|                           | keyPtr                                                                                                                | holds a pointer to the key which has to be used during the Asymmetrical decryption computation                                    |
| <b>Parameters (Inout)</b> | None                                                                                                                  |                                                                                                                                   |
| <b>Parameters (Out)</b>   | None                                                                                                                  |                                                                                                                                   |
| <b>Return Value</b>       | Std_ReturnType                                                                                                        | E_OK: request successful<br>E_NOT_OK: request failed<br>CSM_E_BUSY: request failed, service is still busy                         |

|                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b>              | This interface shall be used to initialize the asymmetrical decrypt service of the CSM module.<br>If the service state is "active", the function shall return with "CSM_E_BUSY". Otherwise, this function shall store the given configuration information which is identified by "cfgId", call the function Cry_<Primitive>Start of the primitive which is identified by the "cfgId" and return the value returned by that function. If Cry_<Primitive>Start returned successfully, the service state has to be set to "active". |
| <b>Preconditions</b>            | CSM Asym interface should have been initialized.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Configuration Dependency</b> | CSM_ASYM_DECRYPT_STATUS should be STD_ON                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

|                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                   |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Function Name</b>            | Csm_AsymDecryptUpdate                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                   |
| <b>Syntax</b>                   | FUNC(Csm_ReturnType, CSM_CODE)<br>Csm_AsymDecryptUpdate(Csm_ConfigIdType cfgId, const uint8* cipherTextPtr, uint32 cipherTextLength, uint8* plainTextPtr, uint32* plainTextLengthPtr)                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                   |
| <b>Service ID</b>               | 0x1a                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                   |
| <b>Sync/Async</b>               | Sync or Async, dependent on configuration                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                   |
| <b>Reentrancy</b>               | Non Re-entrant                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                   |
| <b>Parameters (In)</b>          | cfgId                                                                                                                                                                                                                                                                                                                                                                                                                                         | Holds the identifier of the CSM module configuration that has to be used during the operation.                                                                                                                                                                                    |
|                                 | cipherTextPtr                                                                                                                                                                                                                                                                                                                                                                                                                                 | holds a pointer to the constant cipher text that shall be decrypted.                                                                                                                                                                                                              |
|                                 | cipherTextLength                                                                                                                                                                                                                                                                                                                                                                                                                              | contains the length of the cipher text in bytes.                                                                                                                                                                                                                                  |
| <b>Parameters (Inout)</b>       | plainTextLengthPtr                                                                                                                                                                                                                                                                                                                                                                                                                            | holds a pointer to a memory location in which the length information is stored. On calling this function this parameter shall contain the size of the buffer provided by plainTextPtr. When the request has finished, the amount of data that has been decrypted shall be stored. |
| <b>Parameters (Out)</b>         | plainTextPtr                                                                                                                                                                                                                                                                                                                                                                                                                                  | holds a pointer to the memory location which will hold the decrypted text.                                                                                                                                                                                                        |
| <b>Return Value</b>             | Std_ReturnType                                                                                                                                                                                                                                                                                                                                                                                                                                | E_OK: request successful<br>E_NOT_OK: request failed<br>CSM_E_BUSY: request failed, service is still busy<br>CSM_E_SMALL_BUFFER: the provided buffer is too small to store the result                                                                                             |
| <b>Description</b>              | This interface shall be used to feed the asymmetrical decryption service with the input data. If the service state is "idle", the function has to return with "CSM_E_NOT_OK". Otherwise, this function shall call the function Cry_<Primitive>Update of the primitive which is identified by the stored configuration information and return the value returned by that function. The decryption process is done by the underlying primitive. |                                                                                                                                                                                                                                                                                   |
| <b>Preconditions</b>            | CSM Asym interface should have been initialized.                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                   |
| <b>Configuration Dependency</b> | CSM_ASYM_DECRYPT_STATUS should be STD_ON                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                   |

|                      |                       |
|----------------------|-----------------------|
| <b>Function Name</b> | Csm_AsymDecryptFinish |
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|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Syntax</b>                   | FUNC(Csm_ReturnType, CSM_CODE)<br>Csm_AsymDecryptFinish(Csm_ConfigIdType cfgId, uint8* plainTextPtr, uint32* plainTextLengthPtr )                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                   |
| <b>Service ID</b>               | 0x21                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                   |
| <b>Sync/Async</b>               | Sync or Async, dependent on configuration                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                   |
| <b>Reentrancy</b>               | Non-Re-entrant                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                   |
| <b>Parameters (In)</b>          | cfgId                                                                                                                                                                                                                                                                                                                                                                                                                   | Holds the identifier of the CSM module configuration that has to be used during the operation.                                                                                                                                                                                    |
| <b>Parameters (Inout)</b>       | plainTextLengthPtr                                                                                                                                                                                                                                                                                                                                                                                                      | holds a pointer to a memory location in which the length information is stored. On calling this function this parameter shall contain the size of the buffer provided by plainTextPtr. When the request has finished, the amount of data that has been decrypted shall be stored. |
| <b>Parameters (Out)</b>         | plainTextPtr                                                                                                                                                                                                                                                                                                                                                                                                            | holds a pointer to the memory location which will hold the decrypted text.                                                                                                                                                                                                        |
| <b>Return Value</b>             | Std_ReturnType                                                                                                                                                                                                                                                                                                                                                                                                          | E_OK: request successful<br>E_NOT_OK: request failed<br>CSM_E_BUSY: request failed, service is still busy<br>CSM_E_SMALL_BUFFER: the provided buffer is too small to store the result                                                                                             |
| <b>Description</b>              | This interface shall be used to finish the asymmetrical decryption service. If the service state is "idle", the function has to return with "E_NOT_OK". Otherwise, this function shall call the function Cry_<Primitive>Finish of the primitive which is identified by the stored configuration information and return the value returned by that function. The decryption process is done by the underlying primitive. |                                                                                                                                                                                                                                                                                   |
| <b>Preconditions</b>            | CSM Asym interface should have been initialized.                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                   |
| <b>Configuration Dependency</b> | CSM_ASYNC_DECRYPT_STATUS should be STD_ON                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                   |

### Signature interface Api

|                           |                                                                                                                              |                                                                                                           |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| <b>Function Name</b>      | Csm_SignatureGenerateStart                                                                                                   |                                                                                                           |
| <b>Syntax</b>             | FUNC(Std_ReturnType, CSM_CODE)<br>Csm_SignatureGenerateStart( Csm_ConfigIdType cfgId, const Csm_AsymPrivateKeyType* keyPtr ) |                                                                                                           |
| <b>Service ID</b>         | 0x22                                                                                                                         |                                                                                                           |
| <b>Sync/Async</b>         | Sync or Async, dependent on configuration                                                                                    |                                                                                                           |
| <b>Reentrancy</b>         | Non Re-entrant                                                                                                               |                                                                                                           |
| <b>Parameters (In)</b>    | cfgId                                                                                                                        | holds the identifier of the CSM module configuration which has to be used during the signature generation |
|                           | keyPtr                                                                                                                       | Holds a pointer to the key necessary for the signature generation.                                        |
| <b>Parameters (Inout)</b> | None                                                                                                                         |                                                                                                           |
| <b>Parameters (Out)</b>   | None                                                                                                                         |                                                                                                           |
| <b>Return Value</b>       | Std_ReturnType                                                                                                               | E_OK: request successful<br>E_NOT_OK: request failed<br>CSM_E_BUSY: request failed, service is still busy |

|                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b>              | This interface shall be used to initialize the signature generate service of the CSM module. If the service state is "active", the function shall return with "CSM_E_BUSY". Otherwise, this function shall store the given configuration information which is identified by "cfgld", call the function Cry_<Primitive>Start of the primitive which is identified by the "cfgld" and return the value returned by that function. If Cry_<Primitive>Start returned successfully, the service state has to be set to "active". |
| <b>Preconditions</b>            | CSM Signature interface should have been initialized.                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Configuration Dependency</b> | CSM_SIGNATURE_GENERATE_STATUS should be STD_ON                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

|                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                           |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| <b>Function Name</b>            | Csm_SignatureGenerateUpdate                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                           |
| <b>Syntax</b>                   | FUNC(Std_ReturnType, CSM_CODE)<br>Csm_SignatureGenerateUpdate(Csm_ConfigIdType cfgld, const uint8* dataPtr, uint32 dataLength)                                                                                                                                                                                                                                                                                                            |                                                                                                           |
| <b>Service ID</b>               | 0x23                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                           |
| <b>Sync/Async</b>               | Sync or Async, dependent on configuration                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                           |
| <b>Reentrancy</b>               | Non Re-entrant                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                           |
| <b>Parameters (In)</b>          | cfgld                                                                                                                                                                                                                                                                                                                                                                                                                                     | Holds the identifier of the CSM module configuration that has to be used during the operation.            |
|                                 | dataPtr                                                                                                                                                                                                                                                                                                                                                                                                                                   | holds a pointer to the signature which shall be signed                                                    |
|                                 | dataLength                                                                                                                                                                                                                                                                                                                                                                                                                                | contains the length of the data to be signed                                                              |
| <b>Parameters (Inout)</b>       | None                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                           |
| <b>Parameters (Out)</b>         | None                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                           |
| <b>Return Value</b>             | Std_ReturnType                                                                                                                                                                                                                                                                                                                                                                                                                            | E_OK: request successful<br>E_NOT_OK: request failed<br>CSM_E_BUSY: request failed, service is still busy |
| <b>Description</b>              | This interface shall be used to feed the signature generation service with the input data. If the service state is "idle", the function has to return with "E_NOT_OK". Otherwise, this function shall call the function Cry_<Primitive>Update of the primitive which is identified by the stored configuration information and return the value returned by that function. The signature computation is done by the underlying primitive. |                                                                                                           |
| <b>Preconditions</b>            | CSM Signature interface should have been initialized.                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                           |
| <b>Configuration Dependency</b> | CSM_SIGNATURE_GENERATE_STATUS should be STD_ON                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                           |

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| <b>Function Name</b>      | Csm_SignatureGenerateFinish                                                                                                        |                                                                                                                                                         |
| <b>Syntax</b>             | FUNC(Std_ReturnType, CSM_CODE)<br>Csm_SignatureGenerateFinish( Csm_ConfigIdType cfgld, uint8* resultPtr, uint32* resultLengthPtr ) |                                                                                                                                                         |
| <b>Service ID</b>         | 0x24                                                                                                                               |                                                                                                                                                         |
| <b>Sync/Async</b>         | Sync or Async, dependent on configuration                                                                                          |                                                                                                                                                         |
| <b>Reentrancy</b>         | Non Re-entrant                                                                                                                     |                                                                                                                                                         |
| <b>Parameters (In)</b>    | cfgld                                                                                                                              | Holds the identifier of the CSM module configuration that has to be used during the operation.                                                          |
| <b>Parameters (Inout)</b> | resultLengthPtr                                                                                                                    | holds a pointer to the memory location in which the length information is stored. On calling this function this parameter shall contain the size of the |

|                                 |                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                       |
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|                                 |                                                                                                                                                                                                                                                                                                                                                                                                                         | buffer provided by resultPtr. When the request has finished, the actual length of the computed signature shall be stored                                                              |
| <b>Parameters (Out)</b>         | resultPtr                                                                                                                                                                                                                                                                                                                                                                                                               | Holds a pointer to the memory location which will hold the result of the signature generation.                                                                                        |
| <b>Return Value</b>             | Std_ReturnType                                                                                                                                                                                                                                                                                                                                                                                                          | E_OK: request successful<br>E_NOT_OK: request failed<br>CSM_E_BUSY: request failed, service is still busy<br>CSM_E_SMALL_BUFFER: the provided buffer is too small to store the result |
| <b>Description</b>              | This interface shall be used to finish the signature generation service. If the service state is "idle", the function has to return with "E_NOT_OK". Otherwise, this function shall call the function Cry_<Primitive>Finish of the primitive which is identified by the stored configuration information and return the value returned by that function. The signature computation is done by the underlying primitive. |                                                                                                                                                                                       |
| <b>Preconditions</b>            | CSM Signature interface should have been initialized.                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                       |
| <b>Configuration Dependency</b> | CSM_SIGNATURE_GENERATE_STATUS should be STD_ON                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                       |

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| <b>Function Name</b>            | Csm_SignatureVerifyStart                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                         |
| <b>Syntax</b>                   | FUNC(Std_ReturnType, CSM_CODE) Csm_SignatureVerifyStart( Csm_ConfigIdType cfgId, const Csm_AsymPublicKeyType* keyPtr )                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                         |
| <b>Service ID</b>               | 0x25                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                         |
| <b>Sync/Async</b>               | Sync or Async, dependent on configuration                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                         |
| <b>Reentrancy</b>               | Non Re-entrant                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                         |
| <b>Parameters (In)</b>          | cfgId                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | holds the identifier of the CSM module configuration which has to be used during the signature computation/verification |
|                                 | keyPtr                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Holds a pointer to the key necessary for the signature verification.                                                    |
| <b>Parameters (Inout)</b>       | None                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                         |
| <b>Parameters (Out)</b>         | None                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                         |
| <b>Return Value</b>             | Std_ReturnType                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | E_OK: request successful<br>E_NOT_OK: request failed<br>CSM_E_BUSY: request failed, service is still busy               |
| <b>Description</b>              | This interface shall be used to initialize the signature verify service of the CSM module. If the service state is "active", the function shall return with "CSM_E_BUSY". Otherwise, this function shall store the given configuration information which is identified by "cfgId", call the function Cry_<Primitive>Start of the primitive which is identified by the "cfgId" and return the value returned by that function. If Cry_<Primitive>Start returned successfully, the service state has to be set to "active". |                                                                                                                         |
| <b>Preconditions</b>            | CSM Signature interface should have been initialized.                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                         |
| <b>Configuration Dependency</b> | CSM_SIGNATURE_VERIFY_STATUS should be STD_ON                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                         |

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| <b>Function Name</b> | Csm_SignatureVerifyUpdate |
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|                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                           |
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| <b>Syntax</b>                   | FUNC(Std_ReturnType, CSM_CODE) Csm_SignatureVerifyUpdate(Csm_ConfigIdType cfgId, const uint8* dataPtr, uint32 dataLength)                                                                                                                                                                                                                                                                                                                   |                                                                                                           |
| <b>Service ID</b>               | 0x26                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                           |
| <b>Sync/Async</b>               | Sync or Async, dependent on configuration                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                           |
| <b>Reentrancy</b>               | Non Re-entrant                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                           |
| <b>Parameters (In)</b>          | cfgId                                                                                                                                                                                                                                                                                                                                                                                                                                       | Holds the identifier of the CSM module configuration that has to be used during the operation.            |
|                                 | dataPtr                                                                                                                                                                                                                                                                                                                                                                                                                                     | holds a pointer to the signature which shall be verified                                                  |
|                                 | dataLength                                                                                                                                                                                                                                                                                                                                                                                                                                  | contains the length of the signature to verify in bytes                                                   |
| <b>Parameters (Inout)</b>       | None                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                           |
| <b>Parameters (Out)</b>         | None                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                           |
| <b>Return Value</b>             | Std_ReturnType                                                                                                                                                                                                                                                                                                                                                                                                                              | E_OK: request successful<br>E_NOT_OK: request failed<br>CSM_E_BUSY: request failed, service is still busy |
| <b>Description</b>              | This interface shall be used to feed the signature verification service with the input data. If the service state is "idle", the function has to return with "E_NOT_OK". Otherwise, this function shall call the function Cry_<Primitive>Update of the primitive which is identified by the stored configuration information and return the value returned by that function. The signature computation is done by the underlying primitive. |                                                                                                           |
| <b>Preconditions</b>            | CSM Signature interface should have been initialized.                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                           |
| <b>Configuration Dependency</b> | CSM_SIGNATURE_VERIFY_STATUS should be STD_ON                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                           |

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| <b>Function Name</b>      | Csm_SignatureVerifyFinish                                                                                                                                              |                                                                                                           |
| <b>Syntax</b>             | FUNC(Std_ReturnType, CSM_CODE) Csm_SignatureVerifyFinish( Csm_ConfigIdType cfgId, const uint8* signaturePtr, uint32 signatureLength, Csm_VerifyResultType* resultPtr ) |                                                                                                           |
| <b>Service ID</b>         | 0x27                                                                                                                                                                   |                                                                                                           |
| <b>Sync/Async</b>         | Sync or Async, dependent on configuration                                                                                                                              |                                                                                                           |
| <b>Reentrancy</b>         | Non Re-entrant                                                                                                                                                         |                                                                                                           |
| <b>Parameters (In)</b>    | cfgId                                                                                                                                                                  | Holds the identifier of the CSM module configuration that has to be used during the operation.            |
|                           | signaturePtr                                                                                                                                                           | holds a pointer to the memory location which holds the signature to be verified                           |
|                           | signatureLength                                                                                                                                                        | holds the length of the Signature to be verified                                                          |
| <b>Parameters (Inout)</b> | None                                                                                                                                                                   |                                                                                                           |
| <b>Parameters (Out)</b>   | resultPtr                                                                                                                                                              | Holds a pointer to the memory location which will hold the result of the signature verification.          |
| <b>Return Value</b>       | Std_ReturnType                                                                                                                                                         | E_OK: request successful<br>E_NOT_OK: request failed<br>CSM_E_BUSY: request failed, service is still busy |
| <b>Description</b>        | This interface shall be used to finish the signature verification service. If the service state is "idle", the function has to return with "E_NOT_OK". Otherwise,      |                                                                                                           |

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|                                 | this function shall call the function Cry_<Primitive>Finish of the primitive which is identified by the stored configuration information and return the value returned by that function. The signature computation is done by the underlying primitive. |
| <b>Preconditions</b>            | CSM Signature interface should have been initialized.                                                                                                                                                                                                   |
| <b>Configuration Dependency</b> | CSM_SIGNATURE_VERIFY_STATUS should be STD_ON                                                                                                                                                                                                            |

### Key Derive interface Api

|                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                     |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Function Name</b>            | Csm_KeyDeriveStart                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                     |
| <b>Syntax</b>                   | FUNC(Std_ReturnType, CSM_CODE) Csm_KeyDeriveStart( Csm_ConfigIdType cfgId, uint32 keyLength, uint32 iterations )                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                     |
| <b>Service ID</b>               | 0x2b                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                     |
| <b>Sync/Async</b>               | Sync or Async, dependent on configuration                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                     |
| <b>Reentrancy</b>               | Non Re-entrant                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                     |
| <b>Parameters (In)</b>          | cfgId                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | holds the identifier of the CSM module configuration which has to be used during the key derivation                                                                 |
|                                 | keyLength                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Determines the type of password entered in Csm_KeyDeriveUpdate.<br>0: PSK not used. keyLength value is ignored<br>Non-zero: Use PSK. Use keyLength as PSK key index |
|                                 | iterations                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | holds the number of iterations to be performed by the underlying key derivation primitive                                                                           |
| <b>Parameters (Inout)</b>       | None                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                     |
| <b>Parameters (Out)</b>         | None                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                     |
| <b>Return Value</b>             | Std_ReturnType                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | E_OK: request successful<br>E_NOT_OK: request failed<br>CSM_E_BUSY: request failed, service is still busy                                                           |
| <b>Description</b>              | This interface shall be used to initialize the key derivation service of the CSM module. If the service state is "active", the function shall return with "CSM_E_BUSY". Otherwise, this function shall store the given configuration information which is identified by "cfgId", call the function Cry_<Primitive>Start of the primitive which is identified by the "cfgId" and return the value returned by that function. If Cry_<Primitive>Start returned successfully, the service state has to be set to "active". |                                                                                                                                                                     |
| <b>Preconditions</b>            | CSM Key Derivation interface should have been initialized.                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                     |
| <b>Configuration Dependency</b> | CSM_KEY_DERIVATION_STATUS should be STD_ON                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                     |

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| <b>Function Name</b>   | Csm_KeyDeriveUpdate                                                                                                                                                    |                                                                                                |
| <b>Syntax</b>          | FUNC(Std_ReturnType, CSM_CODE) Csm_KeyDeriveUpdate (Csm_ConfigIdType cfgId, const uint8* passwordPtr, uint32 passwordLength, const uint8* saltPtr, uint32 saltLength ) |                                                                                                |
| <b>Service ID</b>      | 0x2c                                                                                                                                                                   |                                                                                                |
| <b>Sync/Async</b>      | Sync or Async, dependent on configuration                                                                                                                              |                                                                                                |
| <b>Reentrancy</b>      | Non Re-entrant                                                                                                                                                         |                                                                                                |
| <b>Parameters (In)</b> | cfgId                                                                                                                                                                  | Holds the identifier of the CSM module configuration that has to be used during the operation. |

|                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                              |
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|                                 | passwordPtr                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Valid only when keyLength of Csm_KeyDeriveStart is set to 0. If keyLength is anything other than 0, it is ignored.<br>holds a pointer to the password, i.e. the original key, from which to derive a new key |
|                                 | passwordLength                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Valid only when keyLength of Csm_KeyDeriveStart is set to 0. If keyLength is anything other than 0, it is ignored.<br>holds the length of the password in bytes                                              |
|                                 | saltPtr                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | holds a pointer to the cryptographic salt, i.e. a random number, for the underlying primitive                                                                                                                |
|                                 | saltLength                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | holds the length of the salt in bytes                                                                                                                                                                        |
| <b>Parameters (Inout)</b>       | None                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                              |
| <b>Parameters (Out)</b>         | None                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                              |
| <b>Return Value</b>             | Std_ReturnType                                                                                                                                                                                                                                                                                                                                                                                                                                                             | E_OK: request successful<br>E_NOT_OK: request failed<br>CSM_E_BUSY: request failed, service is still busy                                                                                                    |
| <b>Description</b>              | <p>This interface shall be used to feed the key derivation service with the input data.</p> <p>If the service state is "idle", the function has to return with "E_NOT_OK".</p> <p>Otherwise, this function shall call the function Cry_&lt;Primitive&gt;Update of the primitive which is identified by the stored configuration information and return the value returned by that function.</p> <p>The key derivation computation is done by the underlying primitive.</p> |                                                                                                                                                                                                              |
| <b>Preconditions</b>            | CSM Key Derivation interface should have been initialized.                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                              |
| <b>Configuration Dependency</b> | CSM_KEY_DERIVATION_STATUS should be STD_ON                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                              |

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| <b>Function Name</b>      | Csm_KeyDeriveFinish                                                                                                                                          |                                                                                                           |
| <b>Syntax</b>             | FUNC(Std_ReturnType, CSM_CODE) Csm_KeyDeriveFinish ( Csm_ConfigIdType cfgId, Csm_SymKeyType* keyPtr )                                                        |                                                                                                           |
| <b>Service ID</b>         | 0x2d                                                                                                                                                         |                                                                                                           |
| <b>Sync/Async</b>         | Sync or Async, dependent on configuration                                                                                                                    |                                                                                                           |
| <b>Reentrancy</b>         | Non Re-entrant                                                                                                                                               |                                                                                                           |
| <b>Parameters (In)</b>    | cfgId                                                                                                                                                        | Holds the identifier of the CSM module configuration that has to be used during the operation.            |
| <b>Parameters (Inout)</b> | keyPtr                                                                                                                                                       | holds a pointer to the memory location which will hold the result of the key derivation                   |
| <b>Parameters (Out)</b>   | None                                                                                                                                                         |                                                                                                           |
| <b>Return Value</b>       | Std_ReturnType                                                                                                                                               | E_OK: request successful<br>E_NOT_OK: request failed<br>CSM_E_BUSY: request failed, service is still busy |
| <b>Description</b>        | <p>This interface shall be used to finish the key derivation service.</p> <p>If the service state is "idle", the function has to return with "E_NOT_OK".</p> |                                                                                                           |

|                                 |                                                                                                                                                                                                                                                                                |
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|                                 | Otherwise, this function shall call the function Cry_<Primitive>Finish of the primitive which is identified by the stored configuration information and return the value returned by that function.<br><br>The key derivation computation is done by the underlying primitive. |
| <b>Preconditions</b>            | CSM Key Derivation interface should have been initialized.                                                                                                                                                                                                                     |
| <b>Configuration Dependency</b> | CSM_KEY_DERIVATION_STATUS should be STD_ON                                                                                                                                                                                                                                     |

### Key Exchange Calc Secret interface Api

|                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                           |
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| <b>Function Name</b>            | Csm_KeyExchangeCalcSecretStart                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                           |
| <b>Syntax</b>                   | FUNC(Std_ReturnType, CSM_CODE) Csm_KeyExchangeCalcSecretStart( Csm_ConfigIdType cfgId, const Csm_KeyExchangeBaseType* basePtr, const Csm_KeyExchangePrivateType* privateValuePtr )                                                                                                                                                                                                                                                                                                                                    |                                                                                                           |
| <b>Service ID</b>               | 0x2f                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                           |
| <b>Sync/Async</b>               | Sync or Async, dependent on configuration                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                           |
| <b>Reentrancy</b>               | Non Re-entrant                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                           |
| <b>Parameters (In)</b>          | cfgId                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | holds the identifier of the CSM module configuration which has to be used during the key exchange         |
|                                 | basePtr                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | holds a pointer to the base information known to both users of the key exchange protocol                  |
|                                 | privateValuePtr                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | holds a pointer to the private information known only to the current user of the key exchange protocol    |
| <b>Parameters (Inout)</b>       | None                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                           |
| <b>Parameters (Out)</b>         | None                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                           |
| <b>Return Value</b>             | Std_ReturnType                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | E_OK: request successful<br>E_NOT_OK: request failed<br>CSM_E_BUSY: request failed, service is still busy |
| <b>Description</b>              | This interface shall be used to initialize the key exchange service of the CSM module. If the service state is "active", the function shall return with "CSM_E_BUSY". Otherwise, this function shall store the given configuration information which is identified by "cfgId", call the function Cry_<Primitive>Start of the primitive which is identified by the "cfgId" and return the value returned by that function. If Cry_<Primitive>Start returned successfully, the service state has to be set to "active". |                                                                                                           |
| <b>Preconditions</b>            | CSM Key Exchange Calc Secret interface should have been initialized.                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                           |
| <b>Configuration Dependency</b> | CSM_KEY_EXCHANGE_CALC_SECRET_STATUS should be STD_ON                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                           |

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| <b>Function Name</b>   | Csm_KeyExchangeCalcSecretUpdate                                                                                                                              |                                                                                                |
| <b>Syntax</b>          | FUNC(Std_ReturnType, CSM_CODE) Csm_KeyExchangeCalcSecretUpdate (Csm_ConfigIdType cfgId, const uint8* partnerPublicValuePtr, uint32 partnerPublicValueLength) |                                                                                                |
| <b>Service ID</b>      | 0x30                                                                                                                                                         |                                                                                                |
| <b>Sync/Async</b>      | Sync or Async, dependent on configuration                                                                                                                    |                                                                                                |
| <b>Reentrancy</b>      | Non Re-entrant                                                                                                                                               |                                                                                                |
| <b>Parameters (In)</b> | cfgId                                                                                                                                                        | Holds the identifier of the CSM module configuration that has to be used during the operation. |

|                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                           |
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|                                 | partnerPublicValuePtr                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | holds a pointer to the data representing the public value of the key exchange partner                     |
|                                 | partnerPublicValueLength                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | contains the length of the part of the partner value in bytes.                                            |
| <b>Parameters (Inout)</b>       | None                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                           |
| <b>Parameters (Out)</b>         | None                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                           |
| <b>Return Value</b>             | Std_ReturnType                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | E_OK: request successful<br>E_NOT_OK: request failed<br>CSM_E_BUSY: request failed, service is still busy |
| <b>Description</b>              | <p>This interface shall be used to feed the key exchange service with the public value coming from the partner of the key exchange protocol.</p> <p>If the service state is "idle", the function has to return with "E_NOT_OK".</p> <p>Otherwise, this function shall call the function Cry_&lt;Primitive&gt;Update of the primitive which is identified by the stored configuration information and return the value returned by that function.</p> <p>The calculation of the shared secret is done by the underlying primitive.</p> |                                                                                                           |
| <b>Preconditions</b>            | CSM Key Exchange Calc Secret interface should have been initialized.                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                           |
| <b>Configuration Dependency</b> | CSM_KEY_EXCHANGE_CALC_SECRET_STATUS should be STD_ON                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                           |

|                           |                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                     |
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| <b>Function Name</b>      | Csm_KeyExchangeCalcSecretFinish                                                                                                                                                  |                                                                                                                                                                                                                                                                                     |
| <b>Syntax</b>             | FUNC(Std_ReturnType, CSM_CODE) Csm_KeyExchangeCalcSecretFinish<br>( Csm_ConfigIdType cfgId, uint8* sharedSecretPtr, uint32* sharedSecretLengthPtr, boolean TruncationIsAllowed ) |                                                                                                                                                                                                                                                                                     |
| <b>Service ID</b>         | 0x31                                                                                                                                                                             |                                                                                                                                                                                                                                                                                     |
| <b>Sync/Async</b>         | Sync or Async, dependent on configuration                                                                                                                                        |                                                                                                                                                                                                                                                                                     |
| <b>Reentrancy</b>         | Non Re-entrant                                                                                                                                                                   |                                                                                                                                                                                                                                                                                     |
| <b>Parameters (In)</b>    | cfgId                                                                                                                                                                            | Holds the identifier of the CSM module configuration that has to be used during the operation.                                                                                                                                                                                      |
|                           | TruncationIsAllowed                                                                                                                                                              | This parameter states whether a truncation of the result is allowed or not. TRUE: truncation is allowed. FALSE: truncation is not allowed.                                                                                                                                          |
| <b>Parameters (Inout)</b> | sharedSecretLengthPtr                                                                                                                                                            | holds a pointer to the memory location in which the length information is stored. On calling this function this parameter shall contain the size of the buffer provided by sharedSecretPtr. When the request has finished, the actual length of the computed value shall be stored. |
| <b>Parameters (Out)</b>   | sharedSecretPtr                                                                                                                                                                  | holds a pointer to the memory location which will hold the result of the key exchange. If the result does not fit into the given buffer, and truncation is allowed, the result shall be truncated                                                                                   |
| <b>Return Value</b>       | Std_ReturnType                                                                                                                                                                   | E_OK: request successful<br>E_NOT_OK: request failed<br>CSM_E_BUSY: request failed, service is still busy                                                                                                                                                                           |

|                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                  |
|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
|                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                              | CSM_E_SMALL_BUFFER: the result provided buffer is too small to store the result, and truncation was not allowed. |
| <b>Description</b>              | <p>This interface shall be used to finish the key exchange service.</p> <p>If the service state is "idle", the function has to return with "E_NOT_OK".</p> <p>Otherwise, this function shall call the function Cry_&lt;Primitive&gt;Finish of the primitive which is identified by the stored configuration information and return the value returned by that function.</p> <p>The calculation of the shared secret is done by the underlying primitive.</p> |                                                                                                                  |
| <b>Preconditions</b>            | CSM Key Exchange Calc Secret interface should have been initialized.                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                  |
| <b>Configuration Dependency</b> | CSM_KEY_EXCHANGE_CALC_SECRET_STATUS should be STD_ON                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                  |

## 7. Generator

### Generator Option

#### Csm

| Option | Description                       |
|--------|-----------------------------------|
| -S     | Software Component Description 생성 |
| -B     | BSW Module Description 생성         |

### Generator Message

#### Error Message

- 1) **ERR110002: Unexpected Error Found. This error may be due to the incorrect configuration of the element(s) 'Element Name'. If the error is not resolved, then please contact AUTRON AUTOSAR Support System.**
  - This error may occur due to incorrect configuration of the Parameter Name/ Container Name provided in the error message. If the error is not resolved, then contact [AUTRON AUTOSAR Support System](#). For more information refer [Section 4](#).
- 2) **ERR110003: 'CSM' Component is not present in the input file(s).**
  - This error occurs, if any one of CSM component is not present in any of the input ECU Configuration Description File(s).
- 3) **ERR110005: The parameter 'Parameter Name' in the container 'Container Name' should be configured.**
  - This error occurs, if any of the mandatory configuration parameters mentioned below is not configured in ECU Configuration Description File.

| Container Name         | Parameter Name         |
|------------------------|------------------------|
| BSW-IMPLEMENTATION     | AR-RELEASE-VERSION     |
|                        | VENDOR-ID              |
|                        | SW-VERSION             |
| BSW-MODULE-DESCRIPTION | MODULE-ID              |
| CsmGeneral             | CsmDevErrorDetect      |
|                        | CsmMaxAlignScalarType  |
|                        | CsmMaximumBlockingTime |

| Container Name           | Parameter Name                      |
|--------------------------|-------------------------------------|
|                          | CsmUseInterruption                  |
|                          | CsmUseSyncJobProcessing             |
|                          | CsmVersionInfoApi                   |
| CsmSignatureVerify       | CsmSignatureVerifyMaxKeySize        |
| CsmSignatureVerifyConfig | CsmCallbackSignatureVerify          |
|                          | CsmSignatureVerifyInitConfiguration |
|                          | CsmSignatureVerifyPrimitiveName     |

- 4) ERR110006: The value configured for the parameter 'Parameter Name' in the container 'Container Name' should follow the pattern: <Pattern>.

- This error occurs, if the parameter 'Parameter Name' is not configured as per the pattern.

| Parameter Name           | Container Name                      | Pattern                    | Example |
|--------------------------|-------------------------------------|----------------------------|---------|
| AR-RELEASE-VERSION       | BSW-IMPLEMENTATION                  | 4. of [0-9]+. of [0-9]+    | 4.0.3   |
| SW-VERSION               |                                     | 1. of [0-9]+. of [0-9]+    | 1.0.0   |
| CsmMaxAlignScalarType    | CsmGeneral                          | [a-zA-Z] of [a-zA-Z0-9W_]* | Adc     |
| CsmSignatureVerifyConfig | CsmSignatureVerifyInitConfiguration |                            |         |
|                          | CsmSignatureVerifyPrimitiveName     |                            |         |

- 5) ERR110013: The reference path <Reference Path> provided for the parameter 'Parameter Name' in the container 'Container Name', having short name <Container Short Name> is incorrect.

- This error occurs, if incorrect reference is provided for the parameter 'CSM\_E\_INIT\_FAILED' in the container 'CsmDemEventParameterRefs'.

| Container Name           | Parameter Name    |
|--------------------------|-------------------|
| CsmDemEventParameterRefs | CSM_E_INIT_FAILED |

- 6) ERR110051: Value of the parameter 'CsmMaxAlignSCalarType' in the container 'CsmGeneral' should be configured as either <uint8>, <uint16> or <uint32>.

This error occurs, when the parameter 'CsmMaxAlignSCalarType' in the container 'CsmGeneral' is not one of 'uint8', 'uint16' and 'uint32'.

- 7) ERR110053: Value of the parameter 'CsmMainFunctionPeriod' should be configured, since 'UseSyncJobProcessing' container is configured with false.

- This error occurs, when the parameter 'CsmMainFunctionPeriod' in the container 'CsmGeneral' is needed because Csm module works asynchronously.

### Warning Messages

- 1) WRN110020: Value of the parameter 'CsmSeedRefAddr' in the container 'CsmGeneral' should be configured as a valid address.
  - This warning occurs, if the referenced address is not valid.

### Information Messages

7.2.2  
None

7.2.3

## 8. Appendix

### 알고리즘 사용시 유의사항

#### Csm API 유의사항

##### 8.1.1.1 인스턴스 제약 사항

Csm 모듈은 각 서비스에 대해 한번에 하나의 인스턴스만 사용 가능하다

##### CSM0016

The CSM module shall only support processing of a single instance of each service at a time.

예를 들어 SHA1 알고리즘을 Csm\_Hash service 로 사용 중인 경우, 해당 처리가 완전히 완료(Csm\_HashStart → Csm\_HashUpdate → Csm\_HashFinish) 되지 않은 상태에서 SHA256 알고리즘을 사용(Csm\_Hash service 사용)하려 하는 경우 CSM\_BUSY 가 리턴된다.

CSM\_BUSY 가 리턴된 경우, 기존 요청이 완료되지 않은 상태이므로 기존 요청이 완료될 때까지 대기한다.

##### 8.1.1.2 SPC58x HSM AES 기반 알고리즘 제약 사항

SPC58x HSM 알고리즘 중 AES 기반 알고리즘(AES ECB Encrypt/Decrypt, AES CBC Encrypt/Decrypt, AES CTR Encrypt/Decrypt, CMAC-AES Generation/Verification)을 사용할 경우 HW 제약으로 인해 한 번에 하나의 요청만을 처리할 수 있다.

만일 AES 기반 알고리즘 요청 후 해당 알고리즘 처리가 완료되지 않은 상태에서 신규 AES 기반 알고리즘 요청시 E\_NOT\_OK 리턴이 발생한다.

E\_NOT\_OK 오류 발생시 Hsm\_GaaErrLog 변수를 확인하여 오류 원인이 HSM\_CRYPTO\_EXEC\_ERROR 인 경우 AES 기반 알고리즘 중복 요청으로 인한 오류이므로 기존 요청 완료 후 신규 요청을 재시도하도록 한다.

#### Async 동작

##### 8.1.2.1 Async 동작 설정

일반적으로 비대칭키 기반의 알고리즘 혹은 iteration 횟수를 높게 한 키유도 함수 등은 수행시간이 수백 ms에서 길게는 초 단위 이상 걸릴 수 있으므로 사용하고자 하는 알고리즘의 수행시간이 동작 테스트가 OS\_LIMIT 이 발생할 것으로 예상되는 경우 async로 동작하도록 설정해 주어야 한다. 설정 방법은 각 알고리즘 Config 컨테이너의 Callback 파라미터를 설정하면 해당 알고리즘은 async 로 동작한다.

##### 8.1.2.2 Async 동작 사용 유의사항

Async 호출 인자 전달시 버퍼의 경우 Csm 모듈은 API 호출시 해당 버퍼의 포인터 정보만을 보유하고 다른 컨텍스트(Csm\_MainFunction)에서 해당 포인터로 버퍼를 접근하기 때문에 반드시 callback 호출시까지 유지가 되는 버퍼를 전달해야 한다.

#### 알고리즘 호출 방식

##### 1) API 호출 순서

- Csm\_RandomGenerate 을 제외한 나머지 알고리즘은 streaming approach 방식으로 호출한다.

즉, 아래의 호출 순서대로 API를 호출하여 해당 알고리즘 요청을 수행하게 된다.

Csm\_<Service>Start → Csm\_<Service>Update (필요한 횟수만큼 반복) → Csm\_<Service>Finish

EX)

```
Csm_HashStart(CSM_HAESHA256_INDEX);

for (LulIndex = 0; LulIndex < 4; LulIndex++)
{
    Csm_HashUpdate(CSM_HAESHA256_INDEX, &GaaHashData[LulIndex], 64U);
}

Csm_HashFinish(CSM_HAESHA256_INDEX, GaaHashDigest, GulHashDigestLength, FALSE);
```

2) API 첫번째 호출 인자

- 각 API 호출시 사용되는 첫번째 인자 Csm\_ConfigIdType cfgId 는 Csm<Service>Config 컨테이너 설정의 ShortName 에 따라 다음과 같은 macro가 Csm\_Cfg.h에 자동 생성된다.

ShortName → CSM\_<SHORTNAME>\_INDEX

| Container Details - CsmHashConfig |               |               |                    |                |          |                   |  |
|-----------------------------------|---------------|---------------|--------------------|----------------|----------|-------------------|--|
| Index                             | Short Name    | Callback Hash | Init Configurat... | Primitive Name | Use Port | Use External T... |  |
| 0                                 | AuthHsmSha256 | (f)           | (s)                | AuthHsmSha256  | true     | true              |  |

<Ecud\_Csm.arxml>

#define CSM\_AUTHSMSHA256\_INDEX (0)  
<Csm\_Cfg.h>

※ AUTOSAR 는 App 이 Bsw 모듈 API 를 직접 호출하지 않고 Rte 레이어를 통해 간접 호출한다. 이 때 Rte 생성 API 에 Csm\_ConfigIdType cfgId 가 자동 매핑, Rte API 첫번째 인자값으로 자동 입력되므로 Port 연결만 정확히 해주면 해당 인자값은 신경쓰지 않아도 된다.

8.1.4

## Key 사용 방식

HSM 프레임워크이 지원되는 MCU 중 HSM 1.0 (Autron HSM) SPC584Bx / SPC58ECx, RH850 F1K 의 일부 알고리즘의 경우 Raw Key 사용을 지원한다. 사용 가능 여부는 [알고리즘 파라미터 허용 범위](#)를 참고한다.

1) Raw Key 방식

a) 대칭키

대칭키 사용 알고리즘의 경우 Csm\_SymKeyType 구조체를 통해 키 입력을 받는다.

```
typedef struct {
    uint32 length;
    Csm_AlignType data[xxx];
} Csm_SymKeyType;
```

length : Raw key 길이  
data : Raw key 데이터

b) 비대칭키

대칭키 사용 알고리즘의 경우 Csm\_AsymKeyType 구조체를 통해 키 입력을 받는다.

```
typedef struct {
    uint32 length;
```

```
Csm_AlignType data[xxx];
} Csm_AsymPrivateKeyType;
```

대칭키 알고리즘의 경우 알고리즘에 따라 입력 데이터가 다를 수 있다. length, data 의 구체적인 입력 데이터는 [알고리즘 파라미터 허용 범위](#)를 참고하도록 한다.

## 2) Pre-shared Key 방식

Pre-shared Key 는 HSM 프레임워크 내에 사전에 저장되어 있는 키값을 사용한다.  
대칭키, 비대칭키 사용 방식은 동일하다.

Key index 는 사용 HSM 프레임워크에 따라 최대 2 byte 까지 사용가능하다. Key index 가 256 보다 작은 경우 length 를 1 로, 256 이상인 경우 length 를 2로 사용한다.

### 1) $1 \leq \text{Key index} < 256$

length : 1

data[0] : 사용하고자 하는 Pre-shared key 의 index

### 2) $256 \leq \text{Key index} < 65536$

length : 2

data[0] : 사용하고자 하는 Pre-shared key 의 index 중 하위 byte 값

data[1] : 사용하고자 하는 Pre-shared key 의 index 중 상위 byte 값

ex) key index : 401

data[0] = 401 & 0xFF

data[1] = (401 & 0xFF00) >>8

## 8.1.5 App 연결

App에서 알고리즘을 직접 사용하고자 할 경우 Rte 레이어를 통해 Client-Server Interface 연결을 해야 한다.

각 알고리즘 Config 컨테이너의 Use Port 파라미터가 true 로 설정된 경우 Generated / Bsw\_Output / swcd / Swcd\_Bsw\_Csm.arxml 의 Csm 컴포넌트에 해당 알고리즘 Config 컨테이너의 [ShortName]\_<Service> 의 P-Port 가 생성되며 관련된 Port API Option, Runnable, Operation Invoked Event 가 같이 생성된다.

App는 일반적인 Client-Server Interface 연결 방식을 따라 해당 P-Port 와 연계된 R-Port 를 설정, 연결하여 사용하면 된다.

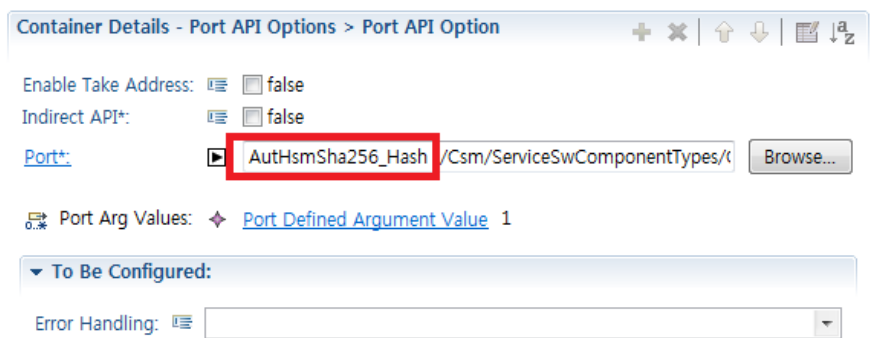
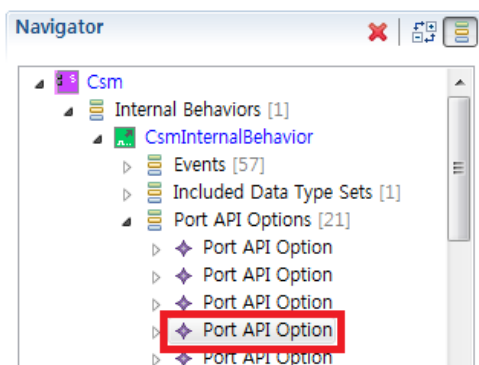
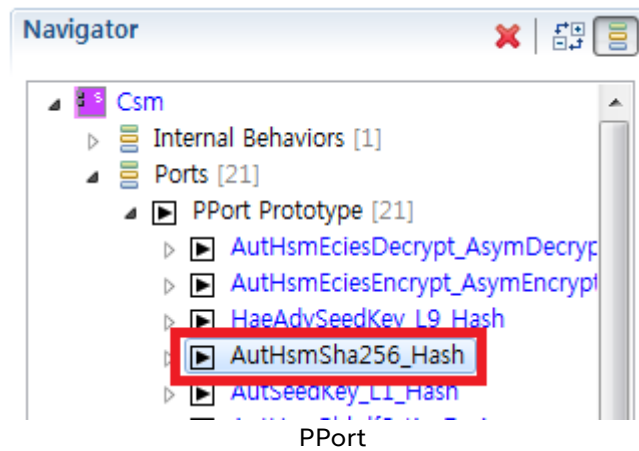
예제) SPC584Bx / SPC58ECx 계열에서 HSM 프레임워크의 Sha256 알고리즘 사용 예  
다른 MCU, 다른 알고리즘의 경우도 연결 방식은 동일하다.

1) 아래와 같이 CsmHash / CsmHashConfig 컨테이너로 'AuthHsmSha256' ShortName, Use Port : true 을 가진 컨테이너를 설정한다.

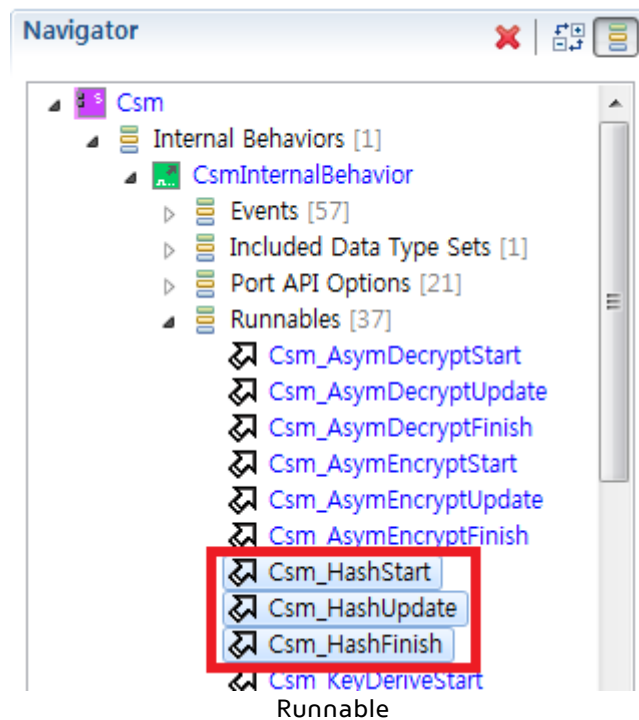
| Index | Short Name       | Callback Hash | Init Configur... | Primitive Name | Use Port | Use External T... |
|-------|------------------|---------------|------------------|----------------|----------|-------------------|
| 0     | AutSeedKey_L1    | (f)           | (S)              | AutSeedKey     | true     | false             |
| 1     | HaeAdvSeedKey_L9 | (f)           | (S)              | HaeAdvSeedKey  | true     | false             |
| 2     | AuthHsmSha256    | (f)           | (S)              | AuthHsmSha256  | true     | true              |

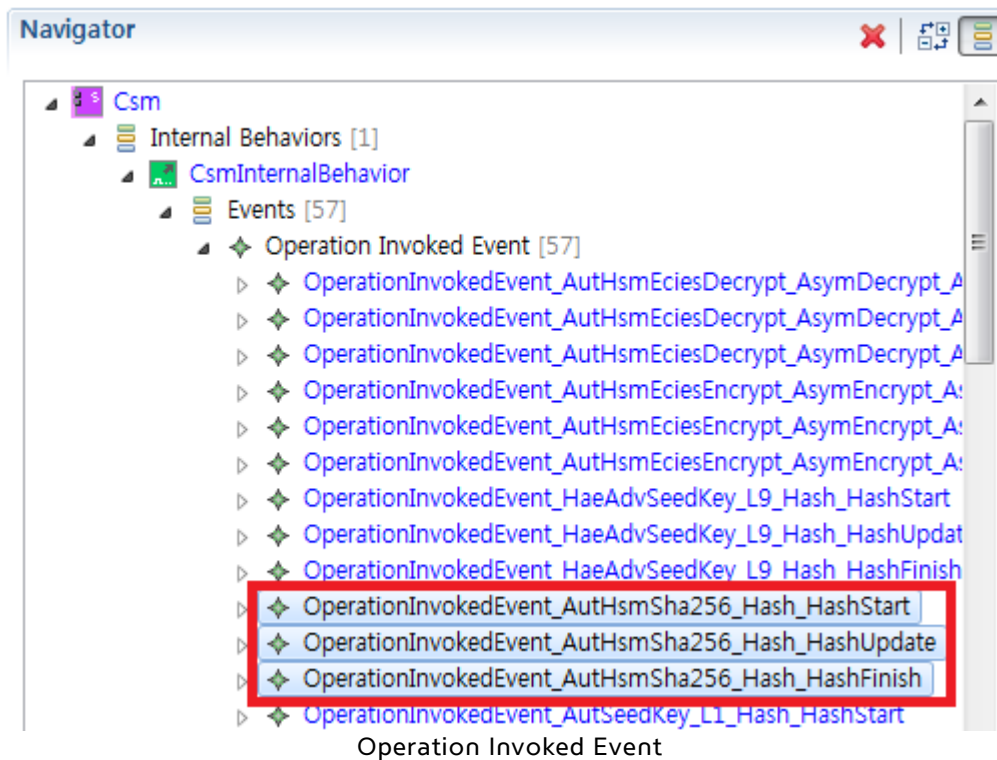
2) Ecu\_Csm.arxml 설정 저장 후 build 수행시 Rte 연결을 위해 필요한 Service 쪽 설정인 Csm ServiceSwComponent 가 Generated\Bsw\_Output\swcd 폴더에 Swcd\_Bsw\_Csm.arxml 파일로 자동 생성된다. 다만, 플랫폼 배포 정책에 따라 해당 파일이 자동 생성되지 않고 Configuration\System\Swcd\_Bsw 폴더에 사전에 저장되어 있을 수 있다.

Csm ServiceSwComponent 는 Port(P-Port, Callback 설정시 R-Port), Port API Option, Runnable, Operation Invoked Event 및 필요 데이터 타입 등의 설정이 들어 있다.



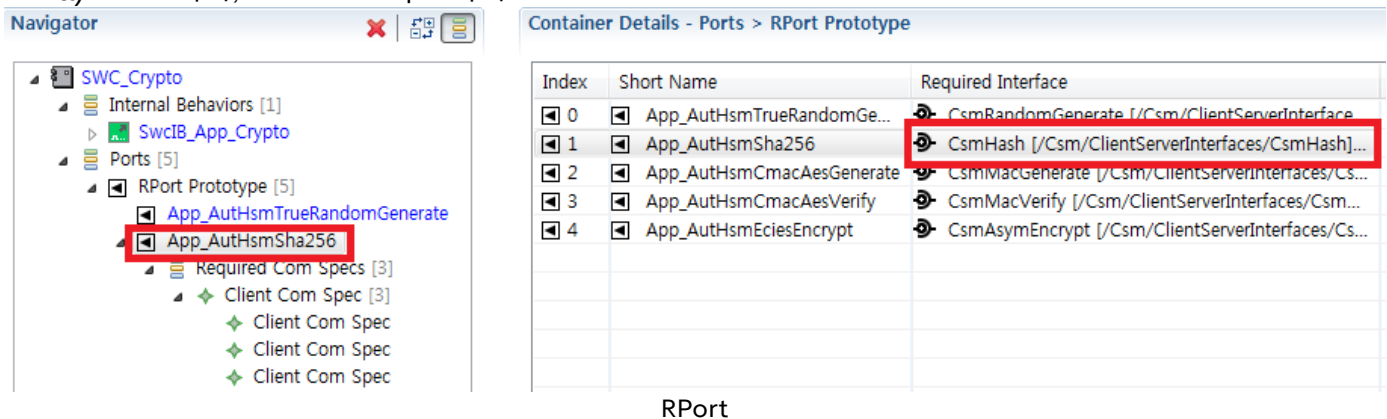
Port API Option

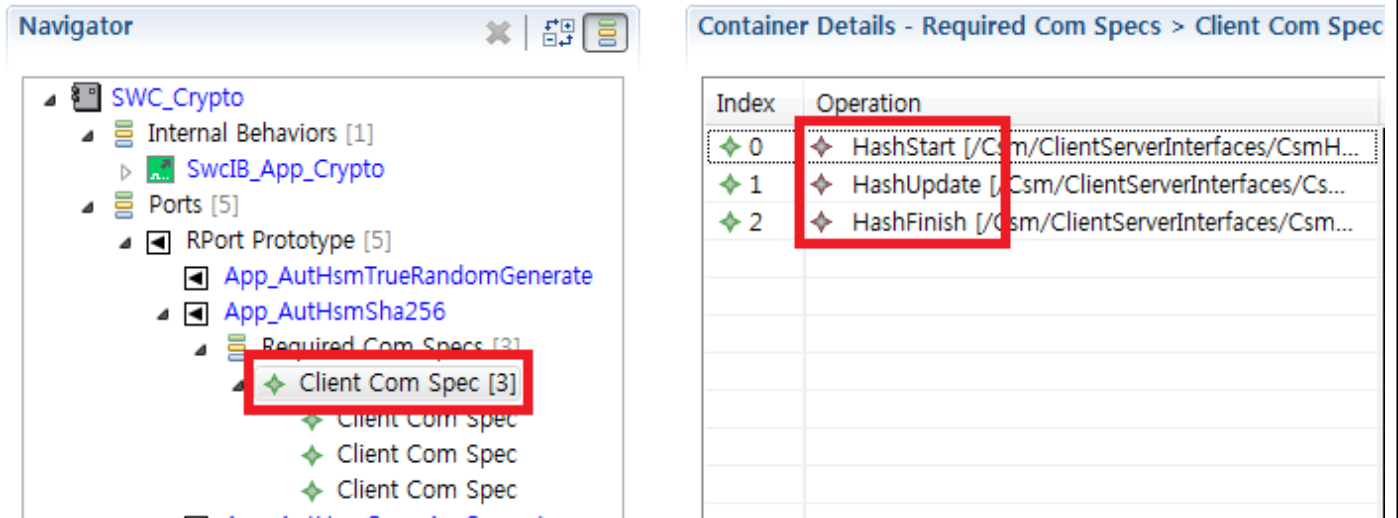




3) 사용자는 알고리즘을 사용하고자 하는 Application component 설계, 설정하고 해당 알고리즘 port 연결을 해야 한다.

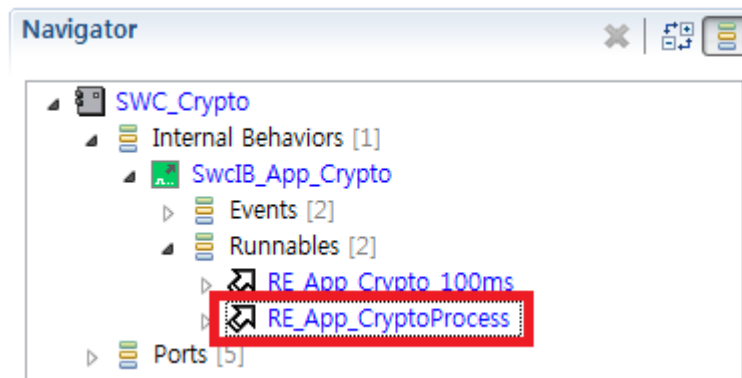
a) RPort 추가, Client Com Spec 추가



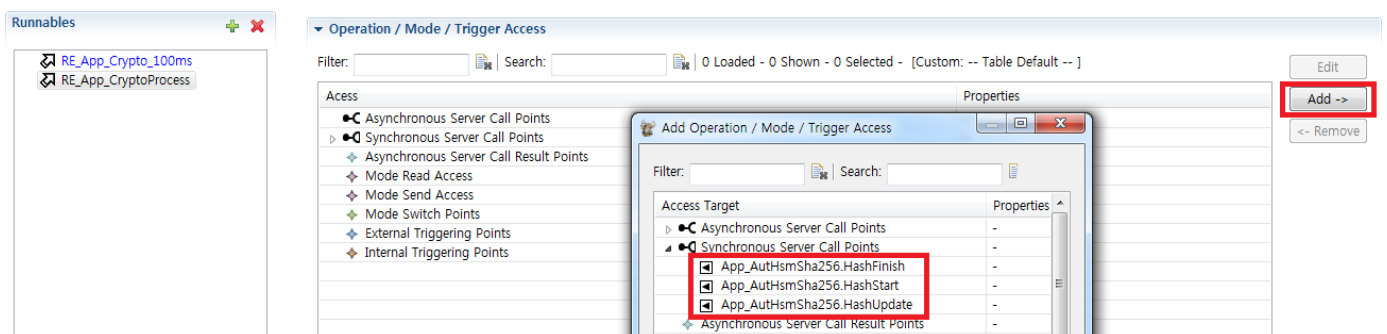


Client Com Spec

b) Runnable 추가, Synchronous Server Call Points 추가

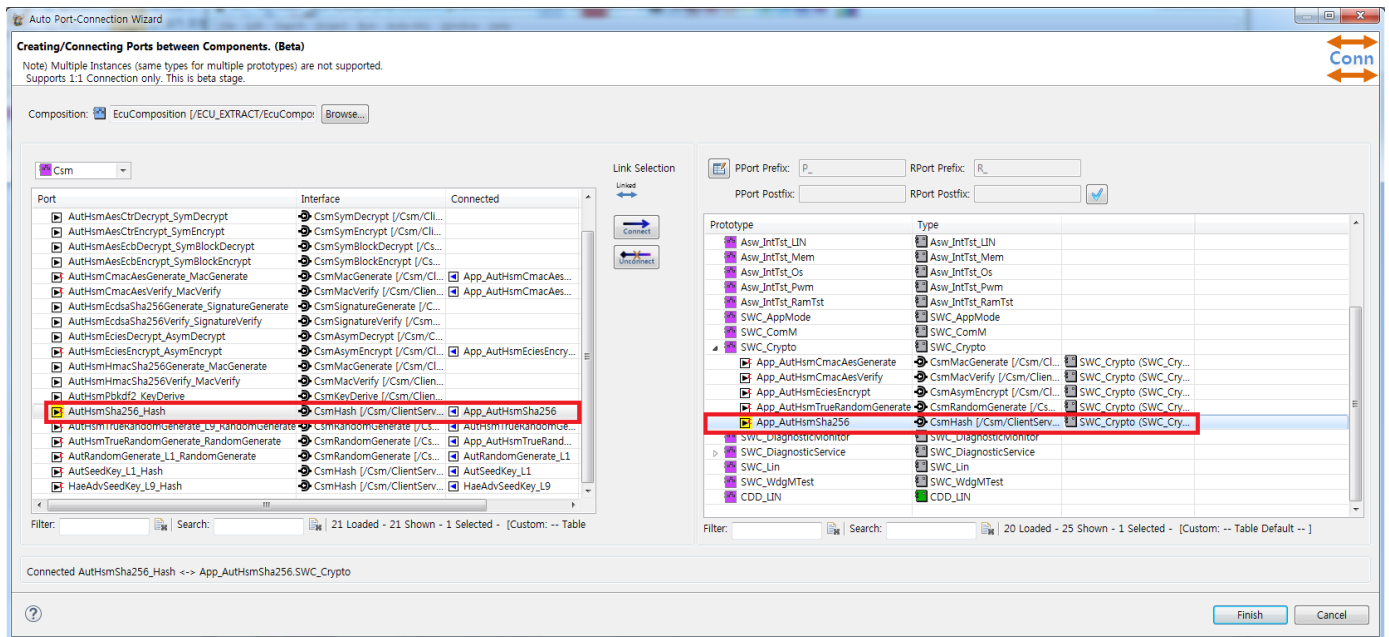
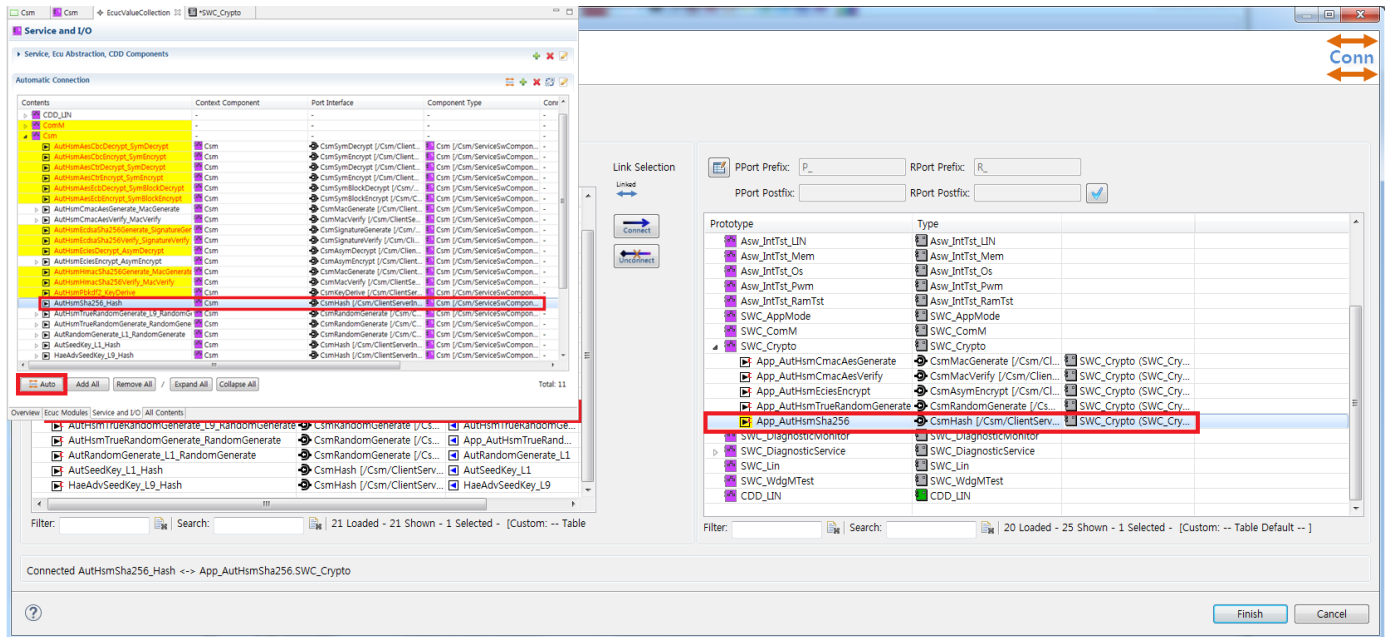


Runnable



Synchronous Server Call Points

4) Service component 의 PPort 와 Application component 의 RPort 를 연결해준다.



5) Build 후 관련 코드가 생성되었는지 확인한다.

아래의 API 는 Port 연결로 생성되는 Rte 코드이며 아래의 규칙으로 생성된다.

Rte\_Call\_<Swc ShortName>\_<RPort ShortName>\_<operation>

```

FUNC(Std_ReturnType, RTE_CODE) Rte_Call_SWC_Crypto_App_AuthHsmSha256_HashFinish(
    OUT P2VAR(uint8, AUTOMATIC, RTE_APPL_DATA) resultBuffer,
    INOUT P2VAR(uint32, AUTOMATIC, RTE_APPL_DATA) resultLength,
    IN boolean TruncationIsAllowed)
{
    VAR(Std_ReturnType, RTE_DATA) LddRetVal = RTE_E_OK;
    LddRetVal = Csm_HashFinish(
        Rte_PDAV_CsmAuthHsmSha256_Hash_1,
        resultBuffer,
        resultLength,
        TruncationIsAllowed);
    return LddRetVal;
}

FUNC(Std_ReturnType, RTE_CODE) Rte_Call_SWC_Crypto_App_AuthHsmSha256_HashStart(void)
{
    VAR(Std_ReturnType, RTE_DATA) LddRetVal = RTE_E_OK;
    LddRetVal = Csm_HashStart(
        Rte_PDAV_CsmAuthHsmSha256_Hash_1);
    return LddRetVal;
}

FUNC(Std_ReturnType, RTE_CODE) Rte_Call_SWC_Crypto_App_AuthHsmSha256_HashUpdate(
    IN P2CONST(uint8, AUTOMATIC, RTE_APPL_DATA) dataBuffer,
    IN uint32 dataLength)
{
    VAR(Std_ReturnType, RTE_DATA) LddRetVal = RTE_E_OK;
    LddRetVal = Csm_HashUpdate(
        Rte_PDAV_CsmAuthHsmSha256_Hash_1,
        dataBuffer,
        dataLength);
    return LddRetVal;
}

```

실제 사용자 코드에서 사용은 Rte\_<Swc ShortName>.h 파일 include 후,  
Rte\_Call\_<RPort ShortName>\_<operation> 매크로를 사용한다.

```

#ifndef Rte_Call_App_AuthHsmSha256_HashFinish
#define Rte_Call_App_AuthHsmSha256_HashFinish Rte_Call_SWC_Crypto_App_AuthHsmSha256_HashFinish
#endif

#ifndef Rte_Call_App_AuthHsmSha256_HashStart
#define Rte_Call_App_AuthHsmSha256_HashStart Rte_Call_SWC_Crypto_App_AuthHsmSha256_HashStart
#endif

#ifndef Rte_Call_App_AuthHsmSha256_HashUpdate
#define Rte_Call_App_AuthHsmSha256_HashUpdate Rte_Call_SWC_Crypto_App_AuthHsmSha256_HashUpdate
#endif

```

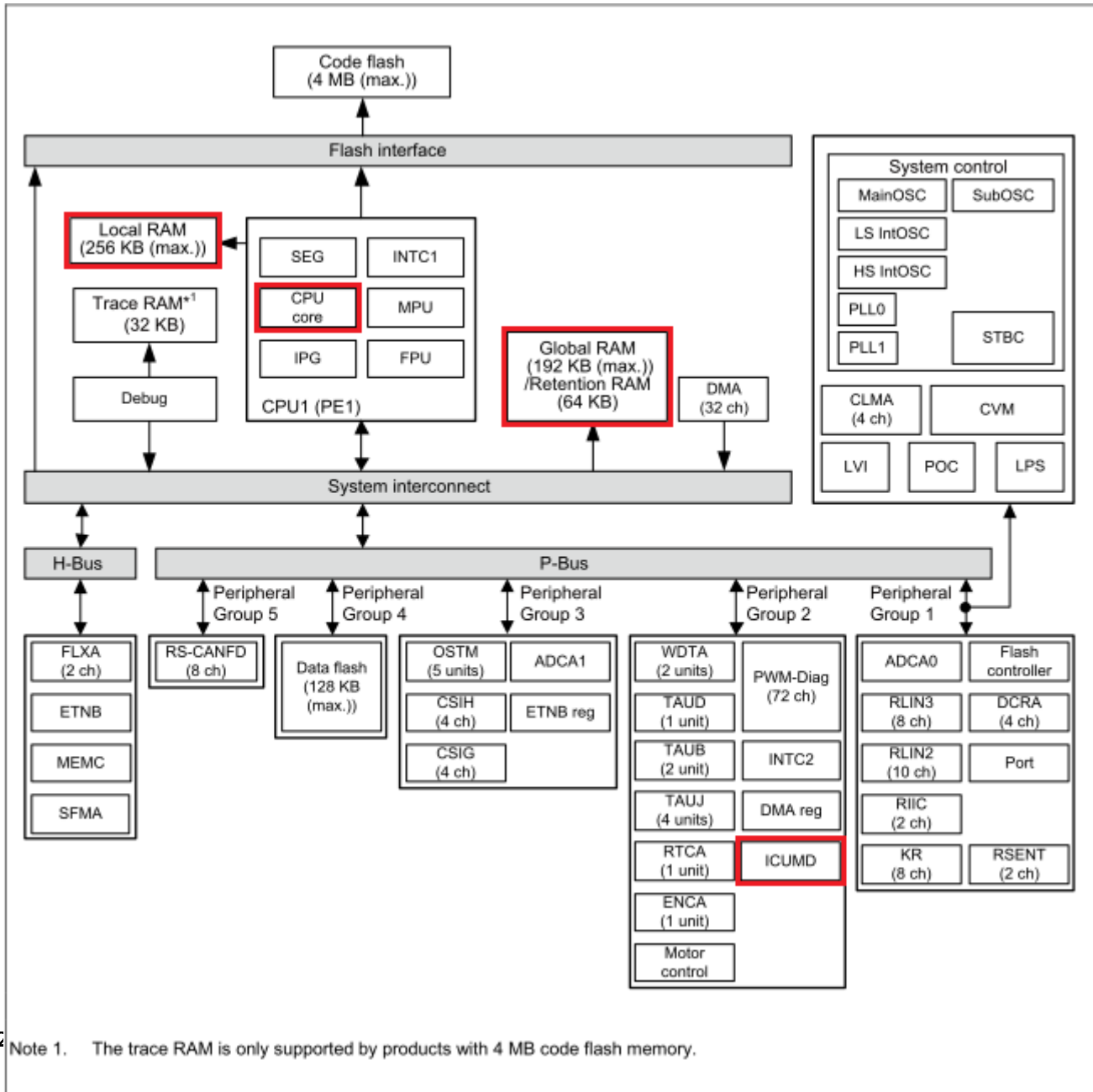
8.1

※ Csm 모듈의 Secure Access(ShortName 의 postfix 가 “\_L1”, “\_L9”, “\_L21”)와 관련된 알고리즘 설정은 변경하지 않도록 유의한다.

### RH850 F1KM HSM 공유 메모리 배치

RH850 F1KM 의 경우 CPU core 와 HSM core(ICUMD) 간에 공유할 수 있는 메모리는 아래 그림과 같이 Global RAM / Retention RAM 만 P-Bus 를 통해 공유 가능하다. 만일 HSM 으로 넘겨주는 메모리(input, output 버퍼)가 Local RAM 인 경우 공유가 불가하기 때문에 HSM API 호출시 HSM\_MEMORY\_FAILURE 에러가 리턴된다.

따라서 **HSM 에 공유할 메모리는 반드시 Global RAM A / B or Retention RAM 에 배치**되도록 메모리 section 지정을 해야 한다.



## 알고리즘 파라미터 허용 범위

각 알고리즘의 API 사용시 아래의 파라미터별 허용 가능한 값 및 함께 명기된 유의사항을 반드시 지킴을 한다.

### Hash

#### 8.2.1.1 AutSeedKey

4 byte 길이의 Seed 값에 대한 4 byte 길이의 Key 값을 유도하는 알고리즘

| API | Parameter | Value |
|-----|-----------|-------|
|-----|-----------|-------|

|                |                     |       |
|----------------|---------------------|-------|
| Csm_HashUpdate | dataLength          | >= 4  |
| Csm_HashFinish | *resultLengthPtr    | >= 4  |
|                | TruncationIsAllowed | FALSE |

#### 8.2.1.2 HaeAdvSeedKey

8 byte 길이의 Seed 값에 대한 8 byte 길이의 Key 값을 유도하는 알고리즘

Variation 을 갖고 있으므로 동일한 Seed 값에 대해서도 플랫폼에 따라 다른 Key 값이 나올 수 있다.

| API            | Parameter           | Value |
|----------------|---------------------|-------|
| Csm_HashUpdate | dataLength          | >= 8  |
| Csm_HashFinish | *resultLengthPtr    | >= 8  |
|                | TruncationIsAllowed | FALSE |

#### 8.2.1.3 HaeSha1

다양한 길이의 data 에 대해 무결성 확인을 위한 20 byte 길이의 Digest 를 생성하며, truncation 이 true 인 경우 20 byte 보다 짧은 길이의 Digest 를 얻을 수 있다.

| API            | Parameter           | Value                                |
|----------------|---------------------|--------------------------------------|
| Csm_HashUpdate | dataLength          | 제한없음                                 |
| Csm_HashFinish | *resultLengthPtr    | >= 20 (TruncationIsAllowed == FALSE) |
|                | TruncationIsAllowed | > 0 (TruncationIsAllowed == TRUE)    |

#### 8.2.1.4 HaeSha256

다양한 길이의 data 에 대해 무결성 확인을 위한 32 byte 길이의 Digest 를 생성하며, truncation 이 true 인 경우 32 byte 보다 짧은 길이의 Digest 를 얻을 수 있다.

| API            | Parameter           | Value                                |
|----------------|---------------------|--------------------------------------|
| Csm_HashUpdate | dataLength          | 제한없음                                 |
| Csm_HashFinish | *resultLengthPtr    | >= 32 (TruncationIsAllowed == FALSE) |
|                | TruncationIsAllowed | > 0 (TruncationIsAllowed == TRUE)    |

#### 8.2.1.5 AuthHsmSha1

다양한 길이의 data 에 대해 무결성 확인을 위한 20 byte 길이의 Digest 를 생성하며, truncation 이 true 인 경우 20 byte 보다 짧은 길이의 Digest 를 얻을 수 있다.

해당 알고리즘은 Autron HSM 프레임워크 내에서 수행된다.

| API            | Parameter           | Value                                |
|----------------|---------------------|--------------------------------------|
| Csm_HashUpdate | dataLength          | 제한없음                                 |
| Csm_HashFinish | *resultLengthPtr    | >= 20 (TruncationIsAllowed == FALSE) |
|                | TruncationIsAllowed | > 0 (TruncationIsAllowed == TRUE)    |

#### 8.2.1.6 AuthHsmSha256

다양한 길이의 data 에 대해 무결성 확인을 위한 32 byte 길이의 Digest 를 생성하며, truncation 이 true 인 경우 32 byte 보다 짧은 길이의 Digest 를 얻을 수 있다.

해당 알고리즘은 Autron HSM 프레임워크 내에서 수행된다.

| API            | Parameter  | Value |
|----------------|------------|-------|
| Csm_HashUpdate | dataLength | 제한없음  |

|                |                     |                                      |
|----------------|---------------------|--------------------------------------|
| Csm_HashFinish | *resultLengthPtr    | >= 32 (TruncationIsAllowed == FALSE) |
|                | TruncationIsAllowed | > 0 (TruncationIsAllowed == TRUE)    |

#### 8.2.1.7 HaeHsmSha1

다양한 길이의 data 에 대해 무결성 확인을 위한 20 byte 길이의 Digest 를 생성하며, truncation 이 true 인 경우 20 byte 보다 짧은 길이의 Digest 를 얻을 수 있다.

해당 알고리즘은 Autoever HSM 프레임워크 내에서 수행된다.

| API            | Parameter           | Value                                |
|----------------|---------------------|--------------------------------------|
| Csm_HashUpdate | dataLength          | 제한없음                                 |
| Csm_HashFinish | *resultLengthPtr    | >= 20 (TruncationIsAllowed == FALSE) |
|                | TruncationIsAllowed | > 0 (TruncationIsAllowed == TRUE)    |

#### 8.2.1.8 HaeHsmSha256

다양한 길이의 data 에 대해 무결성 확인을 위한 32 byte 길이의 Digest 를 생성하며, truncation 이 true 인 경우 32 byte 보다 짧은 길이의 Digest 를 얻을 수 있다.

해당 알고리즘은 Autoever HSM 프레임워크 내에서 수행된다.

| API            | Parameter           | Value                                |
|----------------|---------------------|--------------------------------------|
| Csm_HashUpdate | dataLength          | 제한없음                                 |
| Csm_HashFinish | *resultLengthPtr    | >= 32 (TruncationIsAllowed == FALSE) |
|                | TruncationIsAllowed | > 0 (TruncationIsAllowed == TRUE)    |

### 8.2.2

#### Mac

다양한 길이의 Data 와 Key 값과 수신된 MAC 을 기반으로 수신한 Data의 무결성, 인증을 확인한다.

전송 효율을 위해 MAC은 truncation 될 수 있고 수신한 MAC 은 bit 단위의 길이로 사용한다.

| API                 | Parameter      | Value  |
|---------------------|----------------|--------|
| Csm_MacVerifyStart  | keyPtr->length | 16     |
| Csm_MacVerifyUpdate | dataLength     | 제한없음   |
| Csm_MacVerifyFinish | MacLength      | >= 128 |

#### 8.2.2.1 HaeHmacSha1

다양한 길이의 Data 와 Key 값을 기반으로 무결성, 인증을 확인하기 위한 20 byte 길이의 MAC 을 생성하며, truncation 이 true 인 경우 20 byte 보다 짧은 길이의 MAC 을 얻을 수 있다.

| API                   | Parameter           | Value                                |
|-----------------------|---------------------|--------------------------------------|
| Csm_MacGenerateStart  | keyPtr->length      | 제한없음                                 |
| Csm_MacGenerateUpdate | dataLength          | 제한없음                                 |
| Csm_MacGenerateFinish | *resultLengthPtr    | >= 20 (TruncationIsAllowed == FALSE) |
|                       | TruncationIsAllowed | > 0 (TruncationIsAllowed == TRUE)    |

다양한 길이의 Data 와 Key 값과 수신된 MAC 을 기반으로 수신한 Data의 무결성, 인증을 확인한다.

전송 효율을 위해 MAC은 truncation 될 수 있고 수신한 MAC 은 bit 단위의 길이로 사용한다.

| API | Parameter | Value |
|-----|-----------|-------|
|-----|-----------|-------|

|                     |                |        |
|---------------------|----------------|--------|
| Csm_MacVerifyStart  | keyPtr->length | 제한없음   |
| Csm_MacVerifyUpdate | dataLength     | 제한없음   |
| Csm_MacVerifyFinish | MacLength      | >= 160 |

#### 8.2.2.2 HaeHmacSha256

다양한 길이의 Data 와 Key 값을 기반으로 무결성, 인증을 확인하기 위한 32 byte 길이의 MAC 을 생성하며, truncation 이 true 인 경우 32 byte 보다 짧은 길이의 MAC 을 얻을 수 있다.

| API                   | Parameter           | Value                                |
|-----------------------|---------------------|--------------------------------------|
| Csm_MacGenerateStart  | keyPtr->length      | 제한없음                                 |
| Csm_MacGenerateUpdate | dataLength          | 제한없음                                 |
| Csm_MacGenerateFinish | *resultLengthPtr    | >= 32 (TruncationIsAllowed == FALSE) |
|                       | TruncationIsAllowed | > 0 (TruncationIsAllowed == TRUE)    |

다양한 길이의 Data 와 Key 값과 수신된 MAC 을 기반으로 수신한 Data의 무결성, 인증을 확인한다.

전송 효율을 위해 MAC은 truncation 될 수 있고 수신한 MAC 은 bit 단위의 길이로 사용한다.

| API                 | Parameter      | Value  |
|---------------------|----------------|--------|
| Csm_MacVerifyStart  | keyPtr->length | 제한없음   |
| Csm_MacVerifyUpdate | dataLength     | 제한없음   |
| Csm_MacVerifyFinish | MacLength      | >= 256 |

#### 8.2.2.3 HaeCmacAes

다양한 길이의 Data 와 16 byte 길이의 Key 값을 기반으로 무결성, 인증을 확인하기 위한 16 byte 길이의 MAC 을 생성하며, truncation 이 true 인 경우 16 byte 보다 짧은 길이의 MAC 을 얻을 수 있다.

| API                   | Parameter           | Value                                |
|-----------------------|---------------------|--------------------------------------|
| Csm_MacGenerateStart  | keyPtr->length      | 16                                   |
| Csm_MacGenerateUpdate | dataLength          | 제한없음                                 |
| Csm_MacGenerateFinish | *resultLengthPtr    | >= 16 (TruncationIsAllowed == FALSE) |
|                       | TruncationIsAllowed | > 0 (TruncationIsAllowed == TRUE)    |

다양한 길이의 Data 와 Key 값과 수신된 MAC 을 기반으로 수신한 Data의 무결성, 인증을 확인한다.

전송 효율을 위해 MAC은 truncation 될 수 있고 수신한 MAC 은 bit 단위의 길이로 사용한다.

| API                 | Parameter      | Value  |
|---------------------|----------------|--------|
| Csm_MacVerifyStart  | keyPtr->length | 16     |
| Csm_MacVerifyUpdate | dataLength     | 제한없음   |
| Csm_MacVerifyFinish | MacLength      | >= 128 |

#### 8.2.2.4 AuthHsmHmacSha1

다양한 길이의 Data 와 Key 값을 기반으로 무결성, 인증을 확인하기 위한 20 byte 길이의 MAC 을 생성하며, truncation 이 true 인 경우 20 byte 보다 짧은 길이의 MAC 을 얻을 수 있다.

해당 알고리즘은 Autron HSM 프레임워크 내에서 수행된다.

| API                   | Parameter        | Value                                |
|-----------------------|------------------|--------------------------------------|
| Csm_MacGenerateStart  | keyPtr->length   | 제한없음                                 |
| Csm_MacGenerateUpdate | dataLength       | 제한없음                                 |
| Csm_MacGenerateFinish | *resultLengthPtr | >= 20 (TruncationIsAllowed == FALSE) |

|  |                     |                                         |
|--|---------------------|-----------------------------------------|
|  | TruncationIsAllowed | E)<br>> 0 (TruncationIsAllowed == TRUE) |
|--|---------------------|-----------------------------------------|

다양한 길이의 Data 와 Key 값과 수신된 MAC 을 기반으로 수신한 Data의 무결성, 인증을 확인한다.  
전송 효율을 위해 MAC은 truncation 될 수 있고 수신한 MAC 은 bit 단위의 길이로 사용한다.  
해당 알고리즘은 Autron HSM 프레임워크 내에서 수행된다.

| API                 | Parameter      | Value  |
|---------------------|----------------|--------|
| Csm_MacVerifyStart  | keyPtr->length | 제한없음   |
| Csm_MacVerifyUpdate | dataLength     | 제한없음   |
| Csm_MacVerifyFinish | MacLength      | >= 160 |

#### 8.2.2.5 AuthHsmHmacSha256

다양한 길이의 Data 와 Key 값을 기반으로 무결성, 인증을 확인하기 위한 32 byte 길이의 MAC 을 생성하며, truncation 이 true 인 경우 32 byte 보다 짧은 길이의 MAC 을 얻을 수 있다.  
해당 알고리즘은 Autron HSM 프레임워크 내에서 수행된다.

| API                   | Parameter           | Value                                |
|-----------------------|---------------------|--------------------------------------|
| Csm_MacGenerateStart  | keyPtr->length      | 제한없음                                 |
| Csm_MacGenerateUpdate | dataLength          | 제한없음                                 |
| Csm_MacGenerateFinish | *resultLengthPtr    | >= 32 (TruncationIsAllowed == FALSE) |
|                       | TruncationIsAllowed | > 0 (TruncationIsAllowed == TRUE)    |

다양한 길이의 Data 와 Key 값과 수신된 MAC 을 기반으로 수신한 Data의 무결성, 인증을 확인한다.  
전송 효율을 위해 MAC은 truncation 될 수 있고 수신한 MAC 은 bit 단위의 길이로 사용한다.  
해당 알고리즘은 Autron HSM 프레임워크 내에서 수행된다.

| API                 | Parameter      | Value  |
|---------------------|----------------|--------|
| Csm_MacVerifyStart  | keyPtr->length | 제한없음   |
| Csm_MacVerifyUpdate | dataLength     | 제한없음   |
| Csm_MacVerifyFinish | MacLength      | >= 256 |

#### 8.2.2.6 AuthHsmCmacAes

다양한 길이의 Data 와 16 byte 길이의 Key 값을 기반으로 무결성, 인증을 확인하기 위한 16 byte 길이의 MAC 을 생성하며, truncation 이 true 인 경우 16 byte 보다 짧은 길이의 MAC 을 얻을 수 있다.  
HSM 프레임워크 내에 사전에 저장된 Key 값을 사용할 경우 keyPtr->length 는 1, keyPtr->data 는 Key 값의 Index를 사용한다.  
AuthHsmCmacAes 는 Csm\_MacGenerateUpdate 사용시 반복 호출을 허용하지 않는다. 즉, Csm\_MacGenerateStart, Csm\_MacGenerateUpdate, Csm\_MacGenerateFinish 를 한 번씩만 호출해야 한다.  
해당 알고리즘은 Autron HSM 프레임워크 내에서 수행된다.

| API                   | Parameter      | Value                                                                        |
|-----------------------|----------------|------------------------------------------------------------------------------|
| Csm_MacGenerateStart  | keyPtr->length | 1 or 16<br>※ 1 : Pre-shared Key Index 사용시<br>16 : Raw Key 사용시                |
|                       | keyPtr->data   | Pre-shared Key Index (keyPtr->length == 1)<br>Raw Key (keyPtr->length == 16) |
| Csm_MacGenerateUpdate | dataLength     | 제한없음                                                                         |

|                       |                     |                                      |
|-----------------------|---------------------|--------------------------------------|
| Csm_MacGenerateFinish | *resultLengthPtr    | >= 16 (TruncationIsAllowed == FALSE) |
|                       | TruncationIsAllowed | > 0 (TruncationIsAllowed == TRUE)    |

다양한 길이의 Data 와 Key 값과 수신된 MAC 을 기반으로 수신한 Data의 무결성, 인증을 확인한다.

전송 효율을 위해 MAC은 truncation 될 수 있고 수신한 MAC 은 bit 단위의 길이로 사용한다.

HSM 프레임워크 내에 사전에 저장된 Key 값을 사용할 경우 keyPtr->length 는 1, keyPtr->data 는 Key 값의 Index를 사용한다.

AuthHsmCmacAes 는 Csm\_MacVerifyUpdate 사용시 반복 호출을 허용하지 않는다. 즉, Csm\_MacVerifyStart, Csm\_MacVerifyUpdate, Csm\_MacVerifyFinish 를 한 번씩만 호출해야 한다.

해당 알고리즘은 Autron HSM 프레임워크 내에서 수행된다.

| API                 | Parameter      | Value                                                                        |
|---------------------|----------------|------------------------------------------------------------------------------|
| Csm_MacVerifyStart  | keyPtr->length | 1 or 16<br>※ 1 : Pre-shared Key Index 사용시<br>16 : Raw Key 사용시                |
|                     | keyPtr->data   | Pre-shared Key Index (keyPtr->length == 1)<br>Raw Key (keyPtr->length == 16) |
| Csm_MacVerifyUpdate | dataLength     | 제한없음                                                                         |
| Csm_MacVerifyFinish | MacLength      | >= 128                                                                       |

#### 8.2.2.7 AutIcusCmacAes

다양한 길이의 Data 와 16 byte 길이의 Key 값을 기반으로 무결성, 인증을 확인하기 위한 16 byte 길이의 MAC 을 생성하며, truncation 이 true 인 경우 16 byte 보다 짧은 길이의 MAC 을 얻을 수 있다.

ICUS 내에 사전에 저장된 Key 값을 사용할 경우 keyPtr->length 는 1, keyPtr->data 는 Key 값의 Index를 사용한다. (Key Index 정보는 CryptoLib UM 참조)

해당 알고리즘은 ICUS 내에서 수행된다.

| API                   | Parameter           | Value                                                                        |
|-----------------------|---------------------|------------------------------------------------------------------------------|
| Csm_MacGenerateStart  | keyPtr->length      | 1 or 16<br>※ 1 : Pre-shared Key Index 사용시<br>16 : Raw Key 사용시                |
|                       | keyPtr->data        | Pre-shared Key Index (keyPtr->length == 1)<br>Raw Key (keyPtr->length == 16) |
| Csm_MacGenerateUpdate | dataLength          | 제한없음                                                                         |
| Csm_MacGenerateFinish | *resultLengthPtr    | >= 16 (TruncationIsAllowed == FALSE)                                         |
|                       | TruncationIsAllowed | > 0 (TruncationIsAllowed == TRUE)                                            |

다양한 길이의 Data 와 Key 값과 수신된 MAC 을 기반으로 수신한 Data의 무결성, 인증을 확인한다.

전송 효율을 위해 MAC은 truncation 될 수 있고 수신한 MAC 은 bit 단위의 길이로 사용한다.

ICUS 내에 사전에 저장된 Key 값을 사용할 경우 keyPtr->length 는 1, keyPtr->data 는 Key 값의 Index를 사용한다. (Key Index 정보는 CryptoLib UM 참조)

해당 알고리즘은 ICUS 내에서 수행된다.

| API                | Parameter      | Value   |
|--------------------|----------------|---------|
| Csm_MacVerifyStart | keyPtr->length | 1 or 16 |

|                     |              |                                                                              |
|---------------------|--------------|------------------------------------------------------------------------------|
|                     |              | ※ 1 : Pre-shared Key Index 사용시<br>16 : Raw Key 사용시                           |
|                     | keyPtr->data | Pre-shared Key Index (keyPtr->length == 1)<br>Raw Key (keyPtr->length == 16) |
| Csm_MacVerifyUpdate | dataLength   | 제한없음                                                                         |
| Csm_MacVerifyFinish | MacLength    | >= 128                                                                       |

#### 8.2.2.8 HaeHsmHmacSha256

다양한 길이의 Data 와 Key 값을 기반으로 무결성, 인증을 확인하기 위한 32 byte 길이의 MAC 을 생성하며, truncation 이 true 인 경우 32 byte 보다 짧은 길이의 MAC 을 얻을 수 있다.

Autoever HSM 프레임워크는 사전에 저장된 Key 값만을 사용할 수 있다. key index 입력 방식은 8.1.4 Key 사용 방식을 참조한다.

해당 알고리즘은 Autoever HSM 프레임워크 내에서 수행된다.

| API                   | Parameter           | Value                                |
|-----------------------|---------------------|--------------------------------------|
| Csm_MacGenerateStart  | keyPtr->length      | 1 or 2<br>※ 8.1.4 Key 사용 방식 참조       |
|                       | keyPtr->data        | Pre-shared Key Index                 |
| Csm_MacGenerateUpdate | dataLength          | 제한없음                                 |
| Csm_MacGenerateFinish | *resultLengthPtr    | >= 32 (TruncationIsAllowed == FALSE) |
|                       | TruncationIsAllowed | > 0 (TruncationIsAllowed == TRUE)    |

다양한 길이의 Data 와 Key 값과 수신된 MAC 을 기반으로 수신한 Data의 무결성, 인증을 확인한다.

전송 효율을 위해 MAC은 truncation 될 수 있고 수신한 MAC 은 bit 단위의 길이로 사용한다.

Autoever HSM 프레임워크는 사전에 저장된 Key 값만을 사용할 수 있다. key index 입력 방식은 8.1.4 Key 사용 방식을 참조한다.

해당 알고리즘은 Autoever HSM 프레임워크 내에서 수행된다.

| API                 | Parameter      | Value                          |
|---------------------|----------------|--------------------------------|
| Csm_MacVerifyStart  | keyPtr->length | 1 or 2<br>※ 8.1.4 Key 사용 방식 참조 |
|                     | keyPtr->data   | Pre-shared Key Index           |
| Csm_MacVerifyUpdate | dataLength     | 제한없음                           |
| Csm_MacVerifyFinish | MacLength      | >= 256                         |

#### 8.2.2.9 HaeHsmCmacAes

다양한 길이의 Data 와 16 byte 길이의 Key 값을 기반으로 무결성, 인증을 확인하기 위한 16 byte 길이의 MAC 을 생성하며, truncation 이 true 인 경우 16 byte 보다 짧은 길이의 MAC 을 얻을 수 있다.

Autoever HSM 프레임워크는 사전에 저장된 Key 값만을 사용할 수 있다. key index 입력 방식은 8.1.4 Key 사용 방식을 참조한다.

해당 알고리즘은 Autoever HSM 프레임워크 내에서 수행된다.

| API                   | Parameter      | Value                          |
|-----------------------|----------------|--------------------------------|
| Csm_MacGenerateStart  | keyPtr->length | 1 or 2<br>※ 8.1.4 Key 사용 방식 참조 |
|                       | keyPtr->data   | Pre-shared Key Index           |
| Csm_MacGenerateUpdate | dataLength     | 4의 배수                          |

|                       |                     |                                             |
|-----------------------|---------------------|---------------------------------------------|
|                       |                     | 원본 데이터 길이가 4의 배수가 아닌 경우 App에서 0 패딩을 해줘야 한다. |
| Csm_MacGenerateFinish | *resultLengthPtr    | >= 16 (TruncationIsAllowed == FALSE)        |
|                       | TruncationIsAllowed | > 0 (TruncationIsAllowed == TRUE)           |

다양한 길이의 Data 와 Key 값과 수신된 MAC 을 기반으로 수신한 Data의 무결성, 인증을 확인한다.

전송 효율을 위해 MAC은 truncation 될 수 있고 수신한 MAC 은 bit 단위의 길이로 사용한다.

Autoever HSM 프레임워크는 사전에 저장된 Key 값만을 사용할 수 있다. key index 입력 방식은 8.1.4 Key 사용 방식을 참조한다.

해당 알고리즘은 Autoever HSM 프레임워크 내에서 수행된다.

| API                 | Parameter      | Value                                                |
|---------------------|----------------|------------------------------------------------------|
| Csm_MacVerifyStart  | keyPtr->length | 1 or 2<br>※ 8.1.4 Key 사용 방식 참조                       |
|                     | keyPtr->data   | Pre-shared Key Index                                 |
| Csm_MacVerifyUpdate | dataLength     | 4의 배수<br>원본 데이터 길이가 4의 배수가 아닌 경우 App에서 0 패딩을 해줘야 한다. |
| Csm_MacVerifyFinish | MacLength      | >= 128                                               |

#### 8.2.2.10 AutCsecCmacAes

다양한 길이의 Data 와 16 byte 길이의 Key 값을 기반으로 무결성, 인증을 확인하기 위한 16 byte 길이의 MAC 을 생성하며, truncation 이 true 인 경우 16 byte 보다 짧은 길이의 MAC 을 얻을 수 있다.

CSEC 내에 사전에 저장된 Key 값을 사용할 경우 keyPtr->length 는 1, keyPtr->data 는 Key 값의 Index를 사용한다. Key Index 는 CSEC 기능 설정에 따른 사용 가능한 범위가 정해진다. (Key Index 정보는 CryptoLib UM 참조)

해당 알고리즘은 CSEC 내에서 수행된다.

| API                   | Parameter           | Value                                                                        |
|-----------------------|---------------------|------------------------------------------------------------------------------|
| Csm_MacGenerateStart  | keyPtr->length      | 1 or 16<br>※ 1 : Pre-shared Key Index 사용시<br>16 : Raw Key 사용시                |
|                       | keyPtr->data        | Pre-shared Key Index (keyPtr->length == 1)<br>Raw Key (keyPtr->length == 16) |
| Csm_MacGenerateUpdate | dataLength          | 제한없음                                                                         |
| Csm_MacGenerateFinish | *resultLengthPtr    | >= 16 (TruncationIsAllowed == FALSE)                                         |
|                       | TruncationIsAllowed | > 0 (TruncationIsAllowed == TRUE)                                            |

다양한 길이의 Data 와 Key 값과 수신된 MAC 을 기반으로 수신한 Data의 무결성, 인증을 확인한다.

전송 효율을 위해 MAC은 truncation 될 수 있고 수신한 MAC 은 bit 단위의 길이로 사용한다.

CSEC 내에 사전에 저장된 Key 값을 사용할 경우 keyPtr->length 는 1, keyPtr->data 는 Key 값의 Index를 사용한다. Key Index 는 CSEC 기능 설정에 따른 사용 가능한 범위가 정해진다. (Key Index 정보는 CryptoLib UM 참조)

해당 알고리즘은 CSEC 내에서 수행된다.

| API                | Parameter      | Value                                     |
|--------------------|----------------|-------------------------------------------|
| Csm_MacVerifyStart | keyPtr->length | 1 or 16<br>※ 1 : Pre-shared Key Index 사용시 |

|                     |              |                                                                              |
|---------------------|--------------|------------------------------------------------------------------------------|
|                     |              | 16 : Raw Key 사용시                                                             |
|                     | keyPtr->data | Pre-shared Key Index (keyPtr->length == 1)<br>Raw Key (keyPtr->length == 16) |
| Csm_MacVerifyUpdate | dataLength   | 제한없음                                                                         |
| Csm_MacVerifyFinish | MacLength    | >= 128                                                                       |

## Random

### 8.2.3.1 HaePseudoRandom

의사 난수 생성을 위한 16 byte 길이의 Seed 값을 주입한다.

8.2.3.1 SeedKey, AdvancedSeedKey 알고리즘의 Seed 와는 관계없는 값임을 주의한다.

| API                  | Parameter  | Value |
|----------------------|------------|-------|
| Csm_RandomSeedUpdate | seedLength | 16    |

원하는 길이의 의사 난수를 생성한다.

| API                | Parameter    | Value |
|--------------------|--------------|-------|
| Csm_RandomGenerate | resultLength | 제한없음  |

### 8.2.3.2 AuthHsmTrueRandom

원하는 길이의 진성 난수를 생성한다.

| API                | Parameter    | Value |
|--------------------|--------------|-------|
| Csm_RandomGenerate | resultLength | 제한없음  |

### 8.2.3.3 AuthHsmPseudoRandom

진성 난수를 최초 seed 로 사용하여 원하는 길이의 의사난수를 생성한다.

| API                | Parameter    | Value |
|--------------------|--------------|-------|
| Csm_RandomGenerate | resultLength | 제한없음  |

### 8.2.3.4 HaeHsmTrueRandom

원하는 길이의 진성 난수를 생성한다.

| API                | Parameter    | Value   |
|--------------------|--------------|---------|
| Csm_RandomGenerate | resultLength | <= 1024 |

### 8.2.3.3 HaeHsmTrueRandom

원하는 길이의 의사 난수를 생성한다.

| API                | Parameter    | Value   |
|--------------------|--------------|---------|
| Csm_RandomGenerate | resultLength | <= 1024 |

### 8.2.3.4 AutIcusRandom

진성 난수를 최초 seed 로 사용하여 원하는 길이의 의사난수를 생성한다.

| API                | Parameter    | Value |
|--------------------|--------------|-------|
| Csm_RandomGenerate | resultLength | 제한없음  |

### 8.2.3.5 AutCsecRandom

진성 난수를 최초 seed 로 사용하여 원하는 길이의 의사난수를 생성한다.

| API                | Parameter    | Value |
|--------------------|--------------|-------|
| Csm_RandomGenerate | resultLength | 제한없음  |

## Symmetric Block

### 8.2.4.1 HaeAesEcb

16 byte 배수 길의 평문과 16 byte 길이의 Key 값을 기반으로 입력 평문과 동일한 길이의 AES 128 알고리즘의 ECB 모드를 적용한 암호문을 생성한다.

| API                       | Parameter            | Value              |
|---------------------------|----------------------|--------------------|
| Csm_SymBlockEncryptStart  | keyPtr->length       | 16                 |
| Csm_SymBlockEncryptUpdate | plainTextLength      | 16의 배수             |
|                           | *cipherTextLengthPtr | >= plainTextLength |

16 byte 배수 길의 암호문과 16 byte 길이의 Key 값을 기반으로 입력 암호문과 동일한 길이의 AES 128 알고리즘의 ECB 모드를 적용한 평문을 생성한다.

| API                       | Parameter           | Value               |
|---------------------------|---------------------|---------------------|
| Csm_SymBlockDecryptStart  | keyPtr->length      | 16                  |
| Csm_SymBlockDecryptUpdate | cipherTextLength    | 16의 배수              |
|                           | *plainTextLengthPtr | >= cipherTextLength |

### 8.2.4.2 AuthHsmAesEcb

16 byte 배수 길의 평문과 16 byte 길이의 Key 값을 기반으로 입력 평문과 동일한 길이의 AES 128 알고리즘의 ECB 모드를 적용한 암호문을 생성한다.

HSM 프레임워크 내에 사전에 저장된 Key 값을 사용할 경우 keyPtr->length 는 1, keyPtr->data 는 Key 값의 Index를 사용한다.

해당 알고리즘은 Autron HSM 프레임워크 내에서 수행된다.

| API                       | Parameter            | Value                                                                        |
|---------------------------|----------------------|------------------------------------------------------------------------------|
| Csm_SymBlockEncryptStart  | keyPtr->length       | 1 or 16<br>※ 1 : Pre-shared Key Index 사용시<br>16 : Raw Key 사용시                |
|                           | keyPtr->data         | Pre-shared Key Index (keyPtr->length == 1)<br>Raw Key (keyPtr->length == 16) |
| Csm_SymBlockEncryptUpdate | plainTextLength      | 16의 배수                                                                       |
|                           | *cipherTextLengthPtr | >= plainTextLength                                                           |

16 byte 배수 길의 암호문과 16 byte 길이의 Key 값을 기반으로 입력 암호문과 동일한 길이의 AES 128 알고리즘의 ECB 모드를 적용한 평문을 생성한다.

HSM 프레임워크 내에 사전에 저장된 Key 값을 사용할 경우 keyPtr->length 는 1, keyPtr->data 는 Key 값의 Index를 사용한다.

해당 알고리즘은 Autron HSM 프레임워크 내에서 수행된다.

| API                      | Parameter      | Value                                     |
|--------------------------|----------------|-------------------------------------------|
| Csm_SymBlockDecryptStart | keyPtr->length | 1 or 16<br>※ 1 : Pre-shared Key Index 사용시 |

|                           |                     |                                                                              |
|---------------------------|---------------------|------------------------------------------------------------------------------|
|                           |                     | 16 : Raw Key 사용시                                                             |
|                           | keyPtr->data        | Pre-shared Key Index (keyPtr->length == 1)<br>Raw Key (keyPtr->length == 16) |
| Csm_SymBlockDecryptUpdate | cipherTextLength    | 16의 배수                                                                       |
|                           | *plainTextLengthPtr | >= cipherTextLength                                                          |

#### 8.2.4.3 AutIcusAesEcb

16 byte 배수 길의 평문과 16 byte 길이의 Key 값을 기반으로 입력 평문과 동일한 길이의 AES 128 알고리즘의 ECB 모드를 적용한 암호문을 생성한다.

ICUS 내에 사전에 저장된 Key 값을 사용할 경우 keyPtr->length 는 1, keyPtr->data 는 Key 값의 Index를 사용한다. (Key Index 정보는 CryptoLib UM 참조)

해당 알고리즘은 ICUS 내에서 수행된다.

| API                       | Parameter            | Value                                                                        |
|---------------------------|----------------------|------------------------------------------------------------------------------|
| Csm_SymBlockEncryptStart  | keyPtr->length       | 1 or 16<br>※ 1 : Pre-shared Key Index 사용시<br>16 : Raw Key 사용시                |
|                           | keyPtr->data         | Pre-shared Key Index (keyPtr->length == 1)<br>Raw Key (keyPtr->length == 16) |
| Csm_SymBlockEncryptUpdate | plainTextLength      | 16의 배수                                                                       |
|                           | *cipherTextLengthPtr | >= plainTextLength                                                           |

16 byte 배수 길의 암호문과 16 byte 길이의 Key 값을 기반으로 입력 암호문과 동일한 길이의 AES 128 알고리즘의 ECB 모드를 적용한 평문을 생성한다.

ICUS 내에 사전에 저장된 Key 값을 사용할 경우 keyPtr->length 는 1, keyPtr->data 는 Key 값의 Index를 사용한다. (Key Index 정보는 CryptoLib UM 참조)

해당 알고리즘은 ICUS 내에서 수행된다.

| API                       | Parameter           | Value                                                                        |
|---------------------------|---------------------|------------------------------------------------------------------------------|
| Csm_SymBlockDecryptStart  | keyPtr->length      | 1 or 16<br>※ 1 : Pre-shared Key Index 사용시<br>16 : Raw Key 사용시                |
|                           | keyPtr->data        | Pre-shared Key Index (keyPtr->length == 1)<br>Raw Key (keyPtr->length == 16) |
| Csm_SymBlockDecryptUpdate | cipherTextLength    | 16의 배수                                                                       |
|                           | *plainTextLengthPtr | >= cipherTextLength                                                          |

#### 8.2.4.4 AutCsecAesEcb

16 byte 배수 길의 평문과 16 byte 길이의 Key 값을 기반으로 입력 평문과 동일한 길이의 AES 128 알고리즘의 ECB 모드를 적용한 암호문을 생성한다.

CSEC 내에 사전에 저장된 Key 값을 사용할 경우 keyPtr->length 는 1, keyPtr->data 는 Key 값의 Index를 사용한다. Key Index 는 CSEC 기능 설정에 따른 사용 가능한 범위가 정해진다. (Key Index 정보는 CryptoLib UM 참조)

해당 알고리즘은 CSEC 내에서 수행된다.

| API                      | Parameter      | Value   |
|--------------------------|----------------|---------|
| Csm_SymBlockEncryptStart | keyPtr->length | 1 or 16 |

|                           |                      |                                                                              |
|---------------------------|----------------------|------------------------------------------------------------------------------|
| t                         |                      | ※ 1 : Pre-shared Key Index 사용시<br>16 : Raw Key 사용시                           |
|                           | keyPtr->data         | Pre-shared Key Index (keyPtr->length == 1)<br>Raw Key (keyPtr->length == 16) |
| Csm_SymBlockEncryptUpdate | plainTextLength      | 16의 배수                                                                       |
|                           | *cipherTextLengthPtr | >= plainTextLength                                                           |

16 byte 배수 길의 암호문과 16 byte 길이의 Key 값을 기반으로 입력 암호문과 동일한 길이의 AES 128 알고리즘의 ECB 모드를 적용한 평문을 생성한다.

CSEC 내에 사전에 저장된 Key 값을 사용할 경우 keyPtr->length 는 1, keyPtr->data 는 Key 값의 Index를 사용한다. Key Index 는 CSEC 기능 설정에 따른 사용 가능한 범위가 정해진다. (Key Index 정보는 CryptoLib UM 참조)

해당 알고리즘은 CSEC 내에서 수행된다.

| API                       | Parameter           | Value                                                                        |
|---------------------------|---------------------|------------------------------------------------------------------------------|
| Csm_SymBlockDecryptStart  | keyPtr->length      | 1 or 16<br>※ 1 : Pre-shared Key Index 사용시<br>16 : Raw Key 사용시                |
|                           | keyPtr->data        | Pre-shared Key Index (keyPtr->length == 1)<br>Raw Key (keyPtr->length == 16) |
| Csm_SymBlockDecryptUpdate | cipherTextLength    | 16의 배수                                                                       |
|                           | *plainTextLengthPtr | >= cipherTextLength                                                          |

#### 8.2.4.3 HaeHsmAesEcb

16 byte 배수 길의 평문과 16 byte 길이의 Key 값을 기반으로 입력 평문과 동일한 길이의 AES 128 알고리즘의 ECB 모드를 적용한 암호문을 생성한다. Autoever HSM 2.0의 경우 HSM에 저장된 Key 만을 사용할 수 있다. keyPtr->length 는 1, keyPtr->data 는 Key 값의 Index를 사용한다.

해당 알고리즘은 Autoever HSM 프레임워크 내에서 수행된다.

| API                       | Parameter            | Value                                      |
|---------------------------|----------------------|--------------------------------------------|
| Csm_SymBlockEncryptStart  | keyPtr->length       | 1<br>(Pre-shared Key Index 사용)             |
|                           | keyPtr->data         | Pre-shared Key Index (keyPtr->length == 1) |
| Csm_SymBlockEncryptUpdate | plainTextLength      | 16의 배수                                     |
|                           | *cipherTextLengthPtr | >= plainTextLength                         |

16 byte 배수 길의 암호문과 16 byte 길이의 Key 값을 기반으로 입력 암호문과 동일한 길이의 AES 128 알고리즘의 ECB 모드를 적용한 평문을 생성한다.

Autoever HSM 2.0의 경우 HSM에 저장된 Key 만을 사용할 수 있기 때문에 keyPtr->length 는 1, keyPtr->data 는 Key 값의 Index를 사용한다.

해당 알고리즘은 Autron HSM 프레임워크 내에서 수행된다.

| API                      | Parameter      | Value                          |
|--------------------------|----------------|--------------------------------|
| Csm_SymBlockDecryptStart | keyPtr->length | 1<br>(Pre-shared Key Index 사용) |

|                           |                     |                                            |
|---------------------------|---------------------|--------------------------------------------|
|                           | keyPtr->data        | Pre-shared Key Index (keyPtr->length == 1) |
| Csm_SymBlockDecryptUpdate | cipherTextLength    | 16의 배수                                     |
|                           | *plainTextLengthPtr | >= cipherTextLength                        |

## Symmetric

### 8.2.5.1 HaeAesCbc

16 byte 배수 길의 평문과 16 byte 길이의 initialization vector 와 16 byte 길이의 Key 값을 기반으로 입력 평문과 동일한 길이의 AES 128 알고리즘의 CBC 모드를 적용한 암호문을 생성한다.

| API                  | Parameter            | Value              |
|----------------------|----------------------|--------------------|
| Csm_SymEncryptStart  | keyPtr->length       | 16                 |
|                      | InitVectorLength     | 16                 |
| Csm_SymEncryptUpdate | plainTextLength      | 16의 배수             |
|                      | *cipherTextLengthPtr | >= plainTextLength |

16 byte 배수 길의 암호문과 16 byte 길이의 initialization vector 와 16 byte 길이의 Key 값을 기반으로 입력 암호문과 동일한 길이의 AES 128 알고리즘의 CBC 모드를 적용한 평문을 생성한다.

| API                  | Parameter           | Value               |
|----------------------|---------------------|---------------------|
| Csm_SymDecryptStart  | keyPtr->length      | 16                  |
|                      | InitVectorLength    | 16                  |
| Csm_SymDecryptUpdate | cipherTextLength    | 16의 배수              |
|                      | *plainTextLengthPtr | >= cipherTextLength |

### 8.2.5.2 HaeAesCtr

16 byte 길이의 Nonce + Counter 와 16 byte 길이의 Key 값을 기반으로 입력 평문과 동일한 길이의 AES 128 알고리즘 CTR 모드를 적용한 암호문을 생성한다.

| API                  | Parameter            | Value              |
|----------------------|----------------------|--------------------|
| Csm_SymEncryptStart  | keyPtr->length       | 16                 |
|                      | InitVectorLength     | 16                 |
| Csm_SymEncryptUpdate | plainTextLength      | 길이 제한 없음           |
|                      | *cipherTextLengthPtr | >= plainTextLength |

16 byte 길이의 Nonce + Counter 와 16 byte 길이의 Key 값을 기반으로 입력 암호문과 동일한 길이의 AES 128 알고리즘 CTR 모드를 적용한 평문을 생성한다.

| API                  | Parameter           | Value               |
|----------------------|---------------------|---------------------|
| Csm_SymDecryptStart  | keyPtr->length      | 16                  |
|                      | InitVectorLength    | 16                  |
| Csm_SymDecryptUpdate | cipherTextLength    | 길이 제한 없음            |
|                      | *plainTextLengthPtr | >= cipherTextLength |

### 8.2.5.3 AuthHsmAesCbc

16 byte 배수 길의 평문과 16 byte 길이의 initialization vector 와 16 byte 길이의 Key 값을 기반으로 입력 평문과 동일한 길이의 AES 128 알고리즘의 CBC 모드를 적용한 암호문을 생성한다.

HSM 프레임워크 내에 사전에 저장된 Key 값을 사용할 경우 keyPtr->length 는 1, keyPtr->data 는 Key 값의 Index를 사용한다.

해당 알고리즘은 Autron HSM 프레임워크 내에서 수행된다.

| API                  | Parameter            | Value                                                                        |
|----------------------|----------------------|------------------------------------------------------------------------------|
| Csm_SymEncryptStart  | keyPtr->length       | 1 or 16<br>※ 1 : Pre-shared Key Index 사용시<br>16 : Raw Key 사용시                |
|                      | keyPtr->data         | Pre-shared Key Index (keyPtr->length == 1)<br>Raw Key (keyPtr->length == 16) |
|                      | InitVectorLength     | 16                                                                           |
| Csm_SymEncryptUpdate | plainTextLength      | 16의 배수                                                                       |
|                      | *cipherTextLengthPtr | >= plainTextLength                                                           |

16 byte 배수 길의 암호문과 16 byte 길이의 initialization vector 와 16 byte 길이의 Key 값을 기반으로 입력 암호문과 동일한 길이의 AES 128 알고리즘의 CBC 모드를 적용한 평문을 생성한다.

HSM 프레임워크 내에 사전에 저장된 Key 값을 사용할 경우 keyPtr->length 는 1, keyPtr->data 는 Key 값의 Index를 사용한다.

해당 알고리즘은 Autron HSM 프레임워크 내에서 수행된다.

| API                  | Parameter           | Value                                                                        |
|----------------------|---------------------|------------------------------------------------------------------------------|
| Csm_SymDecryptStart  | keyPtr->length      | 1 or 16<br>※ 1 : Pre-shared Key Index 사용시<br>16 : Raw Key 사용시                |
|                      | keyPtr->data        | Pre-shared Key Index (keyPtr->length == 1)<br>Raw Key (keyPtr->length == 16) |
|                      | InitVectorLength    | 16                                                                           |
| Csm_SymDecryptUpdate | cipherTextLength    | 16의 배수                                                                       |
|                      | *plainTextLengthPtr | >= cipherTextLength                                                          |

#### 8.2.5.4 AuthHsmAesCtr

16 byte 길이의 Nonce + Counter 와 16 byte 길이의 Key 값을 기반으로 입력 평문과 동일한 길이의 AES 128 알고리즘 CTR 모드를 적용한 암호문을 생성한다.

HSM 프레임워크 내에 사전에 저장된 Key 값을 사용할 경우 keyPtr->length 는 1, keyPtr->data 는 Key 값의 Index를 사용한다.

해당 알고리즘은 Autron HSM 프레임워크 내에서 수행된다.

| API                  | Parameter            | Value                                                                        |
|----------------------|----------------------|------------------------------------------------------------------------------|
| Csm_SymEncryptStart  | keyPtr->length       | 1 or 16<br>※ 1 : Pre-shared Key Index 사용시<br>16 : Raw Key 사용시                |
|                      | keyPtr->data         | Pre-shared Key Index (keyPtr->length == 1)<br>Raw Key (keyPtr->length == 16) |
|                      | InitVectorLength     | 16                                                                           |
| Csm_SymEncryptUpdate | plainTextLength      | 길이 제한 없음                                                                     |
|                      | *cipherTextLengthPtr | >= plainTextLength                                                           |

16 byte 길이의 Nonce + Counter 와 16 byte 길이의 Key 값을 기반으로 입력 암호문과 동일한 길이의 AES 128 알고리즘

CTR 모드를 적용한 평문을 생성한다.

HSM 프레임워크 내에 사전에 저장된 Key 값을 사용할 경우 keyPtr->length 는 1, keyPtr->data 는 Key 값의 Index를 사용한다.

해당 알고리즘은 Autron HSM 프레임워크 내에서 수행된다.

| API                  | Parameter           | Value                                                                        |
|----------------------|---------------------|------------------------------------------------------------------------------|
| Csm_SymDecryptStart  | keyPtr->length      | 1 or 16<br>※ 1 : Pre-shared Key Index 사용시<br>16 : Raw Key 사용시                |
|                      | keyPtr->data        | Pre-shared Key Index (keyPtr->length == 1)<br>Raw Key (keyPtr->length == 16) |
|                      | InitVectorLength    | 16                                                                           |
| Csm_SymDecryptUpdate | cipherTextLength    | 길이 제한 없음                                                                     |
|                      | *plainTextLengthPtr | >= cipherTextLength                                                          |

#### 8.2.5.5 AuticusAesCbc

16 byte 배수 길의 평문과 16 byte 길이의 initialization vector 와 16 byte 길이의 Key 값을 기반으로 입력 평문과 동일한 길이의 AES 128 알고리즘의 ECB 모드를 적용한 암호문을 생성한다.

ICUS 내에 사전에 저장된 Key 값을 사용할 경우 keyPtr->length 는 1, keyPtr->data 는 Key 값의 Index를 사용한다. (Key Index 정보는 CryptoLib UM 참조)

해당 알고리즘은 ICUS 내에서 수행된다.

| API                  | Parameter            | Value                                                                        |
|----------------------|----------------------|------------------------------------------------------------------------------|
| Csm_SymEncryptStart  | keyPtr->length       | 1 or 16<br>※ 1 : Pre-shared Key Index 사용시<br>16 : Raw Key 사용시                |
|                      | keyPtr->data         | Pre-shared Key Index (keyPtr->length == 1)<br>Raw Key (keyPtr->length == 16) |
|                      | InitVectorLength     | 16                                                                           |
| Csm_SymEncryptUpdate | plainTextLength      | 16의 배수                                                                       |
|                      | *cipherTextLengthPtr | >= plainTextLength                                                           |

16 byte 배수 길의 암호문과 16 byte 길이의 initialization vector 와 16 byte 길이의 Key 값을 기반으로 입력 암호문과 동일한 길이의 AES 128 알고리즘의 ECB 모드를 적용한 평문을 생성한다.

ICUS 내에 사전에 저장된 Key 값을 사용할 경우 keyPtr->length 는 1, keyPtr->data 는 Key 값의 Index를 사용한다. (Key Index 정보는 CryptoLib UM 참조)

해당 알고리즘은 ICUS 내에서 수행된다.

| API                  | Parameter        | Value                                                                        |
|----------------------|------------------|------------------------------------------------------------------------------|
| Csm_SymDecryptStart  | keyPtr->length   | 1 or 16<br>※ 1 : Pre-shared Key Index 사용시<br>16 : Raw Key 사용시                |
|                      | keyPtr->data     | Pre-shared Key Index (keyPtr->length == 1)<br>Raw Key (keyPtr->length == 16) |
|                      | InitVectorLength | 16                                                                           |
| Csm_SymDecryptUpdate | cipherTextLength | 16의 배수                                                                       |

|  |                     |                     |
|--|---------------------|---------------------|
|  | *plainTextLengthPtr | >= cipherTextLength |
|--|---------------------|---------------------|

### 8.2.5.6 AutCsecAesCbc

16 byte 배수 길의 평문과 16 byte 길이의 initialization vector 와 16 byte 길이의 Key 값을 기반으로 입력 평문과 동일한 길이의 AES 128 알고리즘의 ECB 모드를 적용한 암호문을 생성한다.

CSEC 내에 사전에 저장된 Key 값을 사용할 경우 keyPtr->length 는 1, keyPtr->data 는 Key 값의 Index를 사용한다. Key Index 는 CSEC 기능 설정에 따른 사용 가능한 범위가 정해진다. (Key Index 정보는 CryptoLib UM 참조)

해당 알고리즘은 CSEC 내에서 수행된다.

| API                  | Parameter            | Value                                                                        |
|----------------------|----------------------|------------------------------------------------------------------------------|
| Csm_SymEncryptStart  | keyPtr->length       | 1 or 16<br>※ 1 : Pre-shared Key Index 사용시<br>16 : Raw Key 사용시                |
|                      | keyPtr->data         | Pre-shared Key Index (keyPtr->length == 1)<br>Raw Key (keyPtr->length == 16) |
|                      | InitVectorLength     | 16                                                                           |
| Csm_SymEncryptUpdate | plainTextLength      | 16의 배수                                                                       |
|                      | *cipherTextLengthPtr | >= plainTextLength                                                           |

16 byte 배수 길의 암호문과 16 byte 길이의 initialization vector 와 16 byte 길이의 Key 값을 기반으로 입력 암호문과 동일한 길이의 AES 128 알고리즘의 ECB 모드를 적용한 평문을 생성한다.

CSEC 내에 사전에 저장된 Key 값을 사용할 경우 keyPtr->length 는 1, keyPtr->data 는 Key 값의 Index를 사용한다. Key Index 는 CSEC 기능 설정에 따른 사용 가능한 범위가 정해진다. (Key Index 정보는 CryptoLib UM 참조)

해당 알고리즘은 CSEC 내에서 수행된다.

| API                  | Parameter           | Value                                                                        |
|----------------------|---------------------|------------------------------------------------------------------------------|
| Csm_SymDecryptStart  | keyPtr->length      | 1 or 16<br>※ 1 : Pre-shared Key Index 사용시<br>16 : Raw Key 사용시                |
|                      | keyPtr->data        | Pre-shared Key Index (keyPtr->length == 1)<br>Raw Key (keyPtr->length == 16) |
|                      | InitVectorLength    | 16                                                                           |
| Csm_SymDecryptUpdate | cipherTextLength    | 16의 배수                                                                       |
|                      | *plainTextLengthPtr | >= cipherTextLength                                                          |

### 8.2.5.7 HaeHsmAesCbc

16 byte 배수 길의 평문과 16 byte 길이의 initialization vector 와 16 byte 길이의 Key 값을 기반으로 입력 평문과 동일한 길이의 AES 128 알고리즘의 CBC 모드를 적용한 암호문을 생성한다.

Autoever HSM 프레임워크는 사전에 저장된 Key 값만을 사용할 수 있다. key index 입력 방식은 8.1.4 Key 사용 방식을 참조한다. 해당 알고리즘은 Autoever HSM 프레임워크 내에서 수행된다.

| API                  | Parameter        | Value                          |
|----------------------|------------------|--------------------------------|
| Csm_SymEncryptStart  | keyPtr->length   | 1 or 2<br>※ 8.1.4 Key 사용 방식 참조 |
|                      | keyPtr->data     | Pre-shared Key Index           |
|                      | InitVectorLength | 16                             |
| Csm_SymEncryptUpdate | plainTextLength  | 16의 배수                         |

|  |                      |                    |
|--|----------------------|--------------------|
|  | *cipherTextLengthPtr | >= plainTextLength |
|--|----------------------|--------------------|

16 byte 배수 길의 암호문과 16 byte 길이의 initialization vector 와 16 byte 길이의 Key 값을 기반으로 입력 암호문과 동일한 길이의 AES 128 알고리즘의 CBC 모드를 적용한 평문을 생성한다.

Autoever HSM 프레임워크는 사전에 저장된 Key 값만을 사용할 수 있다. key index 입력 방식은 8.1.4 Key 사용 방식을 참조한다. 해당 알고리즘은 Autoever HSM 프레임워크 내에서 수행된다.

| API                  | Parameter           | Value                          |
|----------------------|---------------------|--------------------------------|
| Csm_SymDecryptStart  | keyPtr->length      | 1 or 2<br>※ 8.1.4 Key 사용 방식 참조 |
|                      | keyPtr->data        | Pre-shared Key Index           |
|                      | InitVectorLength    | 16                             |
| Csm_SymDecryptUpdate | cipherTextLength    | 16의 배수                         |
|                      | *plainTextLengthPtr | >= cipherTextLength            |

#### 8.2.5.8 HaeHsmAesCtr

16 byte 길이의 Nonce + Counter 와 16 byte 길이의 Key 값을 기반으로 입력 평문과 동일한 길이의 AES 128 알고리즘 CTR 모드를 적용한 암호문을 생성한다.

Autoever HSM 프레임워크는 사전에 저장된 Key 값만을 사용할 수 있다. key index 입력 방식은 8.1.4 Key 사용 방식을 참조한다. 해당 알고리즘은 Autoever HSM 프레임워크 내에서 수행된다.

| API                  | Parameter            | Value                          |
|----------------------|----------------------|--------------------------------|
| Csm_SymEncryptStart  | keyPtr->length       | 1 or 2<br>※ 8.1.4 Key 사용 방식 참조 |
|                      | keyPtr->data         | Pre-shared Key Index           |
|                      | InitVectorLength     | 16                             |
| Csm_SymEncryptUpdate | plainTextLength      | 길이 제한 없음                       |
|                      | *cipherTextLengthPtr | >= plainTextLength             |

16 byte 길이의 Nonce + Counter 와 16 byte 길이의 Key 값을 기반으로 입력 암호문과 동일한 길이의 AES 128 알고리즘 CTR 모드를 적용한 평문을 생성한다.

Autoever HSM 프레임워크는 사전에 저장된 Key 값만을 사용할 수 있다. key index 입력 방식은 8.1.4 Key 사용 방식을 참조한다. 해당 알고리즘은 Autoever HSM 프레임워크 내에서 수행된다.

| API                  | Parameter           | Value                          |
|----------------------|---------------------|--------------------------------|
| Csm_SymDecryptStart  | keyPtr->length      | 1 or 2<br>※ 8.1.4 Key 사용 방식 참조 |
|                      | keyPtr->data        | Pre-shared Key Index           |
|                      | InitVectorLength    | 16                             |
| Csm_SymDecryptUpdate | cipherTextLength    | 길이 제한 없음                       |
|                      | *plainTextLengthPtr | >= cipherTextLength            |

### Asymmetric

#### 8.2.6.1 HaeRsa

256 byte 이하 길이의 평문과 260 byte(Modulus 256 byte + Exponent 4 byte) 길이의 Public Key 값을 기반으로 RSA 2048 bit 알고리즘의 256 byte 길이 암호문을 생성한다.

비대칭키 기반의 알고리즘으로 상대적으로 긴 수행시간이 필요하기 때문에 async 동작을 고려해야 한다.

| API                   | Parameter            | Value  |
|-----------------------|----------------------|--------|
| Csm_AsymEncryptStart  | keyPtr->length       | 260    |
| Csm_AsymEncryptUpdate | plainTextLength      | <= 256 |
|                       | *cipherTextLengthPtr | >= 256 |

256 byte 길이의 암호문과 512 byte(Modulus 256 byte + Exponent 256 byte) 길이의 Private Key 값을 기반으로 RSA 2048 bit 알고리즘의 입력받은 byte 길이 평문을 생성한다.

비대칭키 기반의 알고리즘으로 상대적으로 긴 수행시간이 필요하기 때문에 async 동작을 고려해야 한다.

| API                   | Parameter           | Value  |
|-----------------------|---------------------|--------|
| Csm_AsymDecryptStart  | keyPtr->length      | 512    |
| Csm_AsymDecryptUpdate | cipherTextLength    | 256    |
|                       | *plainTextLengthPtr | <= 256 |

### 8.2.6.2 AuthHsmRsa

256 byte 이하 길이의 평문과 256 (Modulus 256 byte) or 260 (Modulus 256 byte + Exponent 4 byte) byte 길이의 Public Key 값을 기반으로 RSA 2048 bit 알고리즘의 256 byte 길이 암호문을 생성한다.

비대칭키 기반의 알고리즘으로 상대적으로 긴 수행시간이 필요하기 때문에 async 동작을 고려해야 한다.

HSM 프레임워크 내에 사전에 저장된 Key 값을 사용할 경우 keyPtr->length 는 1, keyPtr->data 는 Key 값의 Index를 사용한다.

해당 알고리즘은 Autron HSM 프레임워크 내에서 수행된다.

| API                   | Parameter            | Value                                                                                                                                                                                        |
|-----------------------|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Csm_AsymEncryptStart  | keyPtr->length       | 1 or 256 or 260<br>※ 1 : Pre-Shared Key Index 사용시<br>256 : Raw Public Key 사용시<br>modulus 256 bytes, exponent = 0x10001 고정<br>260 : Raw Public Key 사용시<br>modulus 256 bytes, exponent 4 bytes |
|                       | keyPtr->data         | Pre-shared Key Index (keyPtr->length == 1)<br>Raw Key (keyPtr->length == 256 or 260)                                                                                                         |
| Csm_AsymEncryptUpdate | plainTextLength      | <= 256                                                                                                                                                                                       |
|                       | *cipherTextLengthPtr | >= 256                                                                                                                                                                                       |

256 byte 길이의 암호문과 512 byte(Modulus 256 byte + Exponent 256 byte) 길이의 Private Key 값을 기반으로 RSA 2048 bit 알고리즘의 입력받은 byte 길이 평문을 생성한다.

비대칭키 기반의 알고리즘으로 상대적으로 긴 수행시간이 필요하기 때문에 async 동작을 고려해야 한다.

HSM 프레임워크 내에 사전에 저장된 Key 값을 사용할 경우 keyPtr->length 는 1, keyPtr->data 는 Key 값의 Index를 사용한다.

해당 알고리즘은 Autron HSM 프레임워크 내에서 수행된다.

| API                  | Parameter      | Value                                                                  |
|----------------------|----------------|------------------------------------------------------------------------|
| Csm_AsymDecryptStart | keyPtr->length | 1 or 512<br>※ 1 : Pre-Shared Key Index 사용시<br>512: Raw Private Key 사용시 |

|                       |                     |                                                                               |
|-----------------------|---------------------|-------------------------------------------------------------------------------|
|                       |                     | modulus 256 bytes, exponent 256 bytes                                         |
|                       | keyPtr->data        | Pre-shared Key Index (keyPtr->length == 1)<br>Raw Key (keyPtr->length == 512) |
| Csm_AsymDecryptUpdate | cipherTextLength    | 256                                                                           |
|                       | *plainTextLengthPtr | <= 256                                                                        |

### 8.2.6.3 AuthHsmEcies

다양한 길이의 평문과 32 byte 길이의 송신자의 Private Key 값, 64 byte 길이의 송신자의 Public Key 값, 64 byte 길이의 수신자의 Public Key 값을 기반으로 ECC 알고리즘의 32 byte 길이 암호문, 메시지 인증 태그를 생성한다.

Csm\_AsymEncryptStart 함수의 dataPtr 로 전달될 구조체는 다음의 구조체를 Csm\_AsymPublicKeyType 캐스팅하여 사용한다.

```
typedef struct {
    uint8 length[4];          /* 4 byte : 키 길이 정보.
                               Private Key A의 길이(1 byte),
                               Public Key A의 길이(1 byte),
                               Public Key B의 길이(1 byte),
                               reserved(1 byte) */
    uint8 privateKeyA[32];    /* 1 or 32 byte : Private Key A. length[0]가 1인 경우 Pre-shared Key Index. IAU
제어기 특화 기능 사용시 HSM_DATA_CUSTOM_IAU_KEY 인덱스 사용 가능 */
    uint8 publicKeyA[64];     /* 1 or 64 byte : Public Key A. length[1]가 1인 경우 Pre-shared Key Index. IAU
제어기 특화 기능 사용시 HSM_DATA_CUSTOM_IAU_KEY 인덱스 사용 가능 */
    uint8 publicKeyB[64];     /* 1 or 64 byte : Public Key B. length[2]가 1인 경우 Pre-shared Key Index */
} Csm_AuthHsmEciesEncryptKeyType;
```

Csm\_AsymEncryptUpdate 함수의 plainTextPtr 로 전달될 구조체는 다음과 같다.

```
typedef struct {
    const uint8 * pMessage;    /* 암호화할 메시지 */
    const uint8 * pParam1;     /* 키 유도 parameter (인증서 해쉬 1609.2 참고) */
    const uint32 ulMessageLength; /* 암호화할 메시지의 길이 */
    const uint8 ucEncMode;     /* 사용할 암호 알고리즘.
                                XOR_ENC : 1U, AES_CTR_IV0 : 2U, AES_CBC_IV0 : 3U */
} Csm_AuthHsmEciesEncryptDataType;
```

Csm\_AsymEncryptUpdate 함수의 cipherTextPtr 로 전달될 구조체는 다음과 같다.

```
typedef struct {
    uint8 publicKeyX[32];      /* X of Public Key A */
    uint8 publicKeyY[32];      /* Y of Public Key A */
    uint8 encMsg[32];          /* 암호화된 메시지 */
    uint8 macTag[16];          /* 메시지 인증 태그 */
    uint32 encMsgLen;          /* 암호화된 메시지의 길이 */
} Csm_AuthHsmEciesPduType;
```

비대칭키 기반의 알고리즘으로 상대적으로 긴 수행시간이 필요하기 때문에 async 동작을 고려해야 한다.

HSM 프레임워크 내에 사전에 저장된 Key 값을 사용할 경우 keyPtr->length 는 1, keyPtr->data 는 Key 값의 Index를 사용한다.

해당 알고리즘은 Autron HSM 프레임워크 내에서 수행된다.

| API                  | Parameter      | Value                           |
|----------------------|----------------|---------------------------------|
| Csm_AsymEncryptStart | keyPtr->length | 4 + 32 + 64 + 64 = 164          |
|                      | keyPtr->data   | 상기 Csm_AuthHsmEciesEncryptKeyTy |

|                        |                      |                                              |
|------------------------|----------------------|----------------------------------------------|
|                        |                      | pe 구조체 설명 참조                                 |
| Csm_AsymEncryptUpdate  | plainTextPtr         | 상기 Csm_AuthHsmEciesEncryptDataType 구조체 설명 참조 |
|                        | plainTextLength      | 사용하지 않음                                      |
|                        | cipherTextPtr        | 상기 Csm_AuthHsmEciesPduType 구조체 설명 참조         |
|                        | *cipherTextLengthPtr | 사용하지 않음                                      |
| Csm_AsymEncryptFinishi | cipherTextPtr        | 사용하지 않음                                      |
|                        | *cipherTextLengthPtr | 사용하지 않음                                      |

32 길이의 암호문과 32 byte 길이의 수신자의 Private Key 값을 기반으로 ECC 알고리즘의 다양한 길이 평문을 생성한다. Csm\_AsymDecryptUpdate 함수의 cipherTextPtr 로 전달될 구조체는 다음과 같다.

typedef struct {

```

    Csm_AuthHsmEciesPduType * pPdu;    /* Encryption 에 사용된 PDU 구조체 */
    const uint8 * pParam1;             /* 키 유도 parameter (인증서 해쉬 1609.2 참고) */
    const uint8 ucEncMode;             /* 사용할 암호(복호) 알고리즘.
                                        XOR_ENC : 1U, AES_CTR_IV0 : 2U, AES_CBC_IV0 : 3U */

```

} Csm\_AuthHsmEciesDecryptDataType;

비대칭키 기반의 알고리즘으로 상대적으로 긴 수행시간이 필요하기 때문에 async 동작을 고려해야 한다.

HSM 프레임워크 내에 사전에 저장된 Key 값을 사용할 경우 keyPtr->length 는 1, keyPtr->data 는 Key 값의 Index를 사용한다.

해당 알고리즘은 Autron HSM 프레임워크 내에서 수행된다.

| API                    | Parameter           | Value                                                                                                                             |
|------------------------|---------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| Csm_AsymDecryptStart   | keyPtr->length      | 1 or 32<br>※ 1 : Pre-Shared Key Index 사용시<br>IAU 제어기 특화 기능 사용시<br>HSM_DATA_CUSTOM_IAU_KEY<br>인덱스 사용 가능<br>32: Raw Private Key 사용시 |
|                        | keyPtr->data        | Pre-shared Key Index (keyPtr->length == 1)<br>Raw Key (keyPtr->length == 32)                                                      |
| Csm_AsymDecryptUpdate  | cipherTextPtr       | 상기 Csm_AuthHsmEciesDecryptDataType 구조체 설명 참조                                                                                      |
|                        | cipherTextLength    | 사용하지 않음                                                                                                                           |
|                        | *plainTextLengthPtr | 사용하지 않음                                                                                                                           |
| Csm_AsymDecryptFinishi | plainTextPtr        | 사용하지 않음                                                                                                                           |
|                        | *plainTextLengthPtr | 사용하지 않음                                                                                                                           |

## Signature

### 8.2.7.1 AuthHsmPkcs1V15Sha1

다양한 길이의 Data 와 512 byte 길이의 Private Key 값을 기반으로 PKCS#1 V1.5 알고리즘의 무결성, 인증을 확인하기 위한 256 byte 길이의 서명을 생성한다. 서명 생성 과정의 Hash 알고리즘은 Sha1 이 사용된다.

비대칭키 기반의 알고리즘으로 상대적으로 긴 수행시간이 필요하기 때문에 async 동작을 고려해야 한다.

HSM 프레임워크 내에 사전에 저장된 Key 값을 사용할 경우 keyPtr->length 는 1, keyPtr->data 는 Key 값의 Index를 사

용한다.

해당 알고리즘은 Autron HSM 프레임워크 내에서 수행된다.

| API                         | Parameter        | Value                                                                                                           |
|-----------------------------|------------------|-----------------------------------------------------------------------------------------------------------------|
| Csm_SignatureGenerateStart  | keyPtr->length   | 1 or 512<br>※ 1 : Pre-Shared Key Index 사용시<br>512: Raw Private Key 사용시<br>modulus 256 bytes, exponent 256 bytes |
|                             | keyPtr->data     | Pre-shared Key Index (keyPtr->length == 1)<br>Raw Key (keyPtr->length == 512)                                   |
| Csm_SignatureGenerateUpdate | dataLength       | 제한없음                                                                                                            |
| Csm_SignatureGenerateFinish | *resultLengthPtr | 256                                                                                                             |

다양한 길이의 Data 와 256 or 260 byte 길이의 Public Key 값과 수신된 서명을 기반으로 PKCS#1 V1.5 알고리즘의 무결성, 인증을 확인한다. 서명 검증 과정의 Hash 알고리즘은 Sha1 이 사용된다.

비대칭키 기반의 알고리즘으로 상대적으로 긴 수행시간이 필요하기 때문에 async 동작을 고려해야 한다.

HSM 프레임워크 내에 사전에 저장된 Key 값을 사용할 경우 keyPtr->length 는 1, keyPtr->data 는 Key 값의 Index를 사용한다.

해당 알고리즘은 Autron HSM 프레임워크 내에서 수행된다.

| API                       | Parameter       | Value                                                                                                                                                                     |
|---------------------------|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Csm_SignatureVerifyStart  | keyPtr->length  | 1 or 256 or 260<br>※ 1 : Pre-Shared Key Index 사용시<br>256 : Raw Public Key 사용시<br>exponent = 0x10001 고정<br>260 : Raw Public Key 사용시<br>modulus 256 bytes, exponent 4 bytes |
|                           | keyPtr->data    | Pre-shared Key Index (keyPtr->length == 1)<br>Raw Key (keyPtr->length == 256 or 260)                                                                                      |
| Csm_SignatureVerifyUpdate | dataLength      | 제한없음                                                                                                                                                                      |
| Csm_SignatureVerifyFinish | signatureLength | 256                                                                                                                                                                       |

#### 8.2.7.2 AuthHsmPkcs1V15Sha1Seed

다양한 길이의 Data 와 256 or 260 byte 길이의 Public Key 값과 수신된 서명을 기반으로 PKCS#1 V1.5 알고리즘의 무결성, 인증을 확인한다. 서명 검증 과정의 Hash 알고리즘은 생략되며 입력 Data 가 Sha1 Hash 가 적용된 Data 로 간주하여 사용한다.

비대칭키 기반의 알고리즘으로 상대적으로 긴 수행시간이 필요하기 때문에 async 동작을 고려해야 한다.

HSM 프레임워크 내에 사전에 저장된 Key 값을 사용할 경우 keyPtr->length 는 1, keyPtr->data 는 Key 값의 Index를 사용한다.

해당 알고리즘은 Autron HSM 프레임워크 내에서 수행된다.

| API                       | Parameter       | Value                                                                                                                                                            |
|---------------------------|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Csm_SignatureVerifyStart  | keyPtr->length  | 1 or 256 or 260<br>※ 1 : Pre-Shared Key Index<br>256 : Raw Public Key<br>exponent = 0x10001 고정<br>260 : Raw Public Key<br>modulus 256 bytes, exponent<br>4 bytes |
|                           | keyPtr->data    | Pre-shared Key Index (keyPtr->length == 1)<br>Raw Key (keyPtr->length == 256 or 260)                                                                             |
| Csm_SignatureVerifyUpdate | dataLength      | <=20                                                                                                                                                             |
| Csm_SignatureVerifyFinish | signatureLength | 256                                                                                                                                                              |

### 8.2.7.3 AuthHsmPkcs1V15Sha256

다양한 길이의 Data 와 512 byte 길이의 Private Key 값을 기반으로 PKCS#1 V1.5 알고리즘의 무결성, 인증을 확인하기 위한 256 byte 길이의 서명을 생성한다. 서명 생성 과정의 Hash 알고리즘은 Sha256 이 사용된다.

비대칭키 기반의 알고리즘으로 상대적으로 긴 수행시간이 필요하기 때문에 async 동작을 고려해야 한다.

HSM 프레임워크 내에 사전에 저장된 Key 값을 사용할 경우 keyPtr->length 는 1, keyPtr->data 는 Key 값의 Index를 사용한다.

해당 알고리즘은 Autron HSM 프레임워크 내에서 수행된다.

| API                         | Parameter        | Value                                                                                                              |
|-----------------------------|------------------|--------------------------------------------------------------------------------------------------------------------|
| Csm_SignatureGenerateStart  | keyPtr->length   | 1 or 512<br>※ 1 : Pre-Shared Key Index 사용시<br>512: Raw Private Key 사용시<br>modulus 256 bytes, exponent<br>256 bytes |
|                             | keyPtr->data     | Pre-shared Key Index (keyPtr->length == 1)<br>Raw Key (keyPtr->length == 512)                                      |
| Csm_SignatureGenerateUpdate | dataLength       | 제한없음                                                                                                               |
| Csm_SignatureGenerateFinish | *resultLengthPtr | 256                                                                                                                |

다양한 길이의 Data 와 256 or 260 byte 길이의 Public Key 값과 수신된 서명을 기반으로 PKCS#1 V1.5 알고리즘의 무결성, 인증을 확인한다. 서명 검증 과정의 Hash 알고리즘은 Sha256 이 사용된다.

비대칭키 기반의 알고리즘으로 상대적으로 긴 수행시간이 필요하기 때문에 async 동작을 고려해야 한다.

HSM 프레임워크 내에 사전에 저장된 Key 값을 사용할 경우 keyPtr->length 는 1, keyPtr->data 는 Key 값의 Index를 사용한다.

해당 알고리즘은 Autron HSM 프레임워크 내에서 수행된다.

| API                      | Parameter      | Value                                             |
|--------------------------|----------------|---------------------------------------------------|
| Csm_SignatureVerifyStart | keyPtr->length | 1 or 256 or 260<br>※ 1 : Pre-Shared Key Index 사용시 |

|                           |                 |                                                                                                                         |
|---------------------------|-----------------|-------------------------------------------------------------------------------------------------------------------------|
|                           |                 | 256 : Raw Public Key 사용시<br>exponent = 0x10001 고정<br>260 : Raw Public Key 사용시<br>modulus 256 bytes, exponent<br>4 bytes |
|                           | keyPtr->data    | Pre-shared Key Index (keyPtr->length == 1)<br>Raw Key (keyPtr->length == 256 or 260)                                    |
| Csm_SignatureVerifyUpdate | dataLength      | 제한없음                                                                                                                    |
| Csm_SignatureVerifyFinish | signatureLength | 256                                                                                                                     |

#### 8.2.7.4 AuthHsmPkcs1V15Sha256Seed

다양한 길이의 Data 와 256 or 260 byte 길이의 Public Key 값과 수신된 서명을 기반으로 PKCS#1 V1.5 알고리즘의 무결성, 인증을 확인한다. 서명 검증 과정의 Hash 알고리즘은 생략되며 입력 Data 가 Sha256 Hash 가 적용된 Data 로 간주하여 사용한다.

비대칭키 기반의 알고리즘으로 상대적으로 긴 수행시간이 필요하기 때문에 async 동작을 고려해야 한다.

HSM 프레임워크 내에 사전에 저장된 Key 값을 사용할 경우 keyPtr->length 는 1, keyPtr->data 는 Key 값의 Index를 사용한다.

해당 알고리즘은 Autron HSM 프레임워크 내에서 수행된다.

| API                       | Parameter       | Value                                                                                                                                                                        |
|---------------------------|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Csm_SignatureVerifyStart  | keyPtr->length  | 1 or 256 or 260<br>※ 1 : Pre-Shared Key Index 사용시<br>256 : Raw Public Key 사용시<br>exponent = 0x10001 고정<br>260 : Raw Public Key 사용시<br>modulus 256 bytes, exponent<br>4 bytes |
|                           | keyPtr->data    | Pre-shared Key Index (keyPtr->length == 1)<br>Raw Key (keyPtr->length == 256 or 260)                                                                                         |
| Csm_SignatureVerifyUpdate | dataLength      | <=32                                                                                                                                                                         |
| Csm_SignatureVerifyFinish | signatureLength | 256                                                                                                                                                                          |

#### 8.2.7.5 AuthHsmPkcs1PssSha256

다양한 길이의 Data 와 512 byte 길이의 Private Key 값을 기반으로 PKCS#1 PSS 알고리즘의 무결성, 인증을 확인하기 위한 256 byte 길이의 서명을 생성한다. 서명 생성 과정의 Hash 알고리즘은 Sha256 이 사용된다.

Csm\_SignatureGenerateUpdate 함수의 dataPtr 로 전달될 구조체는 다음과 같다.

```
typedef struct {
    const uint8 * pData;    /* 평문의 포인터 */
    const uint8 * pSalt;    /* 서명 생성에 사용될 랜덤 값의 포인터 */
    uint32 ulDataLength;    /* 평문의 길이 */
    uint32 ulSaltLength;    /* 서명 생성에 사용될 랜덤 값의 길이 */
}
```

} Csm\_AutHsmPkcs1PssDataType;

비대칭키 기반의 알고리즘으로 상대적으로 긴 수행시간이 필요하기 때문에 async 동작을 고려해야 한다.

HSM 프레임워크 내에 사전에 저장된 Key 값을 사용할 경우 keyPtr->length 는 1, keyPtr->data 는 Key 값의 Index를 사용한다.

해당 알고리즘은 Autron HSM 프레임워크 내에서 수행된다.

| API                         | Parameter        | Value                                                                                                           |
|-----------------------------|------------------|-----------------------------------------------------------------------------------------------------------------|
| Csm_SignatureGenerateStart  | keyPtr->length   | 1 or 512<br>※ 1 : Pre-Shared Key Index 사용시<br>512: Raw Private Key 사용시<br>modulus 256 bytes, exponent 256 bytes |
|                             | keyPtr->data     | Pre-shared Key Index (keyPtr->length == 1)<br>Raw Key (keyPtr->length == 512)                                   |
| Csm_SignatureGenerateUpdate | dataLength       | 사용하지 않음                                                                                                         |
| Csm_SignatureGenerateFinish | *resultLengthPtr | 256                                                                                                             |

다양한 길이의 Data 와 256 or 260 byte 길이의 Public Key 값과 수신된 서명을 기반으로 PKCS#1 PSS 알고리즘의 무결성, 인증을 확인한다. 서명 검증 과정의 Hash 알고리즘은 Sha256 이 사용된다.

Csm\_SignatureGenerateUpdate 함수의 dataPtr 로 전달될 구조체는 다음과 같다.

typedef struct {

const uint8 \* pData; /\* 평문의 포인터 \*/

const uint8 \* pSalt; /\* 서명 생성에 사용될 랜덤 값의 포인터 \*/

uint32 ulDataLength; /\* 평문의 길이 \*/

uint32 ulSaltLength; /\* 서명 생성에 사용될 랜덤 값의 길이 \*/

} Csm\_AutHsmPkcs1PssDataType;

비대칭키 기반의 알고리즘으로 상대적으로 긴 수행시간이 필요하기 때문에 async 동작을 고려해야 한다.

HSM 프레임워크 내에 사전에 저장된 Key 값을 사용할 경우 keyPtr->length 는 1, keyPtr->data 는 Key 값의 Index를 사용한다.

해당 알고리즘은 Autron HSM 프레임워크 내에서 수행된다.

| API                       | Parameter      | Value                                                                                                                                                                     |
|---------------------------|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Csm_SignatureVerifyStart  | keyPtr->length | 1 or 256 or 260<br>※ 1 : Pre-Shared Key Index 사용시<br>256 : Raw Public Key 사용시<br>exponent = 0x10001 고정<br>260 : Raw Public Key 사용시<br>modulus 256 bytes, exponent 4 bytes |
|                           | keyPtr->data   | Pre-shared Key Index (keyPtr->length == 1)<br>Raw Key (keyPtr->length == 256 or 260)                                                                                      |
| Csm_SignatureVerifyUpdate | dataLength     | 사용하지 않음                                                                                                                                                                   |

|                           |                 |     |
|---------------------------|-----------------|-----|
| Csm_SignatureVerifyFinish | signatureLength | 256 |
|---------------------------|-----------------|-----|

### 8.2.7.6 AuthHsmEcdsaSha256

다양한 길이의 Data 와 32 byte 길이의 Private Key 값을 기반으로 ECC 알고리즘의 무결성, 인증을 확인하기 위한 64 byte 길이의 서명을 생성한다. 서명 생성 과정의 Hash 알고리즘은 Sha256 이 사용된다.

Csm\_SignatureGenerateUpdate 함수의 dataPtr 로 전달될 구조체는 다음과 같다.

```
typedef struct {
    const uint8 * pMessage;          /* 평문의 포인터 */
    const uint8 * pSecretNumber;     /* 서명 생성에 사용할 랜덤 값의 포인터. 길이는 32 byte.
                                     서명의 랜덤성을 확보하기 위한 랜덤수로 서명검증에는 사용되지 않는다. */
    uint32 ulMessageLength;         /* 평문의 길이 */
} Csm_AuthHsmEcdsaDataType;
```

Csm\_SignatureGenerateFinish 함수의 resultPtr 로 전달될 구조체는 다음과 같다.

```
typedef struct {
    uint8 * pSignatureR; /* ECDSA 서명 R */
    uint8 * pSignatureS; /* ECDSA 서명 S */
} Csm_AuthHsmEcdsaSignatureType;
```

비대칭키 기반의 알고리즘으로 상대적으로 긴 수행시간이 필요하기 때문에 async 동작을 고려해야 한다.

HSM 프레임워크 내에 사전에 저장된 Key 값을 사용할 경우 keyPtr->length 는 1, keyPtr->data 는 Key 값의 Index를 사용한다.

해당 알고리즘은 Autron HSM 프레임워크 내에서 수행된다.

| API                         | Parameter        | Value                                                                                                                              |
|-----------------------------|------------------|------------------------------------------------------------------------------------------------------------------------------------|
| Csm_SignatureGenerateStart  | keyPtr->length   | 1 or 32<br>※ 1 : Pre-Shared Key Index 사용시.<br>IAU 제어기 특화 기능 사용시<br>HSM_DATA_CUSTOM_IAU_KEY<br>인덱스 사용 가능<br>32: Raw Private Key 사용시 |
|                             | keyPtr->data     | Pre-shared Key Index (keyPtr->length == 1)<br>Raw Key (keyPtr->length == 32)                                                       |
| Csm_SignatureGenerateUpdate | dataLength       | 사용하지 않음                                                                                                                            |
| Csm_SignatureGenerateFinish | *resultLengthPtr | 64                                                                                                                                 |

다양한 길이의 Data 와 64 byte 길이의 Public Key 값과 수신된 서명을 기반으로 ECC 알고리즘의 무결성, 인증을 확인한다. 서명 검증 과정의 Hash 알고리즘은 Sha256 이 사용된다.

Csm\_SignatureVerifyFinish 함수의 signaturePtr 로 전달될 구조체는 다음과 같다.

```
typedef struct {
    uint8 * pSignatureR; /* ECDSA 서명 R */
    uint8 * pSignatureS; /* ECDSA 서명 S */
} Csm_AuthHsmEcdsaSignatureType;
```

비대칭키 기반의 알고리즘으로 상대적으로 긴 수행시간이 필요하기 때문에 async 동작을 고려해야 한다.

HSM 프레임워크 내에 사전에 저장된 Key 값을 사용할 경우 keyPtr->length 는 1, keyPtr->data 는 Key 값의 Index를 사용한다.

해당 알고리즘은 Autron HSM 프레임워크 내에서 수행된다.

| API | Parameter | Value |
|-----|-----------|-------|
|-----|-----------|-------|

|                           |                 |                                                                                                                                    |
|---------------------------|-----------------|------------------------------------------------------------------------------------------------------------------------------------|
| Csm_SignatureVerifyStart  | keyPtr->length  | 1 or 64<br>※ 1 : Pre-Shared Key Index 사용시.<br>IAU 제어기 특화 기능 사용시<br>HSM_DATA_CUSTOM_IAU_KEY<br>인덱스 사용 가능<br>64 : Raw Public Key 사용시 |
|                           | keyPtr->data    | Pre-shared Key Index (keyPtr->length == 1)<br>Raw Key (keyPtr->length == 64)                                                       |
| Csm_SignatureVerifyUpdate | dataLength      | 제한없음                                                                                                                               |
| Csm_SignatureVerifyFinish | signatureLength | 64                                                                                                                                 |

#### 8.2.7.7 HaeHsmPkcs1V15Sha1

다양한 길이의 Data 와 256 byte 길이의 Public Key 값과 수신된 서명을 기반으로 PKCS#1 V1.5 알고리즘의 무결성, 인증을 확인한다. 서명 검증 과정의 Hash 알고리즘은 Sha1 이 사용된다.

비대칭기 기반의 알고리즘으로 상대적으로 긴 수행시간이 필요하기 때문에 async 동작을 고려해야 한다.

Autoever HSM 프레임워크는 사전에 저장된 Key 값만을 사용할 수 있다. key index 입력 방식은 8.1.4 Key 사용 방식을 참조한다.

해당 알고리즘은 Autoever HSM 프레임워크 내에서 수행된다.

| API                       | Parameter       | Value                          |
|---------------------------|-----------------|--------------------------------|
| Csm_SignatureVerifyStart  | keyPtr->length  | 1 or 2<br>※ 8.1.4 Key 사용 방식 참조 |
| Csm_SignatureVerifyUpdate | dataLength      | 제한없음                           |
| Csm_SignatureVerifyFinish | signatureLength | 256                            |

#### 8.2.7.8 HaeHsmPkcs1V15Sha1Seed

다양한 길이의 Data 와 256 byte 길이의 Public Key 값과 수신된 서명을 기반으로 PKCS#1 V1.5 알고리즘의 무결성, 인증을 확인한다. 서명 검증 과정의 Hash 알고리즘은 생략되며 입력 Data 가 Sha1 Hash 가 적용된 Data 로 간주하여 사용한다. 비대칭기 기반의 알고리즘으로 상대적으로 긴 수행시간이 필요하기 때문에 async 동작을 고려해야 한다.

Autoever HSM 프레임워크는 사전에 저장된 Key 값만을 사용할 수 있다. key index 입력 방식은 8.1.4 Key 사용 방식을 참조한다.

해당 알고리즘은 Autoever HSM 프레임워크 내에서 수행된다.

| API                       | Parameter       | Value                          |
|---------------------------|-----------------|--------------------------------|
| Csm_SignatureVerifyStart  | keyPtr->length  | 1 or 2<br>※ 8.1.4 Key 사용 방식 참조 |
| Csm_SignatureVerifyUpdate | dataLength      | <=20                           |
| Csm_SignatureVerifyFinish | signatureLength | 256                            |

#### 8.2.7.9 HaeHsmPkcs1V15Sha256

다양한 길이의 Data 와 256 byte 길이의 Public Key 값과 수신된 서명을 기반으로 PKCS#1 V1.5 알고리즘의 무결성, 인증을 확인한다. 서명 검증 과정의 Hash 알고리즘은 Sha256 이 사용된다.

비대칭기 기반의 알고리즘으로 상대적으로 긴 수행시간이 필요하기 때문에 async 동작을 고려해야 한다.

Autoever HSM 프레임워크는 사전에 저장된 Key 값만을 사용할 수 있다. key index 입력 방식은 8.1.4 Key 사용 방식을 참조한다.

조한다.

해당 알고리즘은 Autoever HSM 프레임워크 내에서 수행된다.

| API                         | Parameter        | Value                          |
|-----------------------------|------------------|--------------------------------|
| Csm_SignatureGenerateStart  | keyPtr->length   | 1 or 2<br>※ 8.1.4 Key 사용 방식 참조 |
| Csm_SignatureGenerateUpdate | dataLength       | 제한없음                           |
| Csm_SignatureGenerateFinish | *resultLengthPtr | 256                            |

다양한 길이의 Data 와 256 or 260 byte 길이의 Public Key 값과 수신된 서명을 기반으로 무결성, 인증을 확인한다. 서명 검증 과정의 Hash 알고리즘은 Sha256 이 사용된다.

Autoever HSM 프레임워크는 사전에 저장된 Key 값만을 사용할 수 있다. key index 입력 방식은 8.1.4 Key 사용 방식을 참조한다.

해당 알고리즘은 Autoever HSM 프레임워크 내에서 수행된다.

| API                       | Parameter       | Value                          |
|---------------------------|-----------------|--------------------------------|
| Csm_SignatureVerifyStart  | keyPtr->length  | 1 or 2<br>※ 8.1.4 Key 사용 방식 참조 |
| Csm_SignatureVerifyUpdate | dataLength      | 제한없음                           |
| Csm_SignatureVerifyFinish | signatureLength | 256                            |

#### 8.2.7.10 HaeHsmPkcs1V15Sha256Seed

다양한 길이의 Data 와 256 byte 길이의 Public Key 값과 수신된 서명을 기반으로 PKCS#1 V1.5 알고리즘의 무결성, 인증을 확인한다. 서명 검증 과정의 Hash 알고리즘은 생략되며 입력 Data 가 Sha256 Hash 가 적용된 Data 로 간주하여 사용한다.

비대칭키 기반의 알고리즘으로 상대적으로 긴 수행시간이 필요하기 때문에 async 동작을 고려해야 한다.

Autoever HSM 프레임워크는 사전에 저장된 Key 값만을 사용할 수 있다. key index 입력 방식은 8.1.4 Key 사용 방식을 참조한다.

해당 알고리즘은 Autoever HSM 프레임워크 내에서 수행된다.

| API                       | Parameter       | Value                          |
|---------------------------|-----------------|--------------------------------|
| Csm_SignatureVerifyStart  | keyPtr->length  | 1 or 2<br>※ 8.1.4 Key 사용 방식 참조 |
| Csm_SignatureVerifyUpdate | dataLength      | <=32                           |
| Csm_SignatureVerifyFinish | signatureLength | 256                            |

#### 8.2.7.11 HaePkcs1V15Sha1

다양한 길이의 Data 와 512 byte 길이의 Private Key 값을 기반으로 PKCS#1 V1.5 알고리즘의 무결성, 인증을 확인하기 위한 256 byte 길이의 서명을 생성한다. 서명 생성 과정의 Hash 알고리즘은 Sha1 이 사용된다.

비대칭키 기반의 알고리즘으로 상대적으로 긴 수행시간이 필요하기 때문에 async 동작을 고려해야 한다.

| API                         | Parameter      | Value |
|-----------------------------|----------------|-------|
| Csm_SignatureGenerateStart  | keyPtr->length | 512   |
| Csm_SignatureGenerateUpdate | dataLength     | 제한없음  |

|                             |                  |     |
|-----------------------------|------------------|-----|
| Csm_SignatureGenerateFinish | *resultLengthPtr | 256 |
|-----------------------------|------------------|-----|

다양한 길이의 Data 와 260 byte 길이의 Public Key 값과 수신된 서명을 기반으로 PKCS#1 V1.5 알고리즘의 무결성, 인증을 확인한다. 서명 검증 과정의 Hash 알고리즘은 Sha1 이 사용된다.

비대칭키 기반의 알고리즘으로 상대적으로 긴 수행시간이 필요하기 때문에 async 동작을 고려해야 한다.

| API                       | Parameter       | Value |
|---------------------------|-----------------|-------|
| Csm_SignatureVerifyStart  | keyPtr->length  | 260   |
| Csm_SignatureVerifyUpdate | dataLength      | 제한없음  |
| Csm_SignatureVerifyFinish | signatureLength | 256   |

#### 8.2.7.12 HaePkcs1V15Sha1Seed

다양한 길이의 Data 와 260 byte 길이의 Public Key 값과 수신된 서명을 기반으로 PKCS#1 V1.5 알고리즘의 무결성, 인증을 확인한다. 서명 검증 과정의 Hash 알고리즘은 생략되며 입력 Data 가 Sha1 Hash 가 적용된 Data 로 간주하여 사용한다.

비대칭키 기반의 알고리즘으로 상대적으로 긴 수행시간이 필요하기 때문에 async 동작을 고려해야 한다.

| API                       | Parameter       | Value |
|---------------------------|-----------------|-------|
| Csm_SignatureVerifyStart  | keyPtr->length  | 260   |
| Csm_SignatureVerifyUpdate | dataLength      | <=20  |
| Csm_SignatureVerifyFinish | signatureLength | 256   |

#### 8.2.7.13 HaePkcs1V15Sha256

다양한 길이의 Data 와 512 byte 길이의 Private Key 값을 기반으로 PKCS#1 V1.5 알고리즘의 무결성, 인증을 확인하기 위한 256 byte 길이의 서명을 생성한다. 서명 생성 과정의 Hash 알고리즘은 Sha256 이 사용된다.

비대칭키 기반의 알고리즘으로 상대적으로 긴 수행시간이 필요하기 때문에 async 동작을 고려해야 한다.

| API                         | Parameter        | Value |
|-----------------------------|------------------|-------|
| Csm_SignatureGenerateStart  | keyPtr->length   | 512   |
| Csm_SignatureGenerateUpdate | dataLength       | 제한없음  |
| Csm_SignatureGenerateFinish | *resultLengthPtr | 256   |

다양한 길이의 Data 와 260 byte 길이의 Public Key 값과 수신된 서명을 기반으로 PKCS#1 V1.5 알고리즘의 무결성, 인증을 확인한다. 서명 검증 과정의 Hash 알고리즘은 Sha256 이 사용된다.

비대칭키 기반의 알고리즘으로 상대적으로 긴 수행시간이 필요하기 때문에 async 동작을 고려해야 한다.

| API                       | Parameter       | Value |
|---------------------------|-----------------|-------|
| Csm_SignatureVerifyStart  | keyPtr->length  | 260   |
| Csm_SignatureVerifyUpdate | dataLength      | 제한없음  |
| Csm_SignatureVerifyFinish | signatureLength | 256   |

#### 8.2.7.14 HaePkcs1V15Sha256Seed

다양한 길이의 Data 와 260 byte 길이의 Public Key 값과 수신된 서명을 기반으로 PKCS#1 V1.5 알고리즘의 무결성, 인증을 확인한다. 서명 검증 과정의 Hash 알고리즘은 생략되며 입력 Data 가 Sha256 Hash 가 적용된 Data 로 간주하여 사용한다.

다.

비대칭키 기반의 알고리즘으로 상대적으로 긴 수행시간이 필요하기 때문에 async 동작을 고려해야 한다.

| API                       | Parameter       | Value |
|---------------------------|-----------------|-------|
| Csm_SignatureVerifyStart  | keyPtr->length  | 260   |
| Csm_SignatureVerifyUpdate | dataLength      | <=32  |
| Csm_SignatureVerifyFinish | signatureLength | 256   |

#### 8.2.7.15 HaeHsmPkcs1PssSha256

다양한 길이의 Data 와 512 byte 길이의 Private Key 값을 기반으로 PKCS#1 PSS 알고리즘의 무결성, 인증을 확인하기 위한 256 byte 길이의 서명을 생성한다. 서명 생성 과정의 Hash 알고리즘은 Sha256 이 사용된다.

Csm\_SignatureGenerateUpdate 함수의 dataPtr 로 전달될 구조체는 다음과 같다.

```
typedef struct {
    const uint8 * pData;    /* 평문의 포인터 */
    const uint8 * pSalt;    /* 서명 생성에 사용될 랜덤 값의 포인터 */
    uint32 ulDataLength;    /* 평문의 길이 */
    uint32 ulSaltLength;    /* 서명 생성에 사용될 랜덤 값의 길이 */
} Csm_HaeHsmPkcs1PssDataType;
```

비대칭키 기반의 알고리즘으로 상대적으로 긴 수행시간이 필요하기 때문에 async 동작을 고려해야 한다.

Autoever HSM 프레임워크는 사전에 저장된 Key 값만을 사용할 수 있다. key index 입력 방식은 8.1.4 Key 사용 방식을 참조한다. 해당 알고리즘은 Autoever HSM 프레임워크 내에서 수행된다.

| API                         | Parameter        | Value                          |
|-----------------------------|------------------|--------------------------------|
| Csm_SignatureGenerateStart  | keyPtr->length   | 1 or 2<br>※ 8.1.4 Key 사용 방식 참조 |
|                             | keyPtr->data     | Pre-shared Key Index           |
| Csm_SignatureGenerateUpdate | dataLength       | 사용하지 않음                        |
| Csm_SignatureGenerateFinish | *resultLengthPtr | 256                            |

다양한 길이의 Data 와 256 or 260 byte 길이의 Public Key 값과 수신된 서명을 기반으로 PKCS#1 PSS 알고리즘의 무결성, 인증을 확인한다. 서명 검증 과정의 Hash 알고리즘은 Sha256 이 사용된다.

Csm\_SignatureGenerateUpdate 함수의 dataPtr 로 전달될 구조체는 다음과 같다.

```
typedef struct {
    const uint8 * pData;    /* 평문의 포인터 */
    const uint8 * pSalt;    /* 서명 생성에 사용될 랜덤 값의 포인터 */
    uint32 ulDataLength;    /* 평문의 길이 */
    uint32 ulSaltLength;    /* 서명 생성에 사용될 랜덤 값의 길이 */
} Csm_HaeHsmPkcs1PssDataType;
```

비대칭키 기반의 알고리즘으로 상대적으로 긴 수행시간이 필요하기 때문에 async 동작을 고려해야 한다.

Autoever HSM 프레임워크는 사전에 저장된 Key 값만을 사용할 수 있다. key index 입력 방식은 8.1.4 Key 사용 방식을 참조한다. 해당 알고리즘은 Autoever HSM 프레임워크 내에서 수행된다.

| API                       | Parameter      | Value                          |
|---------------------------|----------------|--------------------------------|
| Csm_SignatureVerifyStart  | keyPtr->length | 1 or 2<br>※ 8.1.4 Key 사용 방식 참조 |
|                           | keyPtr->data   | Pre-shared Key Index           |
| Csm_SignatureVerifyUpdate | dataLength     | 사용하지 않음                        |

|                           |                 |     |
|---------------------------|-----------------|-----|
| Csm_SignatureVerifyFinish | signatureLength | 256 |
|---------------------------|-----------------|-----|

### 8.2.7.16 HaeHsmEcdsaSha256

다양한 길이의 Data 와 32 byte 길이의 Private Key 값을 기반으로 ECC 알고리즘의 무결성, 인증을 확인하기 위한 64 byte 길이의 서명을 생성한다. 서명 생성 과정의 Hash 알고리즘은 Sha256 이 사용된다.

Csm\_SignatureGenerateUpdate 함수의 dataPtr 로 전달될 구조체는 다음과 같다.

```
typedef struct {
    const uint8 * pMessage;          /* 평문의 포인터 */
    const uint8 * pSecretNumber;     /* 서명 생성에 사용할 랜덤 값의 포인터. 길이는 32 byte.
                                     서명의 랜덤성을 확보하기 위한 랜덤수로 서명검증에는 사용되지 않는다. */
    uint32 ulMessageLength;         /* 평문의 길이 */
} Csm_HaeHsmEcdsaDataType;
```

Csm\_SignatureGenerateFinish 함수의 resultPtr 로 전달될 구조체는 다음과 같다.

```
typedef struct {
    uint8 * pSignatureR; /* ECDSA 서명 R */
    uint8 * pSignatureS; /* ECDSA 서명 S */
} Csm_HaeHsmEcdsaSignatureType;
```

비대칭키 기반의 알고리즘으로 상대적으로 긴 수행시간이 필요하기 때문에 async 동작을 고려해야 한다.

Autoever HSM 프레임워크는 사전에 저장된 Key 값만을 사용할 수 있다. key index 입력 방식은 8.1.4 Key 사용 방식을 참조한다. 해당 알고리즘은 Autoever HSM 프레임워크 내에서 수행된다.

| API                         | Parameter        | Value                          |
|-----------------------------|------------------|--------------------------------|
| Csm_SignatureGenerateStart  | keyPtr->length   | 1 or 2<br>※ 8.1.4 Key 사용 방식 참조 |
|                             | keyPtr->data     | Pre-shared Key Index           |
| Csm_SignatureGenerateUpdate | dataLength       | 사용하지 않음                        |
| Csm_SignatureGenerateFinish | *resultLengthPtr | 64                             |

다양한 길이의 Data 와 64 byte 길이의 Public Key 값과 수신된 서명을 기반으로 ECC 알고리즘의 무결성, 인증을 확인한다. 서명 검증 과정의 Hash 알고리즘은 Sha256 이 사용된다.

Csm\_SignatureVerifyFinish 함수의 signaturePtr 로 전달될 구조체는 다음과 같다.

```
typedef struct {
    uint8 * pSignatureR; /* ECDSA 서명 R */
    uint8 * pSignatureS; /* ECDSA 서명 S */
} Csm_HaeHsmEcdsaSignatureType;
```

비대칭키 기반의 알고리즘으로 상대적으로 긴 수행시간이 필요하기 때문에 async 동작을 고려해야 한다.

Autoever HSM 프레임워크는 사전에 저장된 Key 값만을 사용할 수 있다. key index 입력 방식은 8.1.4 Key 사용 방식을 참조한다. 해당 알고리즘은 Autoever HSM 프레임워크 내에서 수행된다.

| API                       | Parameter       | Value                          |
|---------------------------|-----------------|--------------------------------|
| Csm_SignatureVerifyStart  | keyPtr->length  | 1 or 2<br>※ 8.1.4 Key 사용 방식 참조 |
|                           | keyPtr->data    | Pre-shared Key Index           |
| Csm_SignatureVerifyUpdate | dataLength      | 제한없음                           |
| Csm_SignatureVerifyFinish | signatureLength | 64                             |

## Key Derive

### 8.2.8.1 AuthHsmPbkdf2

패스워드를 기반으로 내부적으로 사용하는 키를 유도하는 PBKDF2 알고리즘으로 내부 Hash 는 Sha2 을 사용한다.

8.2.8

| API                 | Parameter      | Value                                                                                                                                                  |
|---------------------|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| Csm_KeyDeriveStart  | keyLength      | 0 : PSK 미사용. Csm_KeyDeriveUpdate 시 passwordPtr, passwordLength 사용<br>0 이외의 값 : PSK index 로 사용. Csm_KeyDeriveUpdate 시 passwordPtr, passwordLength 사용 안함 |
|                     | iterations     | 제한없음                                                                                                                                                   |
| Csm_KeyDeriveUpdate | passwordLength | 제한없음                                                                                                                                                   |
|                     | saltLength     | 제한없음                                                                                                                                                   |
| Csm_KeyDeriveFinish | keyPtr->length | 제한없음                                                                                                                                                   |

### 8.2.8.2 HaeHsmPbkdf2

패스워드를 기반으로 내부적으로 사용하는 키를 유도하는 PBKDF2 알고리즘이다.

Autoever HSM 프레임워크는 사전에 저장된 Key 값만을 사용할 수 있다. key index 입력 방식은 8.1.4 Key 사용 방식을 참조한다. 해당 알고리즘은 Autoever HSM 프레임워크 내에서 수행된다.

8.2.9

| API                 | Parameter      | Value                                                                             |
|---------------------|----------------|-----------------------------------------------------------------------------------|
| Csm_KeyDeriveStart  | keyLength      | 0 이외의 값 : PSK index 로 사용. Csm_KeyDeriveUpdate 시 passwordPtr, passwordLength 사용 안함 |
|                     | iterations     | 제한없음                                                                              |
| Csm_KeyDeriveUpdate | passwordLength | 제한없음                                                                              |
|                     | saltLength     | 제한없음                                                                              |
| Csm_KeyDeriveFinish | keyPtr->length | 제한없음                                                                              |

## Key Exchange Calc Secret

### 8.2.9.1 AuthHsmEcdh

상대방의 공개키와 자신의 개인키를 기반으로 공유키를 생성한다.

| API                            | Parameter               | Value                                                                                                                               |
|--------------------------------|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| Csm_KeyExchangeCalcSecretStart | basePtr                 | 사용하지 않음                                                                                                                             |
|                                | privateValuePtr->length | 1 or 32<br>※ 1 : Pre-Shared Key Index 사용시.<br>IAU 제어기 특화 기능 사용시<br>HSM_DATA_CUSTOM_IAU_KEY<br>인덱스 사용 가능<br>32 : Raw Private Key 사용시 |

|                                 |                          |                                                                                                                                    |
|---------------------------------|--------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| Csm_KeyExchangeCalcSecretUpdate | partnerPublicValuePtr    | Pre-shared Key Index (partnerPublicValueLength == 1)<br>Raw Key (partnerPublicValueLength == 64)                                   |
|                                 | partnerPublicValueLength | 1 or 64<br>※ 1 : Pre-Shared Key Index 사용시.<br>IAU 제어기 특화 기능 사용시<br>HSM_DATA_CUSTOM_IAU_KEY<br>인덱스 사용 가능<br>64 : Raw Public Key 사용시 |
| Csm_KeyExchangeCalcSecretFinish | sharedSecretLengthPtr    | >= 32 (TruncationIsAllowed == FALSE)                                                                                               |
|                                 | TruncationIsAllowed      | > 0 (TruncationIsAllowed == TRUE)                                                                                                  |

### 8.2.9.2 HaeHsmEcdh

상대방의 공개키와 자신의 개인키를 기반으로 공유키를 생성한다.

Autoever HSM 프레임워크는 사전에 저장된 Key 값만을 사용할 수 있다. key index 입력 방식은 8.1.4 Key 사용 방식을 참조한다. 해당 알고리즘은 Autoever HSM 프레임워크 내에서 수행된다. PartnerPublicValue 값은 HSM 2.0 특성상 Raw Key만을 입력 할 수 있다.

| API                             | Parameter                | Value                                       |
|---------------------------------|--------------------------|---------------------------------------------|
| Csm_KeyExchangeCalcSecretStart  | basePtr                  | 사용하지 않음                                     |
|                                 | privateValuePtr->length  | 1 or 2<br>※ 8.1.4 Key 사용 방식 참조              |
| Csm_KeyExchangeCalcSecretUpdate | partnerPublicValuePtr    | Raw Key<br>(partnerPublicValueLength == 64) |
|                                 | partnerPublicValueLength | 64<br>(Raw Public Key 사용)                   |
| Csm_KeyExchangeCalcSecretFinish | sharedSecretLengthPtr    | >= 32 (TruncationIsAllowed == FALSE)        |
|                                 | TruncationIsAllowed      | > 0 (TruncationIsAllowed == TRUE)           |

## Secure Boot

Secure Boot 는 MCU 부팅시마다 Host 코어 동작 전에 Host 부트로더의 무결성을 검증하는 보안 기술이다.

Secure Boot 기능이 한번 활성화 되면 부트로더의 MAC을 HSM 고유 보안 저장 영역에 저장하고 이후 MCU 부팅시마다 부트로더의 MAC 을 계산, 저장되어 있는 MAC 값과 비교하여 값이 동일할 경우 Host 코어를 동작시키고 값이 다를 경우 부팅을 중단한다.

해당 기술은 HSM 이 포함된 MCU 중 HSM 프레임워크가 적용된 제어기만 제공 가능하다.

Autoever HSM Framework 사용 시 SecureBoot는 Autoever UM을 참조한다.

## 알고리즘 테스트 케이스

### Hash

#### 8.4.1.1 AutSeedKey

| Seed (4 byte)          | Key (4 byte)           |
|------------------------|------------------------|
| 0x50, 0x00, 0xAC, 0x71 | 0xD7, 0xBD, 0x4C, 0xEC |

8.4

#### 8.4.1.2 HaeAdvSeedKey

동일한 Seed 값에 대해 차종 & 제어기별 다른 Key 값 생성됨.  
배포받은 패키지의 test vector 사용.

#### 8.4.1.3 HaeSha1

| Plain Text                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Digest (20 byte)                                                                                                                           |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| 0x45U, 0x92U, 0x7EU, 0x32U, 0xDDU, 0xF8U, 0x01U, 0xCAU, 0xF3U, 0x5EU, 0x18U, 0xE7U, 0xB5U, 0x07U, 0x8BU, 0x7FU, 0x54U, 0x35U, 0x27U, 0x82U, 0x12U, 0xECU, 0x6BU, 0xB9U, 0x9DU, 0xF8U, 0x84U, 0xF4U, 0x9BU, 0x32U, 0x7CU, 0x64U, 0x86U, 0xFEU, 0xAEU, 0x46U, 0xBAU, 0x18U, 0x7DU, 0xC1U, 0xCCU, 0x91U, 0x45U, 0x12U, 0x1EU, 0x14U, 0x92U, 0xE6U, 0xB0U, 0x6EU, 0x90U, 0x07U, 0x39U, 0x4DU, 0xC3U, 0x3BU, 0x77U, 0x48U, 0xF8U, 0x6AU, 0xC3U, 0x20U, 0x7CU, 0xFEU | 0xA7U, 0x0CU, 0xFBU, 0xFEU, 0x75U, 0x63U, 0xDDU, 0x0EU, 0x66U, 0x5CU, 0x7CU, 0x67U, 0x15U, 0xA9U, 0x6AU, 0x8DU, 0x75U, 0x69U, 0x50U, 0xC0U |

#### 8.4.1.4 HaeSha256

| Plain Text                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Digest (32 byte)                                                                                                                                                                                                               |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0x65U, 0x04U, 0x92U, 0x1AU, 0x97U, 0xCDU, 0x57U, 0xAAU, 0x8FU, 0x38U, 0x63U, 0xDCU, 0x32U, 0xE1U, 0xF2U, 0xD0U, 0xB5U, 0x7AU, 0xFFU, 0x63U, 0x10U, 0x6EU, 0x59U, 0xF6U, 0xAFU, 0xC3U, 0xF9U, 0x72U, 0x6BU, 0x45U, 0x93U, 0x88U, 0xBAU, 0xE1U, 0x6BU, 0x3EU, 0x22U, 0x4FU, 0x6AU, 0xA7U, 0xF4U, 0xF4U, 0x71U, 0xF1U, 0x36U, 0x06U, 0xEDU, 0xA6U, 0xE1U, 0xF1U, 0xACU, 0x2BU, 0x4DU, 0xF9U, 0xEFU, 0x8DU, 0xE9U, 0x21U, 0xC0U, 0x7CU, 0x2FU, 0x4CU, 0x85U, 0x98U, 0xD7U, 0xA3U, 0xD6U, 0xECU, 0x4BU, 0x36U, 0x8CU, 0xB8U, 0x5CU, 0xE6U, 0x1AU, 0x74U, 0x33U, 0x82U, 0x21U, 0x11U, 0x8AU, 0x30U, 0x3EU, 0x82U, 0x1CU, 0x0FU, 0x27U, 0x7BU, 0x59U, 0x1AU, 0xF6U, 0x79U, 0x5FU, 0x50U, 0xC4U, 0x02U, 0x26U, 0x12U, 0x7AU, 0x2EU, 0xFAU, 0xCCU, 0xE4U, 0x66U, 0x2FU, 0xD7U, 0x07U, 0x6CU, 0x10U, 0x9EU, 0xB5U, 0x9BU, 0x18U, 0x00U, 0x5EU, 0x71U, 0x65U, 0xF6U, 0x29U, 0x4AU, 0x69U, 0x76U, 0x43U, 0x6EU, 0xE3U, 0x97U, 0x77U, 0x4EU | 0xBDU, 0xF5U, 0xE6U, 0x4CU, 0x28U, 0xC9U, 0x69U, 0x47U, 0xD2U, 0x1CU, 0x8CU, 0x7EU, 0x85U, 0x0DU, 0xA8U, 0x07U, 0x0CU, 0x90U, 0x3BU, 0x5EU, 0xAEU, 0xD0U, 0x95U, 0xF6U, 0xB8U, 0xFFU, 0x48U, 0x7DU, 0x6DU, 0x9CU, 0xE5U, 0xE9U |

#### 8.4.1.5 AuthHsmSha1

| Plain Text                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Digest (20 byte)                                                                                                                           |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| 0x45U, 0x92U, 0x7EU, 0x32U, 0xDDU, 0xF8U, 0x01U, 0xCAU, 0xF3U, 0x5EU, 0x18U, 0xE7U, 0xB5U, 0x07U, 0x8BU, 0x7FU, 0x54U, 0x35U, 0x27U, 0x82U, 0x12U, 0xECU, 0x6BU, 0xB9U, 0x9DU, 0xF8U, 0x84U, 0xF4U, 0x9BU, 0x32U, 0x7CU, 0x64U, 0x86U, 0xFEU, 0xAEU, 0x46U, 0xBAU, 0x18U, 0x7DU, 0xC1U, 0xCCU, 0x91U, 0x45U, 0x12U, 0x1EU, 0x14U, 0x92U, 0xE6U, 0xB0U, 0x6EU, 0x90U, 0x07U, 0x39U, 0x4DU, 0xC3U, 0x3BU, 0x77U, 0x48U, 0xF8U, 0x6AU, 0xC3U, 0x20U, 0x7CU, 0xFEU | 0xA7U, 0x0CU, 0xFBU, 0xFEU, 0x75U, 0x63U, 0xDDU, 0x0EU, 0x66U, 0x5CU, 0x7CU, 0x67U, 0x15U, 0xA9U, 0x6AU, 0x8DU, 0x75U, 0x69U, 0x50U, 0xC0U |

#### 8.4.1.6 AuthHsmSha256

| Plain Text                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Digest (32 byte)                                                                                                                                                                                                               |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0x65U, 0x04U, 0x92U, 0x1AU, 0x97U, 0xCDU, 0x57U, 0xAAU, 0x8FU, 0x38U, 0x63U, 0xDCU, 0x32U, 0xE1U, 0xF2U, 0xD0U, 0xB5U, 0x7AU, 0xFFU, 0x63U, 0x10U, 0x6EU, 0x59U, 0xF6U, 0xAFU, 0xC3U, 0xF9U, 0x72U, 0x6BU, 0x45U, 0x93U, 0x88U, 0xBAU, 0xE1U, 0x6BU, 0x3EU, 0x22U, 0x4FU, 0x6AU, 0xA7U, 0xF4U, 0xF4U, 0x71U, 0xF1U, 0x36U, 0x06U, 0xEDU, 0xA6U, 0xE1U, 0xF1U, 0xACU, 0x2BU, 0x4DU, 0xF9U, 0xEFU, 0x8DU, 0xE9U, 0x21U, 0xC0U, 0x7CU, 0x2FU, 0x4CU, 0x85U, 0x98U, 0xD7U, 0xA3U, 0xD6U, 0xECU, 0x4BU, 0x36U, 0x8CU, 0xB8U, 0x5CU, 0xE6U, 0x1AU, 0x74U, 0x33U, 0x82U, 0x21U, 0x11U, 0x8AU, 0x30U, 0x3EU, 0x82U, 0x1CU, 0x0FU, 0x27U, 0x7BU, 0x59U, 0x1AU, 0xF6U, 0x79U, 0x5FU, 0x50U, 0xC4U, 0x02U, 0x26U, 0x12U, 0x7AU, 0x2EU, 0xFAU, 0xCCU, 0xE4U, 0x66U, 0x2FU, 0xD7U, 0x07U, 0x6CU, 0x10U, 0x9EU, 0xB5U, 0x9BU, 0x18U, 0x00U, 0x5EU, 0x71U, 0x65U, 0xF6U, 0x29U, 0x4AU, 0x69U, 0x76U, 0x43U, 0x6EU, 0xE3U, 0x97U, 0x77U, 0x4EU | 0xBDU, 0xF5U, 0xE6U, 0x4CU, 0x28U, 0xC9U, 0x69U, 0x47U, 0xD2U, 0x1CU, 0x8CU, 0x7EU, 0x85U, 0x0DU, 0xA8U, 0x07U, 0x0CU, 0x90U, 0x3BU, 0x5EU, 0xAEU, 0xD0U, 0x95U, 0xF6U, 0xB8U, 0xFFU, 0x48U, 0x7DU, 0x6DU, 0x9CU, 0xE5U, 0xE9U |

#### 8.4.1.7 HaeHsmSha1

| Plain Text                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Digest (20 byte)                                                                                                                           |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| 0x45U, 0x92U, 0x7EU, 0x32U, 0xDDU, 0xF8U, 0x01U, 0xCAU, 0xF3U, 0x5EU, 0x18U, 0xE7U, 0xB5U, 0x07U, 0x8BU, 0x7FU, 0x54U, 0x35U, 0x27U, 0x82U, 0x12U, 0xECU, 0x6BU, 0xB9U, 0x9DU, 0xF8U, 0x84U, 0xF4U, 0x9BU, 0x32U, 0x7CU, 0x64U, 0x86U, 0xFEU, 0xAEU, 0x46U, 0xBAU, 0x18U, 0x7DU, 0xC1U, 0xCCU, 0x91U, 0x45U, 0x12U, 0x1EU, 0x14U, 0x92U, 0xE6U, 0xB0U, 0x6EU, 0x90U, 0x07U, 0x39U, 0x4DU, 0xC3U, 0x3BU, 0x77U, 0x48U, 0xF8U, 0x6AU, 0xC3U, 0x20U, 0x7CU, 0xFEU | 0xA7U, 0x0CU, 0xFBU, 0xFEU, 0x75U, 0x63U, 0xDDU, 0x0EU, 0x66U, 0x5CU, 0x7CU, 0x67U, 0x15U, 0xA9U, 0x6AU, 0x8DU, 0x75U, 0x69U, 0x50U, 0xC0U |

#### 8.4.1.8 HaeHsmSha256

| Plain Text                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Digest (32 byte)                                                                                                                                                                                                               |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0x65U, 0x04U, 0x92U, 0x1AU, 0x97U, 0xCDU, 0x57U, 0xAAU, 0x8FU, 0x38U, 0x63U, 0xDCU, 0x32U, 0xE1U, 0xF2U, 0xD0U, 0xB5U, 0x7AU, 0xFFU, 0x63U, 0x10U, 0x6EU, 0x59U, 0xF6U, 0xAFU, 0xC3U, 0xF9U, 0x72U, 0x6BU, 0x45U, 0x93U, 0x88U, 0xBAU, 0xE1U, 0x6BU, 0x3EU, 0x22U, 0x4FU, 0x6AU, 0xA7U, 0xF4U, 0xF4U, 0x71U, 0xF1U, 0x36U, 0x06U, 0xEDU, 0xA6U, 0xE1U, 0xF1U, 0xACU, 0x2BU, 0x4DU, 0xF9U, 0xEFU, 0x8DU, 0xE9U, 0x21U, 0xC0U, 0x7CU, 0x2FU, 0x4CU, 0x85U, 0x98U, 0xD7U, 0xA3U, 0xD6U, 0xECU, 0x4BU, 0x36U, 0x8CU, 0xB8U, 0x5CU, 0xE6U, 0x1AU, 0x74U, 0x33U, 0x82U, 0x21U, 0x11U, 0x8AU, 0x30U, 0x3EU, 0x82U, 0x1CU, 0x0FU, 0x27U, 0x7BU, 0x59U, 0x1AU, 0xF6U, 0x79U, 0x5FU, 0x50U, 0xC4U, 0x02U, 0x26U, 0x12U, 0x7AU, 0x2EU, 0xFAU, 0xCCU, 0xE4U, 0x66U, 0x2FU, 0xD7U, 0x07U, 0x6CU, 0x10U, 0x9EU, 0xB5U, 0x9BU, 0x18U, 0x00U, 0x5EU, 0x71U, 0x65U, 0xF6U, 0x29U, 0x4AU, 0x69U, 0x76U, 0x43U, 0x6EU, 0xE3U, 0x97U, 0x77U, 0x4EU | 0xBDU, 0xF5U, 0xE6U, 0x4CU, 0x28U, 0xC9U, 0x69U, 0x47U, 0xD2U, 0x1CU, 0x8CU, 0x7EU, 0x85U, 0x0DU, 0xA8U, 0x07U, 0x0CU, 0x90U, 0x3BU, 0x5EU, 0xAEU, 0xD0U, 0x95U, 0xF6U, 0xB8U, 0xFFU, 0x48U, 0x7DU, 0x6DU, 0x9CU, 0xE5U, 0xE9U |

8.4.2

Mac

#### 8.4.2.1 HaeHmacSha1

| Key                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Plain Text                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Mac (20 byte)                                                                                                                                       |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| 0xCEU, 0xB9U, 0xAEU, 0xDFU, 0x8DU,<br>0x6EU, 0xFCU, 0xF0U, 0xAEU, 0x52U,<br>0xBEU, 0xA0U, 0xFAU, 0x99U, 0xA9U,<br>0xE2U, 0x6AU, 0xE8U, 0x1BU, 0xACU,<br>0xEAU, 0x0CU, 0xFFU, 0x4DU, 0x5EU,<br>0xECU, 0xF2U, 0x01U, 0xE3U, 0xBCU,<br>0xA3U, 0xC3U, 0x57U, 0x74U, 0x80U,<br>0x62U, 0x1BU, 0x81U, 0x8FU, 0xD7U,<br>0x17U, 0xBAU, 0x99U, 0xD6U, 0xFFU,<br>0x95U, 0x8EU, 0xA3U, 0xD5U, 0x9BU,<br>0x25U, 0x27U, 0xB0U, 0x19U, 0xC3U,<br>0x43U, 0xBBU, 0x19U, 0x9EU, 0x64U,<br>0x80U, 0x90U, 0x22U, 0x58U, 0x67U,<br>0xD9U, 0x94U, 0x60U, 0x79U, 0x62U,<br>0xF5U, 0x86U, 0x6AU, 0xA6U, 0x29U,<br>0x30U, 0xD7U, 0x5BU, 0x58U, 0xF6U | 0x99U, 0x95U, 0x8AU, 0xA4U, 0x59U,<br>0x60U, 0x46U, 0x57U, 0xC7U, 0xBFU,<br>0x6EU, 0x4CU, 0xDFU, 0xCCU, 0x87U,<br>0x85U, 0xF0U, 0ABU, 0xF0U, 0x6FU,<br>0xFEU, 0x63U, 0x6BU, 0x5BU, 0x64U,<br>0xECU, 0xD9U, 0x31U, 0xBDU, 0x8AU,<br>0x45U, 0x63U, 0x05U, 0x59U, 0x24U,<br>0x21U, 0xFCU, 0x28U, 0xDBU, 0xCCU,<br>0xCBU, 0x8AU, 0x82U, 0xACU, 0xEAU,<br>0x2BU, 0xE8U, 0xE5U, 0x41U, 0x61U,<br>0xD7U, 0xA7U, 0x8EU, 0x03U, 0x99U,<br>0xA6U, 0x06U, 0x7EU, 0xBAU, 0xCAU,<br>0x3FU, 0x25U, 0x10U, 0x27U, 0x4DU,<br>0xC9U, 0xF9U, 0x2FU, 0x2CU, 0x8AU,<br>0xE4U, 0x26U, 0x5EU, 0xECU, 0x13U,<br>0xD7U, 0xD4U, 0x2EU, 0x9FU, 0x86U,<br>0x12U, 0xD7U, 0xBCU, 0x25U, 0x8FU,<br>0x91U, 0x3EU, 0xCBU, 0x5AU, 0x3AU,<br>0x5CU, 0x61U, 0x03U, 0x39U, 0xB4U,<br>0x9FU, 0xB9U, 0x0EU, 0x90U, 0x37U,<br>0xB0U, 0x2DU, 0x68U, 0x4FU, 0xC6U,<br>0x0DU, 0xA8U, 0x35U, 0x65U, 0x7CU,<br>0xB2U, 0x4EU, 0xABU, 0x35U, 0x27U,<br>0x50U, 0xC8U, 0xB4U, 0x63U, 0xB1U,<br>0xA8U, 0x49U, 0x46U, 0x60U, 0xD3U,<br>0x6CU, 0x3AU, 0xB2U | 0x4AU, 0xC4U, 0x1AU, 0xB8U, 0x9FU,<br>0x62U, 0x5CU, 0x60U, 0x12U, 0x5EU,<br>0xD6U, 0x5FU, 0xFAU, 0x95U, 0x8CU,<br>0x6BU, 0x49U, 0x0EU, 0xA6U, 0x70U |

#### 8.4.2.2 HaeHmacSha256

| Key (74 byte)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Plain Text (128 byte)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Mac (32 byte)                                                                                                                                                                                                                                    |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0x4BU, 0x7AU, 0xB1U, 0x33U, 0xEFU,<br>0xE9U, 0x9EU, 0x02U, 0xFCU, 0x89U,<br>0xA2U, 0x84U, 0x09U, 0xEEU, 0x18U,<br>0x7DU, 0x57U, 0x9EU, 0x77U, 0x4FU,<br>0x4CU, 0xBAU, 0x6FU, 0xC2U, 0x23U,<br>0xE1U, 0x35U, 0x04U, 0xE3U, 0x51U,<br>0x1BU, 0xEFU, 0x8DU, 0x4FU, 0x63U,<br>0x8BU, 0x9AU, 0xCAU, 0x55U, 0xD4U,<br>0xA4U, 0x3BU, 0x8FU, 0xBDU, 0x64U,<br>0xCFU, 0x9DU, 0x74U, 0xDCU, 0xC8U,<br>0xC9U, 0xE8U, 0xD5U, 0x20U, 0x34U,<br>0x89U, 0x8CU, 0x70U, 0x26U, 0x4EU,<br>0xA9U, 0x11U, 0xA3U, 0xFDU, 0x70U,<br>0x81U, 0x3FU, 0xA7U, 0x3BU, 0x08U,<br>0x33U, 0x71U, 0x28U, 0x9BU | 0x13U, 0x8EU, 0xFCU, 0x83U, 0x2CU,<br>0x64U, 0x51U, 0x3DU, 0x11U, 0xB9U,<br>0x87U, 0x3CU, 0x6FU, 0xD4U, 0xD8U,<br>0xA6U, 0x5DU, 0xBFU, 0x36U, 0x70U,<br>0x92U, 0xA8U, 0x26U, 0xDDU, 0xD5U,<br>0x87U, 0xD1U, 0x41U, 0xB4U, 0x01U,<br>0x58U, 0x0BU, 0x79U, 0x8CU, 0x69U,<br>0x02U, 0x5AU, 0xD5U, 0x10U, 0xCFU,<br>0xF0U, 0x5FU, 0xCFU, 0xBCU, 0xEBU,<br>0x6CU, 0xF0U, 0xBBU, 0x03U, 0x20U,<br>0x1AU, 0xAAU, 0x32U, 0xE4U, 0x23U,<br>0xD5U, 0x20U, 0x09U, 0x25U, 0xBDU,<br>0xDFU, 0xADU, 0xD4U, 0x18U, 0xD8U,<br>0xE3U, 0x0EU, 0x18U, 0x05U, 0x0EU,<br>0xB4U, 0xF0U, 0x61U, 0x8EU, 0xB9U,<br>0x95U, 0x9DU, 0x9FU, 0x78U, 0xC1U,<br>0x15U, 0x7DU, 0x4BU, 0x3EU, 0x02U,<br>0xCDU, 0x59U, 0x61U, 0xF1U, 0x38U,<br>0xAFU, 0xD5U, 0x74U, 0x59U, 0x93U,<br>0x99U, 0x17U, 0xD9U, 0x14U, 0x4CU,<br>0x95U, 0xD8U, 0xE6U, 0xA9U, 0x4CU,<br>0x8FU, 0x6DU, 0x4EU, 0xEFU, 0x34U,<br>0x18U, 0xC1U, 0x7BU, 0x1EU, 0xF0U,<br>0xB4U, 0x6CU, 0x2AU, 0x71U, 0x88U,<br>0x30U, 0x5DU, 0x98U, 0x11U, 0xDCU,<br>0xCBU, 0x3DU, 0x99U | 0x4FU, 0x1EU, 0xE7U, 0xCBU, 0x36U,<br>0xC5U, 0x88U, 0x03U, 0xA8U, 0x72U,<br>0x1DU, 0x4AU, 0xC8U, 0xC4U, 0xCFU,<br>0x8CU, 0xAEU, 0x5DU, 0x88U, 0x32U,<br>0x39U, 0x2EU, 0xEDU, 0x2AU, 0x96U,<br>0xDCU, 0x59U, 0x69U, 0x42U, 0x52U,<br>0x80U, 0x1BU |

#### 8.4.2.3 HaeCmacAes

| Key (16 byte)                                                                                                           | Plain Text (40 byte)                                                                                                                                                                                                                                                                                        | Mac (16 byte)                                                                                                           |
|-------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| 0x2BU, 0x7EU, 0x15U, 0x16U, 0x28U,<br>0xAEU, 0xD2U, 0xA6U, 0xABU, 0xF7U,<br>0x15U, 0x88U, 0x09U, 0xCFU, 0x4FU,<br>0x3CU | 0x6BU, 0xC1U, 0xBEU, 0xE2U, 0x2EU,<br>0x40U, 0x9FU, 0x96U, 0xE9U, 0x3DU,<br>0x7EU, 0x11U, 0x73U, 0x93U, 0x17U,<br>0x2AU, 0xAEU, 0x2DU, 0x8AU, 0x57U,<br>0x1EU, 0x03U, 0xACU, 0x9CU, 0x9EU,<br>0xB7U, 0x6FU, 0xACU, 0x45U, 0xAFU,<br>0x8EU, 0x51U, 0x30U, 0xC8U, 0x1CU,<br>0x46U, 0xA3U, 0x5CU, 0xE4U, 0x11U | 0xDFU, 0xA6U, 0x67U, 0x47U, 0xDEU,<br>0x9AU, 0xE6U, 0x30U, 0x30U, 0xCAU,<br>0x32U, 0x61U, 0x14U, 0x97U, 0xC8U,<br>0x27U |

#### 8.4.2.4 AuthHsmHmacSha1

| Key                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Plain Text                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Mac (20 byte)                                                                                                                                       |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| 0xCEU, 0xB9U, 0xAEU, 0xDFU, 0x8DU,<br>0x6EU, 0xFCU, 0xF0U, 0xAEU, 0x52U,<br>0xBEU, 0xA0U, 0xFAU, 0x99U, 0xA9U,<br>0xE2U, 0x6AU, 0xE8U, 0x1BU, 0xACU,<br>0xEAU, 0x0CU, 0xFFU, 0x4DU, 0x5EU,<br>0xECU, 0xF2U, 0x01U, 0xE3U, 0xBCU,<br>0xA3U, 0xC3U, 0x57U, 0x74U, 0x80U,<br>0x62U, 0x1BU, 0x81U, 0x8FU, 0xD7U,<br>0x17U, 0xBAU, 0x99U, 0xD6U, 0xFFU,<br>0x95U, 0x8EU, 0xA3U, 0xD5U, 0x9BU,<br>0x25U, 0x27U, 0xB0U, 0x19U, 0xC3U,<br>0x43U, 0xBBU, 0x19U, 0x9EU, 0x64U,<br>0x80U, 0x90U, 0x22U, 0x58U, 0x67U,<br>0xD9U, 0x94U, 0x60U, 0x79U, 0x62U,<br>0xF5U, 0x86U, 0x6AU, 0xA6U, 0x29U,<br>0x30U, 0xD7U, 0x5BU, 0x58U, 0xF6U | 0x99U, 0x95U, 0x8AU, 0xA4U, 0x59U,<br>0x60U, 0x46U, 0x57U, 0xC7U, 0xBFU,<br>0x6EU, 0x4CU, 0xDFU, 0xCCU, 0x87U,<br>0x85U, 0xF0U, 0xABU, 0xF0U, 0x6FU,<br>0xFEU, 0x63U, 0x6BU, 0x5BU, 0x64U,<br>0xECU, 0xD9U, 0x31U, 0xBDU, 0x8AU,<br>0x45U, 0x63U, 0x05U, 0x59U, 0x24U,<br>0x21U, 0xFCU, 0x28U, 0xDBU, 0xCCU,<br>0xCBU, 0x8AU, 0x82U, 0xACU, 0xEAU,<br>0x2BU, 0xE8U, 0xE5U, 0x41U, 0x61U,<br>0xD7U, 0xA7U, 0x8EU, 0x03U, 0x99U,<br>0xA6U, 0x06U, 0x7EU, 0xBAU, 0xCAU,<br>0x3FU, 0x25U, 0x10U, 0x27U, 0x4DU,<br>0xC9U, 0xF9U, 0x2FU, 0x2CU, 0x8AU,<br>0xE4U, 0x26U, 0x5EU, 0xECU, 0x13U,<br>0xD7U, 0xD4U, 0x2EU, 0x9FU, 0x86U,<br>0x12U, 0xD7U, 0xBCU, 0x25U, 0x8FU,<br>0x91U, 0x3EU, 0xCBU, 0x5AU, 0x3AU,<br>0x5CU, 0x61U, 0x03U, 0x39U, 0xB4U,<br>0x9FU, 0xB9U, 0x0EU, 0x90U, 0x37U,<br>0xB0U, 0x2DU, 0x68U, 0x4FU, 0xC6U,<br>0x0DU, 0xA8U, 0x35U, 0x65U, 0x7CU,<br>0xB2U, 0x4EU, 0xABU, 0x35U, 0x27U,<br>0x50U, 0xC8U, 0xB4U, 0x63U, 0xB1U,<br>0xA8U, 0x49U, 0x46U, 0x60U, 0xD3U,<br>0x6CU, 0x3AU, 0xB2U | 0x4AU, 0xC4U, 0x1AU, 0xB8U, 0x9FU,<br>0x62U, 0x5CU, 0x60U, 0x12U, 0x5EU,<br>0xD6U, 0x5FU, 0xFAU, 0x95U, 0x8CU,<br>0x6BU, 0x49U, 0x0EU, 0xA6U, 0x70U |

#### 8.4.2.5 AuthHsmHmacSha256

| Key (74 byte)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Plain Text (128 byte)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Mac (32 byte)                                                                                                                                                                                                                                    |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0x4BU, 0x7AU, 0xB1U, 0x33U, 0xEFU,<br>0xE9U, 0x9EU, 0x02U, 0xFCU, 0x89U,<br>0xA2U, 0x84U, 0x09U, 0xEEU, 0x18U,<br>0x7DU, 0x57U, 0x9EU, 0x77U, 0x4FU,<br>0x4CU, 0xBAU, 0x6FU, 0xC2U, 0x23U,<br>0xE1U, 0x35U, 0x04U, 0xE3U, 0x51U,<br>0x1BU, 0xEFU, 0x8DU, 0x4FU, 0x63U,<br>0x8BU, 0x9AU, 0xCAU, 0x55U, 0xD4U,<br>0xA4U, 0x3BU, 0x8FU, 0xBDU, 0x64U,<br>0xCFU, 0x9DU, 0x74U, 0xDCU, 0xC8U,<br>0xC9U, 0xE8U, 0xD5U, 0x20U, 0x34U,<br>0x89U, 0x8CU, 0x70U, 0x26U, 0x4EU,<br>0xA9U, 0x11U, 0xA3U, 0xFDU, 0x70U,<br>0x81U, 0x3FU, 0xA7U, 0x3BU, 0x08U,<br>0x33U, 0x71U, 0x28U, 0x9BU | 0x13U, 0x8EU, 0xFCU, 0x83U, 0x2CU,<br>0x64U, 0x51U, 0x3DU, 0x11U, 0xB9U,<br>0x87U, 0x3CU, 0x6FU, 0xD4U, 0xD8U,<br>0xA6U, 0x5DU, 0xBFU, 0x36U, 0x70U,<br>0x92U, 0xA8U, 0x26U, 0xDDU, 0xD5U,<br>0x87U, 0xD1U, 0x41U, 0xB4U, 0x01U,<br>0x58U, 0x0BU, 0x79U, 0x8CU, 0x69U,<br>0x02U, 0x5AU, 0xD5U, 0x10U, 0xCFU,<br>0xF0U, 0x5FU, 0xCFU, 0xBCU, 0xEBU,<br>0x6CU, 0xF0U, 0xBBU, 0x03U, 0x20U,<br>0x1AU, 0xAAU, 0x32U, 0xE4U, 0x23U,<br>0xD5U, 0x20U, 0x09U, 0x25U, 0xBDU,<br>0xDFU, 0xADU, 0xD4U, 0x18U, 0xD8U,<br>0xE3U, 0x0EU, 0x18U, 0x05U, 0x0EU,<br>0xB4U, 0xF0U, 0x61U, 0x8EU, 0xB9U,<br>0x95U, 0x9DU, 0x9FU, 0x78U, 0xC1U,<br>0x15U, 0x7DU, 0x4BU, 0x3EU, 0x02U, | 0x4FU, 0x1EU, 0xE7U, 0xCBU, 0x36U,<br>0xC5U, 0x88U, 0x03U, 0xA8U, 0x72U,<br>0x1DU, 0x4AU, 0xC8U, 0xC4U, 0xCFU,<br>0x8CU, 0xAEU, 0x5DU, 0x88U, 0x32U,<br>0x39U, 0x2EU, 0xEDU, 0x2AU, 0x96U,<br>0xDCU, 0x59U, 0x69U, 0x42U, 0x52U,<br>0x80U, 0x1BU |

|  |                                                                                                                                                                                                                                                                                                                                     |  |
|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
|  | 0xCDU, 0x59U, 0x61U, 0xF1U, 0x38U,<br>0xAFU, 0xD5U, 0x74U, 0x59U, 0x93U,<br>0x99U, 0x17U, 0xD9U, 0x14U, 0x4CU,<br>0x95U, 0xD8U, 0xE6U, 0xA9U, 0x4CU,<br>0x8FU, 0x6DU, 0x4EU, 0xEFU, 0x34U,<br>0x18U, 0xC1U, 0x7BU, 0x1EU, 0xF0U,<br>0xB4U, 0x6CU, 0x2AU, 0x71U, 0x88U,<br>0x30U, 0x5DU, 0x98U, 0x11U, 0xDCU,<br>0xCBU, 0x3DU, 0x99U |  |
|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|

#### 8.4.2.6 AuthHsmCmacAes

Raw Key 방식 사용시

| Key (16 byte)                                                                                                           | Plain Text (40 byte)                                                                                                                                                                                                                                                                                        | Mac (16 byte)                                                                                                           |
|-------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| 0x2BU, 0x7EU, 0x15U, 0x16U, 0x28U,<br>0xAEU, 0xD2U, 0xA6U, 0xABU, 0xF7U,<br>0x15U, 0x88U, 0x09U, 0xCFU, 0x4FU,<br>0x3CU | 0x6BU, 0xC1U, 0xBEU, 0xE2U, 0x2EU,<br>0x40U, 0x9FU, 0x96U, 0xE9U, 0x3DU,<br>0x7EU, 0x11U, 0x73U, 0x93U, 0x17U,<br>0x2AU, 0xAEU, 0x2DU, 0x8AU, 0x57U,<br>0x1EU, 0x03U, 0xACU, 0x9CU, 0x9EU,<br>0xB7U, 0x6FU, 0xACU, 0x45U, 0xAFU,<br>0x8EU, 0x51U, 0x30U, 0xC8U, 0x1CU,<br>0x46U, 0xA3U, 0x5CU, 0xE4U, 0x11U | 0xDFU, 0xA6U, 0x67U, 0x47U, 0xDEU,<br>0x9AU, 0xE6U, 0x30U, 0x30U, 0xCAU,<br>0x32U, 0x61U, 0x14U, 0x97U, 0xC8U,<br>0x27U |

#### 8.4.2.7 AutIcusCmacAes

Raw Key 방식 사용시

| Key (16 byte)                                                                                                           | Plain Text (40 byte)                                                                                                                                                                                                                                                                                        | Mac (16 byte)                                                                                                           |
|-------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| 0x2BU, 0x7EU, 0x15U, 0x16U, 0x28U,<br>0xAEU, 0xD2U, 0xA6U, 0xABU, 0xF7U,<br>0x15U, 0x88U, 0x09U, 0xCFU, 0x4FU,<br>0x3CU | 0x6BU, 0xC1U, 0xBEU, 0xE2U, 0x2EU,<br>0x40U, 0x9FU, 0x96U, 0xE9U, 0x3DU,<br>0x7EU, 0x11U, 0x73U, 0x93U, 0x17U,<br>0x2AU, 0xAEU, 0x2DU, 0x8AU, 0x57U,<br>0x1EU, 0x03U, 0xACU, 0x9CU, 0x9EU,<br>0xB7U, 0x6FU, 0xACU, 0x45U, 0xAFU,<br>0x8EU, 0x51U, 0x30U, 0xC8U, 0x1CU,<br>0x46U, 0xA3U, 0x5CU, 0xE4U, 0x11U | 0xDFU, 0xA6U, 0x67U, 0x47U, 0xDEU,<br>0x9AU, 0xE6U, 0x30U, 0x30U, 0xCAU,<br>0x32U, 0x61U, 0x14U, 0x97U, 0xC8U,<br>0x27U |

#### 8.4.2.8 HaeHsmHmacSha256

| Key | Plain Text                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Mac (32 byte)                                                                                                                                                                                     |
|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 401 | 0x13, 0x8e, 0xfc, 0x83, 0x2c, 0x64, 0x51, 0x3d, 0x11, 0xb9, 0x87, 0x3c, 0x6f, 0xd4, 0xd8, 0xa6,<br>0x5d, 0xbf, 0x36, 0x70, 0x92, 0xa8, 0x26, 0xdd, 0xd5, 0x87, 0xd1, 0x41, 0xb4, 0x01, 0x58, 0x0b,<br>0x79, 0x8c, 0x69, 0x02, 0x5a, 0xd5, 0x10, 0xcf, 0xf0, 0x5f, 0xcf, 0xbc, 0xeb, 0x6c, 0xf0, 0xbb,<br>0x03, 0x20, 0x1a, 0xaa, 0x32, 0xe4, 0x23, 0xd5, 0x20, 0x09, 0x25, 0xbd, 0xdf, 0xad, 0xd4, 0x18,<br>0xd8, 0xe3, 0x0e, 0x18, 0x05, 0x0e, 0xb4, 0xf0, 0x61, 0x8e, 0xb9, 0x95, 0x9d, 0x9f, 0x78, 0xc1,<br>0x15, 0x7d, 0x4b, 0x3e, 0x02, 0xcd, 0x59, 0x61, 0xf1, 0x38, 0xaf, 0xd5, 0x74, 0x59, 0x93, 0x99,<br>0x17, 0xd9, 0x14, 0x4c, 0x95, 0xd8, 0xe6, 0xa9, 0x4c, 0x8f, 0x6d, 0x4e, 0xef, 0x34, 0x18, 0xc1,<br>0x7b, 0x1e, 0xf0, 0xb4, 0x6c, 0x2a, 0x71, 0x88, 0x30, 0x5d, 0x98, 0x11, 0xdc, 0xcb, 0x3d, 0x99 | 0x66, 0x50, 0x8a, 0xcd, 0x70, 0x98, 0x3c, 0x13, 0x51, 0xe8, 0x14, 0xc6, 0x23, 0x84, 0x2e, 0x4e,<br>0x5d, 0xf6, 0x60, 0x56, 0xbf, 0xcf, 0x05, 0xdb, 0x29, 0x65, 0xda, 0x42, 0x3b, 0x2c, 0x53, 0x07 |

#### 8.4.2.9 HaeHsmCmacAes

| Key | Plain Text                                                                                      | Mac (16 byte)                                                                     |
|-----|-------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| 404 | 0x6b, 0xc1, 0xbe, 0xe2, 0x2e, 0x40, 0x9f, 0x96, 0xe9, 0x3d, 0x7e, 0x11, 0x73, 0x93, 0x17, 0x2a, | 0x51, 0xf0, 0xbe, 0xbf, 0x7e, 0x3b, 0x9d, 0x92, 0xfc, 0x49, 0x74, 0x17, 0x79, 0x3 |

|                                                                                                                                                                                                                                                                                                |               |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| 0xae, 0x2d, 0x8a, 0x57, 0x1e, 0x03, 0xac, 0x9c, 0x9e, 0xb7, 0x6f, 0xac, 0x45, 0xaf, 0x8e, 0x51, 0x30, 0xc8, 0x1c, 0x46, 0xa3, 0x5c, 0xe4, 0x11, 0xe5, 0xfb, 0xc1, 0x19, 0x1a, 0x0a, 0x52, 0xef, 0xf6, 0x9f, 0x24, 0x45, 0xdf, 0x4f, 0x9b, 0x17, 0xad, 0x2b, 0x41, 0x7b, 0xe6, 0x6c, 0x37, 0x10 | 6, 0x3c, 0xfe |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|

#### 8.4.2.10 AutCsecCmacAes

Raw Key 방식 사용시

| Key (16 byte)                                                                                                  | Plain Text (40 byte)                                                                                                                                                                                                                                                                   | Mac (16 byte)                                                                                                  |
|----------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| 0x2BU, 0x7EU, 0x15U, 0x16U, 0x28U, 0xAEU, 0xD2U, 0xA6U, 0xABU, 0xF7U, 0x15U, 0x88U, 0x09U, 0xCFU, 0x4FU, 0x3CU | 0x6BU, 0xC1U, 0xBEU, 0xE2U, 0x2EU, 0x40U, 0x9FU, 0x96U, 0xE9U, 0x3DU, 0x7EU, 0x11U, 0x73U, 0x93U, 0x17U, 0x2AU, 0xAEU, 0x2DU, 0x8AU, 0x57U, 0x1EU, 0x03U, 0xACU, 0x9CU, 0x9EU, 0xB7U, 0x6FU, 0xACU, 0x45U, 0xAFU, 0x8EU, 0x51U, 0x30U, 0xC8U, 0x1CU, 0x46U, 0xA3U, 0x5CU, 0xE4U, 0x11U | 0xDFU, 0xA6U, 0x67U, 0x47U, 0xDEU, 0x9AU, 0xE6U, 0x30U, 0x30U, 0xCAU, 0x32U, 0x61U, 0x14U, 0x97U, 0xC8U, 0x27U |

### Symmetric Block

#### 8.4.3.1 HaeAesEcb

| Key (16 byte)                                                                                                  | Plain Text (160 byte)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Cipher Text (160 byte)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|----------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0xEBU, 0xEAU, 0x9CU, 0x6AU, 0x82U, 0x21U, 0x3AU, 0x00U, 0xACU, 0x1DU, 0x22U, 0xFAU, 0xEAU, 0x22U, 0x11U, 0x6FU | 0x45U, 0x1FU, 0x45U, 0x66U, 0x3BU, 0x44U, 0xFDU, 0x00U, 0x5FU, 0x3CU, 0x28U, 0x8AU, 0xE5U, 0x7BU, 0x38U, 0x38U, 0x83U, 0xF0U, 0x2DU, 0x9AU, 0xD3U, 0xDCU, 0x17U, 0x15U, 0xF9U, 0xE3U, 0xD6U, 0x94U, 0x85U, 0x64U, 0x25U, 0x7BU, 0x9BU, 0x06U, 0xD7U, 0xDDU, 0x51U, 0x93U, 0x5FU, 0xEEU, 0x58U, 0x0AU, 0x96U, 0xBBU, 0xDFU, 0xEFU, 0xB9U, 0x18U, 0xB4U, 0xE6U, 0xB1U, 0xDAU, 0xACU, 0x80U, 0x98U, 0x47U, 0x46U, 0x55U, 0x78U, 0xCBU, 0x8BU, 0x53U, 0x56U, 0xEDU, 0x38U, 0x55U, 0x6FU, 0x80U, 0x1FU, 0xF7U, 0xC1U, 0x1EU, 0xCBU, 0xA9U, 0xCDU, 0xD2U, 0x63U, 0x03U, 0x9CU, 0x15U, 0xD0U, 0x59U, 0x00U, 0xFCU, 0x22U, 0x8EU, 0x1CU, 0xAFU, 0x30U, 0x2DU, 0x26U, 0x1DU, 0x7FU, 0xB5U, 0x6CU, 0xEEU, 0x66U, 0x35U, 0x95U, 0xB9U, 0x6FU, 0x19U, 0x2AU, 0x78U, 0xFFU, 0x44U, 0x55U, 0x39U, 0x3AU, 0x5FU, 0xE8U, 0x16U, 0x21U, 0x70U, 0xA0U, 0x66U, 0xFDU, 0xAEU, 0xACU, 0x35U, 0x01U, 0x94U, 0x69U, 0xF2U, 0x2BU, 0x34U, 0x70U, 0x68U, 0x6BU, 0xCEU, 0xD2U, 0xF0U, 0x07U, 0xA1U, 0xA2U, 0xE4U, 0x3EU, 0x01U, 0xB4U, 0x56U, 0x2CU, 0xAAU, 0xA5U, 0x02U, 0xEDU, 0x54U, 0x1BU, 0x82U, 0x05U, 0x87U, 0x4EU, 0xC1U, 0xFFU, 0xB1U, 0xC8U, 0xB2U, 0x55U, 0x76U, 0x69U, 0x42U | 0x01U, 0x04U, 0x30U, 0x53U, 0xF8U, 0x32U, 0xEFU, 0x9BU, 0x91U, 0x1EU, 0xD3U, 0x87U, 0xBAU, 0x57U, 0x74U, 0x51U, 0xE3U, 0x0DU, 0x51U, 0xD4U, 0xB6U, 0xB1U, 0x1FU, 0x31U, 0x9DU, 0x4CU, 0xD5U, 0x39U, 0xD0U, 0x67U, 0xB7U, 0xF4U, 0xF9U, 0xB4U, 0xF4U, 0x1FU, 0x7FU, 0x3DU, 0x4EU, 0x92U, 0x0CU, 0x57U, 0xCBU, 0xE2U, 0xB5U, 0xE1U, 0x88U, 0x5AU, 0xA6U, 0x62U, 0x03U, 0xAEU, 0x49U, 0x3EU, 0x93U, 0xA1U, 0xDFU, 0x63U, 0x79U, 0x3AU, 0x95U, 0x63U, 0xC1U, 0x76U, 0xBCU, 0x67U, 0x75U, 0xDDU, 0x09U, 0xCCU, 0x91U, 0x61U, 0xE2U, 0x78U, 0xA0U, 0x1BU, 0xEBU, 0x8FU, 0xD8U, 0xA1U, 0x92U, 0x00U, 0x32U, 0x6BU, 0xD9U, 0x5AU, 0xBCU, 0x5FU, 0x71U, 0x67U, 0x68U, 0xE3U, 0x4FU, 0x90U, 0xB5U, 0x05U, 0x23U, 0xD3U, 0x0FU, 0xDAU, 0xBBU, 0x10U, 0x3AU, 0x3BU, 0xC0U, 0x20U, 0xAFU, 0xBBU, 0xB0U, 0xCBU, 0x3BU, 0xD2U, 0xADU, 0x51U, 0x2AU, 0x6FU, 0xEAU, 0x79U, 0xF8U, 0xD6U, 0x4CU, 0xEFU, 0x34U, 0x74U, 0x58U, 0xDEU, 0xC4U, 0x8BU, 0xE8U, 0x94U, 0x51U, 0xCBU, 0x0BU, 0x80U, 0x7DU, 0x73U, 0x59U, 0x3FU, 0x27U, 0x3DU, 0x9FU, 0xC5U, 0x21U, 0xB7U, 0x89U, 0xA7U, 0x75U, 0x24U, 0x40U, 0x4FU, 0x43U, 0xE0U, 0x0FU, 0x20U, 0xB3U, 0xB7U, 0x7BU, 0x93U, 0x8BU, 0x1AU |

#### 8.4.3.2 AuthHsmAesEcb

Raw Key 방식 사용시

| Key (16 byte)                                                                                                           | Plain Text (160 byte)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Cipher Text (160 byte)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0xEBU, 0xEAU, 0x9CU, 0x6AU, 0x82U,<br>0x21U, 0x3AU, 0x00U, 0xACU, 0x1DU,<br>0x22U, 0xFAU, 0xEAU, 0x22U, 0x11U,<br>0x6FU | 0x45U, 0x1FU, 0x45U, 0x66U, 0x3BU,<br>0x44U, 0xFDU, 0x00U, 0x5FU, 0x3CU,<br>0x28U, 0x8AU, 0xE5U, 0x7BU, 0x38U,<br>0x38U, 0x83U, 0xF0U, 0x2DU, 0x9AU,<br>0xD3U, 0xDCU, 0x17U, 0x15U, 0xF9U,<br>0xE3U, 0xD6U, 0x94U, 0x85U, 0x64U,<br>0x25U, 0x7BU, 0x9BU, 0x06U, 0xD7U,<br>0xDDU, 0x51U, 0x93U, 0x5FU, 0xEEU,<br>0x58U, 0x0AU, 0x96U, 0xBBU, 0xDFU,<br>0xEFU, 0xB9U, 0x18U, 0xB4U, 0xE6U,<br>0xB1U, 0xDAU, 0xACU, 0x80U, 0x98U,<br>0x47U, 0x46U, 0x55U, 0x78U, 0xCBU,<br>0x8BU, 0x53U, 0x56U, 0xEDU, 0x38U,<br>0x55U, 0x6FU, 0x80U, 0x1FU, 0xF7U, 0<br>xC1U, 0x1EU, 0xCBU, 0xA9U, 0xCDU, 0<br>xD2U, 0x63U, 0x03U, 0x9CU, 0x15U, 0<br>xD0U, 0x59U, 0x00U, 0xFCU, 0x22U, 0<br>x8EU, 0x1CU, 0xAFU, 0x30U, 0x2DU, 0<br>x26U, 0x1DU, 0x7FU, 0xB5U, 0x6CU, 0<br>xEEU, 0x66U, 0x35U, 0x95U, 0xB9U, 0<br>x6FU, 0x19U, 0x2AU, 0x78U, 0xFFU, 0<br>x44U, 0x55U, 0x39U, 0x3AU, 0x5FU, 0<br>xE8U, 0x16U, 0x21U, 0x70U, 0xA0U, 0<br>x66U, 0xFDU, 0xAEU, 0xACU, 0x35U, 0<br>x01U, 0x94U, 0x69U, 0xF2U, 0x2BU, 0<br>x34U, 0x70U, 0x68U, 0x6BU, 0xCEU, 0<br>xD2U, 0xF0U, 0x07U, 0xA1U, 0xA2U, 0<br>xE4U, 0x3EU, 0x01U, 0xB4U, 0x56U, 0<br>x2CU, 0xAAU, 0xA5U, 0x02U, 0xEDU,<br>0x54U, 0x1BU, 0x82U, 0x05U, 0x87U,<br>0x4EU, 0xC1U, 0xFFU, 0xB1U, 0xC8U,<br>0xB2U, 0x55U, 0x76U, 0x69U, 0x42U | 0x01U, 0x04U, 0x30U, 0x53U, 0xF8U,<br>0x32U, 0xEFU, 0x9BU, 0x91U, 0x1EU,<br>0xD3U, 0x87U, 0xBAU, 0x57U, 0x74U,<br>0x51U, 0xE3U, 0x0DU, 0x51U, 0xD4U,<br>0xB6U, 0xB1U, 0x1FU, 0x31U, 0x9DU,<br>0x4CU, 0xD5U, 0x39U, 0xD0U, 0x67U,<br>0xB7U, 0xF4U, 0xF9U, 0xB4U, 0xF4U,<br>0x1FU, 0x7FU, 0x3DU, 0x4EU, 0x92U,<br>0x0CU, 0x57U, 0xCBU, 0xE2U, 0xB5U,<br>0xE1U, 0x88U, 0x5AU, 0xA6U, 0x62U,<br>0x03U, 0xAEU, 0x49U, 0x3EU, 0x93U,<br>0xA1U, 0xDFU, 0x63U, 0x79U, 0x3AU,<br>0x95U, 0x63U, 0xC1U, 0x76U, 0xBCU,<br>0x67U, 0x75U, 0xDDU, 0x09U, 0xCCU,<br>0x91U, 0x61U, 0xE2U, 0x78U, 0xA0U,<br>0x1BU, 0xEBU, 0x8FU, 0xD8U, 0xA1U,<br>0x92U, 0x00U, 0x32U, 0x6BU, 0xD9U,<br>0x5AU, 0xBCU, 0x5FU, 0x71U, 0x67U,<br>0x68U, 0xE3U, 0x4FU, 0x90U, 0xB5U,<br>0x05U, 0x23U, 0xD3U, 0x0FU, 0xDAU,<br>0xBBU, 0x10U, 0x3AU, 0x3BU, 0xC0U,<br>0x20U, 0xAFU, 0xBBU, 0xB0U, 0xCBU,<br>0x3BU, 0xD2U, 0xADU, 0x51U, 0x2AU,<br>0x6FU, 0xEAU, 0x79U, 0xF8U, 0xD6U,<br>0x4CU, 0xEFU, 0x34U, 0x74U, 0x58U,<br>0xDEU, 0xC4U, 0x8BU, 0xE8U, 0x94U,<br>0x51U, 0xCBU, 0x0BU, 0x80U, 0x7DU,<br>0x73U, 0x59U, 0x3FU, 0x27U, 0x3DU,<br>0x9FU, 0xC5U, 0x21U, 0xB7U, 0x89U,<br>0xA7U, 0x75U, 0x24U, 0x40U, 0x4FU,<br>0x43U, 0xE0U, 0x0FU, 0x20U, 0xB3U,<br>0xB7U, 0x7BU, 0x93U, 0x8BU, 0x1AU |

#### 8.4.3.3 AutIcusAesEcb

Raw Key 방식 사용시

| Key (16 byte)                                                                                                           | Plain Text (160 byte)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Cipher Text (160 byte)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0xEBU, 0xEAU, 0x9CU, 0x6AU, 0x82U,<br>0x21U, 0x3AU, 0x00U, 0xACU, 0x1DU,<br>0x22U, 0xFAU, 0xEAU, 0x22U, 0x11U,<br>0x6FU | 0x45U, 0x1FU, 0x45U, 0x66U, 0x3BU,<br>0x44U, 0xFDU, 0x00U, 0x5FU, 0x3CU,<br>0x28U, 0x8AU, 0xE5U, 0x7BU, 0x38U,<br>0x38U, 0x83U, 0xF0U, 0x2DU, 0x9AU,<br>0xD3U, 0xDCU, 0x17U, 0x15U, 0xF9U,<br>0xE3U, 0xD6U, 0x94U, 0x85U, 0x64U,<br>0x25U, 0x7BU, 0x9BU, 0x06U, 0xD7U,<br>0xDDU, 0x51U, 0x93U, 0x5FU, 0xEEU,<br>0x58U, 0x0AU, 0x96U, 0xBBU, 0xDFU,<br>0xEFU, 0xB9U, 0x18U, 0xB4U, 0xE6U,<br>0xB1U, 0xDAU, 0xACU, 0x80U, 0x98U,<br>0x47U, 0x46U, 0x55U, 0x78U, 0xCBU,<br>0x8BU, 0x53U, 0x56U, 0xEDU, 0x38U,<br>0x55U, 0x6FU, 0x80U, 0x1FU, 0xF7U, 0<br>xC1U, 0x1EU, 0xCBU, 0xA9U, 0xCDU, 0<br>xD2U, 0x63U, 0x03U, 0x9CU, 0x15U, 0<br>xD0U, 0x59U, 0x00U, 0xFCU, 0x22U, 0<br>x8EU, 0x1CU, 0xAFU, 0x30U, 0x2DU, 0<br>x26U, 0x1DU, 0x7FU, 0xB5U, 0x6CU, 0<br>xEEU, 0x66U, 0x35U, 0x95U, 0xB9U, 0<br>x6FU, 0x19U, 0x2AU, 0x78U, 0xFFU, 0 | 0x01U, 0x04U, 0x30U, 0x53U, 0xF8U,<br>0x32U, 0xEFU, 0x9BU, 0x91U, 0x1EU,<br>0xD3U, 0x87U, 0xBAU, 0x57U, 0x74U,<br>0x51U, 0xE3U, 0x0DU, 0x51U, 0xD4U,<br>0xB6U, 0xB1U, 0x1FU, 0x31U, 0x9DU,<br>0x4CU, 0xD5U, 0x39U, 0xD0U, 0x67U,<br>0xB7U, 0xF4U, 0xF9U, 0xB4U, 0xF4U,<br>0x1FU, 0x7FU, 0x3DU, 0x4EU, 0x92U,<br>0x0CU, 0x57U, 0xCBU, 0xE2U, 0xB5U,<br>0xE1U, 0x88U, 0x5AU, 0xA6U, 0x62U,<br>0x03U, 0xAEU, 0x49U, 0x3EU, 0x93U,<br>0xA1U, 0xDFU, 0x63U, 0x79U, 0x3AU,<br>0x95U, 0x63U, 0xC1U, 0x76U, 0xBCU,<br>0x67U, 0x75U, 0xDDU, 0x09U, 0xCCU,<br>0x91U, 0x61U, 0xE2U, 0x78U, 0xA0U,<br>0x1BU, 0xEBU, 0x8FU, 0xD8U, 0xA1U,<br>0x92U, 0x00U, 0x32U, 0x6BU, 0xD9U,<br>0x5AU, 0xBCU, 0x5FU, 0x71U, 0x67U,<br>0x68U, 0xE3U, 0x4FU, 0x90U, 0xB5U,<br>0x05U, 0x23U, 0xD3U, 0x0FU, 0xDAU,<br>0xBBU, 0x10U, 0x3AU, 0x3BU, 0xC0U, |

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                               |
|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | x44U, 0x55U, 0x39U, 0x3AU, 0x5FU, 0<br>xE8U, 0x16U, 0x21U, 0x70U, 0xA0U, 0<br>x66U, 0xFDU, 0xAEU, 0xACU, 0x35U, 0<br>x01U, 0x94U, 0x69U, 0xF2U, 0x2BU, 0<br>x34U, 0x70U, 0x68U, 0x6BU, 0xCEU, 0<br>xD2U, 0xF0U, 0x07U, 0xA1U, 0xA2U, 0<br>xE4U, 0x3EU, 0x01U, 0xB4U, 0x56U, 0<br>x2CU, 0xAAU, 0xA5U, 0x02U, 0xEDU,<br>0x54U, 0x1BU, 0x82U, 0x05U, 0x87U,<br>0x4EU, 0xC1U, 0xFFU, 0xB1U, 0xC8U,<br>0xB2U, 0x55U, 0x76U, 0x69U, 0x42U | 0x20U, 0xAFU, 0xBBU, 0xB0U, 0xCBU,<br>0x3BU, 0xD2U, 0xADU, 0x51U, 0x2AU,<br>0x6FU, 0xEAU, 0x79U, 0xF8U, 0xD6U,<br>0x4CU, 0xEFU, 0x34U, 0x74U, 0x58U,<br>0xDEU, 0xC4U, 0x8BU, 0xE8U, 0x94U,<br>0x51U, 0xCBU, 0x0BU, 0x80U, 0x7DU,<br>0x73U, 0x59U, 0x3FU, 0x27U, 0x3DU,<br>0x9FU, 0xC5U, 0x21U, 0xB7U, 0x89U,<br>0xA7U, 0x75U, 0x24U, 0x40U, 0x4FU,<br>0x43U, 0xE0U, 0x0FU, 0x20U, 0xB3U,<br>0xB7U, 0x7BU, 0x93U, 0x8BU, 0x1AU |
|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

#### 8.4.3.4 AutCsecAesEcb

Raw Key 방식 사용시

| Key (16 byte)                                                                                                           | Plain Text (160 byte)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Cipher Text (160 byte)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0xEBU, 0xEAU, 0x9CU, 0x6AU, 0x82U,<br>0x21U, 0x3AU, 0x00U, 0xACU, 0x1DU,<br>0x22U, 0xFAU, 0xEAU, 0x22U, 0x11U,<br>0x6FU | 0x45U, 0x1FU, 0x45U, 0x66U, 0x3BU,<br>0x44U, 0xFDU, 0x00U, 0x5FU, 0x3CU,<br>0x28U, 0x8AU, 0xE5U, 0x7BU, 0x38U,<br>0x38U, 0x83U, 0xF0U, 0x2DU, 0x9AU,<br>0xD3U, 0xDCU, 0x17U, 0x15U, 0xF9U,<br>0xE3U, 0xD6U, 0x94U, 0x85U, 0x64U,<br>0x25U, 0x7BU, 0x9BU, 0x06U, 0xD7U,<br>0xDDU, 0x51U, 0x93U, 0x5FU, 0xEEU,<br>0x58U, 0x0AU, 0x96U, 0xBBU, 0xDFU,<br>0xEFU, 0xB9U, 0x18U, 0xB4U, 0xE6U,<br>0xB1U, 0xDAU, 0xACU, 0x80U, 0x98U,<br>0x47U, 0x46U, 0x55U, 0x78U, 0xCBU,<br>0x8BU, 0x53U, 0x56U, 0xEDU, 0x38U,<br>0x55U, 0x6FU, 0x80U, 0x1FU, 0xF7U, 0<br>xC1U, 0x1EU, 0xCBU, 0xA9U, 0xCDU, 0<br>xD2U, 0x63U, 0x03U, 0x9CU, 0x15U, 0<br>xD0U, 0x59U, 0x00U, 0xFCU, 0x22U, 0<br>x8EU, 0x1CU, 0xAFU, 0x30U, 0x2DU, 0<br>x26U, 0x1DU, 0x7FU, 0xB5U, 0x6CU, 0<br>xEEU, 0x66U, 0x35U, 0x95U, 0xB9U, 0<br>x6FU, 0x19U, 0x2AU, 0x78U, 0xFFU, 0<br>x44U, 0x55U, 0x39U, 0x3AU, 0x5FU, 0<br>xE8U, 0x16U, 0x21U, 0x70U, 0xA0U, 0<br>x66U, 0xFDU, 0xAEU, 0xACU, 0x35U, 0<br>x01U, 0x94U, 0x69U, 0xF2U, 0x2BU, 0<br>x34U, 0x70U, 0x68U, 0x6BU, 0xCEU, 0<br>xD2U, 0xF0U, 0x07U, 0xA1U, 0xA2U, 0<br>xE4U, 0x3EU, 0x01U, 0xB4U, 0x56U, 0<br>x2CU, 0xAAU, 0xA5U, 0x02U, 0xEDU,<br>0x54U, 0x1BU, 0x82U, 0x05U, 0x87U,<br>0x4EU, 0xC1U, 0xFFU, 0xB1U, 0xC8U,<br>0xB2U, 0x55U, 0x76U, 0x69U, 0x42U | 0x01U, 0x04U, 0x30U, 0x53U, 0xF8U,<br>0x32U, 0xEFU, 0x9BU, 0x91U, 0x1EU,<br>0xD3U, 0x87U, 0xBAU, 0x57U, 0x74U,<br>0x51U, 0xE3U, 0x0DU, 0x51U, 0xD4U,<br>0xB6U, 0xB1U, 0x1FU, 0x31U, 0x9DU,<br>0x4CU, 0xD5U, 0x39U, 0xD0U, 0x67U,<br>0xB7U, 0xF4U, 0xF9U, 0xB4U, 0xF4U,<br>0x1FU, 0x7FU, 0x3DU, 0x4EU, 0x92U,<br>0x0CU, 0x57U, 0xCBU, 0xE2U, 0xB5U,<br>0xE1U, 0x88U, 0x5AU, 0xA6U, 0x62U,<br>0x03U, 0xAEU, 0x49U, 0x3EU, 0x93U,<br>0xA1U, 0xDFU, 0x63U, 0x79U, 0x3AU,<br>0x95U, 0x63U, 0xC1U, 0x76U, 0xBCU,<br>0x67U, 0x75U, 0xDDU, 0x09U, 0xCCU,<br>0x91U, 0x61U, 0xE2U, 0x78U, 0xA0U,<br>0x1BU, 0xEBU, 0x8FU, 0xD8U, 0xA1U,<br>0x92U, 0x00U, 0x32U, 0x6BU, 0xD9U,<br>0x5AU, 0xBCU, 0x5FU, 0x71U, 0x67U,<br>0x68U, 0xE3U, 0x4FU, 0x90U, 0xB5U,<br>0x05U, 0x23U, 0xD3U, 0x0FU, 0xDAU,<br>0xBBU, 0x10U, 0x3AU, 0x3BU, 0xC0U,<br>0x20U, 0xAFU, 0xBBU, 0xB0U, 0xCBU,<br>0x3BU, 0xD2U, 0xADU, 0x51U, 0x2AU,<br>0x6FU, 0xEAU, 0x79U, 0xF8U, 0xD6U,<br>0x4CU, 0xEFU, 0x34U, 0x74U, 0x58U,<br>0xDEU, 0xC4U, 0x8BU, 0xE8U, 0x94U,<br>0x51U, 0xCBU, 0x0BU, 0x80U, 0x7DU,<br>0x73U, 0x59U, 0x3FU, 0x27U, 0x3DU,<br>0x9FU, 0xC5U, 0x21U, 0xB7U, 0x89U,<br>0xA7U, 0x75U, 0x24U, 0x40U, 0x4FU,<br>0x43U, 0xE0U, 0x0FU, 0x20U, 0xB3U,<br>0xB7U, 0x7BU, 0x93U, 0x8BU, 0x1AU |

8.4.4

#### 8.4.3.5 HaeHsmAesEcb

Autoever HSM (HSM\_Framework\_2.x\_UserManual\_Crypto\_Service\_KOR\_vx.xx.pdf) 의 7.Crypto Service API 테스트 케이스 참조

### Symmetric

#### 8.4.4.1 HaeAesCbc

| Key (16 byte) | Initial Vector | Plain Text (160 byte) | Cipher Text (160 byte) |
|---------------|----------------|-----------------------|------------------------|
|---------------|----------------|-----------------------|------------------------|

|                                                                                                                |                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | e)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|----------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0x2CU, 0x14U, 0x41U, 0x37U, 0x51U, 0xC3U, 0x1EU, 0x27U, 0x30U, 0x57U, 0x0BU, 0xA3U, 0x36U, 0x1CU, 0x78U, 0x6BU | 0x1DU, 0xBBU, 0xEBU, 0x2FU, 0x19U, 0xABU, 0xB4U, 0x48U, 0xAFU, 0x84U, 0x97U, 0x96U, 0x24U, 0x4AU, 0x19U, 0xD7U | 0x40U, 0xD9U, 0x30U, 0xF9U, 0xA0U, 0x53U, 0x34U, 0xD9U, 0x81U, 0x6FU, 0xE2U, 0x04U, 0x99U, 0x9CU, 0x3FU, 0x82U, 0xA0U, 0x3FU, 0x6AU, 0x04U, 0x57U, 0xA8U, 0xC4U, 0x75U, 0xC9U, 0x45U, 0x53U, 0xD1U, 0xD1U, 0x16U, 0x69U, 0x3AU, 0xDCU, 0x61U, 0x80U, 0x49U, 0xF0U, 0xA7U, 0x69U, 0xA2U, 0xEEU, 0xD6U, 0xA6U, 0xCBU, 0x14U, 0xC0U, 0x14U, 0x3EU, 0xC5U, 0xCCU, 0xCDU, 0xBCU, 0x8DU, 0xE0U, 0x69U, 0xE5U, 0x60U, 0xCFU, 0xD2U, 0x06U, 0x22U, 0x57U, 0x09U, 0x32U, 0x6DU, 0x4DU, 0xE7U, 0x94U, 0x8EU, 0x54U, 0xD6U, 0x03U, 0xD0U, 0x1BU, 0x12U, 0xD7U, 0xFEU, 0xD7U, 0x52U, 0xFBU, 0x23U, 0xF1U, 0xAAU, 0x44U, 0x94U, 0xFBU, 0xB0U, 0x01U, 0x30U, 0xE9U, 0xDEU, 0xD4U, 0xE7U, 0x7EU, 0x37U, 0xC0U, 0x79U, 0x04U, 0x2DU, 0x82U, 0x80U, 0x40U, 0xC3U, 0x25U, 0xB1U, 0xA5U, 0xEFU, 0xD1U, 0x5FU, 0xC8U, 0x42U, 0xE4U, 0x40U, 0x14U, 0xCAU, 0x43U, 0x74U, 0xBFU, 0x38U, 0xF3U, 0xC3U, 0xFCU, 0x3EU, 0xE3U, 0x27U, 0x73U, 0x3BU, 0x0CU, 0x8AU, 0xEEU, 0x1AU, 0xBCU, 0xD0U, 0x55U, 0x77U, 0x2FU, 0x18U, 0xDCU, 0x04U, 0x60U, 0x3FU, 0x7BU, 0x2CU, 0x1EU, 0xA6U, 0x9FU, 0xF6U, 0x62U, 0x36U, 0x1FU, 0x2BU, 0xE0U, 0xA1U, 0x71U, 0xBBU, 0xDCU, 0xE0U, 0x1EU, 0x5DU, 0x3FU | 0x6BU, 0xE8U, 0xA1U, 0x28U, 0x00U, 0x45U, 0x5AU, 0x32U, 0x05U, 0x38U, 0x85U, 0x3EU, 0x0CU, 0xBAU, 0x31U, 0xBDU, 0x2DU, 0x80U, 0xEAU, 0x0CU, 0x85U, 0x16U, 0x4AU, 0x4CU, 0x5CU, 0x26U, 0x1AU, 0xE4U, 0x85U, 0x41U, 0x7DU, 0x93U, 0xEFU, 0xFEU, 0x2EU, 0xBCU, 0x0DU, 0x0AU, 0x0BU, 0x51U, 0xD6U, 0xEAU, 0x18U, 0x63U, 0x3DU, 0x21U, 0x0CU, 0xF6U, 0x3CU, 0x0CU, 0x4DU, 0xDBU, 0xC2U, 0x76U, 0x07U, 0xF2U, 0xE8U, 0x1EU, 0xD9U, 0x11U, 0x31U, 0x91U, 0xEFU, 0x86U, 0xD5U, 0x6FU, 0x3BU, 0x99U, 0xBEU, 0x6CU, 0x41U, 0x5AU, 0x4EU, 0x50U, 0x29U, 0x9FU, 0xB8U, 0x46U, 0xCEU, 0x71U, 0x60U, 0xB4U, 0x0BU, 0x63U, 0xBAU, 0xF1U, 0x17U, 0x9DU, 0x19U, 0x27U, 0x5AU, 0x2EU, 0x83U, 0x69U, 0x83U, 0x76U, 0xD2U, 0x8BU, 0x92U, 0x54U, 0x8CU, 0x68U, 0xE0U, 0x6EU, 0x6DU, 0x99U, 0x4EU, 0x2CU, 0x15U, 0x01U, 0xEDU, 0x29U, 0x70U, 0x14U, 0xE7U, 0x02U, 0xCDU, 0xEFU, 0xEEU, 0x2FU, 0x65U, 0x64U, 0x47U, 0x70U, 0x60U, 0x09U, 0x61U, 0x4DU, 0x80U, 0x1DU, 0xE1U, 0xCAU, 0xAFU, 0x73U, 0xF8U, 0xB7U, 0xFAU, 0x56U, 0xCFU, 0x1BU, 0xA9U, 0x4BU, 0x63U, 0x19U, 0x33U, 0xBBU, 0xE5U, 0x77U, 0x62U, 0x43U, 0x80U, 0x85U, 0x0FU, 0x11U, 0x74U, 0x35U, 0xA0U, 0x35U, 0x5BU, 0x2BU |

#### 8.4.4.2 HaeAesCtr

| Key (16 byte)                                                                                                  | Counter (16 byte)                                                                                              | Plain Text (64 byte)                                                                                                                                                                                                                                                                                                                                                                                                                                           | Cipher Text (64 byte)                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|----------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0x2BU, 0x7EU, 0x15U, 0x16U, 0x28U, 0xAEU, 0xD2U, 0xA6U, 0xABU, 0xF7U, 0x15U, 0x88U, 0x09U, 0xCFU, 0x4FU, 0x3CU | 0xf0U, 0xf1U, 0xf2U, 0xf3U, 0xf4U, 0xf5U, 0xf6U, 0xf7U, 0xf8U, 0xf9U, 0xfaU, 0xfbU, 0xfcU, 0xfdU, 0xfeU, 0xffU | 0x6BU, 0xC1U, 0xBEU, 0xE2U, 0x2EU, 0x40U, 0x9FU, 0x96U, 0xE9U, 0x3DU, 0x7EU, 0x11U, 0x73U, 0x93U, 0x17U, 0x2AU, 0xAEU, 0x2DU, 0x8AU, 0x57U, 0x1EU, 0x03U, 0xACU, 0x9CU, 0x9EU, 0xB7U, 0x6FU, 0xACU, 0x45U, 0xB7U, 0x8EU, 0x51U, 0x30U, 0xC8U, 0x1CU, 0x46U, 0xA3U, 0x5CU, 0xE4U, 0x11U, 0xE5U, 0xFBU, 0xC1U, 0x19U, 0x1AU, 0x0AU, 0x52U, 0xEFU, 0xF6U, 0x9FU, 0x24U, 0x45U, 0xDFU, 0x4FU, 0x9BU, 0x17U, 0xADU, 0x2BU, 0x41U, 0x7BU, 0xE6U, 0x6CU, 0x37U, 0x10U | 0x87U, 0x4DU, 0x61U, 0x91U, 0xB6U, 0x20U, 0xE3U, 0x26U, 0x1BU, 0xEFU, 0x68U, 0x64U, 0x99U, 0x0DU, 0xB6U, 0xCEU, 0x98U, 0x06U, 0xF6U, 0x6BU, 0x79U, 0x70U, 0xFDU, 0xFFU, 0x86U, 0x17U, 0x18U, 0x7BU, 0xB9U, 0xFFU, 0xFDU, 0xFFU, 0x5AU, 0xE4U, 0xDFU, 0x3EU, 0xDBU, 0xD5U, 0xD3U, 0x5EU, 0x5BU, 0x4FU, 0x09U, 0x02U, 0x0DU, 0xB0U, 0x3EU, 0xABU, 0x1EU, 0x03U, 0x1DU, 0xDAU, 0x2FU, 0xBEU, 0x03U, 0xD1U, 0x79U, 0x21U, 0x70U, 0xA0U, 0xF3U, 0x00U, 0x9CU, 0xEEU |

#### 8.4.4.3 AuthHsmAesCbc

Raw Key 방식 사용시

| Key (16 byte) | Initial Vector | Plain Text (160 byte) | Cipher Text (160 byte) |
|---------------|----------------|-----------------------|------------------------|
|---------------|----------------|-----------------------|------------------------|

|                                                                                                                |                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|----------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0x2CU, 0x14U, 0x41U, 0x37U, 0x51U, 0xC3U, 0x1EU, 0x27U, 0x30U, 0x57U, 0x0BU, 0xA3U, 0x36U, 0x1CU, 0x78U, 0x6BU | 0x1DU, 0xBBU, 0xEBU, 0x2FU, 0x19U, 0xABU, 0xB4U, 0x48U, 0xAFU, 0x84U, 0x97U, 0x96U, 0x24U, 0x4AU, 0x19U, 0xD7U | 0x40U, 0xD9U, 0x30U, 0xF9U, 0xA0U, 0x53U, 0x34U, 0xD9U, 0x81U, 0x6FU, 0xE2U, 0x04U, 0x99U, 0x9CU, 0x3FU, 0x82U, 0xA0U, 0x3FU, 0x6AU, 0x04U, 0x57U, 0xA8U, 0xC4U, 0x75U, 0xC9U, 0x45U, 0x53U, 0xD1U, 0xD1U, 0x16U, 0x69U, 0x3AU, 0xDCU, 0x61U, 0x80U, 0x49U, 0xF0U, 0xA7U, 0x69U, 0xA2U, 0xEEU, 0xD6U, 0xA6U, 0xCBU, 0x14U, 0xC0U, 0x14U, 0x3EU, 0xC5U, 0xCCU, 0xCDU, 0xBCU, 0x8DU, 0xECU, 0x4CU, 0xE5U, 0x60U, 0xCFU, 0xD2U, 0x06U, 0x22U, 0x57U, 0x09U, 0x32U, 0x6DU, 0x4DU, 0xE7U, 0x94U, 0x8EU, 0x54U, 0xD6U, 0x03U, 0xD0U, 0x1BU, 0x12U, 0xD7U, 0xFEU, 0xD7U, 0x52U, 0xFBU, 0x23U, 0xF1U, 0xAAU, 0x44U, 0x94U, 0xFBU, 0xB0U, 0x01U, 0x30U, 0xE9U, 0xDEU, 0xD4U, 0xE7U, 0x7EU, 0x37U, 0xC0U, 0x79U, 0x04U, 0x2DU, 0x82U, 0x80U, 0x40U, 0xC3U, 0x25U, 0xB1U, 0xA5U, 0xEFU, 0xD1U, 0x5FU, 0xC8U, 0x42U, 0xE4U, 0x40U, 0x14U, 0xCAU, 0x43U, 0x74U, 0xBFU, 0x38U, 0xF3U, 0xC3U, 0xFCU, 0x3EU, 0xE3U, 0x27U, 0x73U, 0x3BU, 0x0CU, 0x8AU, 0xEEU, 0x1AU, 0xBCU, 0xD0U, 0x55U, 0x77U, 0x2FU, 0x18U, 0xDCU, 0x04U, 0x60U, 0x3FU, 0x7BU, 0x2CU, 0x1EU, 0xA6U, 0x9FU, 0xF6U, 0x62U, 0x36U, 0x1FU, 0x2BU, 0xE0U, 0xA1U, 0x71U, 0xBBU, 0xDCU, 0xEAU, 0x1EU, 0x5DU, 0x3FU | 0x6BU, 0xE8U, 0xA1U, 0x28U, 0x00U, 0x45U, 0x5AU, 0x32U, 0x05U, 0x38U, 0x85U, 0x3EU, 0x0CU, 0xBAU, 0x31U, 0xBDU, 0x2DU, 0x80U, 0xEAU, 0x0CU, 0x85U, 0x16U, 0x4AU, 0x4CU, 0x5CU, 0x26U, 0x1AU, 0xE4U, 0x85U, 0x41U, 0x7DU, 0x93U, 0xEFU, 0xFEU, 0x2EU, 0x8CU, 0x0DU, 0x0AU, 0x0BU, 0x51U, 0xD6U, 0xEAU, 0x18U, 0x63U, 0x3DU, 0x21U, 0x0CU, 0xF6U, 0x3CU, 0x0CU, 0x4DU, 0xDBU, 0xC2U, 0x76U, 0x07U, 0xF2U, 0xE8U, 0x1EU, 0xD9U, 0x11U, 0x31U, 0x91U, 0xEFU, 0x86U, 0xD5U, 0x6FU, 0x3BU, 0x99U, 0x8EU, 0x6CU, 0x41U, 0x5AU, 0x41U, 0x50U, 0x29U, 0x9FU, 0xB8U, 0x46U, 0xCEU, 0x71U, 0x60U, 0xB4U, 0x0BU, 0x63U, 0xBAU, 0xF1U, 0x17U, 0x9DU, 0x19U, 0x27U, 0x5AU, 0x2EU, 0x83U, 0x69U, 0x83U, 0x76U, 0xD2U, 0x8BU, 0x92U, 0x54U, 0x8CU, 0x68U, 0xE0U, 0x6EU, 0x6DU, 0x99U, 0x4EU, 0x2CU, 0x15U, 0x01U, 0xEDU, 0x29U, 0x70U, 0x14U, 0xE7U, 0x02U, 0xCDU, 0xEFU, 0xEEU, 0x2FU, 0x65U, 0x64U, 0x47U, 0x70U, 0x60U, 0x09U, 0x61U, 0x4DU, 0x80U, 0x1DU, 0xE1U, 0xCAU, 0xAFU, 0x73U, 0xF8U, 0xB7U, 0xFAU, 0x56U, 0xCFU, 0x1BU, 0xA9U, 0x4BU, 0x63U, 0x19U, 0x33U, 0xBBU, 0xE5U, 0x77U, 0x62U, 0x43U, 0x80U, 0x85U, 0x0FU, 0x11U, 0x74U, 0x35U, 0xA0U, 0x35U, 0x5BU, 0x2BU |
|----------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

#### 8.4.4.4 AuthHsmAesCtr

Raw Key 방식 사용시

| Key (16 byte)                                                                                                  | Counter (16 byte)                                                                                              | Plain Text (64 byte)                                                                                                                                                                                                                                                                                                                                                                                                                                           | Cipher Text (64 byte)                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|----------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0x2BU, 0x7EU, 0x15U, 0x16U, 0x28U, 0xAEU, 0xD2U, 0xA6U, 0xABU, 0xF7U, 0x15U, 0x88U, 0x09U, 0xCFU, 0x4FU, 0x3CU | 0xf0U, 0xf1U, 0xf2U, 0xf3U, 0xf4U, 0xf5U, 0xf6U, 0xf7U, 0xf8U, 0xf9U, 0xfaU, 0xfbU, 0xfcU, 0xfdU, 0xfeU, 0xffU | 0x6BU, 0xC1U, 0xBEU, 0xE2U, 0x2EU, 0x40U, 0x9FU, 0x96U, 0xE9U, 0x3DU, 0x7EU, 0x11U, 0x73U, 0x93U, 0x17U, 0x2AU, 0xAEU, 0x2DU, 0x8AU, 0x57U, 0x1EU, 0x03U, 0xACU, 0x9CU, 0x9EU, 0xB7U, 0x6FU, 0xACU, 0x45U, 0xAFU, 0x8EU, 0x51U, 0x30U, 0xC8U, 0x1CU, 0x46U, 0xA3U, 0x5CU, 0xE4U, 0x11U, 0xE5U, 0xFBU, 0xC1U, 0x19U, 0x1AU, 0x0AU, 0x52U, 0xEFU, 0xF6U, 0x9FU, 0x24U, 0x45U, 0xDFU, 0x4FU, 0x9BU, 0x17U, 0xADU, 0x2BU, 0x41U, 0x7BU, 0xE6U, 0x6CU, 0x37U, 0x10U | 0x87U, 0x4DU, 0x61U, 0x91U, 0xB6U, 0x20U, 0xE3U, 0x26U, 0x1BU, 0xEFU, 0x68U, 0x64U, 0x99U, 0x0DU, 0xB6U, 0xCEU, 0x98U, 0x06U, 0xF6U, 0x6BU, 0x79U, 0x70U, 0xFDU, 0xFFU, 0x86U, 0x17U, 0x18U, 0x7BU, 0xB9U, 0xFFU, 0xFDU, 0xFFU, 0x5AU, 0xE4U, 0xDFU, 0x3EU, 0xDBU, 0xD5U, 0xD3U, 0x5EU, 0x5BU, 0x4FU, 0x09U, 0x02U, 0x0DU, 0xB0U, 0x3EU, 0xABU, 0x1EU, 0x03U, 0x1DU, 0xDAU, 0x2FU, 0xBEU, 0x03U, 0xD1U, 0x79U, 0x21U, 0x70U, 0xA0U, 0xF3U, 0x00U, 0x9CU, 0xEEU |

#### 8.4.4.5 AutIcusAesCbc

Raw Key 방식 사용시

| Key (16 byte)                 | Initial Vector                | Plain Text (160 byte)         | Cipher Text (160 byte)        |
|-------------------------------|-------------------------------|-------------------------------|-------------------------------|
| 0x2CU, 0x14U, 0x41U, 0x37U, 0 | 0x1DU, 0xBBU, 0xEBU, 0x2FU, 0 | 0x40U, 0xD9U, 0x30U, 0xF9U, 0 | 0x6BU, 0xE8U, 0xA1U, 0x28U, 0 |

|                                                                                   |                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| x51U, 0xC3U, 0x1EU, 0x27U, 0x30U, 0x57U, 0x0BU, 0xA3U, 0x36U, 0x1CU, 0x78U, 0x6BU | x19U, 0xABU, 0xB4U, 0x48U, 0xAFU, 0x84U, 0x97U, 0x96U, 0x24U, 0x4AU, 0x19U, 0xD7U | xA0U, 0x53U, 0x34U, 0xD9U, 0x81U, 0x6FU, 0xE2U, 0x04U, 0x99U, 0x9CU, 0x3FU, 0x82U, 0xA0U, 0x3FU, 0x6AU, 0x04U, 0x57U, 0xA8U, 0xC4U, 0x75U, 0xC9U, 0x45U, 0x53U, 0xD1U, 0xD1U, 0x16U, 0x69U, 0x3AU, 0xDCU, 0x61U, 0x80U, 0x49U, 0xF0U, 0xA7U, 0x69U, 0xA2U, 0xEEU, 0xD6U, 0xA6U, 0xCBU, 0x14U, 0xC0U, 0x14U, 0x3EU, 0xC5U, 0xCCU, 0xCDU, 0xBCU, 0x8DU, 0xE0U, 0x4CU, 0xE5U, 0x60U, 0xCFU, 0xD2U, 0x06U, 0x22U, 0x57U, 0x09U, 0x32U, 0x6DU, 0x4DU, 0xE7U, 0x94U, 0x8EU, 0x54U, 0xD6U, 0x03U, 0xD0U, 0x1BU, 0x12U, 0xD7U, 0xFEU, 0xD7U, 0x52U, 0xFBU, 0x23U, 0xF1U, 0xAAU, 0x44U, 0x94U, 0xFBU, 0xB0U, 0x01U, 0x30U, 0xE9U, 0xDEU, 0xD4U, 0xE7U, 0x7EU, 0x37U, 0xC0U, 0x79U, 0x04U, 0x2DU, 0x82U, 0x80U, 0x40U, 0xC3U, 0x25U, 0xB1U, 0xA5U, 0xEFU, 0xD1U, 0x5FU, 0xC8U, 0x42U, 0xE4U, 0x40U, 0x14U, 0xCAU, 0x43U, 0x74U, 0xBFU, 0x38U, 0xF3U, 0xC3U, 0xFCU, 0x3EU, 0xE3U, 0x27U, 0x73U, 0x3BU, 0x0CU, 0x8AU, 0xEEU, 0x1AU, 0xBCU, 0xD0U, 0x55U, 0x77U, 0x2FU, 0x18U, 0xDCU, 0x04U, 0x60U, 0x3FU, 0x7BU, 0x2CU, 0x1EU, 0xA6U, 0x9FU, 0xF6U, 0x62U, 0x36U, 0x1FU, 0x2BU, 0xE0U, 0xA1U, 0x71U, 0xBBU, 0xDCU, 0xE0U, 0x1EU, 0x5DU, 0x3FU | x00U, 0x45U, 0x5AU, 0x32U, 0x05U, 0x38U, 0x85U, 0x3EU, 0x0CU, 0xBAU, 0x31U, 0xBDU, 0x2DU, 0x80U, 0xEAU, 0x0CU, 0x85U, 0x16U, 0x4AU, 0x4CU, 0x5CU, 0x26U, 0x1AU, 0xE4U, 0x85U, 0x41U, 0x7DU, 0x93U, 0xEFU, 0xFEU, 0x2EU, 0xBCU, 0x0DU, 0x0AU, 0x0BU, 0x51U, 0xD6U, 0xEAU, 0x18U, 0x63U, 0x3DU, 0x21U, 0x0CU, 0xF6U, 0x3CU, 0x0CU, 0x4DU, 0xDBU, 0xC2U, 0x76U, 0x07U, 0xF2U, 0xE8U, 0x1EU, 0xD9U, 0x11U, 0x31U, 0x91U, 0xEFU, 0x86U, 0xD5U, 0x6FU, 0x3BU, 0x99U, 0xBEU, 0x6CU, 0x41U, 0x5AU, 0x41U, 0x50U, 0x29U, 0x9FU, 0xB8U, 0x46U, 0xCEU, 0x71U, 0x60U, 0xB4U, 0x0BU, 0x63U, 0xBAU, 0xF1U, 0x17U, 0x9DU, 0x19U, 0x27U, 0x5AU, 0x2EU, 0x83U, 0x69U, 0x83U, 0x76U, 0xD2U, 0x8BU, 0x92U, 0x54U, 0x8CU, 0x68U, 0xE0U, 0x6EU, 0x6DU, 0x99U, 0x4EU, 0x2CU, 0x15U, 0x01U, 0xEDU, 0x29U, 0x70U, 0x14U, 0xE7U, 0x02U, 0xCDU, 0xEFU, 0xEEU, 0x2FU, 0x65U, 0x64U, 0x47U, 0x70U, 0x60U, 0x09U, 0x61U, 0x4DU, 0x80U, 0x1DU, 0xE1U, 0xCAU, 0xAFU, 0x73U, 0xF8U, 0xB7U, 0xFAU, 0x56U, 0xCFU, 0x1BU, 0xA9U, 0x4BU, 0x63U, 0x19U, 0x33U, 0xBBU, 0xE5U, 0x77U, 0x62U, 0x43U, 0x80U, 0x85U, 0x0FU, 0x11U, 0x74U, 0x35U, 0xA0U, 0x35U, 0x5BU, 0x2BU |
|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

#### 8.4.4.6 AutCsecAesCbc

Raw Key 방식 사용시

| Key (16 byte)                                                                                                  | Initial Vector                                                                                                 | Plain Text (160 byte)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Cipher Text (160 byte)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|----------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0x2CU, 0x14U, 0x41U, 0x37U, 0x51U, 0xC3U, 0x1EU, 0x27U, 0x30U, 0x57U, 0x0BU, 0xA3U, 0x36U, 0x1CU, 0x78U, 0x6BU | 0x1DU, 0xBBU, 0xEBU, 0x2FU, 0x19U, 0xABU, 0xB4U, 0x48U, 0xAFU, 0x84U, 0x97U, 0x96U, 0x24U, 0x4AU, 0x19U, 0xD7U | 0x40U, 0xD9U, 0x30U, 0xF9U, 0xA0U, 0x53U, 0x34U, 0xD9U, 0x81U, 0x6FU, 0xE2U, 0x04U, 0x99U, 0x9CU, 0x3FU, 0x82U, 0xA0U, 0x3FU, 0x6AU, 0x04U, 0x57U, 0xA8U, 0xC4U, 0x75U, 0xC9U, 0x45U, 0x53U, 0xD1U, 0xD1U, 0x16U, 0x69U, 0x3AU, 0xDCU, 0x61U, 0x80U, 0x49U, 0xF0U, 0xA7U, 0x69U, 0xA2U, 0xEEU, 0xD6U, 0xA6U, 0xCBU, 0x14U, 0xC0U, 0x14U, 0x3EU, 0xC5U, 0xCCU, 0xCDU, 0xBCU, 0x8DU, 0xE0U, 0x4CU, 0xE5U, 0x60U, 0xCFU, 0xD2U, 0x06U, 0x22U, 0x57U, 0x09U, 0x32U, 0x6DU, 0x4DU, 0xE7U, 0x94U, 0x8EU, 0x54U, 0xD6U, 0x03U, 0xD0U, 0x1BU, 0x12U, 0xD7U, 0xFEU, 0xD7U, 0x52U, 0xFBU, 0x23U, 0xF1U, 0xAAU, 0x44U, 0x94U, 0xFBU, 0xB0U, 0x01U, 0x30U, 0xE9U, 0xDEU, 0xD4U, 0xE7U, 0x7EU, 0x37U, 0xC0U, 0x79U, 0x04U, 0x2DU, 0x82U, 0x80U, 0x40U, 0xC3U, | 0x6BU, 0xE8U, 0xA1U, 0x28U, 0x00U, 0x45U, 0x5AU, 0x32U, 0x05U, 0x38U, 0x85U, 0x3EU, 0x0CU, 0xBAU, 0x31U, 0xBDU, 0x2DU, 0x80U, 0xEAU, 0x0CU, 0x85U, 0x16U, 0x4AU, 0x4CU, 0x5CU, 0x26U, 0x1AU, 0xE4U, 0x85U, 0x41U, 0x7DU, 0x93U, 0xEFU, 0xFEU, 0x2EU, 0xBCU, 0x0DU, 0x0AU, 0x0BU, 0x51U, 0xD6U, 0xEAU, 0x18U, 0x63U, 0x3DU, 0x21U, 0x0CU, 0xF6U, 0x3CU, 0x0CU, 0x4DU, 0xDBU, 0xC2U, 0x76U, 0x07U, 0xF2U, 0xE8U, 0x1EU, 0xD9U, 0x11U, 0x31U, 0x91U, 0xEFU, 0x86U, 0xD5U, 0x6FU, 0x3BU, 0x99U, 0xBEU, 0x6CU, 0x41U, 0x5AU, 0x41U, 0x50U, 0x29U, 0x9FU, 0xB8U, 0x46U, 0xCEU, 0x71U, 0x60U, 0xB4U, 0x0BU, 0x63U, 0xBAU, 0xF1U, 0x17U, 0x9DU, 0x19U, 0x27U, 0x5AU, 0x2EU, 0x83U, 0x69U, 0x83U, 0x76U, 0xD2U, 0x8BU, 0x92U, 0x54U, 0x8CU, 0x68U, 0xE0U, |

|  |  |                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                               |
|--|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  |  | 0x25U, 0xB1U, 0xA5U, 0xEFU, 0xD1U, 0x5FU, 0xC8U, 0x42U, 0xE4U, 0x40U, 0x14U, 0xCAU, 0x43U, 0x74U, 0xBFU, 0x38U, 0xF3U, 0xC3U, 0xFCU, 0x3EU, 0xE3U, 0x27U, 0x73U, 0x3BU, 0x0CU, 0x8AU, 0xEEU, 0x1AU, 0xBCU, 0xD0U, 0x55U, 0x77U, 0x2FU, 0x18U, 0xDCU, 0x04U, 0x60U, 0x3FU, 0x7BU, 0x2CU, 0x1EU, 0xA6U, 0x9FU, 0xF6U, 0x62U, 0x36U, 0x1FU, 0x2BU, 0xE0U, 0xA1U, 0x71U, 0xBBU, 0xDCU, 0xEAU, 0x1EU, 0x5DU, 0x3FU | 0x6EU, 0x6DU, 0x99U, 0x4EU, 0x2CU, 0x15U, 0x01U, 0xEDU, 0x29U, 0x70U, 0x14U, 0xE7U, 0x02U, 0xCDU, 0xEFU, 0xEEU, 0x2FU, 0x65U, 0x64U, 0x47U, 0x70U, 0x60U, 0x09U, 0x61U, 0x4DU, 0x80U, 0x1DU, 0xE1U, 0xCAU, 0xAFU, 0x73U, 0xF8U, 0xB7U, 0xFAU, 0x56U, 0xCFU, 0x1BU, 0xA9U, 0x4BU, 0x63U, 0x19U, 0x33U, 0xBBU, 0xE5U, 0x77U, 0x62U, 0x43U, 0x80U, 0x85U, 0x0FU, 0x11U, 0x74U, 0x35U, 0xA0U, 0x35U, 0x5BU, 0x2BU |
|--|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

#### 8.4.4.7 HaeHsmAesCbc

Autoever HSM (HSM\_Framework\_2.x\_UserManual\_Crypto\_Service\_KOR\_vx.xx.pdf) 의 7.Crypto Service API 테스트 케이스 참조

#### 8.4.4.8 HaeHsmAesCtr

Autoever HSM (HSM\_Framework\_2.x\_UserManual\_Crypto\_Service\_KOR\_vx.xx.pdf) 의 7.Crypto Service API 테스트 케이스 참조

### 8.4.5 Asymmetric

#### 8.4.5.1 HaeRsa

##### 1) Encryption

| Public Key                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Plain Text (28 byte)                                                                                                                                                                                | Cipher Text (256 byte)              |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|
| /* modulus : 256 byte */<br>0xE0U, 0xB1U, 0x4BU, 0x99U, 0xCDU, 0x61U, 0xCDU, 0x3DU, 0xB9U, 0xC2U, 0x07U, 0x66U, 0x68U, 0x84U, 0x13U, 0x24U, 0xFAU, 0x31U, 0x74U, 0xF3U, 0x3CU, 0xE6U, 0x6FU, 0xFDU, 0x51U, 0x43U, 0x94U, 0xD3U, 0x41U, 0x78U, 0xD2U, 0x9AU, 0x49U, 0x49U, 0x32U, 0x76U, 0xB6U, 0x77U, 0x72U, 0x33U, 0xE7U, 0xD4U, 0x6AU, 0x3EU, 0x68U, 0xBCU, 0x7CU, 0xA7U, 0xE8U, 0x99U, 0xE9U, 0x01U, 0xD5U, 0x4FU, 0x6DU, 0xEEU, 0x07U, 0x49U, 0xC3U, 0xE4U, 0x8DU, 0xDFU, 0x68U, 0x68U, 0x58U, 0x67U, 0xEEU, 0x2AU, 0xE6U, 0x6DU, 0xF8U, 0x8EU, 0xB5U, 0x63U, 0xF6U, 0xDBU, 0x13U, 0x7AU, 0x9FU, 0x6BU, 0x17U, 0x5AU, 0x11U, 0x2EU, 0x0EU, 0xDAU, 0x83U, 0x68U, 0xE8U, 0x8EU, 0x45U, 0xEFU, 0xE1U, 0xCEU, 0x14U, 0xBCU, 0x60U, 0x16U, 0xD5U, 0x26U, 0x39U, 0x62U, 0x70U, 0x66U, 0xAFU, 0x18U, 0x72U, 0xC7U, 0x2FU, 0x60U, 0xB9U, 0x16U, 0x1CU, 0x1DU, 0x23U, 0x7EU, 0xEBU, 0x34U, 0x80U, 0xF8U, 0x41U, 0xB3U, 0xF0U, 0x89U, 0x6FU, 0x9FU, 0xE0U, 0xE1U, 0x6BU, 0x0FU, 0x74U, 0x35U, 0x2DU, 0x10U, 0x12U, 0x92U, 0xCCU, 0x46U, 0x4AU, 0x7EU, 0x78U, 0x61U, 0xBBU, 0xEBU, 0x86U, 0xF6U, 0xDFU, 0x61U, 0x51U, 0xCBU, 0x26U, 0x54U, 0x17U, 0xC6U, 0x6CU, 0x56U, 0x5EU, 0xD8U, 0x97U, 0x4BU, 0xD8U, 0xFCU, 0x98U, 0x4DU, 0x5DU, 0xDFU, 0xD4U, 0xEBU, 0x91U, 0xA3U, 0xD5U, 0x23U, 0x4CU, 0xE1U, 0xB5U, 0x46U, 0x7FU, 0x3AU, 0xDEU, 0x37U, 0x5FU, 0x80U, 0x2EU, 0xC0U, 0x72U, 0x93U, 0xF1U, 0x23U, 0x6EU, 0xFAU, 0x30U, 0x68U, 0xBCU, 0x91U, 0xB1U, 0x58U, 0x55U, 0x1CU, 0x87U, 0x5CU, 0x5DU, 0xC0U, 0xA9U, 0xD6U, 0xFAU, 0x32U, 0x1BU, 0xF9U, 0x42U, 0x1FU, 0x08U, 0xDEU, 0xACU, 0x91U, 0x0EU, 0x35U, 0xC1U, 0xC2U, 0x85U, 0x49U, 0xEEU, 0x8EU, 0xEDU, 0x83U, 0x30U, 0xCFU, 0x70U, 0x59U, 0x5FU, 0xF7U, 0x0BU, 0x94U, 0xB4U, 0x99U, 0x07U, 0xE2U, 0x76U, 0x98U, 0xA9U, 0xD9U, 0x11U, 0xF7U, 0xACU, 0x07U, 0x06U, 0xAFU, 0xCBU, 0x1AU, 0x4AU, 0x39U, 0xFEU, 0xB3U, 0x8BU, 0x0AU, 0x80U, 0x49U<br><br>/* exponent : 4 bytes */<br>0x00U, 0x01U, 0x00U, 0x01U | 0x8BU, 0xBAU, 0x6BU, 0xF8U, 0x2AU, 0x6CU, 0x0FU, 0x86U, 0xD5U, 0xF1U, 0x75U, 0x6EU, 0x97U, 0x95U, 0x68U, 0x70U, 0xB0U, 0x89U, 0x53U, 0xB0U, 0x6BU, 0x4EU, 0xB2U, 0x05U, 0xBCU, 0x16U, 0x94U, 0xEEU, | 암호화 연산 중 랜덤 값이 삽입되므로, 암호화문은 일정하지 않음 |

##### 2) Decryption

| Private Key                                                                          | Cipher Text (256 byte)                                                                                            | Plain Text (28 byte)                                                                                              |
|--------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| /* modulus : 256 bytes */<br>0xE0U, 0xB1U, 0x4BU, 0x99U, 0xCDU, 0x61U, 0xCDU, 0x3DU, | 0x9AU, 0x36U, 0x90U, 0x1CU, 0xDCU, 0x9CU, 0x7BU, 0x4FU, 0xBAU, 0x81U, 0xAFU, 0x93U, 0x6BU, 0x42U, 0x5FU, 0xE1U, 0 | 0x8BU, 0xBAU, 0x6BU, 0xF8U, 0x2AU, 0x6CU, 0x0FU, 0x86U, 0xD5U, 0xF1U, 0x75U, 0x6EU, 0x97U, 0x95U, 0x68U, 0x70U, 0 |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        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                                                                                                                                                                                                                                                     |                                                                                           |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| <p>0xB9U, 0xC2U, 0x07U, 0x66U, 0x68U, 0x84U, 0x13U, 0x24U, 0xFAU, 0x31U, 0x74U, 0xF3U, 0x3CU, 0xE6U, 0x6FU, 0xFDU, 0x51U, 0x43U, 0x94U, 0xD3U, 0x41U, 0x78U, 0xD2U, 0x9AU, 0x49U, 0x49U, 0x32U, 0x76U, 0xB6U, 0x77U, 0x72U, 0x33U, 0xE7U, 0xD4U, 0x6AU, 0x3EU, 0x68U, 0x8CU, 0x7CU, 0xA7U, 0xE8U, 0x99U, 0xE9U, 0x01U, 0xD5U, 0x4FU, 0x6DU, 0xEEU, 0x07U, 0x49U, 0xC3U, 0xE4U, 0x8DU, 0xDFU, 0x68U, 0x68U, 0x58U, 0x67U, 0xEEU, 0x2AU, 0xE6U, 0x6DU, 0xF8U, 0x8EU, 0xB5U, 0x63U, 0xF6U, 0xDBU, 0x13U, 0x7AU, 0x9FU, 0x6BU, 0x17U, 0x5AU, 0x11U, 0x2EU, 0x0EU, 0xDAU, 0x83U, 0x68U, 0xE8U, 0x8EU, 0x45U, 0xEFU, 0xE1U, 0xCEU, 0x14U, 0x8CU, 0x60U, 0x16U, 0xD5U, 0x26U, 0x39U, 0x62U, 0x70U, 0x66U, 0xAFU, 0x18U, 0x72U, 0xC7U, 0x2FU, 0x60U, 0x89U, 0x16U, 0x1CU, 0x1DU, 0x23U, 0x7EU, 0xEBU, 0x34U, 0x80U, 0xF8U, 0x41U, 0xB3U, 0xF0U, 0x89U, 0x6FU, 0x9FU, 0xE0U, 0xE1U, 0x68U, 0x0FU, 0x74U, 0x35U, 0x2DU, 0x10U, 0x12U, 0x92U, 0xCCU, 0x46U, 0x4AU, 0x7EU, 0x78U, 0x61U, 0x8BU, 0xEBU, 0x86U, 0xF6U, 0xDFU, 0x61U, 0x51U, 0xCBU, 0x26U, 0x54U, 0x17U, 0xC6U, 0x6CU, 0x56U, 0x5EU, 0xD8U, 0x97U, 0x4BU, 0xD8U, 0xFCU, 0x98U, 0x4DU, 0x5DU, 0xDFU, 0xD4U, 0xEBU, 0x91U, 0xA3U, 0xD5U, 0x23U, 0x4CU, 0xE1U, 0x85U, 0x46U, 0x7FU, 0x3AU, 0xDEU, 0x37U, 0x5FU, 0x80U, 0x2EU, 0xC0U, 0x72U, 0x93U, 0xF1U, 0x23U, 0x6EU, 0xFAU, 0x30U, 0x68U, 0xBCU, 0x91U, 0xB1U, 0x58U, 0x55U, 0x1CU, 0x87U, 0x5CU, 0x5DU, 0xC0U, 0xA9U, 0xD6U, 0x8FU, 0x32U, 0x18U, 0xF9U, 0x42U, 0x1FU, 0x08U, 0xDEU, 0xAEU, 0x91U, 0x0EU, 0x35U, 0xC1U, 0xC2U, 0x85U, 0x49U, 0x8CU, 0xEDU, 0x83U, 0x30U, 0xCFU, 0x70U, 0x59U, 0x5FU, 0xF7U, 0x08U, 0x94U, 0xB4U, 0x99U, 0x07U, 0xE2U, 0x76U, 0x98U, 0xA9U, 0xD9U, 0x11U, 0xF7U, 0xACU, 0x07U, 0x06U, 0xAFU, 0xCBU, 0x1AU, 0x4AU, 0x39U, 0xFEU, 0xB3U, 0x8BU, 0x0AU, 0x80U, 0x49U, 0x49U, /* exponent : 256 bytes */ 0x1DU, 0xB8U, 0xA9U, 0x2EU, 0x42U, 0x45U, 0xC2U, 0xD5U, 0x7BU, 0xFBU, 0xA7U, 0x62U, 0x10U, 0xCCU, 0x06U, 0x02U, 0x9BU, 0x50U, 0x27U, 0x53U, 0x87U, 0xC8U, 0x21U, 0xA3U, 0x2BU, 0x79U, 0x9FU, 0x8DU, 0x33U, 0xC9U, 0x8BU, 0x49U, 0xDBU, 0x10U, 0x22U, 0x6BU, 0x1EU, 0xACU, 0x01U, 0x43U, 0xC8U, 0x57U, 0x4EU, 0xF6U, 0x52U, 0x83U, 0x38U, 0x96U, 0x37U, 0x4DU, 0x03U, 0x4EU, 0xF8U, 0x4DU, 0xAAU, 0x55U, 0x59U, 0xC6U, 0x93U, 0xF3U, 0xF0U, 0x28U, 0xD4U, 0x97U, 0x16U, 0x8BU, 0x2EU, 0x87U, 0xA3U, 0xF6U, 0x82U, 0xF2U, 0x54U, 0x24U, 0x56U, 0x3BU, 0xD9U, 0x40U, 0x9DU, 0xCFU, 0x9DU, 0x08U, 0x11U, 0x05U, 0x00U, 0xF7U, 0x3FU, 0x74U, 0x07U, 0x6FU, 0x28U, 0xE7U, 0x5EU, 0x01U, 0x99U, 0xB1U, 0xF2U, 0x9FU, 0xA2U, 0xF7U, 0x0BU, 0x9AU, 0x31U, 0x19U, 0x0DU, 0xECU, 0x54U, 0xE8U, 0x72U, 0xA7U, 0x40U, 0xE7U, 0xA1U, 0xB1U, 0xE3U, 0x8CU, 0x3DU, 0x11U, 0xBCU, 0xA8U, 0x26U, 0x7DU, 0xEBU, 0x84U, 0x2CU, 0xEFU, 0x42U, 0x62U, 0x23U, 0x7AU, 0xC8U, 0x75U, 0x72U, 0x50U, 0x68U, 0xF3U, 0x25U, 0x63U, 0xB4U, 0x78U, 0xACU, 0xA8U, 0xD6U, 0xA9U, 0x9FU, 0x34U, 0xCBU, 0x88U, 0x76U, 0xB9U, 0x71U, 0x45U, 0xB2U, 0xE8U, 0x52U, 0x9EU, 0xC8U, 0xADU, 0xEAU, 0x83U, 0xEAU, 0xD4U, 0xECU, 0x63U, 0xE3U, 0xFFU, 0x2DU, 0x17U, 0xA2U, 0xFFU, 0xEFU, 0xB0U, 0x5CU, 0x90U, 0x2CU, 0xA7U, 0xA9U, 0x21U, 0x68U, 0x37U, 0x8CU, 0x89U, 0xF7U, 0x5CU, 0x92U, 0x8FU, 0xC4U, 0xF0U, 0x70U, 0x7EU, 0x43U, 0x48U, 0x7AU, 0x4FU, 0x47U, 0xDFU, 0x70U, 0xCAU, 0xE8U, 0x7EU, 0x24U, 0x27U, 0x2CU, 0x13U, 0x6DU, 0x3EU, 0x98U, 0xCFU, 0x59U, 0x06U, 0x6DU, 0x41U, 0xA3U, 0xD0U, 0x38U, 0x85U, 0x7DU, 0x07U, 0x3DU, 0x8BU, 0x4DU, 0x2CU, 0x27U, 0xB8U, 0xF0U, 0xEAU, 0x6BU, 0xFAU, 0x50U, 0xD2U, 0x63U, 0x09U, 0x1AU, 0x44U, 0x18U, 0xC6U, 0x3FU, 0x44U, 0x6BU, 0xC9U, 0xA6U, 0x1EU, 0x8CU, 0x4AU, 0x68U, 0x83U, 0x47U, 0xB2U, 0x43U, 0x5EU, 0xC8U, 0xE7U, 0x2EU, 0xDDU, 0xAEU, 0xA7U,</p> | <p>xE0U, 0x7EU, 0x24U, 0x98U, 0x4BU, 0xA0U, 0xBAU, 0x17U, 0xA9U, 0x6FU, 0x41U, 0x4FU, 0x29U, 0xD4U, 0x47U, 0x48U, 0x46U, 0x46U, 0xA8U, 0xB3U, 0xCBU, 0x84U, 0x70U, 0xD5U, 0xC3U, 0xE1U, 0x25U, 0xABU, 0x72U, 0xD7U, 0xD5U, 0x70U, 0xC5U, 0x17U, 0x8AU, 0xAFU, 0x3BU, 0x81U, 0x3BU, 0x27U, 0xDCU, 0xF8U, 0xAFU, 0xDBU, 0xBFU, 0x8CU, 0x2FU, 0xB2U, 0xF9U, 0x93U, 0x96U, 0x84U, 0x8AU, 0xD7U, 0xB1U, 0x9CU, 0x9AU, 0xE0U, 0x4CU, 0x0DU, 0xFEU, 0x67U, 0x3EU, 0x94U, 0x97U, 0xAAU, 0x03U, 0x9BU, 0x12U, 0xEFU, 0x73U, 0x55U, 0xEEU, 0xE3U, 0xD0U, 0x50U, 0x96U, 0x71U, 0xD5U, 0xCBU, 0x81U, 0xDCU, 0xC7U, 0xC9U, 0x8CU, 0x25U, 0x31U, 0xC7U, 0x1CU, 0xF4U, 0xC2U, 0xF3U, 0xB1U, 0xE7U, 0x05U, 0xCDU, 0xB2U, 0xBEU, 0x77U, 0x5FU, 0xCDU, 0xE1U, 0xDDU, 0x9EU, 0xCAU, 0xE7U, 0x28U, 0xF9U, 0x00U, 0x28U, 0x6EU, 0x69U, 0x3FU, 0x2EU, 0x74U, 0xE3U, 0x8CU, 0xD7U, 0xC1U, 0x5CU, 0x93U, 0x33U, 0x4AU, 0xA5U, 0xBAU, 0x88U, 0xA7U, 0x2CU, 0xDU, 0x4BU, 0xF2U, 0x93U, 0xDFU, 0x8BU, 0x09U, 0x05U, 0xD0U, 0x02U, 0xD5U, 0x19U, 0x1AU, 0xC7U, 0x86U, 0xFEU, 0x58U, 0x62U, 0x8EU, 0x5FU, 0xB9U, 0x82U, 0x22U, 0xC5U, 0xFFU, 0xA4U, 0x06U, 0xACU, 0x79U, 0x43U, 0x6AU, 0xB5U, 0x18U, 0x04U, 0x1EU, 0x98U, 0x18U, 0x8CU, 0xB4U, 0x63U, 0x4DU, 0xDAU, 0x02U, 0xA7U, 0x76U, 0x69U, 0xF7U, 0xFBU, 0xA8U, 0xEFU, 0x03U, 0xE4U, 0xB6U, 0x05U, 0x27U, 0xA0U, 0x92U, 0x56U, 0x9CU, 0xA3U, 0x1CU, 0x81U, 0xB6U, 0xE3U, 0xE4U, 0xCAU, 0x48U, 0x09U, 0xD0U, 0xC7U, 0x20U, 0x67U, 0x6FU, 0xDDU, 0x18U, 0x5AU, 0xBCU, 0x5AU, 0x79U, 0x26U, 0x9AU, 0x80U, 0xFFU, 0x57U, 0x09U, 0xD9U, 0xF1U, 0xEDU, 0x01U, 0xB9U, 0x9BU, 0x8BU, 0x6BU, 0x82U, 0xBBU, 0x13U, 0xE5U, 0xB3U, 0xA7U, 0xD7U, 0x27U, 0xC3U, 0x89U, 0x76U, 0xD9U, 0xB7U, 0xE1U, 0x25U, 0x46U, 0xF2U, 0xD0U, 0x8AU</p> | <p>x80U, 0x89U, 0x53U, 0x80U, 0x6BU, 0x4EU, 0xB2U, 0x05U, 0xBCU, 0x16U, 0x94U, 0xEEU,</p> |
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## 8.4.5.2 AuthHsmRsa

### Raw Key 방식 사용시

#### 1) Encryption

| Public Key                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Plain Text (28 byte)                                                                                                                                                                                       | Cipher Text (256 byte)                     |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|
| <p>/* modulus : 256 byte */<br/>0xE0U, 0xB1U, 0x4BU, 0x99U, 0xCDU, 0x61U, 0xCDU, 0x3DU, 0xB9U, 0xC2U, 0x07U, 0x66U, 0x68U, 0x84U, 0x13U, 0x24U, 0xFAU, 0x31U, 0x74U, 0xF3U, 0x3CU, 0xE6U, 0x6FU, 0xFDU, 0x51U, 0x43U, 0x94U, 0xD3U, 0x41U, 0x78U, 0xD2U, 0x9AU, 0x49U, 0x49U, 0x32U, 0x76U, 0xB6U, 0x77U, 0x72U, 0x33U, 0xE7U, 0xD4U, 0x6AU, 0x3EU, 0x68U, 0x8CU, 0x7CU, 0xA7U, 0xE8U, 0x99U, 0xE9U, 0x01U, 0xD5U, 0x4FU, 0x6DU, 0xEEU, 0x07U, 0x49U, 0xC3U, 0xE4U, 0x8DU, 0xDFU, 0x68U, 0x68U, 0x58U, 0x67U, 0xEEU, 0x2AU, 0xE6U, 0x6DU, 0xF8U, 0x8EU, 0xB5U, 0x63U, 0xF6U, 0xDBU, 0x13U, 0x7AU, 0x9FU, 0x6BU, 0x17U, 0x5AU, 0x11U, 0x2EU, 0x0EU, 0xDAU, 0x83U, 0x68U, 0xE8U, 0x8EU, 0x45U, 0xEFU, 0xE1U, 0xCEU, 0x14U, 0xB8U, 0x60U, 0x16U, 0xD5U, 0x26U, 0x39U, 0x62U, 0x70U, 0x66U, 0xAFU, 0x18U, 0x72U, 0xC7U, 0x2FU, 0x60U, 0x89U, 0x16U, 0x1CU, 0x1DU, 0x23U, 0x7EU, 0xEBU, 0x34U, 0x80U, 0xF8U, 0x41U, 0xB3U, 0xF0U, 0x89U, 0x6FU, 0x9FU, 0xE0U, 0xE1U, 0x68U, 0x0FU, 0x74U, 0x35U, 0x2DU, 0x10U, 0x12U, 0x92U, 0xCCU, 0x46U, 0x4AU, 0x7EU, 0x78U, 0x61U, 0x8BU, 0xEBU, 0x86U, 0xF6U, 0xDFU, 0x61U, 0x51U, 0xCBU, 0x26U, 0x54U, 0x17U, 0xC6U, 0x6CU, 0x56U, 0x5EU, 0xD8U, 0x97U, 0x4BU, 0xD8U, 0xFCU, 0x98U, 0x4DU, 0x5DU, 0xDFU, 0xD4U, 0xEBU, 0x91U, 0xA3U, 0xD5U, 0x23U, 0x4CU, 0xE1U, 0x85U, 0x46U, 0x7FU, 0x3AU, 0xDEU, 0x37U, 0x5FU, 0x80U, 0x2EU, 0xC0U, 0x72U, 0x93U, 0xF1U, 0x23U, 0x6EU, 0xFAU, 0x30U, 0x68U, 0xBCU,</p> | <p>0x8BU, 0xBAU, 0x6BU, 0xF8U, 0x2AU, 0x6CU, 0x0FU, 0x86U, 0xD5U, 0xF1U, 0x75U, 0x6EU, 0x97U, 0x95U, 0x68U, 0x70U, 0x80U, 0x89U, 0x53U, 0x80U, 0x6BU, 0x4EU, 0xB2U, 0x05U, 0xBCU, 0x16U, 0x94U, 0xEEU,</p> | <p>암호화 연산 중 랜덤 값이 삽입되므로, 암호화문은 일정하지 않음</p> |

U, 0x91U, 0xB1U, 0x58U, 0x55U, 0x1CU, 0x87U, 0x5CU, 0x5D  
U, 0xC0U, 0xA9U, 0xD6U, 0xFAU, 0x32U, 0x1BU, 0xF9U, 0x42  
U, 0x1FU, 0x08U, 0xDEU, 0xACU, 0x91U, 0x0EU, 0x35U, 0xC1  
U, 0xC2U, 0x85U, 0x49U, 0xEEU, 0x8EU, 0xEDU, 0x83U, 0x30U,  
0xCFU, 0x70U, 0x59U, 0x5FU, 0xF7U, 0x0BU, 0x94U, 0xB4U,  
0x99U, 0x07U, 0xE2U, 0x76U, 0x98U, 0xA9U, 0xD9U, 0x11U, 0  
xF7U, 0xACU, 0x07U, 0x06U, 0xAFU, 0xCBU, 0x1AU, 0x4AU, 0  
x39U, 0xFEU, 0xB3U, 0x8BU, 0x0AU, 0x80U, 0x49U

/\* exponent : 4 bytes \*/  
0x00U, 0x01U, 0x00U, 0x01U

## 2) Decryption

| Private Key                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            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                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Plain Text (28 byte)                                                                                                                                                                                                     |
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| <p>/* modulus : 256 bytes */<br/>0xE0U, 0xB1U, 0x4BU, 0x99U, 0xC0U, 0x61U, 0xCDU, 0x3DU,<br/>0xB9U, 0xC2U, 0x07U, 0x66U, 0x68U, 0x84U, 0x13U, 0x24U, 0<br/>xFAU, 0x31U, 0x74U, 0xF3U, 0x3CU, 0xE6U, 0x6FU, 0xFDU, 0x<br/>51U, 0x43U, 0x94U, 0xD3U, 0x41U, 0x78U, 0xD2U, 0x9AU, 0x<br/>49U, 0x49U, 0x32U, 0x76U, 0xB6U, 0x77U, 0x72U, 0x33U, 0xE<br/>7U, 0xD4U, 0x6AU, 0x3EU, 0x68U, 0x8CU, 0x7CU, 0xA7U, 0xE<br/>8U, 0x99U, 0xE9U, 0x01U, 0xD5U, 0x4FU, 0x6DU, 0xEEU, 0x07<br/>U, 0x49U, 0xC3U, 0xE4U, 0x8DU, 0xDFU, 0x68U, 0x68U, 0x58<br/>U, 0x67U, 0xEEU, 0x2AU, 0xE6U, 0x6DU, 0xF8U, 0x8EU, 0xB5<br/>U, 0x63U, 0xF6U, 0xDBU, 0x13U, 0x7AU, 0x9FU, 0x6BU, 0x17<br/>U, 0x5AU, 0x11U, 0x2EU, 0x0EU, 0xDAU, 0x83U, 0x68U, 0xE8<br/>U, 0x8EU, 0x45U, 0xEFU, 0xE1U, 0xCEU, 0x14U, 0xB8U, 0x60<br/>U, 0x16U, 0xD5U, 0x26U, 0x39U, 0x62U, 0x70U, 0x66U, 0xAF<br/>U, 0x18U, 0x72U, 0xC7U, 0x2FU, 0x60U, 0xB9U, 0x16U, 0x1C<br/>U, 0x1DU, 0x23U, 0x7EU, 0xEBU, 0x34U, 0x80U, 0xF8U, 0x41<br/>U, 0xB3U, 0xF0U, 0x89U, 0x6FU, 0x9FU, 0xE0U, 0xE1U, 0x68<br/>U, 0x0FU, 0x74U, 0x35U, 0x2DU, 0x10U, 0x12U, 0x92U, 0xCC<br/>U, 0x46U, 0x4AU, 0x7EU, 0x78U, 0x61U, 0xB8U, 0xEBU, 0x86<br/>U, 0xF6U, 0xDFU, 0x61U, 0x51U, 0xCBU, 0x26U, 0x54U, 0x17<br/>U, 0xC6U, 0x6CU, 0x56U, 0x5EU, 0xD8U, 0x97U, 0x4BU, 0xD8<br/>U, 0xFCU, 0x98U, 0x40U, 0x5DU, 0xDFU, 0xD4U, 0xEBU, 0x91<br/>U, 0xA3U, 0xD5U, 0x23U, 0x4CU, 0xE1U, 0xB5U, 0x46U, 0x7F<br/>U, 0x3AU, 0xDEU, 0x37U, 0x5FU, 0x80U, 0x2EU, 0xC0U, 0x72<br/>U, 0x93U, 0xF1U, 0x23U, 0x6EU, 0xFAU, 0x30U, 0x68U, 0xBC<br/>U, 0x91U, 0xB1U, 0x58U, 0x55U, 0x1CU, 0x87U, 0x5CU, 0x5D<br/>U, 0xC0U, 0xA9U, 0xD6U, 0xFAU, 0x32U, 0x1BU, 0xF9U, 0x42<br/>U, 0x1FU, 0x08U, 0xDEU, 0xACU, 0x91U, 0x0EU, 0x35U, 0xC1<br/>U, 0xC2U, 0x85U, 0x49U, 0xEEU, 0x8EU, 0xEDU, 0x83U, 0x30<br/>U, 0xCFU, 0x70U, 0x59U, 0x5FU, 0xF7U, 0x0BU, 0x94U, 0xB4<br/>U, 0x99U, 0x07U, 0xE2U, 0x76U, 0x98U, 0xA9U, 0xD9U, 0x11<br/>U, 0xF7U, 0x07U, 0x06U, 0xAFU, 0xCBU, 0x1AU, 0x4AU, 0x4<br/>U, 0x39U, 0xFEU, 0xB3U, 0x8BU, 0x0AU, 0x80U, 0x49U,</p> <p>/* exponent : 256 bytes */<br/>0x1DU, 0xB8U, 0xA9U, 0x2EU, 0x42U, 0x45U, 0xC2U, 0xD5U,<br/>0x7BU, 0xFBU, 0xA7U, 0x62U, 0x10U, 0xC2U, 0x06U, 0x02U,<br/>0x9BU, 0x50U, 0x27U, 0x53U, 0x87U, 0xC8U, 0x21U, 0xA3U,<br/>0x2BU, 0x79U, 0x9FU, 0xBDU, 0x33U, 0xC9U, 0x8BU, 0x49U,<br/>0xDBU, 0x10U, 0x22U, 0x6BU, 0x1EU, 0x4CU, 0x01U, 0x43U,<br/>0xC8U, 0x57U, 0x4EU, 0xF6U, 0x52U, 0x83U, 0x3BU, 0x96U, 0<br/>x37U, 0x4DU, 0x03U, 0x4EU, 0xF8U, 0x4DU, 0xAAU, 0x55U, 0<br/>x59U, 0xC6U, 0x93U, 0xF3U, 0xF0U, 0x28U, 0xD4U, 0x97U, 0x<br/>16U, 0x8BU, 0x2EU, 0x87U, 0xA3U, 0xF6U, 0x82U, 0xF2U, 0x5<br/>4U, 0x24U, 0x56U, 0x3BU, 0xD9U, 0x40U, 0x9DU, 0xCFU, 0x9<br/>DU, 0x08U, 0x11U, 0x05U, 0x00U, 0xF7U, 0x3FU, 0x74U, 0x07<br/>U, 0x6FU, 0x28U, 0xE7U, 0x5EU, 0x01U, 0x99U, 0xB1U, 0xF2<br/>U, 0x9FU, 0xA2U, 0xF7U, 0x0BU, 0x9AU, 0x31U, 0x19U, 0x0D<br/>U, 0xECU, 0x54U, 0xE8U, 0x72U, 0xA7U, 0x40U, 0xE7U, 0xA1<br/>U, 0xB1U, 0xE3U, 0x8CU, 0x3DU, 0x11U, 0x8CU, 0xA8U, 0x26<br/>U, 0x7DU, 0xEBU, 0x84U, 0x2CU, 0xEFU, 0x42U, 0x62U, 0x23<br/>U, 0x7AU, 0xC8U, 0x75U, 0x72U, 0x50U, 0x68U, 0xF3U, 0x25<br/>U, 0x63U, 0xB4U, 0x78U, 0x4CU, 0xA8U, 0xD6U, 0xA9U, 0x9F<br/>U, 0x34U, 0xC8U, 0x8BU, 0x76U, 0xB9U, 0x71U, 0x45U, 0xB2<br/>U, 0xE8U, 0x52U, 0x9EU, 0xC8U, 0xADU, 0xEAU, 0x83U, 0xEA<br/>U, 0xD4U, 0xECU, 0x63U, 0xE3U, 0xFFU, 0x2DU, 0x17U, 0xA2<br/>U, 0xFFU, 0xEFU, 0xB0U, 0x5CU, 0x90U, 0x2CU, 0xA7U, 0xA9<br/>U, 0x21U, 0x68U, 0x37U, 0x8CU, 0x89U, 0xF7U, 0x5CU, 0x92<br/>U, 0x8FU, 0xC4U, 0xF0U, 0x70U, 0x7EU, 0x43U, 0x48U, 0x7A<br/>U, 0x4FU, 0x47U, 0xDFU, 0x70U, 0x4AU, 0xE8U, 0x7EU, 0x24<br/>U, 0x27U, 0x2CU, 0x13U, 0x6DU, 0x3EU, 0x98U, 0xCFU, 0x59<br/>U, 0x06U, 0x6DU, 0x41U, 0xA3U, 0xD0U, 0x38U, 0x85U, 0x7D<br/>U, 0x07U, 0x3DU, 0x8BU, 0x4DU, 0x2CU, 0x27U, 0x8BU, 0xF0<br/>U, 0xEAU, 0x6BU, 0xFAU, 0x50U, 0xD2U, 0x63U, 0x09U, 0x1A<br/>U, 0x4AU, 0x18U, 0xC6U, 0x3FU, 0x44U, 0x6BU, 0xC9U, 0xA6<br/>U, 0x1EU, 0x8CU, 0x4AU, 0x68U, 0x83U, 0x47U, 0xB2U, 0x43<br/>U, 0x5EU, 0xC8U, 0xE7U, 0x2EU, 0xDDU, 0xAEU, 0xA7U,</p> | <p>0x9AU, 0x36U, 0x90U, 0x1CU, 0xDCU, 0x9CU, 0x7BU, 0x4FU,<br/>0xBAU, 0x81U, 0xAFU, 0x93U, 0x6BU, 0x42U, 0x5FU, 0xE1U, 0<br/>xE0U, 0x7EU, 0x24U, 0x98U, 0x4BU, 0xA0U, 0xBAU, 0x17U, 0<br/>xA9U, 0x6FU, 0x41U, 0x4FU, 0x29U, 0xD4U, 0x47U, 0x48U, 0x<br/>46U, 0x46U, 0xA8U, 0xB3U, 0xCBU, 0x84U, 0x70U, 0xD5U, 0x<br/>C3U, 0xE1U, 0x25U, 0xABU, 0x72U, 0xD7U, 0xD5U, 0x70U, 0x<br/>C5U, 0x17U, 0x8AU, 0xAFU, 0x3BU, 0x81U, 0x3BU, 0x27U, 0x<br/>DCU, 0xF8U, 0xAFU, 0xDBU, 0xBFU, 0x8CU, 0x2FU, 0xB2U, 0x<br/>F9U, 0x93U, 0x96U, 0x84U, 0x8AU, 0xD7U, 0xB1U, 0x9CU, 0x<br/>9AU, 0xE0U, 0x4CU, 0x0DU, 0xFEU, 0x67U, 0x3EU, 0x94U, 0x<br/>97U, 0xAAU, 0x03U, 0x9BU, 0x12U, 0xEFU, 0x73U, 0x55U, 0x<br/>EEU, 0xE3U, 0xD0U, 0x50U, 0x96U, 0x71U, 0xD5U, 0xCBU, 0x<br/>81U, 0xDCU, 0xC7U, 0xC9U, 0x8CU, 0x25U, 0x31U, 0xC7U, 0x<br/>1CU, 0xF4U, 0xC2U, 0xF3U, 0xB1U, 0xE7U, 0x05U, 0xCDU, 0x<br/>B2U, 0xBEU, 0x77U, 0x5FU, 0xCDU, 0xE1U, 0xDDU, 0x9EU, 0x<br/>CAU, 0xE7U, 0x28U, 0xF9U, 0x00U, 0x28U, 0x6EU, 0x69U, 0x3<br/>FU, 0x2EU, 0x74U, 0xE3U, 0x8CU, 0xD7U, 0xC1U, 0x5CU, 0x9<br/>3U, 0x33U, 0x4AU, 0xA5U, 0xBAU, 0x8BU, 0x8U, 0x2CU, 0xD<br/>EU, 0x4BU, 0xF2U, 0x93U, 0xDFU, 0x8BU, 0x9U, 0x05U, 0xD<br/>0U, 0x02U, 0xD5U, 0x19U, 0x1AU, 0xC7U, 0x86U, 0xFEU, 0x5<br/>8U, 0x62U, 0x8EU, 0x5FU, 0xB9U, 0x82U, 0x22U, 0xC5U, 0xFF<br/>U, 0xA4U, 0x06U, 0x4CU, 0x79U, 0x43U, 0x6AU, 0xB5U, 0x18<br/>U, 0x04U, 0x1EU, 0x98U, 0x1BU, 0x8CU, 0xB4U, 0x63U, 0x4D<br/>U, 0xDAU, 0x02U, 0xA7U, 0x76U, 0x69U, 0xF7U, 0xFBU, 0xA8<br/>U, 0xEFU, 0x03U, 0xE4U, 0xB6U, 0x05U, 0x27U, 0xA0U, 0x92<br/>U, 0x56U, 0x9CU, 0xA3U, 0x1CU, 0x81U, 0xB6U, 0xE3U, 0xE4<br/>U, 0xCAU, 0x48U, 0x09U, 0xD0U, 0xC7U, 0x20U, 0x67U, 0x6F<br/>U, 0xDDU, 0x18U, 0x5AU, 0x8CU, 0x5AU, 0x79U, 0x26U, 0x9A<br/>U, 0x80U, 0xFFU, 0x57U, 0x09U, 0xD9U, 0xF1U, 0xEDU, 0x01<br/>U, 0xB9U, 0x9BU, 0x8BU, 0x6BU, 0x82U, 0x8BU, 0x13U, 0xE5<br/>U, 0xB3U, 0xA7U, 0xD7U, 0x27U, 0xC3U, 0x89U, 0x76U, 0xD9<br/>U, 0xB7U, 0xE1U, 0x25U, 0x46U, 0xF2U, 0xD0U, 0x8AU</p> | <p>0x8BU, 0xBAU, 0x6BU, 0xF8U, 0x2AU, 0x6CU, 0x0FU, 0x86U,<br/>0xD5U, 0xF1U, 0x75U, 0x6EU, 0x97U, 0x95U, 0x68U, 0x70U, 0<br/>x80U, 0x89U, 0x53U, 0x80U, 0x6BU, 0x4EU, 0xB2U, 0x05U, 0x<br/>BCU, 0x16U, 0x94U, 0xEEU,</p> |

### 8.4.5.3 AuthHsmEcies

#### Raw Key 방식 사용시

#### 1) Encryption

| Sender Private Key                                                                                                                                                                                                                                | Sender Public Key                                                                                                                                                                                                                                                          | Receiver Public Key                                                                                                                                                                                                                                                        |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>0x13U, 0x84U, 0xC3U, 0x1DU, 0x69U, 0x82U, 0xD5U, 0x2BU,<br/>0xCAU, 0x3BU, 0xEDU, 0x8AU, 0x7EU, 0x60U, 0xF5U, 0x2FU,<br/>0xECU, 0xDAU, 0x84U, 0x4EU, 0x5CU, 0x0EU, 0xA1U, 0x66U,<br/>0x81U, 0x5AU, 0x81U, 0x59U, 0xE0U, 0x9FU, 0xFBU, 0x42U</p> | <p>/* x */<br/>0xF4U, 0x5AU, 0x99U, 0x13U, 0x7BU, 0x1BU, 0xB2U, 0xC1U,<br/>0x50U, 0xD6U, 0xD8U, 0xCFU, 0x72U, 0x92U, 0xCAU, 0x07U,<br/>0xDAU, 0x68U, 0xC0U, 0x03U, 0xDAU, 0xA7U, 0x66U, 0xA9U,<br/>0xAFU, 0x7FU, 0x67U, 0xF5U, 0xEEU, 0x91U, 0x68U, 0x28U,<br/>/* y */</p> | <p>/* x */<br/>0x8CU, 0x5EU, 0x20U, 0xFEU, 0x31U, 0x93U, 0x5FU, 0x6FU,<br/>0xA6U, 0x82U, 0xA1U, 0xF6U, 0xD4U, 0x6EU, 0x44U, 0x68U,<br/>0x53U, 0x4FU, 0xFEU, 0xA1U, 0xA6U, 0x98U, 0x81U, 0x48U,<br/>0x0BU, 0x12U, 0x51U, 0x3EU, 0xEDU, 0x8DU, 0xEBU, 0x11U,<br/>/* y */</p> |

|  |                                                                                                                                                                                                                                |                                                                                                                                                                                                                                |
|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | 0xF6U, 0xA2U, 0x52U, 0x16U, 0xF4U, 0x4CU, 0xB6U, 0x4AU, 0x96U, 0xC2U, 0x29U, 0xAEU, 0x00U, 0xB4U, 0x79U, 0x85U, 0x7BU, 0x3BU, 0x81U, 0xC1U, 0x31U, 0x9FU, 0xB2U, 0xADU, 0xF0U, 0xE8U, 0xDBU, 0x26U, 0x81U, 0x76U, 0x97U, 0x29U | 0x12U, 0x70U, 0xFEU, 0xC2U, 0x42U, 0x7EU, 0x6AU, 0x15U, 0x4DU, 0xFCU, 0xAEU, 0x33U, 0x68U, 0x58U, 0x43U, 0x96U, 0xC8U, 0x25U, 0x1AU, 0x04U, 0xE2U, 0xAEU, 0x7DU, 0x87U, 0xB0U, 0x16U, 0xFFU, 0x65U, 0xD2U, 0x2DU, 0x6FU, 0x9EU |
|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

| Plain Text                                                                                                     | Parameter 1                                                                                                                                                                                                                    | Cipher Text                                                                                    |
|----------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| 0x91U, 0x69U, 0x15U, 0x5BU, 0x08U, 0xB0U, 0x76U, 0x74U, 0xCB0, 0xADU, 0xF7U, 0x5FU, 0xB4U, 0x6AU, 0x7BU, 0x0DU | 0xA6U, 0xB7U, 0xB5U, 0x25U, 0x54U, 0xB4U, 0x20U, 0x3FU, 0x7EU, 0x3AU, 0xCFU, 0xDBU, 0x3AU, 0x3EU, 0xD8U, 0x67U, 0x4EU, 0xE0U, 0x86U, 0xCEU, 0x59U, 0x06U, 0xA7U, 0xCAU, 0xC2U, 0xF8U, 0xA3U, 0x98U, 0x30U, 0x6DU, 0x3BU, 0xE9U | 0xA6, 0x34, 0x20, 0x13, 0xD6, 0x23, 0xAD, 0x6C, 0x5F, 0x68, 0x82, 0x46, 0x96, 0x73, 0xAE, 0x33 |

## 2) Decryption

| Reciever Private Key                                                                                                                                                                                                           | Parameter 1                                                                                                                                                                                                                    | Sender Public Key                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0x06U, 0x0EU, 0x41U, 0x44U, 0x0AU, 0x4EU, 0x35U, 0x15U, 0x4CU, 0xA0U, 0xEFU, 0xC8U, 0x52U, 0x41U, 0x21U, 0x45U, 0x83U, 0x6AU, 0xD0U, 0x32U, 0x83U, 0x3EU, 0x6BU, 0xC7U, 0x81U, 0xE5U, 0x33U, 0xBFU, 0x14U, 0x85U, 0x10U, 0x85U | 0xA6U, 0xB7U, 0xB5U, 0x25U, 0x54U, 0xB4U, 0x20U, 0x3FU, 0x7EU, 0x3AU, 0xCFU, 0xDBU, 0x3AU, 0x3EU, 0xD8U, 0x67U, 0x4EU, 0xE0U, 0x86U, 0xCEU, 0x59U, 0x06U, 0xA7U, 0xCAU, 0xC2U, 0xF8U, 0xA3U, 0x98U, 0x30U, 0x6DU, 0x3BU, 0xE9U | $/* x */$<br>0xF4U, 0x5AU, 0x99U, 0x13U, 0x7BU, 0x18U, 0xB2U, 0xC1U, 0x50U, 0xD6U, 0xD8U, 0xCFU, 0x72U, 0x92U, 0xCAU, 0x07U, 0xDAU, 0x68U, 0xC0U, 0x03U, 0xDAU, 0xA7U, 0x66U, 0xA9U, 0xAFU, 0x7FU, 0x67U, 0xF5U, 0xEEU, 0x91U, 0x68U, 0x28U, $/* y */$<br>0xF6U, 0xA2U, 0x52U, 0x16U, 0xF4U, 0x4CU, 0xB6U, 0x4AU, 0x96U, 0xC2U, 0x29U, 0xAEU, 0x00U, 0xB4U, 0x79U, 0x85U, 0x7BU, 0x3BU, 0x81U, 0xC1U, 0x31U, 0x9FU, 0xB2U, 0xADU, 0xF0U, 0xE8U, 0xDBU, 0x26U, 0x81U, 0x76U, 0x97U, 0x29U |

| Cipher Text                                                                                    | Tag                                                                                            | Plain Text                                                                                                     |
|------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| 0xA6, 0x34, 0x20, 0x13, 0xD6, 0x23, 0xAD, 0x6C, 0x5F, 0x68, 0x82, 0x46, 0x96, 0x73, 0xAE, 0x33 | 0x80, 0xE1, 0xD8, 0x5D, 0x30, 0xF1, 0xBA, 0xE4, 0xEC, 0xF1, 0xA5, 0x34, 0xA8, 0x9A, 0x07, 0x86 | 0x91U, 0x69U, 0x15U, 0x5BU, 0x08U, 0xB0U, 0x76U, 0x74U, 0xCB0, 0xADU, 0xF7U, 0x5FU, 0xB4U, 0x6AU, 0x7BU, 0x0DU |

## Signature

### 8.4.6

#### 8.4.6.1 AuthHsmPkcs1V15Sha1

##### Raw Key 방식 사용시

##### Generation, Verification

| Private Key                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            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| $/* \text{modulus} : 256 \text{ bytes} */$<br>0xc2U, 0xc2U, 0xe6U, 0x41U, 0xf6U, 0x84U, 0xbeU, 0xe9U, 0xb3U, 0x03U, 0xc6U, 0x8aU, 0xb8U, 0x15U, 0xe7U, 0x08U, 0x17U, 0x50U, 0xcbU, 0x90U, 0xe1U, 0x59U, 0x42U, 0xcFU, 0x17U, 0x63U, 0x5aU, 0x54U, 0xdbU, 0xb2U, 0x62U, 0x77U, 0x70U, 0x38U, 0xb7U, 0x93U, 0x13U, 0x1aU, 0x85U, 0x01U, 0x16U, 0x0eU, 0x7dU, 0xeeU, 0xc1U, 0xe8U, 0xa7U, 0xf7U, 0x24U, 0xf1U, 0x84U, 0x23U, 0x5bU, 0xf2U, 0x5aU, 0xa3U, 0xc5U, 0x04U, 0x20U, 0x75U, 0xa9U, 0x9cU, 0x14U, 0x9bU, 0xaaU, 0x00U, 0x2dU, 0xb1U, 0x80U, 0xa6U, 0x68U, 0xb1U, 0x40U, 0xc5U, 0xe4U, 0x6aU, 0xf2U, 0x9fU, 0xf5U, 0xc5U, 0x69U, 0xdfU, 0x1bU, 0x1eU, 0x84U, 0x1eU, 0xdfU, 0xd9U, 0x52U, 0xc3U, 0xbbU, 0x96U, 0x6fU, 0xe8U, 0xf1U, 0x02U, 0xc5U, 0x1cU, 0x81U, 0x24U, 0x8eU, 0xd6U, 0x0eU, 0x5aU, 0x57U, 0xe5U, 0x29U, 0xb8U, 0x37U, 0x0eU, 0x23U, 0x4eU, 0x8fU, 0xdaU, 0xb0U, 0x7dU, 0xcaU, 0x04U, 0xe2U, 0xebU, 0x8fU, 0x79U, 0xe4U, 0xd8U, 0xadU, 0x8eU, 0x02U, 0x69U, 0x87U, 0x48U, 0x19U, 0x2bU, 0x01U, 0x9eU, 0xabU, 0x8fU, 0x83U, 0x55U, 0x46U, 0x1aU, 0x7cU, 0x3cU, 0xa7U, 0xb4U, 0xc9U, 0x90U, 0xc3U, 0x49U, 0xd2U, 0x0dU, 0xbcU, 0xd7U, 0x51U, 0x1aU, 0xccU, 0x60U, 0x9dU, 0x9bU, 0x0cU, 0xe1U, 0x18U, 0x10U, 0x19U, 0xd2U, 0x48U, 0x1bU, 0xf1U, 0xfeU, 0x50U, 0x89U, 0x49U, 0xfeU, 0xa1U, 0x82U, 0xd9U, 0x7aU, 0x42U, 0x5eU, 0x6aU, 0x0eU, 0x26U, 0x5cU, 0x61U, 0x36U, 0xd2U, 0x67U, 0x8eU, 0x55U, 0xeeU, 0xa2U, 0xbfbU, 0xe9U, 0xfaU, 0xaeU, 0xddU, 0x1fU, 0x0eU, 0xadU, 0x4eU, 0xd3U, 0x5bU, 0x61U, 0x02U, 0x0fU, 0x8dU, 0x80U, 0xb6U, 0xfeU, 0x12U, 0x7dU, 0x95U, 0x25U, 0x76U, 0x95U, 0x4dU, 0x6aU, 0x73U, 0x41U, 0x05U, 0x62U, 0xcbU, 0x72U, 0x40U, 0x0eU, 0x01U, 0x44U, 0x20U, 0xaeU, 0x51U, 0x04U, 0x57U, 0x97U, 0x72U, 0xf7U, 0x53U, 0x99U, 0xcdU, 0x61U, 0x02U, 0x9FU, 0x4aU, 0x3fU, 0xe0U, 0x0eU, 0xa0U, 0x41U, 0x44U, 0xd7U, 0x77U, 0xbbU, 0xb0U, 0x94U, 0x85U, 0xe0U, 0x1cU, 0x59U, $/* \text{exponent} : 256 \text{ bytes} */$<br>0x3fU, 0xd6U, 0x34U, 0x34U, 0xc0U, 0xc1U, 0xa7U, 0xf5U, 0xdfU, 0xc8U, 0x48U, 0x06U, 0x65U, 0xaeU, 0x3fU, 0xecU, 0xa5U, 0xffU, 0xc4U, 0x89U, 0x7bU, 0x63U, 0x8fU, 0x83U, 0x63U, 0x8eU, 0x1cU, 0x28U, 0x9eU, 0x4eU, 0x2cU, 0xfaU, 0x2bU, 0x1dU, 0x5cU, 0x06U, 0x68U, 0x07U, 0x94U, 0x3aU, 0xd3U, 0xe8U, 0xf2U, 0x1cU, 0x15U, 0x10U, 0xd6U, 0x25U, 0x2aU, 0xc5U, 0xc2U, 0xe2U, 0xf8U, 0x1bU, 0x4aU, 0xdcdU, 0x4dU, 0x12U, 0x5aU, 0x35U, 0x56U, 0xdfU, 0xa2U, 0xddU, 0x96U, 0xeeU, 0x32U, 0x0bU, 0xf1U, 0x38U, 0x1aU, 0x2eU, 0xa7U, 0x25U, 0x76U, 0x06U, 0x1bU, 0xfaU, 0x94U, 0x4bU, 0x1dU, 0xb5U, 0x94U, 0x24U, 0xffU, 0x77U, 0x43U, 0x35U, 0xefU, 0x6bU, 0xdaU, 0x8bU, 0xa7U, 0xa8U, 0xb9U, 0x7eU, 0x87U, 0xefU, 0x1cU, 0xe1U, 0x57U, 0x70U, 0x04U, 0x37U, 0xa6U, 0x81U, 0xa5U, 0x98U, 0x0dU, 0x2FU, 0xd1U, 0x70U, 0xc9U, 0xb4U, 0x0bU, 0xacU, 0xdfU, 0x39U, 0x94U, 0xcbU, | $/* \text{modulus} : 256 \text{ bytes} */$<br>0xc2U, 0xc2U, 0xe6U, 0x41U, 0xf6U, 0x84U, 0xbeU, 0xe9U, 0xb3U, 0x03U, 0xc6U, 0x8aU, 0xb8U, 0x15U, 0xe7U, 0x08U, 0x17U, 0x50U, 0xcbU, 0x90U, 0xe1U, 0x59U, 0x42U, 0xcFU, 0x17U, 0x63U, 0x5aU, 0x54U, 0xdbU, 0xb2U, 0x62U, 0x77U, 0x70U, 0x38U, 0xb7U, 0x93U, 0x13U, 0x1aU, 0x85U, 0x01U, 0x16U, 0x0eU, 0x7dU, 0xeeU, 0xc1U, 0xe8U, 0xa7U, 0xf7U, 0x24U, 0xf1U, 0x84U, 0x23U, 0x5bU, 0xf2U, 0x5aU, 0xa3U, 0xc5U, 0x04U, 0x20U, 0x75U, 0xa9U, 0x9cU, 0x14U, 0x9bU, 0xaaU, 0x00U, 0x2dU, 0xb1U, 0x80U, 0xa6U, 0x68U, 0xb1U, 0x40U, 0xc5U, 0xe4U, 0x6aU, 0xf2U, 0x9fU, 0xf5U, 0xc5U, 0x69U, 0xdfU, 0x1bU, 0x1eU, 0x84U, 0x1eU, 0xdfU, 0xd9U, 0x52U, 0xc3U, 0xbbU, 0x96U, 0x6fU, 0xe8U, 0xf1U, 0x02U, 0xc5U, 0x1cU, 0x81U, 0x24U, 0x8eU, 0xd6U, 0x0eU, 0x5aU, 0x57U, 0xe5U, 0x29U, 0xb8U, 0x37U, 0x0eU, 0x23U, 0x4eU, 0x8fU, 0xdaU, 0xb0U, 0x7dU, 0xcaU, 0x04U, 0xe2U, 0xebU, 0x8fU, 0x79U, 0xe4U, 0xd8U, 0xadU, 0x8eU, 0x02U, 0x69U, 0x87U, 0x48U, 0x19U, 0x2bU, 0xc0U, 0x9eU, 0xabU, 0x8fU, 0x83U, 0x55U, 0x46U, 0x1aU, 0x7cU, 0x3cU, 0xa7U, 0xb4U, 0xc9U, 0x90U, 0xc3U, 0x49U, 0xd2U, 0x0dU, 0xbcbU, 0xd7U, 0x51U, 0x1aU, 0xccU, 0x60U, 0x9dU, 0x9bU, 0x0dU, 0xe1U, 0x18U, 0x10U, 0x19U, 0xd2U, 0x48U, 0x1bU, 0xf1U, 0xfeU, 0x50U, 0x89U, 0x49U, 0xfeU, 0xa1U, 0x82U, 0xd9U, 0x7aU, 0x42U, 0x5eU, 0x6aU, 0xd6U, 0x26U, 0x5cU, 0x61U, 0x36U, 0xd2U, 0x67U, 0x8eU, 0x55U, 0xeeU, 0xa2U, 0xbfbU, 0xe9U, 0xfaU, 0xaeU, 0xddU, 0x1fU, 0x0eU, 0xadU, 0x4eU, 0xd3U, 0x5bU, 0x61U, 0x02U, 0x0fU, 0x8dU, 0x80U, 0xb6U, 0xfeU, 0x12U, 0x7dU, 0x95U, 0x25U, 0x76U, 0x95U, 0x4dU, 0x6aU, 0x73U, 0x41U, 0x05U, 0x62U, 0xcbU, 0x72U, 0x40U, 0x0eU, 0x01U, 0x44U, 0x20U, 0xaeU, 0x51U, 0x04U, 0x57U, 0x97U, 0x72U, 0xf7U, 0x53U, 0x99U, 0xcdU, 0x61U, 0x02U, 0x9FU, 0x4aU, 0x3fU, 0xe0U, 0x0eU, 0xa0U, 0x41U, 0x44U, 0xd7U, 0x77U, 0xbbU, 0xb0U, 0x94U, 0x85U, 0xe0U, 0x1cU, 0x59U, $/* \text{exponent} : 4 \text{ bytes} */$<br>0x00U, 0x01U, 0x00U, 0x01U, | 0x61U, 0x62U, 0x63U, 0x64U, 0x62U, 0x63U, 0x64U, 0x65U, 0x63U, 0x64U, 0x65U, 0x66U, 0x64U, 0x65U, 0x66U, 0x67U, 0x65U, 0x66U, 0x67U, 0x68U, 0x66U, 0x67U, 0x68U, 0x69U, 0x67U, 0x68U, 0x69U, 0x6AU, 0x68U, 0x69U, 0x6AU, 0x68U, 0x69U, 0x6AU, 0x68U, 0x6CU, 0x6AU, 0x68U, 0x6CU, 0x6DU, 0x68U, 0x6CU, 0x6DU, 0x6EU, 0x6CU, 0x6DU, 0x6EU, 0x6CU, 0x6DU, 0x6EU, 0x6FU, 0x6DU, 0x6EU, 0x6FU, 0x70U, 0x6EU, 0x6FU, 0x70U, 0x71U                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            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|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        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                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 0x18U, 0x39U, 0x03U, 0x0CU, 0x21U, 0xF4U, 0xE3U, 0x0CU, 0x53U, 0x13U, 0xF5U, 0x8DU, 0xC2U, 0xF6U, 0x2DU, 0x0DU, 0xD2U, 0xD0U, 0x5DU, 0x12U, 0xFFU, 0x37U, 0x3AU, 0x5BU, 0x36U, 0xDEU, 0xADU, 0x6DU, 0xBAU, 0x1AU, 0xCFU, 0xA4U, 0x1DU, 0x3BU, 0x4CU, 0x93U, 0xAU, 0xFDU, 0x0FU, 0x69U, 0xDDU, 0x79U, 0x3BU, 0x68U, 0xADU, 0xE1U, 0x8FU, 0xB5U, 0x44U, 0x28U, 0x97U, 0x50U, 0x0AU, 0x0AU, 0x61U, 0x67U, 0x94U, 0x78U, 0x31U, 0x2EU, 0x0DU, 0x13U, 0xDDU, 0xDAU, 0x6DU, 0x03U, 0x53U, 0xC5U, 0xB4U, 0xDAU, 0xA9U, 0x34U, 0x0EU, 0xF7U, 0xE1U, 0x5FU, 0x0AU, 0x81U, 0x59U, 0xCBU, 0xB2U, 0x4BU, 0xEU, 0xE9U, 0xD4U, 0x7AU, 0xD1U, 0xDCU, 0x8CU, 0x3EU, 0xD8U, 0xD8U, 0xDAU, 0xCDU, 0xEFU, 0xF2U, 0x15U, 0x57U, 0x24U, 0x5EU, 0x8BU, 0xC2U, 0x91U, 0x80U, 0x71U, 0xE0U, 0x40U, 0xE8U, 0xD6U, 0x39U, 0xD7U, 0xD3U, 0xCEU, 0xA2U, 0x00U, 0xF2U, 0x3BU, 0x42U, 0xD6U, 0x30U, 0x16U, 0x5CU, 0x65U, 0x20U, 0xC2U, 0x49U, 0x53U, 0x78U, 0xE8U, 0x0BU, 0xF2U, 0xC2U, 0x83U, 0x9FU, 0x61U, 0xB5U, 0xC8U, 0xA2U, 0xCAU, 0xF1U, 0x11U, 0xC7U, 0x46U, 0x5BU, 0x6EU, 0x75U, 0xD2U, 0x91U, 0x68U, 0xAEU, 0xB2U, 0xB7U, 0xC5U, 0x60U, 0x6FU, 0xBDU, 0xD2U, 0x28U, 0x17U, 0x45U, 0x68U, 0x7EU, 0xA7U, 0x24U, 0x82U, 0x28U, 0x7BU, 0x71U, 0x26U, 0x6DU, 0xFBU, 0x5BU, 0x53U, 0xD6U, 0xACU, 0xF8U, 0x51U, 0xF8U, 0x6CU, 0xB6U, 0x55U, 0x32U, 0x90U, 0x43U, 0x95U, 0x6EU, 0xE1U, 0x8DU, 0x1U, 0x6FU, 0x8EU, 0xF0U, 0x8BU, 0x09U, 0x49U, 0x06U, 0xF1U, 0x87U, 0xE0U, 0x69U, 0x5FU, 0x45U, 0x13U, 0x28U, 0xE8U, 0xF0U, 0x5FU, 0x3BU, 0x41U, 0xF1U, 0x34U, 0xB4U, 0xCBU, 0x71U, 0x89U, 0x95U, 0xD3U, 0xC6U, 0x00U, 0x99U, 0xFEU, 0x45U, 0xE2U, |

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| 0x53U, 0x1eU, 0xc8U, 0x04U, 0xc1U, 0x76U, 0x3FU, 0x91U,<br>0x96U, 0x32U, 0x35U, 0x32U, 0x12U, 0x73U, 0x42U, 0x7eU,<br>0x52U, 0x17U, 0xcdU, 0x87U, 0x2FU, 0xbeU, 0xebU, 0xd1U,<br>0x8FU, 0x80U, 0x2bU, 0x8aU, 0xfdU, 0xe1U, 0x15U, 0x9FU,<br>0xaaU, 0x6FU, 0xc4U, 0x6cU, 0x0FU, 0xeeU, 0xc6U, 0xe5U,<br>0xb6U, 0x4eU, 0x07U, 0xa2U, 0xa1U, 0xfdU, 0x89U, 0x82U,<br>0xdFU, 0x33U, 0x82U, 0x19U, 0x3eU, 0x68U, 0x46U, 0xcFU,<br>0x66U, 0xeaU, 0x98U, 0x50U, 0xaaU, 0x10U, 0x2dU, 0x31U,<br>0x3dU, 0x63U, 0xf7U, 0x22U, 0x70U, 0x7FU, 0xbcU, 0x61U,<br>0x5aU, 0x41U, 0x22U, 0xb4U, 0xe6U, 0xfeU, 0x0bU, 0xddU,<br>0x09U, 0x42U, 0x14U, 0xabU, 0xb4U, 0x04U, 0xe1U, 0x23U,<br>0xb7U, 0xaeU, 0xf7U, 0x5FU, 0x23U, 0x43U, 0x1bU, 0x7eU,<br>0x65U, 0x21U, 0xfbU, 0xe9U, 0x2dU, 0x8eU, 0x01U, 0xaaU,<br>0xf9U, 0x7FU, 0xdcU, 0x85U, 0x01U, 0x1dU, 0xe3U, 0x21U,<br>0x4aU, 0xefU, 0xeaU, 0x9aU, 0x72U, 0x8cU, 0xf6U, 0x69U,<br>0x29U, 0x44U, 0xb9U, 0xdcU, 0xf9U, 0xd2U, 0xc9U, 0xcbU,<br>0xdfU, 0x97U, 0x82U, 0x96U, 0x5cU, 0x69U, 0x77U, 0x91U | 0x0CU, 0xE2U, 0x22U, 0x9CU, 0x3CU, 0xF0U, 0x12U, 0x17U,<br>0xC2U, 0x35U, 0x0EU, 0x93U, 0xB2U, 0x8FU, 0x91U, 0x37U,<br>0x3EU, 0x0CU, 0xE6U, 0xA5U, 0x51U, 0x7AU, 0x1EU, 0x31U,<br>0x27U, 0xADU, 0xE2U, 0x10U, 0x3BU, 0x9BU, 0xCFU, 0x51U |
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#### 8.4.6.2 AutHsmPkcs1V15Sha1Seed

## Raw Key 방식 사용시

## Verification

| Public Key                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Hash(Seed)                                                                                                                                          | Signature                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
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| /* modulus : 256 bytes */<br>0xc2U, 0xc2U, 0xe6U, 0x41U, 0xf6U, 0x84U, 0xbeU, 0xe9U,<br>0xb3U, 0x03U, 0xc6U, 0x8aU, 0xb8U, 0x15U, 0xe7U, 0x08U,<br>0x17U, 0x50U, 0xcbU, 0x90U, 0xe1U, 0x59U, 0x42U, 0xcfcU,<br>0x17U, 0x63U, 0x5aU, 0x54U, 0xd0U, 0xb2U, 0x62U, 0x77U,<br>0x70U, 0x38U, 0xb7U, 0x93U, 0x13U, 0x1aU, 0x85U, 0x07U,<br>0x16U, 0x0eU, 0x7dU, 0xeeU, 0xc1U, 0xe8U, 0xa7U, 0xf7U,<br>0x24U, 0xf1U, 0x84U, 0x23U, 0x5bU, 0xf2U, 0x5aU, 0xa3U,<br>0xc5U, 0x04U, 0x20U, 0x75U, 0xa9U, 0x9cU, 0x14U, 0x9bU,<br>0xaaU, 0x00U, 0x2dU, 0xb1U, 0x80U, 0xa6U, 0x68U, 0xb1U,<br>0x40U, 0xc5U, 0xe4U, 0x6aU, 0xf2U, 0x9fU, 0xf5U, 0xc5U,<br>0x69U, 0xdfU, 0x1bU, 0x1eU, 0x84U, 0x1eU, 0xdfU, 0xd9U,<br>0x52U, 0xc3U, 0xbbU, 0x96U, 0x6fU, 0xe8U, 0xf1U, 0x02U,<br>0xc5U, 0x1cU, 0x81U, 0x24U, 0xbeU, 0xd6U, 0xdelU, 0x5aU,<br>0x57U, 0xe5U, 0x29U, 0xb8U, 0x37U, 0xdelU, 0x23U, 0x4eU,<br>0x8fU, 0xdaU, 0x08U, 0xd7U, 0xcaU, 0x04U, 0xe2U, 0xebU,<br>0x8fU, 0x79U, 0x94U, 0xd8U, 0xadU, 0x8eU, 0x02U, 0x69U,<br>0x87U, 0x48U, 0x19U, 0x2bU, 0x01U, 0x9eU, 0xabU, 0x8fU,<br>0x83U, 0x55U, 0x46U, 0x1aU, 0x7cU, 0x3cU, 0xa7U, 0xb4U,<br>0xc9U, 0x90U, 0x03U, 0x49U, 0xd2U, 0xd0U, 0xbcU, 0xd7U,<br>0x51U, 0x1aU, 0x9cU, 0x60U, 0x9dU, 0x9bU, 0x0dU, 0xe1U,<br>0x86U, 0x10U, 0x19U, 0xd2U, 0x48U, 0x1bU, 0xf1U, 0xf6U,<br>0x50U, 0x89U, 0x49U, 0xf6U, 0xa1U, 0x82U, 0xd9U, 0x7aU,<br>0x42U, 0x5eU, 0x6aU, 0xdelU, 0x26U, 0x5cU, 0x61U, 0x36U,<br>0xd2U, 0x67U, 0x8eU, 0x55U, 0xeeU, 0xa2U, 0xbfU, 0xe9U,<br>0xfauU, 0xaeU, 0xddU, 0x1fuU, 0xdelU, 0xadU, 0x4eU, 0xd3U,<br>0x5bU, 0x61U, 0x02U, 0x0fU, 0x8dU, 0x80U, 0xb6U, 0xf6U,<br>0x12U, 0x7dU, 0x95U, 0x25U, 0x76U, 0x95U, 0x4dU, 0x6aU,<br>0x73U, 0x41U, 0x05U, 0x62U, 0xcbU, 0x72U, 0x40U, 0x0eU,<br>0x01U, 0x44U, 0x20U, 0xaeU, 0x51U, 0x04U, 0x57U, 0x97U,<br>0x72U, 0xf7U, 0x53U, 0x99U, 0xcdU, 0x61U, 0x02U, 0x9fU,<br>0x4aU, 0x3fU, 0xe0U, 0xe0U, 0xa0U, 0x41U, 0x44U, 0xd7U,<br>0x77U, 0xbbbU, 0xb0U, 0x94U, 0x85U, 0xe0U, 0x1cU, 0x59U,<br>/* exponent : 4 bytes */<br>0x00U, 0x01U, 0x00U, 0x01U, | 0x84U, 0x98U, 0x3eU, 0x44U, 0x1cU,<br>0x3bU, 0xd2U, 0x6eU, 0xbaU, 0xaeU,<br>0x4aU, 0xa1U, 0xf9U, 0x51U, 0x29U,<br>0xe5U, 0xe5U, 0x46U, 0x70U, 0xf1U | 0x18U, 0x39U, 0x03U, 0x0CU, 0x21U, 0xF4U, 0xE3U, 0x0CU,<br>0x53U, 0x13U, 0xF5U, 0x8DU, 0xC2U, 0xF6U, 0xD2U, 0xD0U,<br>0xD2U, 0xDDU, 0x5DU, 0x12U, 0xFFU, 0x37U, 0x34U, 0x5BU,<br>0x36U, 0xDEU, 0xADU, 0x6DU, 0x8AU, 0x1AU, 0xCFU, 0xA4U,<br>0x1DU, 0x3BU, 0x4CU, 0x93U, 0x8AU, 0xFDU, 0x0FU, 0x69U,<br>0xDDU, 0x79U, 0x3BU, 0x68U, 0xADU, 0xE1U, 0x8FU, 0xB5U,<br>0x44U, 0x28U, 0x97U, 0x87U, 0x50U, 0x0AU, 0x0AU, 0x61U,<br>0x67U, 0x94U, 0x78U, 0x31U, 0x2EU, 0xD0U, 0x13U, 0xDDU,<br>0xDAU, 0x6DU, 0x03U, 0x53U, 0xC5U, 0xB4U, 0xDAU, 0xA9U,<br>0x34U, 0x0EU, 0xF7U, 0xE1U, 0x5FU, 0x0AU, 0x81U, 0x59U,<br>0xCBU, 0xB2U, 0x4BU, 0xECU, 0x8DU, 0x7AU, 0xD1U,<br>0xDCU, 0x8CU, 0x3EU, 0xD8U, 0xD8U, 0xDAU, 0xCDU, 0xEFU,<br>0xF2U, 0x15U, 0x57U, 0x24U, 0x5EU, 0x88U, 0xC2U, 0x91U,<br>0x80U, 0x71U, 0xE0U, 0x40U, 0xE8U, 0xD6U, 0x39U, 0xD7U,<br>0xD3U, 0xCEU, 0xA2U, 0x00U, 0xF2U, 0x3BU, 0x42U, 0xD6U,<br>0x30U, 0x16U, 0x5CU, 0x65U, 0x20U, 0xC2U, 0x49U, 0x53U,<br>0x78U, 0xE8U, 0x0BU, 0xF2U, 0xC2U, 0x83U, 0x9FU, 0x61U,<br>0x85U, 0xC8U, 0xA2U, 0xCAU, 0xF1U, 0x11U, 0xC7U, 0x46U,<br>0x5BU, 0x6EU, 0x75U, 0xd2U, 0x91U, 0x68U, 0xAEU, 0xB2U,<br>0x87U, 0xC5U, 0x60U, 0x6FU, 0xD8U, 0xD2U, 0x8U, 0x17U,<br>0x45U, 0x68U, 0x7EU, 0xA7U, 0x24U, 0xB2U, 0x28U, 0x7BU,<br>0x71U, 0x26U, 0x6DU, 0xFBU, 0x5BU, 0x53U, 0xD6U, 0xACU,<br>0xF8U, 0x51U, 0xF8U, 0x6CU, 0xB6U, 0x55U, 0x32U, 0x90U,<br>0x43U, 0x95U, 0x6EU, 0xE1U, 0x8DU, 0x21U, 0x6FU, 0x8EU,<br>0xF0U, 0x8BU, 0x09U, 0x49U, 0x06U, 0xF1U, 0x87U, 0xE0U,<br>0x69U, 0x5FU, 0x45U, 0x13U, 0x28U, 0xE8U, 0xF0U, 0x5FU,<br>0x3BU, 0x41U, 0xF1U, 0x34U, 0x84U, 0xCBU, 0x71U, 0x89U,<br>0x95U, 0xD3U, 0xC6U, 0x00U, 0x99U, 0xFEU, 0x45U, 0xE2U,<br>0x0CU, 0xE2U, 0x22U, 0x9CU, 0x3CU, 0xF0U, 0x12U, 0x17U,<br>0xC2U, 0x35U, 0x0EU, 0x93U, 0x82U, 0x8FU, 0x91U, 0x37U,<br>0x3EU, 0x0CU, 0xE6U, 0x45U, 0x51U, 0x7AU, 0x1EU, 0x31U,<br>0x27U, 0xADU, 0xE2U, 0x10U, 0x3BU, 0x9BU, 0xCFU, 0x51U |

#### 8.4.6.3 AutHsmPkcs1V15Sha256

## Raw Key 방식 사용시

## Generation, Verification

| Private Key                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Public Key                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Plain Text                                                                                                                                                                                                                                                                                                                                                                                                               |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| /* modulus : 256 bytes */<br>0xc2U, 0xc2U, 0xe6U, 0x41U, 0xf6U, 0x84U, 0x8bU, 0xe9U,<br>0xb3U, 0x03U, 0xc6U, 0x8aU, 0xb8U, 0x15U, 0xe7U, 0x08U,<br>0x17U, 0x50U, 0xcbbU, 0x90U, 0xe1U, 0x59U, 0x42U, 0xcFU,<br>0x17U, 0x63U, 0x5aU, 0x54U, 0xdbU, 0xb2U, 0x62U, 0x77U,<br>0x70U, 0x38U, 0xb7U, 0x93U, 0x13U, 0x1aU, 0x85U, 0x01U,<br>0x16U, 0x0eU, 0x7dU, 0xeeU, 0x1U, 0xe8U, 0xa7U, 0xf7U,<br>0x24U, 0xf1U, 0x84U, 0x23U, 0x5bU, 0xf2U, 0x5aU, 0xa3U,<br>0x5U, 0x04U, 0x20U, 0x75U, 0xa9U, 0x9cU, 0x14U, 0x9bU,<br>0xaaU, 0x00U, 0x2dU, 0xb1U, 0x80U, 0xa6U, 0x68U, 0xb1U,<br>0x40U, 0xc5U, 0xe4U, 0x6aU, 0xf2U, 0x9FU, 0xf5U, 0xc5U,<br>0x69U, 0xdfU, 0x1bU, 0x1eU, 0x84U, 0x1eU, 0xdfU, 0xd9U,<br>0x52U, 0xc3U, 0xbbU, 0x96U, 0x6FU, 0xe8U, 0xf1U, 0x02U,<br>0xc5U, 0x1cU, 0x81U, 0x24U, 0xbeU, 0xd6U, 0xd8U, 0x5aU,<br>0x57U, 0xe5U, 0x29U, 0xb8U, 0x37U, 0xdeU, 0x23U, 0x4eU,<br>0x8FU, 0xdaU, 0xb0U, 0xd7U, 0xcaU, 0x04U, 0x2U, 0xebU,<br>0x8FU, 0x79U, 0xe4U, 0xd8U, 0xadU, 0x8eU, 0x02U, 0x69U,<br>0x87U, 0x48U, 0x19U, 0x2bU, 0x01U, 0x9eU, 0xabU, 0x8FU,<br>0x83U, 0x5U, 0x46U, 0x1aU, 0x7cU, 0x3cU, 0xa7U, 0xb4U,<br>0xc9U, 0x90U, 0xc3U, 0x49U, 0x2U, 0xd0U, 0xbcU, 0xd7U, | /* modulus : 256 bytes */<br>0xc2U, 0xc2U, 0xe6U, 0x41U, 0xf6U, 0x84U, 0x8bU, 0xe9U,<br>0xb3U, 0x03U, 0xc6U, 0x8aU, 0xb8U, 0x15U, 0xe7U, 0x08U,<br>0x17U, 0x50U, 0xcbbU, 0x90U, 0xe1U, 0x59U, 0x42U, 0xcFU,<br>0x17U, 0x63U, 0x5aU, 0x54U, 0xdbU, 0xb2U, 0x62U, 0x77U,<br>0x70U, 0x38U, 0xb7U, 0x93U, 0x13U, 0x1aU, 0x85U, 0x01U,<br>0x16U, 0x0eU, 0x7dU, 0xeeU, 0xc1U, 0xe8U, 0xa7U, 0xf7U,<br>0x24U, 0xf1U, 0x84U, 0x23U, 0x5bU, 0xf2U, 0x5aU, 0xa3U,<br>0x5U, 0x04U, 0x20U, 0x75U, 0xa9U, 0x9cU, 0x14U, 0x9bU,<br>0xaaU, 0x00U, 0x2dU, 0xb1U, 0x80U, 0xa6U, 0x68U, 0xb1U,<br>0x40U, 0xc5U, 0xe4U, 0x6aU, 0xf2U, 0x9FU, 0xf5U, 0xc5U,<br>0x69U, 0xdfU, 0x1bU, 0x1eU, 0x84U, 0x1eU, 0xdfU, 0xd9U,<br>0x52U, 0xc3U, 0xbbU, 0x96U, 0x6FU, 0xe8U, 0xf1U, 0x02U,<br>0x5U, 0x1cU, 0x81U, 0x24U, 0xbeU, 0xd6U, 0xd8U, 0x5aU,<br>0x57U, 0xe5U, 0x29U, 0xb8U, 0x37U, 0xdeU, 0x23U, 0x4eU,<br>0x8FU, 0xdaU, 0xb0U, 0xd7U, 0xcaU, 0x04U, 0x2U, 0xebU,<br>0x8FU, 0x79U, 0xe4U, 0xd8U, 0xadU, 0x8eU, 0x02U, 0x69U,<br>0x87U, 0x48U, 0x19U, 0x2bU, 0x01U, 0x9eU, 0xabU, 0x8FU,<br>0x83U, 0x55U, 0x46U, 0x1aU, 0x7cU, 0x3cU, 0xa7U, 0xb4U,<br>0xc9U, 0x90U, 0xc3U, 0x49U, 0x2U, 0xd0U, 0xbcU, 0xd7U, | 0x61U, 0x62U, 0x63U, 0x64U, 0x62U, 0x63U, 0x64U, 0x65U,<br>0x63U, 0x64U, 0x65U, 0x66U, 0x64U, 0x65U, 0x66U, 0x67U,<br>0x65U, 0x66U, 0x67U, 0x68U, 0x66U, 0x67U, 0x68U, 0x69U,<br>0x67U, 0x68U, 0x69U, 0x6AU, 0x68U, 0x69U, 0x6AU, 0x68U,<br>0x69U, 0x6AU, 0x68U, 0x6CU, 0x6AU, 0x6BU, 0x6CU, 0x6DU,<br>0x6BU, 0x6CU, 0x6DU, 0x6EU, 0x6CU, 0x6DU, 0x6EU, 0x6FU,<br>0x6DU, 0x6EU, 0x6FU, 0x70U, 0x6EU, 0x6FU, 0x70U, 0x71U |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Signature                                                                                                                                                                                                                                                                                                                                                                                                                |

0x51U, 0x1aU, 0xccU, 0x60U, 0x9dU, 0x9bU, 0x0dU, 0xe1U, 0x18U, 0x10U, 0x19U, 0xd2U, 0x48U, 0x1bU, 0xf1U, 0xfeU, 0x50U, 0x89U, 0x49U, 0xfeU, 0xa1U, 0x82U, 0xd9U, 0x7aU, 0x42U, 0x5eU, 0x6aU, 0xdeU, 0x26U, 0x5cU, 0x61U, 0x36U, 0xd2U, 0x67U, 0x8eU, 0x55U, 0xeeU, 0xa2U, 0xbfU, 0xe9U, 0xfaU, 0xaeU, 0xddU, 0x1fU, 0xdeU, 0xadU, 0x4eU, 0xd3U, 0x5bU, 0x61U, 0x02U, 0x0fU, 0x8dU, 0x80U, 0xb6U, 0xfeU, 0x12U, 0x7dU, 0x95U, 0x25U, 0x76U, 0x95U, 0x4dU, 0x6aU, 0x73U, 0x41U, 0x05U, 0x62U, 0xc0U, 0x72U, 0x40U, 0x0eU, 0x01U, 0x44U, 0x20U, 0xaeU, 0x51U, 0x04U, 0x57U, 0x97U, 0x72U, 0xf7U, 0x53U, 0x99U, 0xcdU, 0x61U, 0x02U, 0x9fU, 0x4aU, 0x3fU, 0xe0U, 0x0eU, 0xa0U, 0x41U, 0x44U, 0xd7U, 0x77U, 0xbbU, 0xb0U, 0x94U, 0x85U, 0xe0U, 0x1cU, 0x59U, /\* exponent : 256 bytes \*/  
0x3fU, 0xd6U, 0x34U, 0x34U, 0xc0U, 0xc1U, 0xa7U, 0xf5U, 0xdfU, 0xc8U, 0x48U, 0x06U, 0x65U, 0xeaU, 0x3fU, 0xecU, 0xa5U, 0xffU, 0xc4U, 0x89U, 0x7bU, 0x63U, 0x8fU, 0x83U, 0x63U, 0x8eU, 0x1cU, 0x28U, 0x9eU, 0x4eU, 0x2cU, 0xfaU, 0x2bU, 0x1dU, 0x5cU, 0x06U, 0x68U, 0x07U, 0x94U, 0x3aU, 0xd3U, 0xe8U, 0xf2U, 0x1cU, 0x15U, 0x10U, 0xd6U, 0x25U, 0x2aU, 0xc5U, 0xc2U, 0xe2U, 0xf8U, 0x1bU, 0x4aU, 0xdcU, 0x4dU, 0x12U, 0x5aU, 0x35U, 0x56U, 0xdfU, 0xa2U, 0xddU, 0x96U, 0xeeU, 0x32U, 0x0bU, 0xf1U, 0x38U, 0x1aU, 0x2eU, 0xa7U, 0x25U, 0x76U, 0x06U, 0x1bU, 0xfaU, 0x94U, 0x4bU, 0x1dU, 0xb5U, 0x94U, 0x24U, 0xffU, 0x77U, 0x43U, 0x35U, 0xefU, 0x6bU, 0xdaU, 0x8bU, 0xa7U, 0xa8U, 0xb9U, 0x7eU, 0x87U, 0xefU, 0x1cU, 0xe1U, 0x57U, 0x70U, 0x04U, 0x37U, 0xa6U, 0x81U, 0xa5U, 0x98U, 0x0dU, 0x2fU, 0xd1U, 0x70U, 0xc9U, 0xb4U, 0x0bU, 0xacU, 0xdfU, 0x39U, 0x94U, 0xc0U, 0x53U, 0x1eU, 0xc8U, 0x04U, 0xc1U, 0x76U, 0x3fU, 0x91U, 0x96U, 0x32U, 0x35U, 0x32U, 0x12U, 0x73U, 0x42U, 0x7eU, 0x52U, 0x17U, 0xcdU, 0x87U, 0x2fU, 0xbeU, 0xebU, 0xd1U, 0x8fU, 0x80U, 0x2bU, 0x8aU, 0xfdU, 0xe1U, 0x15U, 0x9fU, 0xaaU, 0x6fU, 0xc4U, 0x6cU, 0x0fU, 0xeeU, 0xc6U, 0xe5U, 0xb6U, 0x4eU, 0x07U, 0xa2U, 0xa1U, 0xfdU, 0x89U, 0x82U, 0xdfU, 0x33U, 0x82U, 0x19U, 0x3eU, 0x68U, 0x46U, 0xc7U, 0x66U, 0xeaU, 0x98U, 0x50U, 0xaaU, 0x10U, 0x2dU, 0x31U, 0x3dU, 0x63U, 0xf7U, 0x22U, 0x70U, 0x7fU, 0x0bU, 0x61U, 0x5aU, 0x41U, 0x22U, 0xb4U, 0xe6U, 0xfeU, 0x0bU, 0xddU, 0x09U, 0x42U, 0x14U, 0xabU, 0xb4U, 0x04U, 0xe1U, 0x23U, 0xb7U, 0xaeU, 0xf7U, 0x5fU, 0x23U, 0x43U, 0x1bU, 0x7eU, 0x65U, 0x21U, 0xfbU, 0xe9U, 0x2dU, 0x8eU, 0x01U, 0xaaU, 0xf9U, 0x7fU, 0xdcU, 0x85U, 0x01U, 0x1dU, 0xe3U, 0x21U, 0x4aU, 0xefU, 0xeaU, 0x9aU, 0x72U, 0x8cU, 0xf6U, 0x69U, 0x29U, 0x44U, 0xb9U, 0xdcU, 0xf9U, 0xd2U, 0xc9U, 0xcbU, 0xdfU, 0x97U, 0x82U, 0x96U, 0x5cU, 0x69U, 0x77U, 0x91U

0x51U, 0x1aU, 0xccU, 0x60U, 0x9dU, 0x9bU, 0x0dU, 0xe1U, 0x18U, 0x10U, 0x19U, 0xd2U, 0x48U, 0x1bU, 0xf1U, 0xfeU, 0x50U, 0x89U, 0x49U, 0xfeU, 0xa1U, 0x82U, 0xd9U, 0x7aU, 0x42U, 0x5eU, 0x6aU, 0xdeU, 0x26U, 0x5cU, 0x61U, 0x36U, 0xd2U, 0x67U, 0x8eU, 0x55U, 0xeeU, 0xa2U, 0xbfU, 0xe9U, 0xfaU, 0xaeU, 0xddU, 0x1fU, 0xdeU, 0xadU, 0x4eU, 0xd3U, 0x5bU, 0x61U, 0x02U, 0x0fU, 0x8dU, 0x80U, 0xb6U, 0xfeU, 0x12U, 0x7dU, 0x95U, 0x25U, 0x76U, 0x95U, 0x4dU, 0x6aU, 0x73U, 0x41U, 0x05U, 0x62U, 0xc0U, 0x72U, 0x40U, 0x0eU, 0x01U, 0x44U, 0x20U, 0xaeU, 0x51U, 0x04U, 0x57U, 0x97U, 0x72U, 0xf7U, 0x53U, 0x99U, 0xcdU, 0x61U, 0x02U, 0x9fU, 0x4aU, 0x3fU, 0xe0U, 0x0eU, 0xa0U, 0x41U, 0x44U, 0xd7U, 0x77U, 0xbbU, 0xb0U, 0x94U, 0x85U, 0xe0U, 0x1cU, 0x59U, /\* exponent : 4 bytes \*/  
0x00U, 0x01U, 0x00U, 0x01U,

0x54U, 0xA2U, 0x4EU, 0x58U, 0xF3U, 0x28U, 0xA0U, 0xF5U, 0xF8U, 0x2CU, 0x4FU, 0xBFU, 0x7BU, 0x84U, 0x16U, 0xF3U, 0xDCU, 0xB2U, 0x16U, 0x6CU, 0xD4U, 0x01U, 0x7CU, 0xA0U, 0xB6U, 0x0BU, 0x1AU, 0x4DU, 0x99U, 0xB6U, 0x96U, 0xA4U, 0x2FU, 0x91U, 0xC5U, 0xEFU, 0xC7U, 0xDBU, 0xCFU, 0x5CU, 0x7AU, 0x92U, 0x26U, 0x04U, 0x23U, 0x3EU, 0x3AU, 0x0DU, 0x01U, 0xD1U, 0x71U, 0xC4U, 0x94U, 0x0DU, 0xA2U, 0xF7U, 0x49U, 0xA2U, 0x21U, 0x5BU, 0x1FU, 0xF8U, 0x3EU, 0xADU, 0x2CU, 0xB2U, 0xADU, 0xDBU, 0xCDU, 0x38U, 0x6CU, 0xE9U, 0x61U, 0xE8U, 0x4AU, 0xE8U, 0x5EU, 0x61U, 0x66U, 0x8FU, 0x26U, 0xAFU, 0x84U, 0x63U, 0x8AU, 0x1AU, 0x18U, 0x3CU, 0xC4U, 0x36U, 0x35U, 0x29U, 0x0DU, 0x97U, 0x81U, 0x2BU, 0xF4U, 0x7AU, 0x4AU, 0xE0U, 0xA1U, 0x62U, 0x43U, 0xECU, 0x36U, 0x9FU, 0xAEU, 0xDBU, 0x0DU, 0x64U, 0x41U, 0x8FU, 0x65U, 0x5BU, 0x44U, 0xB9U, 0x29U, 0x1EU, 0xA6U, 0xA5U, 0x54U, 0x1DU, 0x98U, 0x53U, 0xA9U, 0xAAU, 0x08U, 0x80U, 0x9FU, 0xE1U, 0x2EU, 0xF7U, 0x72U, 0xD3U, 0x2EU, 0xABU, 0x39U, 0xE8U, 0x3DU, 0x65U, 0x42U, 0x19U, 0x33U, 0x3BU, 0xB6U, 0x5DU, 0x8EU, 0x62U, 0x93U, 0x0DU, 0x01U, 0x48U, 0x42U, 0xABU, 0xD3U, 0x01U, 0x26U, 0xA4U, 0x0DU, 0xE9U, 0x0DU, 0x1FU, 0xA1U, 0xF4U, 0x80U, 0xFBU, 0x52U, 0xF7U, 0xB6U, 0x49U, 0x60U, 0xE3U, 0x43U, 0x85U, 0x0BU, 0x4U, 0xC7U, 0xC0U, 0x1CU, 0x7EU, 0xD2U, 0xF7U, 0x65U, 0xBBU, 0x59U, 0x48U, 0xA4U, 0x81U, 0xFEU, 0x6AU, 0x3CU, 0x43U, 0x1DU, 0xF7U, 0x64U, 0x78U, 0xC5U, 0x7EU, 0xFCU, 0x98U, 0x8CU, 0xD5U, 0xCAU, 0x2EU, 0x9CU, 0xB7U, 0x1BU, 0x57U, 0xCDU, 0xB7U, 0xECU, 0xC1U, 0x6AU, 0x1BU, 0xC7U, 0x2FU, 0xADU, 0x15U, 0x90U, 0x47U, 0xF2U, 0x5DU, 0x84U, 0xCB, 0x15U, 0xFFU, 0x28U, 0x58U, 0x1FU, 0x8CU, 0xAEU, 0x62U, 0x71U, 0xD0U, 0xE2U, 0x4AU, 0x80U, 0x56U, 0xC7U, 0xB2U, 0xFAU, 0xABU, 0x36U, 0xB1U, 0x5DU, 0x8FU, 0x58U, 0xF1U, 0x1FU, 0xB7U, 0x10U, 0xA5U, 0xFBU, 0xBBU, 0x4AU

#### 8.4.6.4 AuthHsmPkcs1V15Sha256Seed

##### Raw Key 방식 사용시

##### Verification

| Public Key                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Hash(Seed)                                                                                                                                                                                                                     | Signature                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| /* modulus : 256 bytes */<br>0xc2U, 0xc2U, 0xe6U, 0x41U, 0xf6U, 0x84U, 0xbeU, 0xe9U, 0xb3U, 0x03U, 0xc6U, 0x8aU, 0xb8U, 0x15U, 0xe7U, 0x08U, 0x17U, 0x50U, 0xcbU, 0x90U, 0xe1U, 0x59U, 0x42U, 0xc7U, 0x17U, 0x63U, 0x5aU, 0x54U, 0xdbU, 0xb2U, 0x62U, 0x77U, 0x70U, 0x38U, 0xb7U, 0x93U, 0x13U, 0x1aU, 0x85U, 0x01U, 0x16U, 0x0eU, 0x7dU, 0xeeU, 0xc1U, 0xe8U, 0xa7U, 0xf7U, 0x24U, 0xf1U, 0x84U, 0x23U, 0x5bU, 0xf2U, 0x5aU, 0xa3U, 0xc5U, 0x04U, 0x20U, 0x75U, 0xa9U, 0x9cU, 0x14U, 0x9bU, 0xaaU, 0x00U, 0x2dU, 0xb1U, 0x80U, 0xa6U, 0x68U, 0xb1U, 0x40U, 0xc5U, 0xe4U, 0x6aU, 0xf2U, 0x9fU, 0xf5U, 0xc5U, 0x69U, 0xdfU, 0x1bU, 0x1eU, 0x84U, 0x1eU, 0xdfU, 0xd9U, 0x52U, 0xc3U, 0xbbU, 0x96U, 0x6fU, 0xe8U, 0xf1U, 0x02U, 0xc5U, 0x1cU, 0x81U, 0x24U, 0xbeU, 0xd6U, 0xdeU, 0x5aU, 0x57U, 0xe5U, 0x29U, 0xb8U, 0x37U, 0xdeU, 0x23U, 0x4eU, 0x8fU, 0xdaU, 0xb0U, 0xd7U, 0xcaU, 0x04U, 0xe2U, 0xebU, 0x8fU, 0x79U, 0x4eU, 0xd8U, 0xadU, 0x8eU, 0x02U, 0x69U, 0x87U, 0x48U, 0x19U, 0x2bU, 0x01U, 0x9eU, 0xabU, 0x8fU, 0x83U, 0x55U, 0x46U, 0x1aU, 0x7cU, 0x3cU, 0xa7U, 0xb4U, 0xc9U, 0x90U, 0xc3U, 0x49U, 0xd2U, 0x0dU, 0xb0U, 0xd7U, 0x51U, 0x1aU, 0xccU, 0x60U, 0x9dU, 0x9bU, 0x0dU, 0xe1U, 0x18U, 0x10U, 0x19U, 0xd2U, 0x48U, 0x1bU, 0xf1U, 0xfeU, 0x50U, 0x89U, 0x49U, 0xfeU, 0xa1U, 0x82U, 0xd9U, 0x7aU, 0x42U, 0x5eU, 0x6aU, 0xdeU, 0x26U, 0x5cU, 0x61U, 0x36U, 0xd2U, 0x67U, 0x8eU, 0x55U, 0xeeU, 0xa2U, 0xbfU, 0xe9U, 0xfaU, 0xaeU, 0xddU, 0x1fU, 0xdeU, 0xadU, 0x4eU, 0xd3U, 0x5bU, 0x61U, 0x02U, 0x0fU, 0x8dU, 0x80U, 0xb6U, 0xfeU, 0x12U, 0x7dU, 0x95U, 0x25U, 0x76U, 0x95U, 0x4dU, 0x6aU, 0x73U, 0x41U, 0x05U, 0x62U, 0xc0U, 0x72U, 0x40U, 0x0eU, 0x01U, 0x44U, 0x20U, 0xaeU, 0x51U, 0x04U, 0x57U, 0x97U, 0x72U, 0xf7U, 0x53U, 0x99U, 0xcdU, 0x61U, 0x02U, 0x9fU, 0x4aU, 0x3fU, 0xe0U, 0x0eU, 0xa0U, 0x41U, 0x44U, 0xd7U, 0x77U, 0xbbU, 0xb0U, 0x94U, 0x85U, 0xe0U, 0x1cU, 0x59U, /* exponent : 4 bytes */<br>0x00U, 0x01U, 0x00U, 0x01U, | 0x24U, 0x8dU, 0x6aU, 0x61U, 0xd2U, 0x06U, 0x38U, 0xb8U, 0xe5U, 0xc0U, 0x26U, 0x93U, 0x0cU, 0x3eU, 0x60U, 0x39U, 0xa3U, 0x3cU, 0xe4U, 0x59U, 0x64U, 0xffU, 0x21U, 0x67U, 0xf6U, 0xecU, 0xedU, 0xd4U, 0x19U, 0xdbU, 0x06U, 0xc1U | 0x54U, 0xA2U, 0x4EU, 0x58U, 0xF3U, 0x28U, 0xA0U, 0xF5U, 0xF8U, 0x2CU, 0x4FU, 0xBFU, 0x7BU, 0x84U, 0x16U, 0xF3U, 0xDCU, 0xB2U, 0x16U, 0x6CU, 0xD4U, 0x01U, 0x7CU, 0xA0U, 0xB6U, 0x0BU, 0x1AU, 0x4DU, 0x99U, 0xB6U, 0x96U, 0xA4U, 0x2FU, 0x91U, 0xC5U, 0xEFU, 0xC7U, 0xDBU, 0xCFU, 0x5CU, 0x7AU, 0x92U, 0x26U, 0x04U, 0x23U, 0x3EU, 0x3AU, 0x0DU, 0x01U, 0xD1U, 0x71U, 0xC4U, 0x94U, 0x0DU, 0xA2U, 0xF7U, 0x49U, 0xA2U, 0x21U, 0x5BU, 0x1FU, 0xF8U, 0x3EU, 0xADU, 0x2CU, 0xB2U, 0xADU, 0xDBU, 0xCDU, 0x38U, 0x6CU, 0xE9U, 0x61U, 0xE8U, 0x4AU, 0xE8U, 0x5EU, 0x61U, 0x66U, 0x8FU, 0x26U, 0xAFU, 0x84U, 0x63U, 0x8AU, 0x1AU, 0x18U, 0x3CU, 0xC4U, 0x36U, 0x35U, 0x29U, 0x0DU, 0x97U, 0x81U, 0x2BU, 0xF4U, 0x7AU, 0x4AU, 0xE0U, 0xA1U, 0x62U, 0x43U, 0xECU, 0x36U, 0x9FU, 0xAEU, 0xDBU, 0x0DU, 0x64U, 0x41U, 0x8FU, 0x65U, 0x5BU, 0x44U, 0xB9U, 0x29U, 0x1EU, 0xA6U, 0xA5U, 0x54U, 0x1DU, 0x98U, 0x53U, 0xA9U, 0xAAU, 0x08U, 0x80U, 0x9FU, 0xE1U, 0x2EU, 0xF7U, 0x72U, 0xD3U, 0x2EU, 0xABU, 0x39U, 0xE8U, 0x3DU, 0x65U, 0x42U, 0x19U, 0x33U, 0x3BU, 0xB6U, 0x5DU, 0x8EU, 0x62U, 0x93U, 0x0DU, 0x01U, 0x48U, 0x42U, 0xABU, 0xD3U, 0x01U, 0x26U, 0xA4U, 0x0DU, 0xE9U, 0x0DU, 0x1FU, 0xA1U, 0xF4U, 0x80U, 0xFBU, 0x52U, 0xF7U, 0xB6U, 0x49U, 0x60U, 0xE3U, 0x43U, 0x85U, 0x0BU, 0x27U, 0xC7U, 0xC0U, 0x1CU, 0x7EU, 0xD2U, 0xF7U, 0x65U, 0xBBU, 0x59U, 0x48U, 0xA4U, 0x81U, 0xFEU, 0x6AU, 0x3CU, 0x43U, 0x1DU, 0xF7U, 0x64U, 0x78U, 0xC5U, 0x7EU, 0xFCU, 0x98U, 0x8CU, 0xD5U, 0xCAU, 0x2EU, 0x9CU, 0xB7U, 0x1BU, 0x57U, 0xCDU, 0xB7U, 0xECU, 0xC1U, 0x6AU, 0x1BU, 0xC7U, 0x2FU, 0xADU, 0x15U, 0x90U, 0x47U, 0xF2U, 0x5DU, 0x84U, 0xCB, 0x15U, 0xFFU, 0x28U, 0x58U, 0x1FU, 0x8CU, 0x8DU, 0xAEU, 0x62U, 0x71U, 0xD0U, 0xE2U, 0x4AU, 0x80U, 0x56U, 0xC7U, 0xB2U, 0xFAU, 0xABU, 0x36U, 0xB1U, 0x5DU, 0x8FU, 0x58U, 0xF1U, 0x1FU, 0xB7U, 0x10U, 0xA5U, 0xFBU, 0xBBU, 0x4AU |

#### 8.4.6.5 AuthHsmPkcs1PssSha256

##### Raw Key 방식 사용시

##### Generation, Verification

| Private Key                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            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| /* modulus : 256 bytes */<br>0xd9U, 0x5bU, 0x71U, 0xc9U, 0xdfU, 0xeeU, 0x45U, 0x3bU,<br>0xa1U, 0xb1U, 0xa7U, 0xdeU, 0x2cU, 0x1fU, 0x0bU, 0x0aU,<br>0x67U, 0x57U, 0x9eU, 0xe9U, 0x1dU, 0x1dU, 0x3aU, 0xd9U,<br>0x7eU, 0x48U, 0x18U, 0x29U, 0xb8U, 0x6eU, 0xdaU, 0xc7U,<br>0x50U, 0xc4U, 0x8eU, 0x12U, 0xa8U, 0xcdU, 0xb0U, 0x26U,<br>0xc8U, 0x2fU, 0x27U, 0x3dU, 0xafU, 0x2U, 0x22U, 0x00U,<br>0x9fU, 0x0dU, 0xb3U, 0xb0U, 0x8bU, 0x2dU, 0xb1U, 0x0aU,<br>0x69U, 0xc4U, 0xb2U, 0xddU, 0xdaU, 0xaeU, 0xceU, 0xacU,<br>0x1bU, 0x0cU, 0x86U, 0x26U, 0x82U, 0xeeU, 0xf2U, 0x94U,<br>0xe5U, 0x79U, 0xf5U, 0x5aU, 0xabU, 0x87U, 0x1bU, 0xc0U,<br>0xa7U, 0xeeU, 0xabU, 0xc9U, 0x23U, 0xc9U, 0xe8U, 0x0dU,<br>0xddU, 0x2U, 0x2eU, 0xc0U, 0xa2U, 0x70U, 0x02U, 0xaeU,<br>0xe6U, 0xa5U, 0xbaU, 0x66U, 0x39U, 0x7fU, 0x41U, 0x2bU,<br>0xbaU, 0xf5U, 0xfbU, 0x4eU, 0xafU, 0x66U, 0xa1U, 0xa0U,<br>0xf8U, 0x2eU, 0xafU, 0x68U, 0x27U, 0x19U, 0x8cU, 0xafU,<br>0x49U, 0xb3U, 0x47U, 0x25U, 0x8bU, 0x12U, 0x83U, 0xe8U,<br>0xcbU, 0xb1U, 0x0dU, 0xa2U, 0x83U, 0x7fU, 0x6eU, 0xccU,<br>0x34U, 0x90U, 0xc7U, 0x28U, 0xf6U, 0x92U, 0x7fU, 0x44U,<br>0x45U, 0x5aU, 0x6fU, 0x19U, 0x4fU, 0x37U, 0x76U, 0xbfU,<br>0x79U, 0x15U, 0x1dU, 0x9aU, 0xb7U, 0xe2U, 0xdaU, 0xf7U,<br>0x70U, 0xb3U, 0x7dU, 0x12U, 0x62U, 0x7cU, 0xc0U, 0xc5U,<br>0xfbU, 0x62U, 0x48U, 0x4fU, 0x46U, 0x25U, 0x8dU, 0x9cU,<br>0xe2U, 0xc1U, 0x1bU, 0x26U, 0x25U, 0x6dU, 0x09U, 0xcbU,<br>0x41U, 0x2fU, 0x8dU, 0x8fU, 0xf6U, 0x1fU, 0x9eU, 0x1bU,<br>0xb9U, 0x4aU, 0xc2U, 0x7dU, 0xe6U, 0xd2U, 0x6aU, 0x83U,<br>0xa8U, 0x43U, 0x9eU, 0x51U, 0xb3U, 0x5dU, 0xbeU, 0xe4U,<br>0x6bU, 0x3bU, 0x8fU, 0x9U, 0x91U, 0xd6U, 0x67U, 0xbbU,<br>0x53U, 0xeeU, 0xeeU, 0x85U, 0xffU, 0x16U, 0x52U, 0xc8U,<br>0x98U, 0x1fU, 0x14U, 0x1dU, 0x47U, 0xc8U, 0x20U, 0x57U,<br>0x91U, 0xceU, 0xf5U, 0xb3U, 0x2dU, 0x71U, 0x8dU, 0xcdU,<br>0x08U, 0x2eU, 0xd0U, 0xddU, 0x54U, 0x28U, 0x26U, 0x41U,<br>0x6bU, 0x22U, 0x71U, 0x06U, 0x4eU, 0xf4U, 0x37U, 0xa9U,<br>/* exponent : 256 bytes */<br>0x2fU, 0x21U, 0xb0U, 0x1bU, 0xe9U, 0x4dU, 0xdeU, 0x7fU,<br>0x5eU, 0xc1U, 0x8aU, 0x38U, 0x17U, 0xf3U, 0x27U, 0x4eU,<br>0xbbU, 0x37U, 0xf9U, 0xc2U, 0x6cU, 0xc8U, 0xc0U, 0xd1U,<br>0x16U, 0x9cU, 0x05U, 0x79U, 0x4eU, 0x7fU, 0xe3U, 0x3aU,<br>0xe3U, 0x1dU, 0xabU, 0xfdu, 0x09U, 0xd3U, 0x88U, 0x45U,<br>0x0U, 0x94U, 0xa0U, 0xfaU, 0xb4U, 0x58U, 0xf1U, 0x4cU,<br>0x97U, 0x30U, 0xbeU, 0x6dU, 0x22U, 0xd0U, 0xe6U, 0x99U,<br>0xeeU, 0x73U, 0x73U, 0xa1U, 0xbdU, 0xe0U, 0xb7U, 0xfaU,<br>0x03U, 0xe7U, 0x84U, 0x53U, 0x67U, 0x82U, 0xeeU, 0xe1U,<br>0x30U, 0x9dU, 0x70U, 0x81U, 0x97U, 0xbeU, 0x35U, 0x5bU,<br>0x62U, 0x4eU, 0xd3U, 0xbbU, 0x4aU, 0xe2U, 0x66U, 0x4aU,<br>0x53U, 0x72U, 0x0deU, 0xf6U, 0x70U, 0x82U, 0xbfU, 0x62U,<br>0x33U, 0xabU, 0x6eU, 0x2eU, 0xaeU, 0x7aU, 0xd8U, 0xa3U,<br>0xe5U, 0xe7U, 0x9eU, 0xf5U, 0xe1U, 0xfcU, 0x9cU, 0x41U,<br>0x5eU, 0x6fU, 0x9aU, 0x23U, 0x79U, 0x8fU, 0x05U, 0xbdU,<br>0xa0U, 0xcaU, 0x9aU, 0x3bU, 0x0eU, 0xdbU, 0x45U, 0xf4U,<br>0xd7U, 0x81U, 0xefU, 0x1aU, 0x4fU, 0x50U, 0x75U, 0xcdU,<br>0x9bU, 0xb3U, 0x99U, 0x63U, 0x5dU, 0xa3U, 0xe9U, 0xa6U,<br>0x88U, 0x0eU, 0xd0U, 0x21U, 0xa7U, 0x50U, 0xbCU, 0x98U,<br>0x06U, 0xafU, 0x81U, 0xfbU, 0xffU, 0xcdU, 0x4aU, 0xceU,<br>0xafU, 0x80U, 0x4eU, 0x7U, 0x68U, 0x08U, 0xaeU, 0x18U,<br>0x67U, 0x15U, 0xc7U, 0x72U, 0xcaU, 0xa9U, 0x61U, 0xa8U,<br>0x62U, 0x99U, 0x1cU, 0x67U, 0xcaU, 0x8bU, 0xffU, 0xefU,<br>0x6bU, 0x34U, 0x08U, 0x7bU, 0x44U, 0xdbU, 0x5bU, 0x59U,<br>0xabU, 0xceU, 0x09U, 0x31U, 0x77U, 0x47U, 0xfcU, 0x75U,<br>0x25U, 0x2fU, 0x17U, 0x05U, 0x26U, 0x0bU, 0x13U, 0xddU,<br>0x62U, 0xccU, 0xbCU, 0x74U, 0x50U, 0x91U, 0xf3U, 0xc1U,<br>0xb6U, 0x4fU, 0x59U, 0x03U, 0x1dU, 0x34U, 0x0cU, 0x73U,<br>0x62U, 0xa0U, 0xe1U, 0x06U, 0x6aU, 0xb0U, 0x55U, 0x4dU,<br>0x46U, 0x6fU, 0x20U, 0x9aU, 0x3cU, 0xf5U, 0x1bU, 0xc6U,<br>0x4bU, 0x3cU, 0x70U, 0xc3U, 0xceU, 0x52U, 0xf4U, 0x13U,<br>0xd8U, 0x1bU, 0x22U, 0x8fU, 0xa3U, 0x1dU, 0x9eU, 0xfdu | /* modulus : 256 bytes */<br>0xd9U, 0x5bU, 0x71U, 0xc9U, 0xdfU, 0xeeU, 0x45U, 0x3bU,<br>0xa1U, 0xb1U, 0xa7U, 0xdeU, 0x2cU, 0x1fU, 0x0bU, 0x0aU,<br>0x67U, 0x57U, 0x9eU, 0xe9U, 0x1dU, 0x1dU, 0x3aU, 0xd9U,<br>0x7eU, 0x48U, 0x18U, 0x29U, 0xb8U, 0x6eU, 0xdaU, 0xc7U,<br>0x50U, 0xc4U, 0x8eU, 0x12U, 0xa8U, 0xcdU, 0xb0U, 0x26U,<br>0xc8U, 0x2fU, 0x27U, 0x3dU, 0xafU, 0x2U, 0x22U, 0x00U,<br>0x9fU, 0x0dU, 0xb3U, 0xb0U, 0x8bU, 0x2dU, 0xb1U, 0x0aU,<br>0x69U, 0xc4U, 0xb2U, 0xddU, 0xdaU, 0xaeU, 0xceU, 0xacU,<br>0x1bU, 0x0cU, 0x86U, 0x26U, 0x82U, 0xeeU, 0xf2U, 0x94U,<br>0xe5U, 0x79U, 0xf5U, 0x5aU, 0xabU, 0x87U, 0x1bU, 0xc0U,<br>0xa7U, 0xeeU, 0xabU, 0xc9U, 0x23U, 0xc9U, 0xe8U, 0x0dU,<br>0xddU, 0x2U, 0x2eU, 0xc0U, 0xa2U, 0x70U, 0x02U, 0xaeU,<br>0xe6U, 0xa5U, 0xbaU, 0x66U, 0x39U, 0x7fU, 0x41U, 0x2bU,<br>0xbaU, 0xf5U, 0xfbU, 0x4eU, 0xafU, 0x66U, 0xa1U, 0xa0U,<br>0xf8U, 0x2eU, 0xafU, 0x68U, 0x27U, 0x19U, 0x8cU, 0xafU,<br>0x49U, 0xb3U, 0x47U, 0x25U, 0x8bU, 0x12U, 0x83U, 0xe8U,<br>0xcbU, 0xb1U, 0x0dU, 0xa2U, 0x83U, 0x7fU, 0x6eU, 0xccU,<br>0x34U, 0x90U, 0xc7U, 0x28U, 0xf6U, 0x92U, 0x7fU, 0x44U,<br>0x45U, 0x5aU, 0x6fU, 0x19U, 0x4fU, 0x37U, 0x76U, 0xbfU,<br>0x79U, 0x15U, 0x1dU, 0x9aU, 0xb7U, 0xe2U, 0xdaU, 0xf7U,<br>0x70U, 0xb3U, 0x7dU, 0x12U, 0x62U, 0x7cU, 0xc0U, 0xc5U,<br>0xfbU, 0x62U, 0x48U, 0x4fU, 0x46U, 0x25U, 0x8dU, 0x9cU,<br>0xe2U, 0xc1U, 0x1bU, 0x26U, 0x25U, 0x6dU, 0x09U, 0xcbU,<br>0x41U, 0x2fU, 0x8dU, 0x8fU, 0xf6U, 0x1fU, 0x9eU, 0x1bU,<br>0xb9U, 0x4aU, 0xc2U, 0x7dU, 0xe6U, 0xd2U, 0x6aU, 0x83U,<br>0xa8U, 0x43U, 0x9eU, 0x51U, 0xb3U, 0x5dU, 0xbeU, 0xe4U,<br>0x6bU, 0x3bU, 0x8fU, 0xf9U, 0x91U, 0xd6U, 0x67U, 0xbbU,<br>0x53U, 0xeeU, 0xeeU, 0x85U, 0xffU, 0x16U, 0x52U, 0xc8U,<br>0x98U, 0x1fU, 0x14U, 0x1dU, 0x47U, 0xc8U, 0x20U, 0x57U,<br>0x91U, 0xceU, 0xf5U, 0xb3U, 0x2dU, 0x71U, 0x8dU, 0xcdU,<br>0x08U, 0x2eU, 0xd0U, 0xddU, 0x54U, 0x28U, 0x26U, 0x41U,<br>0x6bU, 0x22U, 0x71U, 0x06U, 0x4eU, 0xf4U, 0x37U, 0xa9U,<br>/* exponent : 4 bytes */<br>0x00U, 0x01U, 0x00U, 0x01U, | 0xcdU, 0x74U, 0xaeU, 0x61U, 0x52U, 0xd5U, 0xfeU, 0x5cU,<br>0xe3U, 0xd9U, 0x07U, 0x3cU, 0x92U, 0x1eU, 0x86U, 0x1aU,<br>0x24U, 0x20U, 0x8fU, 0x0cU, 0x68U, 0x47U, 0x7fU, 0x49U,<br>0xc8U, 0x25U, 0x33U, 0x8eU, 0x1eU, 0xf8U, 0x77U, 0xc0U,<br>0xc9U, 0x77U, 0xc1U, 0xd2U, 0xffU, 0xc8U, 0x20U, 0xe9U,<br>0x64U, 0xdbU, 0x6fU, 0xbeU, 0xcdU, 0xccU, 0xceU, 0x44U,<br>0x9eU, 0xc8U, 0x53U, 0x8cU, 0x8bU, 0xffU, 0xfcU, 0x5eU,<br>0xbdU, 0xecU, 0xe8U, 0x47U, 0x62U, 0xdaU, 0xc7U, 0xf2U,<br>0xcbU, 0xa6U, 0x90U, 0x52U, 0xc0U, 0xc6U, 0x72U, 0x26U,<br>0x17U, 0x8aU, 0x0cU, 0xe1U, 0x85U, 0xa2U, 0xe0U, 0x50U,<br>0xb3U, 0xe1U, 0x05U, 0x7eU, 0x94U, 0x41U, 0x1dU, 0xd5U,<br>0xf7U, 0x26U, 0x87U, 0x85U, 0x58U, 0xe7U, 0xd6U, 0x2aU,<br>0xfcU, 0x8aU, 0x81U, 0xa9U, 0x3dU, 0xcFU, 0xdbU, 0x5aU,<br>0x22U, 0x71U, 0x46U, 0x6dU, 0x32U, 0xa8U, 0xa4U, 0x86U,<br>0x8aU, 0xf2U, 0x0fU, 0xabU, 0x2eU, 0x13U, 0xcaU, 0x60U,<br>0x9dU, 0x5aU, 0x77U, 0x10U, 0x8aU, 0x17U, 0x8aU, 0xafU                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        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|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        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                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 0x63U, 0x75U, 0x75U, 0x5eU, 0xffU, 0x8dU, 0x48U, 0xafU,<br>0xb3U, 0x26U, 0x3bU, 0x3bU, 0x96U, 0x98U, 0x8aU, 0x2aU,<br>0xfdu, 0x18U, 0x1bU, 0xa0U, 0x61U, 0x79U, 0x3eU, 0xa0U,<br>0x09U, 0x78U, 0x3bU, 0xb1U, 0x59U, 0x9dU, 0x03U, 0x94U,<br>0x4dU, 0x98U, 0x76U, 0x20U, 0xa2U, 0x66U, 0x8aU, 0xc9U,<br>0x71U, 0x4dU, 0x6fU, 0x2aU, 0x21U, 0xf7U, 0xe5U, 0x20U,<br>0x0dU, 0x63U, 0x92U, 0x3fU, 0x42U, 0xc8U, 0x32U, 0xe6U,<br>0x33U, 0x01U, 0xc8U, 0xdeU, 0x58U, 0xc7U, 0x0aU, 0x20U,<br>0x39U, 0x10U, 0x64U, 0x0dU, 0xa9U, 0x67U, 0xd0U, 0x3fU,<br>0x4fU, 0x62U, 0x92U, 0xf6U, 0xc8U, 0x19U, 0x97U, 0x59U,<br>0x82U, 0x27U, 0x90U, 0xc0U, 0xc5U, 0xbCU, 0xfbu, 0x1dU,<br>0x4fU, 0xaaU, 0x59U, 0x46U, 0x5cU, 0x3dU, 0xb2U, 0xeaU,<br>0x1fU, 0xffU, 0xd5U, 0x5eU, 0x43U, 0x33U, 0x56U, 0x32U,<br>0xb7U, 0x47U, 0x45U, 0xbfU, 0x1eU, 0x18U, 0x47U, 0x3cU,<br>0x0aU, 0x8bU, 0x4aU, 0x89U, 0xdeU, 0xf6U, 0xb2U, 0x7eU,<br>0xdfU, 0x0dU, 0x7dU, 0x73U, 0x5eU, 0xe1U, 0x3fU, 0x88U,<br>0x70U, 0x41U, 0xc9U, 0xd8U, 0xa9U, 0x1eU, 0x62U, 0x18U,<br>0x6aU, 0x9aU, 0x1eU, 0x0bU, 0x1aU, 0xfbU, 0x48U, 0xe5U,<br>0x77U, 0xf6U, 0x88U, 0x7cU, 0xa6U, 0x1bU, 0x7cU, 0x1bU,<br>0xb2U, 0x6bU, 0x4aU, 0x8eU, 0x2cU, 0xc4U, 0x64U, 0xa9U,<br>0xafU, 0x03U, 0x44U, 0x4bU, 0x3dU, 0xa5U, 0xbeU, 0xd0U,<br>0x8bU, 0x73U, 0xf1U, 0x26U, 0x2bU, 0xd3U, 0xd6U, 0x1fU,<br>0x4cU, 0x78U, 0xf4U, 0x9fU, 0xacU, 0x6aU, 0x3bU, 0xfcU,<br>0x9eU, 0x85U, 0x48U, 0xb4U, 0xb8U, 0xe6U, 0x4cU, 0xceU,<br>0x6aU, 0x60U, 0x90U, 0xfcU, 0x48U, 0x0eU, 0xfdu, 0x1fU,<br>0x36U, 0xc1U, 0x8cU, 0x10U, 0xbCU, 0x09U, 0xbeU, 0x9dU,<br>0x95U, 0x7aU, 0x79U, 0xf7U, 0x07U, 0xa1U, 0x05U, 0x77U,<br>0xa1U, 0xbfU, 0x6eU, 0x9eU, 0x2dU, 0x48U, 0x49U, 0x69U,<br>0x3fU, 0xa5U, 0x8dU, 0x88U, 0x77U, 0xc8U, 0xf1U, 0xe5U,<br>0x51U, 0x81U, 0x95U, 0x5dU, 0x6cU, 0x2bU, 0x94U, 0xb1U,<br>0xd6U, 0xd9U, 0x40U, 0x1bU, 0x5fU, 0xb8U, 0x0cU, 0xc3U,<br>0x2bU, 0x35U, 0x89U, 0x34U, 0xfeU, 0xc2U, 0xaeU, 0xdbU |

#### 8.4.6.6 AuthHsmEcdsaSha256

##### Raw Key 방식 사용시

##### Generation, Verification

| Private Key                                                                                                                                                                                                                                                   | Plain Text                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| /* privatekey d */<br>0xe2U, 0x83U, 0x87U, 0x12U, 0x39U, 0x83U, 0x7eU, 0x13U,<br>0xb9U, 0x5fU, 0x78U, 0x9eU, 0x6eU, 0x1aU, 0xf6U, 0x3bU,<br>0xf6U, 0x1cU, 0x91U, 0x8cU, 0x99U, 0x2eU, 0x62U, 0xbCU,<br>0xa0U, 0x40U, 0xd6U, 0x4cU, 0xadU, 0x1fU, 0xc2U, 0xf6U | 0x40U, 0x41U, 0x42U, 0x43U, 0x44U, 0x45U, 0x46U, 0x47U,<br>0x48U, 0x49U, 0x4AU, 0x4BU, 0x4CU, 0x4DU, 0x4EU, 0x4FU,<br>0x70U, 0x71U, 0x72U, 0x73U, 0x74U, 0x75U, 0x76U, 0x77U,<br>0x78U, 0x79U, 0x7AU, 0x7BU, 0x7CU, 0x7DU, 0x7EU, 0x7FU,<br>0x30U, 0x31U, 0x32U, 0x33U, 0x34U, 0x35U, 0x36U, 0x37U,<br>0x38U, 0x39U, 0x3AU, 0x3BU, 0x3CU, 0x3DU, 0x3EU, 0x3FU,<br>0x00U, 0x11U, 0x22U, 0x33U, 0x44U, 0x55U, 0x66U, 0x77U,<br>0x88U, 0x99U, 0xAAU, 0xBBU, 0xCCU, 0xDDU, 0xEEU, 0xFFU,<br>0xA0U, 0xA1U, 0xA2U, 0xA3U, 0xA4U, 0xA5U, 0xA6U, 0xA7U,<br>0xA8U, 0xA9U, 0xAAU, 0xABU, 0xACU, 0xADU, 0xAEU, 0xAFU, |
| Public Key                                                                                                                                                                                                                                                    | Secret Number                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

/\* publickey Qx \*/

0x74U, 0xccU, 0xd8U, 0xa6U, 0x2fU, 0xbaU, 0x0eU, 0x66U,  
0x7cU, 0x50U, 0x92U, 0x9aU, 0x53U, 0xf7U, 0x8cU, 0x21U,  
0xb8U, 0xffU, 0x0cU, 0x3cU, 0x73U, 0x7bU, 0x0bU, 0x40U,  
0xb1U, 0x75U, 0x0bU, 0x23U, 0x02U, 0xb0U, 0xbdU, 0xe8U,  
/\* publickey Qy \*/  
0x29U, 0x07U, 0x4eU, 0x21U, 0xf3U, 0xa0U, 0xefU, 0x88U,  
0xb9U, 0xefU, 0xdfU, 0x10U, 0xd0U, 0x6aU, 0xa4U, 0xc2U,  
0x95U, 0xccU, 0x16U, 0x71U, 0xf7U, 0x58U, 0xcaU, 0x0eU,  
0x4cU, 0xd1U, 0x08U, 0x80U, 0x3dU, 0x0fU, 0x26U, 0x14U

0xadU, 0x5eU, 0x88U, 0x7eU, 0xb2U, 0xb3U, 0x80U, 0xb8U,  
0xd8U, 0x28U, 0x0aU, 0xd6U, 0xe5U, 0xffU, 0x8aU, 0x60U,  
0xf4U, 0xd2U, 0x62U, 0x43U, 0xe0U, 0x12U, 0x4cU, 0x2fU,  
0x31U, 0xa2U, 0x97U, 0xb5U, 0xd0U, 0x83U, 0x5dU, 0xe2U

### Signature

/\* Signature R \*/

0x35U, 0xfbU, 0x60U, 0xf5U, 0xcaU, 0x0fU, 0x3cU, 0xa0U,  
0x85U, 0x42U, 0xfbU, 0x3cU, 0xc6U, 0x41U, 0xc8U, 0x26U,  
0x3aU, 0x2cU, 0xabU, 0x7aU, 0x90U, 0xeeU, 0x6aU, 0x5eU,  
0x15U, 0x83U, 0xfaU, 0xc2U, 0xbbU, 0x6fU, 0x6bU, 0xd1U

/\* Signature S \*/

0x8AU, 0x9DU, 0xCBU, 0x72U, 0xACU, 0xEDU, 0x69U, 0x92U,  
0x18U, 0x08U, 0xB4U, 0x09U, 0xD8U, 0x42U, 0xC7U, 0xECU,  
0xD5U, 0xBEU, 0x47U, 0xDCU, 0x46U, 0x2AU, 0x8FU, 0xBAU,  
0xD0U, 0x2EU, 0x1FU, 0xDEU, 0xF6U, 0xF4U, 0x75U, 0x7DU

### 8.4.6.7 HaeHsmPkcs1V15Sha1

Verification 테스트 벡터 이미지

### 8.4.6.8 HaeHsmPkcs1V15Sha1Seed

Verification

| Publi<br>c Ke<br>y<br>Inde<br>x | Hash(Seed)                                                                                                             | Signature                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 401                             | 0x07, 0xc9, 0xe2, 0xf4, 0x38, 0x6b, 0xc8, 0x5f, 0x2c, 0x41, 0x72, 0x89, 0x48, 0xb8, 0x2a, 0x55, 0x91, 0x01, 0x1d, 0xb7 | 0xd0, 0xd0, 0x3c, 0xcb, 0x3b, 0x30, 0xb7, 0xc9, 0xc4, 0xd6, 0xee, 0xee, 0x2e, 0xc2, 0x6d, 0x06, 0x92, 0x46, 0xe0, 0x19, 0xfb, 0x8f, 0xa2, 0xf3, 0xa9, 0xb7, 0x2c, 0x9b, 0xbe, 0x23, 0x1d, 0x93, 0xce, 0x05, 0x3d, 0xf8, 0x05, 0xa0, 0x45, 0xe2, 0xef, 0x6b, 0xd8, 0xd0, 0x8b, 0xfb, 0x0c, 0x36, 0x92, 0x2e, 0x5a, 0x6f, 0x10, 0xb9, 0x47, 0xb2, 0x60, 0x7f, 0x59, 0x6b, 0x6c, 0xbd, 0x3c, 0x9e, 0xef, 0xef, 0x56, 0xf5, 0x39, 0x68, 0x05, 0xe8, 0xb2, 0x8b, 0x1c, 0xa1, 0x82, 0xc7, 0x8c, 0x0b, 0x12, 0xb9, 0x79, 0x6a, 0xa8, 0x56, 0xaf, 0x69, 0xc3, 0x55, 0x04, 0xf8, 0xac, 0xc7, 0xaf, 0xa7, 0x4b, 0xc0, 0xf7, 0x7a, 0x1d, 0x61, 0xda, 0x94, 0x94, 0x40, 0x57, 0xa9, 0xee, 0x72, 0xd2, 0xf0, 0xa9, 0x6c, 0xba, 0xa2, 0xf6, 0x46, 0x76, 0xf5, 0x31, 0x8b, 0x71, 0xe5, 0x6f, 0x51, 0x9d, 0x0d, 0xa1, 0xce, 0x8f, 0x42, 0xdb, 0x0e, 0xbe, 0x50, 0x45, 0xfc, 0xc7, 0x26, 0xe3, 0x9f, 0xb0, 0x03, 0x2f, 0x22, 0x87, 0x91, 0x8f, 0x91, 0x90, 0xf3, 0xfb, 0x3d, 0x4d, 0xe5, 0x42, 0x03, 0x04, 0x41, 0xf6, 0x73, 0x6c, 0x62, 0x05, 0xa2, 0xbc, 0xd2, 0x45, 0x0e, 0xb4, 0x11, 0x08, 0x53, 0x11, 0xc7, 0x32, 0x0b, 0xaa, 0x42, 0x68, 0xfd, 0x2f, 0xd8, 0xbc, 0xc8, 0xeb, 0xfd, 0xdb, 0xb6, 0x07, 0x40, 0xcf, 0xf0, 0xb3, 0xb0, 0x0f, 0x61, 0x87, 0x77, 0xeb, 0xcf, 0xb3, 0x46, 0x8f, 0x30, 0x9d, 0x92, 0x3c, 0x95, 0x7c, 0x81, 0x70, 0x72, 0x7a, 0x54, 0x58, 0xac, 0x2c, 0x90, 0x70, 0xf9, 0x3c, 0xfc, 0x37, 0xd3, 0x1c, 0xf9, 0xf1, 0xa3, 0x5d, 0x0c, 0xc3, 0xab, 0xf2, 0x5a, 0xf8, 0xdc, 0x9e, 0x15, 0x90, 0xce, 0x59, 0xab, 0x39, 0xd0, 0x1c, 0xf0, 0xc1, 0x54, 0xab, 0x8d, 0x06, 0x35, 0xc5, 0xe9 |

### 8.4.6.9 HaeHsmPkcs1V15Sha256

Generation, Verification 테스트 벡터 이미지

#### 8.4.6.10 HaeHsmPkcs1V15Sha256Seed Verification

| Public Key Index | Hash(Seed)                                                                                                                                                                                     | Signature                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 401              | 0xbd, 0xf5, 0xe6, 0x4c, 0x28, 0xc9, 0x69, 0x47, 0xd2, 0x1c, 0x8c, 0x7e, 0x85, 0x0d, 0xa8, 0x07, 0x0c, 0x90, 0x3b, 0x5e, 0xae, 0xd0, 0x95, 0xf6, 0xb8, 0xff, 0x48, 0x7d, 0x6d, 0x9c, 0xe5, 0xe9 | 0x33, 0x5f, 0xfa, 0xdc, 0x0b, 0x1b, 0x8b, 0xd2, 0xb1, 0xeb, 0x67, 0x0d, 0xd2, 0x46, 0xe7, 0x6d, 0xcc, 0xcd, 0xc9, 0x55, 0xa1, 0x68, 0x7a, 0x15, 0xf7, 0x4a, 0xa3, 0xe1, 0x59, 0x6e, 0xbd, 0x43, 0xe6, 0x07, 0xc6, 0x40, 0x52, 0x5f, 0x89, 0xdd, 0xa9, 0x58, 0x09, 0xcf, 0xd0, 0x65, 0xf1, 0xbe, 0x4e, 0x4a, 0x24, 0x94, 0x77, 0xd2, 0x4f, 0x40, 0x0d, 0x4d, 0x4c, 0x94, 0x38, 0xa0, 0xaf, 0x95, 0xb2, 0x6b, 0x28, 0xb4, 0x16, 0xe4, 0x2a, 0xa9, 0x50, 0xe2, 0xa5, 0x28, 0x51, 0xb5, 0x21, 0x32, 0x04, 0x8f, 0x1b, 0x1c, 0xe9, 0x44, 0x32, 0x2f, 0xc9, 0x9c, 0x1a, 0xab, 0xb4, 0x9b, 0x7f, 0xae, 0x4c, 0x2f, 0x0f, 0xef, 0x67, 0x4b, 0x50, 0xad, 0xee, 0x3b, 0xbb, 0x5c, 0x6c, 0x33, 0x82, 0x2b, 0x60, 0x8e, 0x4b, 0x95, 0x77, 0x27, 0x5c, 0xa2, 0x0c, 0x71, 0x0a, 0xf9, 0xfc, 0x41, 0xb1, 0xc0, 0x1d, 0x9c, 0x0f, 0xf6, 0xf0, 0xd8, 0x32, 0x4d, 0xc0, 0x8e, 0x1a, 0x76, 0xe2, 0x32, 0xd8, 0xfe, 0xaa, 0x06, 0xc7, 0x3b, 0xbf, 0x64, 0x05, 0x3b, 0xea, 0x35, 0xf1, 0xc5, 0x28, 0xb2, 0x72, 0x27, 0x64, 0x82, 0x2e, 0xf1, 0xff, 0x06, 0x24, 0x6e, 0x75, 0xa9, 0xa2, 0x2a, 0x10, 0xda, 0x4e, 0xa8, 0x4f, 0xc2, 0x44, 0x1b, 0xea, 0x24, 0xb3, 0x55, 0x06, 0xf8, 0x44, 0x7f, 0xcf, 0x69, 0x09, 0x3c, 0x5d, 0x21, 0xab, 0x03, 0x05, 0xcc, 0xe2, 0xc7, 0xea, 0x9f, 0xfa, 0xc3, 0x57, 0xc6, 0x64, 0xb4, 0x91, 0xfc, 0x55, 0xf2, 0x91, 0x9e, 0xc4, 0x90, 0xc3, 0x8a, 0xcc, 0xba, 0xb3, 0x78, 0xc2, 0x52, 0xac, 0x2d, 0xf3, 0x84, 0x5a, 0xcf, 0xf5, 0x75, 0xec, 0x75, 0x24, 0xcd, 0x2f, 0x58, 0x6c, 0xca, 0x14, 0x97, 0xc7, 0x4f, 0x24, 0xb2, 0x99, 0xd6, 0xd6, 0x25, 0x4c, 0x8c, 0xdb, 0x1d, 0x22, 0x7d |

#### 8.4.6.11 HaePkcs1V15Sha1 Raw Key 방식 사용시 Generation, Verification

| Private Key                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Public Key                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Plain Text                                                                                                                                                                                                                                                                                                                                                                                                                                |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| /* modulus : 256 bytes */<br>0xc2U, 0xc2U, 0xe6U, 0x41U, 0xf6U, 0x84U, 0xbeU, 0xe9U, 0xb3U, 0x03U, 0xc6U, 0x8aU, 0xb8U, 0x15U, 0xe7U, 0x08U, 0x17U, 0x50U, 0xc6U, 0x90U, 0xe1U, 0x59U, 0x42U, 0xcfU, 0x17U, 0x63U, 0x5aU, 0x54U, 0xdbU, 0xb2U, 0x62U, 0x77U, 0x70U, 0x38U, 0xb7U, 0x93U, 0x13U, 0x1aU, 0x85U, 0x01U, 0x16U, 0x0eU, 0x7dU, 0xeeU, 0xc1U, 0xe8U, 0xa7U, 0xf7U, 0x24U, 0xf1U, 0x84U, 0x23U, 0x5bU, 0xf2U, 0x5aU, 0xa3U, 0xc5U, 0x04U, 0x20U, 0x75U, 0xa9U, 0x9cU, 0x14U, 0x9bU, 0xaaU, 0x00U, 0x2dU, 0xb1U, 0x80U, 0xa6U, 0x68U, 0xb1U, 0x40U, 0xc5U, 0xe4U, 0x6aU, 0xf2U, 0x9fU, 0xf5U, 0xc5U, 0x69U, 0xdfU, 0x1bU, 0x1eU, 0x84U, 0x1eU, 0xdfU, 0xd9U, 0x52U, 0xc3U, 0xbbU, 0x96U, 0xf6U, 0xe8U, 0xf1U, 0x02U, 0xc5U, 0x1cU, 0x81U, 0x24U, 0xbeU, 0xd6U, 0xdeU, 0x5aU, 0x57U, 0xe5U, 0x29U, 0xb8U, 0x37U, 0xdeU, 0x23U, 0x4eU, 0x8fU, 0xdaU, 0xb0U, 0xd7U, 0xcaU, 0x04U, 0xe2U, 0xebU, 0x8fU, 0x79U, 0xe4U, 0xd8U, 0xadU, 0x8eU, 0x02U, 0x69U, 0x87U, 0x48U, 0x19U, 0x2bU, 0x01U, 0x9eU, 0xabU, 0x8fU, 0x83U, 0x55U, 0x46U, 0x1aU, 0x7cU, 0x3cU, 0xa7U, 0xb4U, 0xc9U, 0x90U, 0xc3U, 0x49U, 0xd2U, 0x0dU, 0xbCU, 0xd7U, | /* modulus : 256 bytes */<br>0xc2U, 0xc2U, 0xe6U, 0x41U, 0xf6U, 0x84U, 0xbeU, 0xe9U, 0xb3U, 0x03U, 0xc6U, 0x8aU, 0xb8U, 0x15U, 0xe7U, 0x08U, 0x17U, 0x50U, 0xc6U, 0x90U, 0xe1U, 0x59U, 0x42U, 0xcfU, 0x17U, 0x63U, 0x5aU, 0x54U, 0xdbU, 0xb2U, 0x62U, 0x77U, 0x70U, 0x38U, 0xb7U, 0x93U, 0x13U, 0x1aU, 0x85U, 0x01U, 0x16U, 0x0eU, 0x7dU, 0xeeU, 0xc1U, 0xe8U, 0xa7U, 0xf7U, 0x24U, 0xf1U, 0x84U, 0x23U, 0x5bU, 0xf2U, 0x5aU, 0xa3U, 0xc5U, 0x04U, 0x20U, 0x75U, 0xa9U, 0x9cU, 0x14U, 0x9bU, 0xaaU, 0x00U, 0x2dU, 0xb1U, 0x80U, 0xa6U, 0x68U, 0xb1U, 0x40U, 0xc5U, 0xe4U, 0x6aU, 0xf2U, 0x9fU, 0xf5U, 0xc5U, 0x69U, 0xdfU, 0x1bU, 0x1eU, 0x84U, 0x1eU, 0xdfU, 0xd9U, 0x52U, 0xc3U, 0xbbU, 0x96U, 0xf6U, 0xe8U, 0xf1U, 0x02U, 0xc5U, 0x1cU, 0x81U, 0x24U, 0xbeU, 0xd6U, 0xdeU, 0x5aU, 0x57U, 0xe5U, 0x29U, 0xb8U, 0x37U, 0xdeU, 0x23U, 0x4eU, 0x8fU, 0xdaU, 0xb0U, 0xd7U, 0xcaU, 0x04U, 0xe2U, 0xebU, 0x8fU, 0x79U, 0xe4U, 0xd8U, 0xadU, 0x8eU, 0x02U, 0x69U, 0x87U, 0x48U, 0x19U, 0x2bU, 0x01U, 0x9eU, 0xabU, 0x8fU, 0x83U, 0x55U, 0x46U, 0x1aU, 0x7cU, 0x3cU, 0xa7U, 0xb4U, 0xc9U, 0x90U, 0xc3U, 0x49U, 0xd2U, 0x0dU, 0xbCU, 0xd7U, | 0x61U, 0x62U, 0x63U, 0x64U, 0x62U, 0x63U, 0x64U, 0x65U, 0x63U, 0x64U, 0x65U, 0x66U, 0x64U, 0x65U, 0x67U, 0x65U, 0x66U, 0x67U, 0x68U, 0x66U, 0x67U, 0x68U, 0x69U, 0x67U, 0x68U, 0x69U, 0x6AU, 0x68U, 0x69U, 0x6BU, 0x67U, 0x68U, 0x69U, 0x6AU, 0x68U, 0x69U, 0x6BU, 0x69U, 0x6AU, 0x68U, 0x6CU, 0x6AU, 0x68U, 0x6CU, 0x6DU, 0x6BU, 0x6CU, 0x6DU, 0x6EU, 0x6CU, 0x6DU, 0x6EU, 0x6FU, 0x6DU, 0x6EU, 0x6FU, 0x70U, 0x6EU, 0x6FU, 0x70U, 0x71U |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Signature                                                                                                                                                                                                                                                                                                                                                                                                                                 |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        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                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------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| <p>0x51U, 0x1aU, 0xccU, 0x60U, 0x9dU, 0x9bU, 0x0dU, 0xe1U, 0x18U, 0x10U, 0x19U, 0xd2U, 0x48U, 0x1bU, 0xf1U, 0xfeU, 0x50U, 0x89U, 0x49U, 0xfeU, 0xa1U, 0x82U, 0xd9U, 0x7aU, 0x42U, 0x5eU, 0x6aU, 0xdeU, 0x26U, 0x5cU, 0x61U, 0x36U, 0xd2U, 0x67U, 0x8eU, 0x55U, 0xeeU, 0xa2U, 0xbfu, 0xe9U, 0xfaU, 0xaeU, 0xddU, 0x1fU, 0xdeU, 0xadU, 0x4eU, 0xd3U, 0x5bU, 0x61U, 0x02U, 0x0fU, 0x8dU, 0x80U, 0xb6U, 0xfeU, 0x12U, 0x7dU, 0x95U, 0x25U, 0x76U, 0x95U, 0x4dU, 0x6aU, 0x73U, 0x41U, 0x05U, 0x62U, 0xc0U, 0x72U, 0x40U, 0x0eU, 0x01U, 0x44U, 0x20U, 0xaeU, 0x51U, 0x04U, 0x57U, 0x97U, 0x72U, 0xf7U, 0x53U, 0x99U, 0xcdU, 0x61U, 0x02U, 0x9fU, 0x4aU, 0x3fU, 0xe0U, 0x0eU, 0xa0U, 0x41U, 0x44U, 0xd7U, 0x77U, 0xbbU, 0xb0U, 0x94U, 0x85U, 0xe0U, 0x1cU, 0x59U, /* exponent : 256 bytes */</p> <p>0x3fU, 0xd6U, 0x34U, 0x34U, 0xc0U, 0xc1U, 0xa7U, 0xf5U, 0xdfU, 0xc8U, 0x48U, 0x06U, 0x65U, 0xeaU, 0x3fU, 0xecU, 0xa5U, 0xffU, 0xc4U, 0x89U, 0x7bU, 0x63U, 0x8fU, 0x83U, 0x63U, 0x8eU, 0x1cU, 0x28U, 0x9eU, 0x4eU, 0x2cU, 0xfaU, 0x2bU, 0x1dU, 0x5cU, 0x06U, 0x68U, 0x07U, 0x94U, 0x3aU, 0xd3U, 0xe8U, 0xf2U, 0x1cU, 0x15U, 0x10U, 0xd6U, 0x25U, 0x2aU, 0xc5U, 0xc2U, 0xe2U, 0xf8U, 0x1bU, 0x4aU, 0xdcU, 0x4dU, 0x12U, 0x5aU, 0x35U, 0x56U, 0xdfU, 0xa2U, 0xddU, 0x96U, 0xeeU, 0x32U, 0x0bU, 0xf1U, 0x38U, 0x1aU, 0x2eU, 0xa7U, 0x25U, 0x76U, 0x06U, 0x1bU, 0xfaU, 0x94U, 0x4bU, 0x1dU, 0xb5U, 0x94U, 0x24U, 0xffU, 0x77U, 0x43U, 0x35U, 0xefU, 0x6bU, 0xdaU, 0x8bU, 0xa7U, 0xa8U, 0xb9U, 0x7eU, 0x87U, 0xefU, 0x1cU, 0xe1U, 0x57U, 0x70U, 0x04U, 0x37U, 0xa6U, 0x81U, 0xa5U, 0x98U, 0x0dU, 0x2fU, 0xd1U, 0x70U, 0xc9U, 0xb4U, 0x0bU, 0xacU, 0xdfU, 0x39U, 0x94U, 0xc0U, 0x53U, 0x1eU, 0xc8U, 0x04U, 0xc1U, 0x76U, 0x3fU, 0x91U, 0x96U, 0x32U, 0x35U, 0x32U, 0x12U, 0x73U, 0x42U, 0x7eU, 0x52U, 0x17U, 0xcdU, 0x87U, 0x2fU, 0xbeU, 0xebU, 0xd1U, 0x8fU, 0x80U, 0x2bU, 0x8aU, 0xfdfU, 0xe1U, 0x15U, 0x9fU, 0xaaU, 0x6fU, 0xc4U, 0x6cU, 0x0fU, 0xeeU, 0xc6U, 0xe5U, 0xb6U, 0x4eU, 0x07U, 0xa2U, 0xa1U, 0xfdfU, 0x89U, 0x82U, 0xdfU, 0x33U, 0x82U, 0x19U, 0x3eU, 0x68U, 0x46U, 0xc4U, 0x66U, 0xeaU, 0x98U, 0x50U, 0xaaU, 0x10U, 0x2dU, 0x31U, 0x3dU, 0x63U, 0xf7U, 0x22U, 0x70U, 0x7fU, 0xb0U, 0x61U, 0x5aU, 0x41U, 0x22U, 0xb4U, 0xe6U, 0xfeU, 0x0bU, 0xddU, 0x09U, 0x42U, 0x14U, 0xabU, 0xb4U, 0x04U, 0xe1U, 0x23U, 0xb7U, 0xaeU, 0xf7U, 0x5fU, 0x23U, 0x43U, 0x1bU, 0x7eU, 0x65U, 0x21U, 0xfbU, 0xe9U, 0x2dU, 0x8eU, 0x01U, 0xaaU, 0xf9U, 0x7fU, 0xdcU, 0x85U, 0x01U, 0x1dU, 0xe3U, 0x21U, 0x4aU, 0xefU, 0xeaU, 0x9aU, 0x72U, 0x8cU, 0xf6U, 0x69U, 0x29U, 0x44U, 0xb9U, 0xdcU, 0xf9U, 0xd2U, 0xc9U, 0xcbU, 0xdfU, 0x97U, 0x82U, 0x96U, 0x5cU, 0x69U, 0x77U, 0x91U</p> | <p>0x51U, 0x1aU, 0xccU, 0x60U, 0x9dU, 0x9bU, 0x0dU, 0xe1U, 0x18U, 0x10U, 0x19U, 0xd2U, 0x48U, 0x1bU, 0xf1U, 0xfeU, 0x50U, 0x89U, 0x49U, 0xfeU, 0xa1U, 0x82U, 0xd9U, 0x7aU, 0x42U, 0x5eU, 0x6aU, 0xdeU, 0x26U, 0x5cU, 0x61U, 0x36U, 0xd2U, 0x67U, 0x8eU, 0x55U, 0xeeU, 0xa2U, 0xbfu, 0xe9U, 0xfaU, 0xaeU, 0xddU, 0x1fU, 0xdeU, 0xadU, 0x4eU, 0xd3U, 0x5bU, 0x61U, 0x02U, 0x0fU, 0x8dU, 0x80U, 0xb6U, 0xfeU, 0x12U, 0x7dU, 0x95U, 0x25U, 0x76U, 0x95U, 0x4dU, 0x6aU, 0x73U, 0x41U, 0x05U, 0x62U, 0xc0U, 0x72U, 0x40U, 0x0eU, 0x01U, 0x44U, 0x20U, 0xaeU, 0x51U, 0x04U, 0x57U, 0x97U, 0x72U, 0xf7U, 0x53U, 0x99U, 0xcdU, 0x61U, 0x02U, 0x9fU, 0x4aU, 0x3fU, 0xe0U, 0x0eU, 0xa0U, 0x41U, 0x44U, 0xd7U, 0x77U, 0xbbU, 0xb0U, 0x94U, 0x85U, 0xe0U, 0x1cU, 0x59U, /* exponent : 4 bytes */</p> <p>0x00U, 0x01U, 0x00U, 0x01U,</p> | <p>0x18U, 0x39U, 0x03U, 0x0CU, 0x21U, 0xf4U, 0xe3U, 0x0CU, 0x53U, 0x13U, 0xf5U, 0x8DU, 0xC2U, 0xf6U, 0xd2U, 0xd0U, 0xd2U, 0xDDU, 0x5DU, 0x12U, 0xFFU, 0x37U, 0x3AU, 0x5BU, 0x36U, 0xDEU, 0xADU, 0x6DU, 0x8AU, 0x1AU, 0xCFU, 0xA4U, 0x1DU, 0x3BU, 0x4CU, 0x93U, 0xEAU, 0xFDU, 0x0FU, 0x69U, 0xDDU, 0x79U, 0x3BU, 0x68U, 0xADU, 0xE1U, 0x8FU, 0xB5U, 0x44U, 0x28U, 0x97U, 0x87U, 0x50U, 0x0AU, 0x0AU, 0x61U, 0x67U, 0x94U, 0x78U, 0x31U, 0x2EU, 0x0DU, 0x13U, 0xDDU, 0xDAU, 0x6DU, 0x03U, 0x53U, 0xC5U, 0xB4U, 0xDAU, 0xA9U, 0x34U, 0x0EU, 0xF7U, 0xE1U, 0x5FU, 0x0AU, 0x81U, 0x59U, 0xCB, 0xB2U, 0x4BU, 0xE1U, 0x5FU, 0x0AU, 0x7AU, 0xD1U, 0xDCU, 0x8CU, 0x3EU, 0xD8U, 0xD8U, 0xDAU, 0xCDU, 0xEFU, 0xF2U, 0x15U, 0x57U, 0x24U, 0x5EU, 0x8BU, 0xC2U, 0x91U, 0x80U, 0x71U, 0xE0U, 0x40U, 0xE8U, 0xD6U, 0x39U, 0xD7U, 0xD3U, 0xCEU, 0xA2U, 0x00U, 0xF2U, 0x3BU, 0x42U, 0xD6U, 0x30U, 0x16U, 0x5CU, 0x65U, 0x20U, 0xC2U, 0x49U, 0x53U, 0x78U, 0xE8U, 0x0BU, 0xF2U, 0xC2U, 0x83U, 0x9FU, 0x61U, 0xB5U, 0xC8U, 0xA2U, 0xCAU, 0xF1U, 0x11U, 0xC7U, 0x46U, 0x5BU, 0x6EU, 0x75U, 0xD2U, 0x91U, 0x68U, 0xAEU, 0xB2U, 0xB7U, 0xC5U, 0x60U, 0x6FU, 0x8DU, 0xD2U, 0x28U, 0x17U, 0x45U, 0x68U, 0x7EU, 0xA7U, 0x24U, 0x82U, 0x28U, 0x7BU, 0x71U, 0x26U, 0x6DU, 0xFBU, 0x5BU, 0x13U, 0x06U, 0xACU, 0xF8U, 0x51U, 0xF8U, 0x6CU, 0xB6U, 0x55U, 0x32U, 0x90U, 0x43U, 0x95U, 0x6EU, 0xE1U, 0x8DU, 0x21U, 0x6FU, 0x8EU, 0xF0U, 0x8BU, 0x09U, 0x49U, 0x06U, 0xF1U, 0x87U, 0xE0U, 0x69U, 0x5FU, 0x45U, 0x13U, 0x28U, 0xE8U, 0xF0U, 0x5FU, 0x3BU, 0x41U, 0xF1U, 0x34U, 0xB4U, 0xCB, 0x71U, 0x89U, 0x95U, 0xD3U, 0xC6U, 0x00U, 0x99U, 0xFEU, 0x45U, 0xE2U, 0x0CU, 0xE2U, 0x2U, 0x9CU, 0x3CU, 0xF0U, 0x12U, 0x17U, 0xC2U, 0x35U, 0x0EU, 0x93U, 0xB2U, 0x8FU, 0x91U, 0x37U, 0x3EU, 0x0CU, 0xE6U, 0xA5U, 0x51U, 0x7AU, 0x1EU, 0x31U, 0x27U, 0xADU, 0xE2U, 0x10U, 0x3BU, 0x9BU, 0xCFU, 0x51U</p> |
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## 8.4.6.12 HaePkcs1V15Sha1Seed

### Raw Key 방식 사용자

#### Verification

| Public Key                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Hash(Seed)                                                                                                                                        | Signature                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
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| <p>/* modulus : 256 bytes */</p> <p>0xc2U, 0xc2U, 0xe6U, 0x41U, 0xf6U, 0x84U, 0xbeU, 0xe9U, 0xb3U, 0x03U, 0xc6U, 0x8aU, 0xb8U, 0x15U, 0xe7U, 0x08U, 0x17U, 0x50U, 0xc0U, 0x90U, 0xe1U, 0x59U, 0x42U, 0xc4U, 0x17U, 0x63U, 0x5aU, 0x54U, 0xdbU, 0xb2U, 0x62U, 0x77U, 0x70U, 0x38U, 0xb7U, 0x93U, 0x13U, 0x1aU, 0x85U, 0x01U, 0x16U, 0x0eU, 0x7dU, 0xeeU, 0xc1U, 0xe8U, 0xa7U, 0xf7U, 0x24U, 0xf1U, 0x84U, 0x23U, 0x5bU, 0xf2U, 0x5aU, 0xa3U, 0xc5U, 0x04U, 0x20U, 0x75U, 0xa9U, 0x9cU, 0x14U, 0x9bU, 0xaaU, 0x00U, 0x2dU, 0xb1U, 0x80U, 0xa6U, 0x68U, 0xb1U, 0x40U, 0xc5U, 0xe4U, 0x6aU, 0xf2U, 0x9fU, 0xf5U, 0xc5U, 0x69U, 0xdfU, 0x1bU, 0x1eU, 0x84U, 0x1eU, 0xdfU, 0xd9U, 0x52U, 0xc3U, 0xbbU, 0x96U, 0x6fU, 0xe8U, 0xf1U, 0x02U, 0xc5U, 0x1cU, 0x81U, 0x24U, 0xbeU, 0xd6U, 0xdeU, 0x5aU, 0x57U, 0xe5U, 0x29U, 0xb8U, 0x37U, 0xdeU, 0x23U, 0x4eU, 0x8fU, 0xdaU, 0xb0U, 0xd7U, 0xcaU, 0x04U, 0xe2U, 0xebU, 0x8fU, 0x79U, 0xe4U, 0xd8U, 0xadU, 0x8eU, 0x02U, 0x69U, 0x87U, 0x48U, 0x19U, 0x2bU, 0x01U, 0x9eU, 0xabU, 0x8fU, 0x83U, 0x55U, 0x46U, 0x1aU, 0x7cU, 0x3cU, 0xa7U, 0xb4U, 0xc9U, 0x90U, 0xc3U, 0x49U, 0xd2U, 0x0dU, 0xb0U, 0xd7U, 0x51U, 0x1aU, 0xccU, 0x60U, 0x9dU, 0x9bU, 0x0dU, 0xe1U, 0x18U, 0x10U, 0x19U, 0xd2U, 0x48U, 0x1bU, 0xf1U, 0xfeU, 0x50U, 0x89U, 0x49U, 0xfeU, 0xa1U, 0x82U, 0xd9U, 0x7aU, 0x42U, 0x5eU, 0x6aU, 0xdeU, 0x26U, 0x5cU, 0x61U, 0x36U, 0xd2U, 0x67U, 0x8eU, 0x55U, 0xeeU, 0xa2U, 0xbfu, 0xe9U, 0xfaU, 0xaeU, 0xddU, 0x1fU, 0xdeU, 0xadU, 0x4eU, 0xd3U, 0x5bU, 0x61U, 0x02U, 0x0fU, 0x8dU, 0x80U, 0xb6U, 0xfeU, 0x12U, 0x7dU, 0x95U, 0x25U, 0x76U, 0x95U, 0x4dU, 0x6aU, 0x73U, 0x41U, 0x05U, 0x62U, 0xc0U, 0x72U, 0x40U, 0x0eU, 0x01U, 0x44U, 0x20U, 0xaeU, 0x51U, 0x04U, 0x57U, 0x97U, 0x72U, 0xf7U, 0x53U, 0x99U, 0xcdU, 0x61U, 0x02U, 0x9fU, 0x4aU, 0x3fU, 0xe0U, 0x0eU, 0xa0U, 0x41U, 0x44U, 0xd7U, 0x77U, 0xbbU, 0xb0U, 0x94U, 0x85U, 0xe0U, 0x1cU, 0x59U, /* exponent : 4 bytes */</p> <p>0x00U, 0x01U, 0x00U, 0x01U,</p> | <p>0x84U, 0x98U, 0x3eU, 0x44U, 0x1cU, 0x3bU, 0xd2U, 0x6eU, 0xbaU, 0xaeU, 0x4aU, 0xa1U, 0xf9U, 0x51U, 0x29U, 0xe5U, 0xe5U, 0x46U, 0x70U, 0xf1U</p> | <p>0x18U, 0x39U, 0x03U, 0x0CU, 0x21U, 0xf4U, 0xe3U, 0x0CU, 0x53U, 0x13U, 0xf5U, 0x8DU, 0xC2U, 0xf6U, 0xd2U, 0xd0U, 0xd2U, 0xDDU, 0x5DU, 0x12U, 0xFFU, 0x37U, 0x3AU, 0x5BU, 0x36U, 0xDEU, 0xADU, 0x6DU, 0x8AU, 0x1AU, 0xCFU, 0xA4U, 0x1DU, 0x3BU, 0x4CU, 0x93U, 0xEAU, 0xFDU, 0x0FU, 0x69U, 0xDDU, 0x79U, 0x3BU, 0x68U, 0xADU, 0xE1U, 0x8FU, 0xB5U, 0x44U, 0x28U, 0x97U, 0x87U, 0x50U, 0x0AU, 0x0AU, 0x61U, 0x67U, 0x94U, 0x78U, 0x31U, 0x2EU, 0x0DU, 0x13U, 0xDDU, 0xDAU, 0x6DU, 0x03U, 0x53U, 0xC5U, 0xB4U, 0xDAU, 0xA9U, 0x34U, 0x0EU, 0xF7U, 0xE1U, 0x5FU, 0x0AU, 0x81U, 0x59U, 0xCB, 0xB2U, 0x4BU, 0xE1U, 0x5FU, 0x0AU, 0x7AU, 0xD1U, 0xDCU, 0x8CU, 0x3EU, 0xD8U, 0xD8U, 0xDAU, 0xCDU, 0xEFU, 0xF2U, 0x15U, 0x57U, 0x24U, 0x5EU, 0x8BU, 0xC2U, 0x91U, 0x80U, 0x71U, 0xE0U, 0x40U, 0xE8U, 0xD6U, 0x39U, 0xD7U, 0xD3U, 0xCEU, 0xA2U, 0x00U, 0xF2U, 0x3BU, 0x42U, 0xD6U, 0x30U, 0x16U, 0x5CU, 0x65U, 0x20U, 0xC2U, 0x49U, 0x53U, 0x78U, 0xE8U, 0x0BU, 0xF2U, 0xC2U, 0x83U, 0x9FU, 0x61U, 0xB5U, 0xC8U, 0xA2U, 0xCAU, 0xF1U, 0x11U, 0xC7U, 0x46U, 0x5BU, 0x6EU, 0x75U, 0xD2U, 0x91U, 0x68U, 0xAEU, 0xB2U, 0xB7U, 0xC5U, 0x60U, 0x6FU, 0x8DU, 0xD2U, 0x28U, 0x17U, 0x45U, 0x68U, 0x7EU, 0xA7U, 0x24U, 0x82U, 0x28U, 0x7BU, 0x71U, 0x26U, 0x6DU, 0xFBU, 0x5BU, 0x13U, 0x06U, 0xACU, 0xF8U, 0x51U, 0xF8U, 0x6CU, 0xB6U, 0x55U, 0x32U, 0x90U, 0x43U, 0x95U, 0x6EU, 0xE1U, 0x8DU, 0x21U, 0x6FU, 0x8EU, 0xF0U, 0x8BU, 0x09U, 0x49U, 0x06U, 0xF1U, 0x87U, 0xE0U, 0x69U, 0x5FU, 0x45U, 0x13U, 0x28U, 0xE8U, 0xF0U, 0x5FU, 0x3BU, 0x41U, 0xF1U, 0x34U, 0xB4U, 0xCB, 0x71U, 0x89U, 0x95U, 0xD3U, 0xC6U, 0x00U, 0x99U, 0xFEU, 0x45U, 0xE2U, 0x0CU, 0xE2U, 0x2U, 0x9CU, 0x3CU, 0xF0U, 0x12U, 0x17U, 0xC2U, 0x35U, 0x0EU, 0x93U, 0xB2U, 0x8FU, 0x91U, 0x37U, 0x3EU, 0x0CU, 0xE6U, 0xA5U, 0x51U, 0x7AU, 0x1EU, 0x31U, 0x27U, 0xADU, 0xE2U, 0x10U, 0x3BU, 0x9BU, 0xCFU, 0x51U</p> |

## 8.4.6.13 HaePkcs1V15Sha256

### Raw Key 방식 사용자

## Generation, Verification

| Private Key                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            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| /* modulus : 256 bytes */<br>0xc2U, 0xc2U, 0xe6U, 0x41U, 0xf6U, 0x84U, 0xbeU, 0xe9U,<br>0xb3U, 0x03U, 0xc6U, 0x8aU, 0xb8U, 0x15U, 0xe7U, 0x08U,<br>0x17U, 0x50U, 0xc6U, 0x90U, 0xe1U, 0x59U, 0x42U, 0xcFU,<br>0x17U, 0x63U, 0x5aU, 0x54U, 0xdbU, 0xb2U, 0x62U, 0x77U,<br>0x70U, 0x38U, 0xb7U, 0x93U, 0x13U, 0x1aU, 0x85U, 0x01U,<br>0x16U, 0x0eU, 0x7dU, 0xeeU, 0xc1U, 0xe8U, 0xa7U, 0xf7U,<br>0x24U, 0xf1U, 0x84U, 0x23U, 0x5bU, 0xf2U, 0x5aU, 0xa3U,<br>0xc5U, 0x04U, 0x20U, 0x75U, 0xa9U, 0x9cU, 0x14U, 0x9bU,<br>0xaaU, 0x00U, 0x2dU, 0xb1U, 0x80U, 0xa6U, 0x68U, 0xb1U,<br>0x40U, 0xc5U, 0xe4U, 0x6aU, 0xf2U, 0x9fU, 0xf5U, 0xc5U,<br>0x69U, 0xdfU, 0x1bU, 0x1eU, 0x84U, 0x1eU, 0xdfU, 0xd9U,<br>0x52U, 0xc3U, 0xbbU, 0x96U, 0x6fU, 0xe8U, 0xf1U, 0x02U,<br>0xc5U, 0x1cU, 0x81U, 0x24U, 0xbeU, 0xd6U, 0xdeU, 0x5aU,<br>0x57U, 0xe5U, 0x29U, 0xb8U, 0x37U, 0xdeU, 0x23U, 0x4eU,<br>0x8fU, 0xdaU, 0xb0U, 0xd7U, 0xcaU, 0x04U, 0xe2U, 0xebU,<br>0x8fU, 0x79U, 0xe4U, 0xd8U, 0xadU, 0x8eU, 0x02U, 0x69U,<br>0x87U, 0x48U, 0x19U, 0x2bU, 0x01U, 0x9eU, 0xabU, 0x8fU,<br>0x83U, 0x55U, 0x46U, 0x1aU, 0x7cU, 0x3cU, 0xa7U, 0xb4U,<br>0xc9U, 0x90U, 0xc3U, 0x49U, 0xd2U, 0x0dU, 0xbcU, 0xd7U,<br>0x51U, 0x1aU, 0xccU, 0x60U, 0x9dU, 0x9bU, 0x0dU, 0xe1U,<br>0x18U, 0x10U, 0x19U, 0xd2U, 0x48U, 0x1bU, 0xf1U, 0xfeU,<br>0x50U, 0x89U, 0x49U, 0xfeU, 0xa1U, 0x82U, 0xd9U, 0x7aU,<br>0x42U, 0x5eU, 0x6aU, 0x55U, 0x2eU, 0x5cU, 0x61U, 0x36U,<br>0xd2U, 0x67U, 0x8eU, 0x8eU, 0xeeU, 0xa2U, 0xbfU, 0xe9U,<br>0xfaU, 0xaeU, 0xddU, 0x1fU, 0xdeU, 0xadU, 0x4eU, 0xd3U,<br>0x5bU, 0x61U, 0x02U, 0xf0U, 0x8dU, 0x80U, 0xb6U, 0xfeU,<br>0x12U, 0x7dU, 0x95U, 0x25U, 0x76U, 0x95U, 0x4dU, 0x6aU,<br>0x73U, 0x41U, 0x05U, 0x62U, 0xc6U, 0x72U, 0x40U, 0x0eU,<br>0x01U, 0x44U, 0x20U, 0xaeU, 0x51U, 0x04U, 0x57U, 0x97U,<br>0x72U, 0xf7U, 0x53U, 0x99U, 0xcdU, 0x61U, 0x02U, 0x9fU,<br>0x4aU, 0x3fU, 0xe0U, 0x0eU, 0xa0U, 0x41U, 0x44U, 0xd7U,<br>0x77U, 0xbbU, 0xb0U, 0x94U, 0x85U, 0xe0U, 0x1cU, 0x59U,<br>/* exponent : 256 bytes */<br>0x3fU, 0xd6U, 0x34U, 0x34U, 0xc0U, 0x1cU, 0xa7U, 0xf5U,<br>0xdfU, 0x88U, 0x48U, 0x06U, 0x65U, 0xeaU, 0x3fU, 0xecU,<br>0xa5U, 0xffU, 0xc4U, 0x89U, 0x7bU, 0x63U, 0x8fU, 0x83U,<br>0x63U, 0x8eU, 0x1cU, 0x28U, 0x9eU, 0x4eU, 0x2cU, 0xfaU,<br>0x2bU, 0x1dU, 0x5cU, 0x06U, 0x68U, 0x07U, 0x94U, 0x3aU,<br>0xd3U, 0xe8U, 0xf2U, 0x1cU, 0x15U, 0x10U, 0xd6U, 0x25U,<br>0x2aU, 0xc5U, 0xc2U, 0xe2U, 0xf8U, 0x1bU, 0x4aU, 0xdcdU,<br>0x4dU, 0x12U, 0x5aU, 0x35U, 0x56U, 0xdfU, 0xa2U, 0xddU,<br>0x96U, 0xeeU, 0x32U, 0x0bU, 0xf1U, 0x38U, 0x1aU, 0x2eU,<br>0x7aU, 0x25U, 0x76U, 0x06U, 0x1bU, 0xfaU, 0x94U, 0x4bU,<br>0x1dU, 0xb5U, 0x94U, 0x24U, 0xffU, 0x77U, 0x43U, 0x35U,<br>0xefU, 0x6bU, 0xdaU, 0x8bU, 0xa7U, 0xa8U, 0xb9U, 0x7eU,<br>0x87U, 0xefU, 0x1cU, 0xe1U, 0x57U, 0x70U, 0x04U, 0x37U,<br>0xa6U, 0x81U, 0xa5U, 0x98U, 0x0dU, 0x2fU, 0xd1U, 0x70U,<br>0xc9U, 0xb4U, 0x0bU, 0xacU, 0xdfU, 0x39U, 0x94U, 0xc6U,<br>0x53U, 0x1eU, 0xc8U, 0x04U, 0xc1U, 0x76U, 0x3fU, 0x91U,<br>0x96U, 0x32U, 0x35U, 0x32U, 0x12U, 0x73U, 0x42U, 0x7eU,<br>0x52U, 0x17U, 0xcdU, 0x87U, 0x2fU, 0xbeU, 0xebU, 0xd1U,<br>0x8fU, 0x80U, 0x2bU, 0x8aU, 0xfdfU, 0xe1U, 0x15U, 0x9fU,<br>0xaaU, 0x6fU, 0xc4U, 0x6cU, 0x0fU, 0xeeU, 0xc6U, 0xe5U,<br>0xb6U, 0x4eU, 0x07U, 0xa2U, 0xa1U, 0xfdfU, 0x89U, 0x82U,<br>0xdfU, 0x33U, 0x82U, 0x19U, 0x3eU, 0x68U, 0x46U, 0xcFU,<br>0x66U, 0xeaU, 0x98U, 0x50U, 0xaaU, 0x10U, 0x2dU, 0x31U,<br>0x3dU, 0x63U, 0xf7U, 0x22U, 0x70U, 0x7fU, 0x5cU, 0x61U,<br>0x5aU, 0x41U, 0x22U, 0xb4U, 0xe6U, 0xfeU, 0x0bU, 0xddU,<br>0x09U, 0x42U, 0x14U, 0xabU, 0xb4U, 0x04U, 0xe1U, 0x23U,<br>0xb7U, 0xaeU, 0xf7U, 0x5fU, 0x23U, 0x43U, 0x1bU, 0x7eU,<br>0x65U, 0x21U, 0xfbfU, 0xe9U, 0x2dU, 0x8eU, 0x01U, 0xaaU,<br>0xf9U, 0x7fU, 0xdcdU, 0x85U, 0x01U, 0x1dU, 0xe3U, 0x21U,<br>0x4aU, 0xefU, 0xeaU, 0x9aU, 0x72U, 0x8cU, 0xf6U, 0x69U,<br>0x29U, 0x44U, 0xb9U, 0xdcdU, 0xf9U, 0x2dU, 0xc9U, 0xc6U,<br>0xdfU, 0x97U, 0x82U, 0x96U, 0x5cU, 0x69U, 0x77U, 0x91U | /* modulus : 256 bytes */<br>0xc2U, 0xc2U, 0xe6U, 0x41U, 0xf6U, 0x84U, 0xbeU, 0xe9U,<br>0xb3U, 0x03U, 0xc6U, 0x8aU, 0xb8U, 0x15U, 0xe7U, 0x08U,<br>0x17U, 0x50U, 0xc6U, 0x90U, 0xe1U, 0x59U, 0x42U, 0xcFU,<br>0x17U, 0x63U, 0x5aU, 0x54U, 0xdbU, 0xb2U, 0x62U, 0x77U,<br>0x70U, 0x38U, 0xb7U, 0x93U, 0x13U, 0x1aU, 0x85U, 0x01U,<br>0x16U, 0x0eU, 0x7dU, 0xeeU, 0xc1U, 0xe8U, 0xa7U, 0xf7U,<br>0x24U, 0xf1U, 0x84U, 0x23U, 0x5bU, 0xf2U, 0x5aU, 0xa3U,<br>0xc5U, 0x04U, 0x20U, 0x75U, 0xa9U, 0x9cU, 0x14U, 0x9bU,<br>0xaaU, 0x00U, 0x2dU, 0xb1U, 0x80U, 0xa6U, 0x68U, 0xb1U,<br>0x40U, 0xc5U, 0xe4U, 0x6aU, 0xf2U, 0x9fU, 0xf5U, 0xc5U,<br>0x69U, 0xdfU, 0x1bU, 0x1eU, 0x84U, 0x1eU, 0xdfU, 0xd9U,<br>0x52U, 0xc3U, 0xbbU, 0x96U, 0x6fU, 0xe8U, 0xf1U, 0x02U,<br>0xc5U, 0x1cU, 0x81U, 0x24U, 0xbeU, 0xd6U, 0xdeU, 0x5aU,<br>0x57U, 0xe5U, 0x29U, 0xb8U, 0x37U, 0xdeU, 0x23U, 0x4eU,<br>0x8fU, 0xdaU, 0xb0U, 0xd7U, 0xcaU, 0x04U, 0xe2U, 0xebU,<br>0x8fU, 0x79U, 0xe4U, 0xd8U, 0xadU, 0x8eU, 0x02U, 0x69U,<br>0x87U, 0x48U, 0x19U, 0x2bU, 0x01U, 0x9eU, 0xabU, 0x8fU,<br>0x83U, 0x55U, 0x46U, 0x1aU, 0x7cU, 0x3cU, 0xa7U, 0xb4U,<br>0xc9U, 0x90U, 0xc3U, 0x49U, 0xd2U, 0x0dU, 0xbcU, 0xd7U,<br>0x51U, 0x1aU, 0xccU, 0x60U, 0x9dU, 0x9bU, 0x0dU, 0xe1U,<br>0x18U, 0x10U, 0x19U, 0xd2U, 0x48U, 0x1bU, 0xf1U, 0xfeU,<br>0x50U, 0x89U, 0x49U, 0xfeU, 0xa1U, 0x82U, 0xd9U, 0x7aU,<br>0x42U, 0x5eU, 0x6aU, 0x55U, 0x2eU, 0x5cU, 0x61U, 0x36U,<br>0xd2U, 0x67U, 0x8eU, 0x8eU, 0xeeU, 0xa2U, 0xbfU, 0xe9U,<br>0xfaU, 0xaeU, 0xddU, 0x1fU, 0xdeU, 0xadU, 0x4eU, 0xd3U,<br>0x5bU, 0x61U, 0x02U, 0xf0U, 0x8dU, 0x80U, 0xb6U, 0xfeU,<br>0x12U, 0x7dU, 0x95U, 0x25U, 0x76U, 0x95U, 0x4dU, 0x6aU,<br>0x73U, 0x41U, 0x05U, 0x62U, 0xc6U, 0x72U, 0x40U, 0x0eU,<br>0x01U, 0x44U, 0x20U, 0xaeU, 0x51U, 0x04U, 0x57U, 0x97U,<br>0x72U, 0xf7U, 0x53U, 0x99U, 0xcdU, 0x61U, 0x02U, 0x9fU,<br>0x4aU, 0x3fU, 0xe0U, 0x0eU, 0xa0U, 0x41U, 0x44U, 0xd7U,<br>0x77U, 0xbbU, 0xb0U, 0x94U, 0x85U, 0xe0U, 0x1cU, 0x59U,<br>/* exponent : 4 bytes */<br>0x00U, 0x01U, 0x00U, 0x01U, | 0x61U, 0x62U, 0x63U, 0x64U, 0x62U, 0x63U, 0x64U, 0x65U,<br>0x63U, 0x64U, 0x65U, 0x66U, 0x64U, 0x65U, 0x66U, 0x67U,<br>0x65U, 0x66U, 0x67U, 0x68U, 0x66U, 0x67U, 0x68U, 0x69U,<br>0x67U, 0x68U, 0x69U, 0x6AU, 0x68U, 0x69U, 0x6AU, 0x68U,<br>0x69U, 0x6AU, 0x68U, 0x6CU, 0x6AU, 0x68U, 0x6CU, 0x68U,<br>0x68U, 0x6CU, 0x6DU, 0x6EU, 0x6CU, 0x6DU, 0x6EU, 0x6FU,<br>0x6DU, 0x6EU, 0x6FU, 0x70U, 0x6EU, 0x6FU, 0x70U, 0x71U<br><br>Signature<br><br>0x54U, 0xA2U, 0x4EU, 0x58U, 0xF3U, 0x28U, 0xA0U, 0xF5U,<br>0xF8U, 0x2CU, 0x4FU, 0xBFU, 0x7BU, 0x84U, 0x16U, 0xF3U,<br>0xDCU, 0xB2U, 0x16U, 0x6CU, 0xD4U, 0x01U, 0x7CU, 0xA0U,<br>0xB6U, 0x0BU, 0x1AU, 0x4DU, 0x99U, 0xB6U, 0x96U, 0xA4U,<br>0x2FU, 0x91U, 0xC5U, 0xEFU, 0xC7U, 0xDBU, 0xCFU, 0x5CU,<br>0x7AU, 0x92U, 0x26U, 0x04U, 0x23U, 0x3EU, 0x3AU, 0x0DU,<br>0x01U, 0xD1U, 0x71U, 0xC4U, 0x94U, 0x0DU, 0xA2U, 0xF7U,<br>0x49U, 0xA2U, 0x21U, 0x5BU, 0x1FU, 0xF8U, 0x3EU, 0xADU,<br>0x2CU, 0xB2U, 0xADU, 0xDBU, 0xCDU, 0x38U, 0x6CU, 0xE9U,<br>0x61U, 0xE8U, 0x4AU, 0xEBU, 0x5EU, 0x61U, 0x66U, 0x8FU,<br>0x26U, 0xAFU, 0x84U, 0x63U, 0x8AU, 0x1AU, 0x18U, 0x3CU,<br>0xC4U, 0x36U, 0x35U, 0x29U, 0x0DU, 0x97U, 0xB1U, 0x2BU,<br>0xF4U, 0x7AU, 0x4AU, 0xE0U, 0xA1U, 0x62U, 0xA3U, 0xECU,<br>0x36U, 0x9FU, 0xAEU, 0xDBU, 0x0DU, 0x64U, 0x41U, 0x8FU,<br>0x65U, 0x5BU, 0x44U, 0xB9U, 0x29U, 0x1EU, 0xA6U, 0xA5U,<br>0x54U, 0x1DU, 0x98U, 0x53U, 0xA9U, 0xAAU, 0x08U, 0x80U,<br>0x9FU, 0xE1U, 0x2EU, 0xF7U, 0x72U, 0xD3U, 0x2EU, 0xABU,<br>0x39U, 0xEBU, 0x3DU, 0x65U, 0x42U, 0x19U, 0x33U, 0x3BU,<br>0xB6U, 0x5DU, 0x8EU, 0x62U, 0x93U, 0x0DU, 0x01U, 0x48U,<br>0x42U, 0xABU, 0xD3U, 0x01U, 0x26U, 0xA4U, 0x5DU, 0xE9U,<br>0x0DU, 0x1FU, 0xA1U, 0xF4U, 0x80U, 0xFBU, 0x52U, 0xFEU,<br>0xB6U, 0x49U, 0x60U, 0xE3U, 0x43U, 0x85U, 0x0BU, 0x27U,<br>0xC7U, 0xC0U, 0x1CU, 0x7EU, 0xD2U, 0xF7U, 0x65U, 0xBBU,<br>0x59U, 0x48U, 0xA4U, 0x81U, 0xFEU, 0x6AU, 0x3CU, 0x43U,<br>0x1DU, 0xF7U, 0x64U, 0x78U, 0xC5U, 0x7EU, 0xFCU, 0x98U,<br>0x8CU, 0xD5U, 0xCAU, 0x2EU, 0x9CU, 0xB7U, 0x1BU, 0x57U,<br>0xCDU, 0xB7U, 0xECU, 0xC1U, 0xF6U, 0x1BU, 0xC7U, 0x2FU,<br>0xADU, 0x15U, 0x90U, 0x47U, 0xF2U, 0x5DU, 0xB4U, 0xCBU,<br>0x15U, 0xFFU, 0x28U, 0x58U, 0x1FU, 0x8CU, 0x8DU, 0xAEU,<br>0x62U, 0x71U, 0x0DU, 0xE2U, 0x4AU, 0x80U, 0x56U, 0xC7U,<br>0xB2U, 0xFAU, 0xABU, 0x36U, 0xB1U, 0x5DU, 0xBFU, 0x58U,<br>0xF1U, 0x1FU, 0xB7U, 0x10U, 0xA5U, 0xFBU, 0xBBU, 0x4AU |

### 8.4.6.14 HaePkcs1V15Sha256Seed

#### Raw Key 방식 사용시

#### Verification

| Public Key | Hash(Seed) | Signature |
|------------|------------|-----------|
|------------|------------|-----------|

/\* modulus : 256 bytes \*/

0xc2U, 0xc2U, 0xe6U, 0x41U, 0xf6U, 0x84U, 0xbeU, 0xe9U,  
0xb3U, 0x03U, 0xc6U, 0x8aU, 0xb8U, 0x15U, 0xe7U, 0x08U,  
0x17U, 0x50U, 0xc6U, 0x90U, 0xe1U, 0x59U, 0x42U, 0xcFU,  
0x17U, 0x63U, 0x5aU, 0x54U, 0xdbU, 0xb2U, 0x62U, 0x77U,  
0x70U, 0x38U, 0xb7U, 0x93U, 0x13U, 0x1aU, 0x85U, 0x01U,  
0x16U, 0x0eU, 0x7dU, 0xeeU, 0xc1U, 0xe8U, 0xa7U, 0xf7U,  
0x24U, 0xf1U, 0x84U, 0x23U, 0x5bU, 0xf2U, 0x5aU, 0xa3U,  
0x5U, 0x04U, 0x20U, 0x75U, 0xa9U, 0x9cU, 0x14U, 0x9bU,  
0xaaU, 0x00U, 0x2dU, 0xb1U, 0x80U, 0xa6U, 0x68U, 0xb1U,  
0x40U, 0xc5U, 0xe4U, 0x6aU, 0xf2U, 0x9fU, 0xf5U, 0xc5U,  
0x69U, 0xdfU, 0x1bU, 0x1eU, 0x84U, 0x1eU, 0xdfU, 0xd9U,  
0x52U, 0xc3U, 0xbbU, 0x96U, 0x6fU, 0xe8U, 0xf1U, 0x02U,  
0xc5U, 0x1cU, 0x81U, 0x24U, 0xbeU, 0xd6U, 0xdeU, 0x5aU,  
0x57U, 0xe5U, 0x29U, 0xb8U, 0x37U, 0xdeU, 0x23U, 0x4eU,  
0x8fU, 0xdaU, 0xb0U, 0xd7U, 0xcaU, 0x04U, 0xe2U, 0xebU,  
0x8fU, 0x79U, 0xe4U, 0xd8U, 0xadU, 0x8eU, 0x02U, 0x69U,  
0x87U, 0x48U, 0x19U, 0x2bU, 0x01U, 0x9eU, 0xabU, 0x8fU,  
0x83U, 0x55U, 0x46U, 0x1aU, 0x7cU, 0x3cU, 0xa7U, 0xb4U,  
0xc9U, 0x90U, 0xc3U, 0x49U, 0xd2U, 0x0dU, 0xbCU, 0xd7U,  
0x51U, 0x1aU, 0xccU, 0x60U, 0x9dU, 0x9bU, 0x0dU, 0xe1U,  
0x18U, 0x10U, 0x19U, 0xd2U, 0x48U, 0x1bU, 0xf1U, 0xfeU,  
0x50U, 0x89U, 0x49U, 0xfeU, 0xa1U, 0x82U, 0xd9U, 0x7aU,  
0x42U, 0x5eU, 0x6aU, 0xdeU, 0x26U, 0x5cU, 0x61U, 0x36U,  
0xd2U, 0x67U, 0x8eU, 0x55U, 0xeeU, 0xa2U, 0xbFU, 0xe9U,  
0xfaU, 0xaeU, 0xddU, 0x1fU, 0xdeU, 0xadU, 0x4eU, 0xd3U,  
0x5bU, 0x61U, 0x02U, 0x0fU, 0x8dU, 0x80U, 0xb6U, 0xfeU,  
0x12U, 0x7dU, 0x95U, 0x25U, 0x76U, 0x95U, 0x4dU, 0x6aU,  
0x73U, 0x41U, 0x05U, 0x62U, 0xcbU, 0x72U, 0x40U, 0x0eU,  
0x01U, 0x44U, 0x20U, 0xaeU, 0x51U, 0x04U, 0x57U, 0x97U,  
0x72U, 0xf7U, 0x53U, 0x99U, 0xcdU, 0x61U, 0x02U, 0x9fU,  
0x4aU, 0x3fU, 0xe0U, 0x0eU, 0xa0U, 0x41U, 0x44U, 0xd7U,  
0x77U, 0xbbU, 0xb0U, 0x94U, 0x85U, 0xe0U, 0x1cU, 0x59U,  
/\* exponent : 4 bytes \*/  
0x00U, 0x01U, 0x00U, 0x01U,

0x24U, 0x8dU, 0x6aU, 0x61U, 0xd2U, 0x06U, 0x38U, 0xb8U,  
0xe5U, 0xc0U, 0x26U, 0x93U, 0x0cU, 0x3eU, 0x60U, 0x39U,  
0xa3U, 0x3cU, 0xe4U, 0x59U, 0x64U, 0xffU, 0x21U, 0x67U,  
0xf6U, 0xecU, 0xedU, 0xd4U, 0x19U, 0xdbU, 0x06U, 0xc1U

0x54U, 0xA2U, 0x4EU, 0x58U, 0xF3U, 0x28U, 0xA0U, 0xF5U,  
0xF8U, 0x2CU, 0x4FU, 0xBFU, 0x7BU, 0x84U, 0x16U, 0xF3U,  
0xDCU, 0xB2U, 0x16U, 0x6CU, 0xD4U, 0x01U, 0x7CU, 0xA0U,  
0xB6U, 0x0BU, 0x1AU, 0x4DU, 0x99U, 0xB6U, 0x96U, 0xA4U,  
0x2FU, 0x91U, 0xC5U, 0xEFU, 0xC7U, 0xDBU, 0xCFU, 0x5CU,  
0x7AU, 0x92U, 0x26U, 0x04U, 0x23U, 0x3EU, 0x3AU, 0x0DU,  
0x01U, 0xD1U, 0x71U, 0xC4U, 0x94U, 0x0DU, 0xA2U, 0xF7U,  
0x49U, 0xA2U, 0x21U, 0x5BU, 0x1FU, 0xF8U, 0x3EU, 0xADU,  
0x2CU, 0xB2U, 0xADU, 0xDBU, 0xCDU, 0x38U, 0x6CU, 0xE9U,  
0x61U, 0xE8U, 0x4AU, 0xE8U, 0x5EU, 0x61U, 0x66U, 0x8FU,  
0x26U, 0xAFU, 0x84U, 0x63U, 0x8AU, 0x1AU, 0x18U, 0x3CU,  
0xC4U, 0x36U, 0x35U, 0x29U, 0x0DU, 0x97U, 0x81U, 0x2BU,  
0xF4U, 0x7AU, 0x4AU, 0xE0U, 0xA1U, 0x62U, 0xA3U, 0xECU,  
0x36U, 0x9FU, 0xAEU, 0xDBU, 0x0DU, 0x64U, 0x41U, 0x8FU,  
0x65U, 0x5BU, 0x44U, 0xB9U, 0x29U, 0x1EU, 0xA6U, 0xA5U,  
0x54U, 0x1DU, 0x98U, 0x53U, 0xA9U, 0xAAU, 0x08U, 0x80U,  
0x9FU, 0xE1U, 0x2EU, 0xF7U, 0x72U, 0xD3U, 0x2EU, 0xABU,  
0x39U, 0xE8U, 0x3DU, 0x65U, 0x42U, 0x19U, 0x33U, 0x3BU,  
0xB6U, 0x5DU, 0x8EU, 0x62U, 0x93U, 0x0DU, 0x01U, 0x48U,  
0x42U, 0xABU, 0xD3U, 0x01U, 0x26U, 0xA4U, 0xD0U, 0xE9U,  
0x0DU, 0x1FU, 0xA1U, 0xF4U, 0x80U, 0xFBU, 0x52U, 0xEFU,  
0xB6U, 0x49U, 0x60U, 0xE3U, 0x43U, 0x85U, 0x0BU, 0x27U,  
0xC7U, 0xC0U, 0x1CU, 0x7EU, 0xD2U, 0xF7U, 0x65U, 0xBBU,  
0x59U, 0x48U, 0xA4U, 0x81U, 0xFEU, 0x6AU, 0x3CU, 0x43U,  
0x1DU, 0xF7U, 0x64U, 0x78U, 0xC5U, 0x7EU, 0xFCU, 0x98U,  
0x8CU, 0xD5U, 0xCAU, 0x2EU, 0x9CU, 0xB7U, 0x1BU, 0x57U,  
0xCDU, 0xB7U, 0xECU, 0xC1U, 0x6AU, 0x1BU, 0xC7U, 0x2FU,  
0xADU, 0x15U, 0x90U, 0x47U, 0xF2U, 0x5DU, 0xB4U, 0xCBU,  
0x15U, 0xFFU, 0x28U, 0x58U, 0x1FU, 0x8CU, 0x8DU, 0xAEU,  
0x62U, 0x71U, 0xD0U, 0xE2U, 0x4AU, 0x80U, 0x56U, 0xC7U,  
0xB2U, 0xFAU, 0xABU, 0x36U, 0xB1U, 0x5DU, 0xBFU, 0x58U,  
0xF1U, 0x1FU, 0xB7U, 0x10U, 0xA5U, 0xFBU, 0xBBU, 0x4AU

#### 8.4.6.15 HaeHsmPkcs1PssSha256

Autoever HSM (HSM\_Framework\_2.x\_UserManual\_Crypto\_Service\_KOR\_vx.xx.pdf) 의 7.Crypto Service API 테스트 케이스 참조

#### 8.4.6.16 HaeHsmEcdsaSha256

Autoever HSM (HSM\_Framework\_2.x\_UserManual\_Crypto\_Service\_KOR\_vx.xx.pdf) 의 7.Crypto Service API 테스트 케이스 참조

### Key Derive

#### 8.4.7.1 AuthHsmPbkdf2

##### \* Raw Key(Password) 사용

| KeyLength                              | Derived Key                                                                                                                                          |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0                                      | 0x34, 0x8C, 0x89, 0xDB, 0xCB, 0xD3, 0x2B, 0x2F, 0x32, 0xD8, 0x14, 0xB8, 0x11, 0x6E, 0x84, 0xCF, 0x2B, 0x17, 0x34, 0x7E, 0xBC, 0x18, 0x00, 0x18, 0x1C |
| Password                               |                                                                                                                                                      |
| "passwordPASSWORDpassword"             |                                                                                                                                                      |
| Salt                                   |                                                                                                                                                      |
| "saltSALTsaltSALTsaltSALTsaltSALTsalt" | Derived Key Length                                                                                                                                   |
| Iteration                              |                                                                                                                                                      |
| 4096                                   | 25                                                                                                                                                   |

##### \* PSK(10번 Key) 사용

| KeyLength | Derived Key                                     |
|-----------|-------------------------------------------------|
| 10        | 0x2A, 0xC8, 0xA8, 0xC1, 0x01, 0xDF, 0x89, 0x74, |

|           |                                                |
|-----------|------------------------------------------------|
| Password  | 0x2C, 0x90, 0x32, 0xB1, 0x6C, 0x62, 0xCF, 0xD8 |
| 미사용       |                                                |
| Salt      |                                                |
| "saW0lt"  |                                                |
| Iteration | Derived Key Length                             |
| 4096      | 16                                             |

#### 8.4.7.2 HaeHsmPbkdf2

AutHsmPbkdf2와 동일

### Key Exchange Calc Secret

#### 8.4.8.1 AuthHsmEcdh

8.4.8

\* PSK ECC (ECC Private / Public PSK 1번)

| privateValue                                                                                                                                                                                                                    | publicValue                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0x51U, 0x9BU, 0x42U, 0x3DU, 0x71U, 0x5FU, 0x8BU, 0x58U, 0x1FU, 0x4FU, 0xA8U, 0xEEU, 0x59U, 0xF4U, 0x77U, 0x1AU, 0x5BU, 0x44U, 0xC8U, 0x13U, 0x0BU, 0x4EU, 0x3EU, 0xACU, 0xCAU, 0x54U, 0xA5U, 0x6DU, 0xDAU, 0x72U, 0xB4U, 0x64U, | 0x1CU, 0xCBU, 0xE9U, 0x1CU, 0x07U, 0x5FU, 0xC7U, 0xF4U, 0xF0U, 0x33U, 0xBFU, 0xA2U, 0x48U, 0xDBU, 0x8FU, 0xCCU, 0xD3U, 0x56U, 0x5DU, 0xE9U, 0x4BU, 0xBFU, 0xB1U, 0x2FU, 0x3CU, 0x59U, 0xFFU, 0x46U, 0xC2U, 0x71U, 0xBFU, 0x83U, 0xCEU, 0x40U, 0x14U, 0xC6U, 0x88U, 0x11U, 0xF9U, 0xA2U, 0x1AU, 0x1FU, 0xDBU, 0x2CU, 0x0EU, 0x61U, 0x13U, 0xE0U, 0x6DU, 0xB7U, 0xCAU, 0x93U, 0xB7U, 0x40U, 0x4EU, 0x78U, 0xDCU, 0x7CU, 0xCDU, 0x5CU, 0xA8U, 0x9AU, 0x4CU, 0xA9U, |
| sharedSecret                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 0xA8U, 0xBDU, 0x0AU, 0x00U, 0x9AU, 0x11U, 0x75U, 0x33U, 0x1DU, 0x5BU, 0x81U, 0xACU, 0xA8U, 0xE7U, 0x6EU, 0x82U, 0xA1U, 0x23U, 0xB8U, 0x36U, 0xBCU, 0x9AU, 0xEAU, 0xB8U, 0x34U, 0x98U, 0x4BU, 0xCDU, 0x3FU, 0x44U, 0xFCU, 0x92U  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

\* Raw Key

| privateValue                                                                                                                                                                                   | publicValue                                                                                                                                                                                                                                                                                                                                                                                    |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0xA3, 0xD2, 0xD3, 0xB7, 0x59, 0x6F, 0x65, 0x92, 0xCE, 0x98, 0xB4, 0xBF, 0xE1, 0x0D, 0x41, 0x83, 0x7F, 0x10, 0x02, 0x7A, 0x90, 0xD7, 0xBB, 0x75, 0x34, 0x94, 0x90, 0x01, 0x8C, 0xF7, 0x2D, 0x07 | 0x1B, 0xCE, 0xC4, 0x57, 0x0E, 0x1E, 0xC2, 0x43, 0x65, 0x96, 0xB8, 0xDE, 0xD5, 0x8F, 0x60, 0xC3, 0xB1, 0xEB, 0xC6, 0xA4, 0x03, 0xBC, 0x55, 0x43, 0x04, 0x0B, 0xA8, 0x29, 0x63, 0x05, 0x72, 0x44, 0x8A, 0xF6, 0x2A, 0x4C, 0x68, 0x3F, 0x09, 0x6B, 0x28, 0x55, 0x83, 0x20, 0x73, 0x7B, 0xF8, 0x3B, 0x99, 0x59, 0xA4, 0x6A, 0xD2, 0x52, 0x10, 0x04, 0xEF, 0x74, 0xCF, 0x85, 0xE6, 0x74, 0x94, 0xE1 |
| sharedSecret                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                |
| 0xDB, 0x48, 0x65, 0x94, 0xA9, 0x91, 0x9E, 0x42, 0x83, 0x1A, 0xB3, 0xC8, 0x04, 0x2A, 0xC5, 0xB9, 0x24, 0x6F, 0x52, 0x89, 0x71, 0xEE, 0xB0, 0x4D, 0x35, 0x02, 0x03, 0x56, 0xAF, 0x38, 0xBB, 0x50 |                                                                                                                                                                                                                                                                                                                                                                                                |

#### 8.4.8.2 HaeHsmEcdh

Autoever HSM (HSM\_Framework\_2.x\_UserManual\_Crypto\_Service\_KOR\_vx.xx.pdf) 의 7.Crypto Service API 테스트 케이스 참조

### Autoever HSM 2.x 포팅

R4.0.3 Csm에서는 Crypto Service (암호 알고리즘)만을 표준화/추상화 하기 때문에 HSM Secure Application (ex. SecureBoot 등) 이나 HSM 포팅 관련 사항은 유저가 직접 설정하여야 한다. 따라서 Csm 설정 이외의 HSM Init, HSM ISR 관련 설정은 ASW 유저가 오토에버 HSM 모듈에 포함되어있는 UM 가이드를 참조하여 사용한다.

8.5

HSM2.0 가이드 :

[HSM\_2.0spec]HSM\_Framework\_UserManual\_Target\_Guide\_MCU\_vx.x.pdf

[HSM\_2.0spec]HSM\_Framework\_UserManual\_Secure\_Application\_Guide\_vx.x.pdf